SECANT OBJECTS FOR WEIL CLASSES ON ABELIAN VARIETIES

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Abstract. We determine what closing the algebraic locus of the Weil classes of an abelian variety of Weil type requires of a single object at a single point: a perfect complex at one point of complex multiplication whose Chern character is a Weil class plus a polynomial in the polarisation and whose semiregularity map is injective makes the Weil classes algebraic on the whole family; the criterion is never met when the Chern character is exactly a Weil class, for any CM field, and with powers of the polarisation added it becomes one number, computed exactly for quartic CM fields; and the secant sheaves of the split member, sums and convolutions of line bundles, and the supports of a complex meeting the criterion are each excluded at a precise point.

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1. Introduction

Let X be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}^p_X \colon \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) \coloneqq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

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This paper and its companions [I, III] together replace the earlier paper *Algebraic loci of Weil classes on abelian varieties* by the authors and Priyabrata Mandal (archive [BMB26]); every result of that paper appears in one of the three with its proof. The preprint *Relative secant cycles and Hodge classes* [BB26] contains errors and incomplete proofs and is replaced by them.

Every proof is given in the text. Programs that repeat much of the finite arithmetic, and a Lean 4 certificate of part of it, are archived as described in the statement on data availability at the end of the paper; no statement rests on them.

is surjective for every p. Beyond degree two the best understood Hodge classes are those of abelian varieties, and the first of them that divisors do not produce are the Weil classes, which Weil exhibited on abelian varieties with an action of an imaginary quadratic field whose two eigenspaces on the tangent space have the same dimension [Wei77]; the construction works for every CM field F. Markman proved that the Weil classes are algebraic on the abelian sixfolds of split Weil type for every imaginary quadratic field, and, through a product argument of Schoen [Sch98, Proposition 10], on every abelian fourfold of Weil type [Mar25a, Mar25b]. His sheaves are built from secant sheaves on an abelian variety of half the dimension, and they extend along the family because they satisfy an equivariant form of semiregularity.

The companion paper [I] studies the set of members of a family on which a flat Hodge class is algebraic, its *algebraic locus*, and determines it for the Weil classes of every CM field and every dimension at once. Fix a CM field F of degree $2m$, an integer $n \geq 2$ and a discriminant δ. The abelian varieties of (F, n, δ)-Weil type are the members of one family over a connected period domain $\mathcal{D}_F$ of dimension mn^2, their Weil classes form a flat local system of rank $2m$ over it, and the algebraic locus

$$\Sigma_F = \{\, s \in \mathcal{D}_F : \text{the Weil classes of } A_s \text{ are algebraic} \,\}$$

contains an explicit point of complex multiplication, is dense, is the preimage of a countable union of Zariski closed subsets of the Shimura variety of the family, and is either all of $\mathcal{D}_F$ or meagre; so the following are equivalent: Σ_F is closed, it has an interior point, the Weil classes are algebraic on every member, and the Hodge conjecture holds for the very general member ([I, Theorem 1.1], which we call the main theorem throughout). The notation of that paper is kept here: the standard model of [I, Section 4], the integral basis ω_1, ω_2 of the Weil line of [I, Theorem 4.9], the base points of [I, Theorem 15.27] and [I, Theorem 17.5], and the strata of [I, Theorem 16.4].

This paper determines what closing Σ_F requires of a single object at a single point, computes that requirement exactly, and examines the objects that the known constructions supply, showing in each case where they fail. The third paper [III] places the Weil classes inside the whole conjecture.

1.1. **Closing the algebraic locus.** By [I, Theorem 1.1(iv)] the Hodge conjecture for the very general member of a Weil family is the closedness of Σ_F, and by (iii) it is enough to show that one stratum has an interior point. Two statements would do this, and the paper determines what each of them asks.

The first bounds the Hilbert data. If at every point of the orbit of the base point the Weil class is represented by a cycle whose Hilbert data comes from one finite set, then finitely many strata cover a dense set and one of them is everything ([I, Theorem 16.9]). Read on an arbitrary representative this bound is equivalent to its conclusion ([I, Theorem 16.17]); read on the transported cycle, the one representative that can be named, it fails, because the transport multiplies the degree of a component by c^{2n} and divides it only by the order of a stabiliser ([I, Theorem 16.15]). So the bound restates the problem rather than reducing it. The bound is needed only on a small set: by the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture [PST21],[1] it suffices along one sequence of CM points that leaves every proper special subvariety, and such a sequence exists inside the orbit ([I, Theorem 17.14]). Every member at which this paper exhibits algebraic Weil

[1] In the form used here: an irreducible closed subvariety of a Shimura variety of abelian type that contains a Zariski dense set of CM points is a special subvariety. Hence a sequence of CM points that leaves every proper special subvariety is Zariski dense, and so is every infinite subsequence of it.

classes has endomorphisms beyond the field, so it lies in the locus of members with smaller Hodge group. That locus contains the base point, is dense, is stable under $G_F(\mathbb{Q})$ and is a countable union of closed algebraic strata, exactly as Σ_F is, and it is meagre ([I, Proposition 17.12], [I, Corollary 17.13]). The properties listed in [I, Theorem 1.1] therefore do not decide the dichotomy by themselves, and a proof that Σ_F is everything has to reach a member with full Hodge group.

The second is semiregularity. By the theorem of Buchweitz and Flenner, in the form of Pridham for perfect complexes [BF03, Pri24],[2] a perfect complex at one base point whose Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation, and whose semiregularity map is injective, extends along the whole family and makes Σ_F everything (Theorem 2.3, [I, Theorem 17.7(v)]). The paper computes the injectivity condition exactly. The annihilator of the Weil class in $H^{0,2}$ is carried by the polarisation isomorphically onto the tangent space of the family ([I, Lemma 9.6], [I, Theorem 9.17]), so no object whose Chern character lies on the Weil line and on which $H^2(\mathcal{O}_A)$ acts faithfully is semiregular ([I, Theorem 9.7]). When the Chern character is exactly a Weil class the criterion is never met, for any CM field and any $n \geq 2$: a semiregular object would extend to the very general Weil torus of the field, a complex torus that is not algebraic, on which every coherent sheaf has vanishing Chern character strictly between degree zero and the top degree (Proposition 3.4, Theorem 3.5, Theorem 5.4). With powers of the polarisation added the criterion becomes one number: for a Chern character $N\omega + \sum_k c_k \theta^k$ and $n \geq 3$ every object has $\dim \operatorname{Ext}^2(E,E) \geq (4+\rho)n^2 - 2n$, where $\rho \leq 3$ is the rank of the Hankel matrix of the numbers $k!\, c_k$, and an object attaining this bound meets the criterion (Theorem 3.9).[3] A flatness condition on the deformations of the object suffices in place of injectivity (Proposition 3.15). By descent and scalar extension the statement is needed only for the split families of the CM fields of degree at least four ([III, Proposition 2.13]), and for one family it is equivalent to the Lefschetz standard conjecture for the total space of the family over a curve through a base point ([III, Corollary 2.31]).

For a quartic CM field and $n = 2$ we compute the rank exactly for every shape. It takes fifteen values; the least, 80, is taken only by the pure shape, which the Weil tori exclude, so an object meeting the criterion has $\dim \operatorname{Ext}^2(E,E) \geq 88$ and a polynomial part that is exponential or linear at each real place (Theorem 5.14, Corollary 5.15). The Hodge locus of every such character is determined: it is the Weil tori of the field only for the characters of θ^4 shape, which are therefore excluded at rank 88, and the polarised family itself for every other character of that rank (Proposition 5.17, Corollary 5.18).

1.2. **Secant objects.** The objects that a secant construction supplies are examined in Section 4. On the split member $X \times \widehat{X}$ of [I, Section 14] the secant plane is written down in every dimension, and a secant sheaf has rank a, $c_1 = bt$, $c_2 = \frac{ad+b^2}{2}t^2$ and Bogomolov discriminant $\mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d})\, t^2$ ([I, Theorem 14.5], [I, Theorem 14.13]). No object whose Chern character is an effective combination of exponentials of divisor classes contributes to the Weil line, by a positivity argument with the norm form (Theorem 4.5), and no object

[2] The semiregularity map of a perfect complex E on X sends $\operatorname{Ext}^2(E,E)$ to $\bigoplus_q H^{q+2}(X, \Omega_X^q)$ by the trace of the product with powers of the Atiyah class of E; it carries the obstruction to deforming E along a first order deformation of X to the obstruction to keeping $\operatorname{ch}(E)$ of Hodge type. If it is injective, E deforms, formally and then along the whole family, in every direction in which $\operatorname{ch}(E)$ stays a Hodge class. Bloch introduced it for local complete intersection subvarieties.

[3] That is, ρ is the rank of the $(2n-1) \times 3$ matrix with entries μ_{i+j}, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$, where $\mu_k = k!\, c_k$ (Setup 3.8). A vector in its kernel is a linear recurrence of order at most two among the μ_k, and ρ does not change when the character is multiplied by $e^{t\theta}$.

with a line bundle summand off the polarisation line meets the criterion, so no sum of line bundles does in any dimension ([I, Theorem 9.20], Corollary 4.21). Convolutions of line bundles, their iterated cones, are cut down by the maps between the pieces. Fewer than $2n$ pieces with nondegenerate differences never meet the criterion (Theorem 4.26). On E_0^{2n}, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, the signed Chern characters of $2 \cdot 4^{n-1}$ line bundles add up to a real Weil class (Proposition 4.27), and with three multiples of the polarisation no convolution of these pieces meets the criterion at $n = 3$ (Theorem 4.35), nor at $n = 4$ when the shifts of the pieces lie in six consecutive values (Corollary 4.48). The weakened criterion of [Mar25b, Question 11.4] reduces, for the transform of $\mathcal{F} \boxtimes \mathcal{F}$, to the single identity $\mathrm{At}(\mathcal{F})^2 = -d\Theta^2 \cdot \mathrm{id}$ for the sheaf $\mathcal{F}$ on the fourfold (Theorem 4.56); a local computation at an isolated double point of the support decides it in the negative for Markman's candidate in dimension eight (Theorem 4.61), and at a quartic CM field a curve on which a complex is locally free is tangent to at most one of the two eigenplanes of the real multiplication, which decides Markman's explicit pair [Mar25c, Example 11.2.7] in the same way (Theorem 5.28, Corollary 5.30).

The supports are constrained as well. A Cohen–Macaulay support carries no local obstruction, while two smooth branches of codimension two meeting transversally at a point carry one (Corollary 4.77). For a smooth support, a surface S in the fourfold, Grothendieck–Riemann–Roch fixes every numerical invariant, and the Bogomolov–Miyaoka–Yau inequality gives $d \leq b^2$ for the twist $b\Theta$, which leaves only $d \in \{1, 3, 5, 7\}$ for the twist 3Θ of Markman's candidate (Theorem 4.67, Corollary 4.68). Complete intersections are excluded for every b and d (Theorem 4.81), two-term complexes of powers of the polarisation at every rank, twist and discriminant (Theorem 4.89), and a Hilbert–Burch resolution of a secant support has rank at least two (Proposition 4.86); the secant class is an integer combination of classes of line bundles exactly when $d \equiv 15$ or $23 \pmod{24}$ (Theorem 4.83).

The paper is organised as follows. Section 2 proves that one semiregular object at a base point closes Σ_F and draws the consequences of injectivity for the object. Section 3 shows that the pure form of the criterion is never met and computes the polarised form as a number. Section 4 examines the secant objects of the split member, the sums and convolutions of line bundles, the weakened criterion for Markman's candidates and the supports. Section 5 extends the exclusion of the pure form to every CM field and computes the polarised criterion for quartic fields and the counts for sextic fields.

Every proof is written to be checked by hand. Where an argument rests on a finite calculation, the calculation is given where it is used, as an explicit identity, a determinant or a case analysis. The archive [BMB26] holds programs that repeat much of this arithmetic and a Lean 4 certificate of part of it; no statement of the paper rests on them.

2. The semiregularity criterion at one point

2.1. **The second criterion: a rank at one point.** The other way from a dense algebraic locus to an open one is to make the cycle move, and the classical tool for that is semiregularity. The theorem below adds nothing to the theory of semiregularity itself. Its content is that, after [I, Theorem 4.14], [I, Theorem 15.27] and [I, Theorem 16.4], the semiregularity of a single object at a single point implies the Hodge conjecture for the whole family, with no further propagation argument. We begin with the object: it exists, and its Chern character is concentrated in one degree and is a flat Hodge class on the whole family.

Proposition 2.1 (The K-theory class at a base point)**.** *Let $s_0 \in \Sigma$ be a base point of* [I, Theorem 15.27] *and write $A_0 = A_{s_0}$. There are an integer $N \geq 1$ and a perfect complex E on A_0 with*

$$\mathrm{ch}_k(E) = 0 \quad (k \neq n), \qquad \mathrm{ch}_n(E) = N\,\omega_1.$$

The tuple $v = \mathrm{ch}(E)$ extends to a flat section of $\bigoplus_k R^{2k} f_* \mathbb{Q}$ over $\mathcal{S}$ which is of type (k,k) at every point.

Proof. By [I, Theorem 15.27] there is $\xi \in \mathrm{CH}^n(A_0)_{\mathbb{Q}}$ with $\mathrm{cl}(\xi) = \omega_1$. For a smooth quasi-projective variety over a field the Chern character $\mathrm{ch}\colon K_0(A_0)_{\mathbb{Q}} \to \mathrm{CH}^*(A_0)_{\mathbb{Q}}$ is a ring isomorphism [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so there is a unique $u \in K_0(A_0)_{\mathbb{Q}}$ with $\mathrm{ch}(u) = \xi$, and its Chern character has no component outside degree n. Choose $N \geq 1$ such that Nu is the image of a class in $K_0(A_0)$; that class is a difference of classes of vector bundles, hence the class of a perfect complex E, and $\mathrm{ch}(E) = N\xi$. Passing to cohomology gives the displayed identities.

For the last sentence, $\mathrm{ch}(E)$ has a single nonzero component, equal to $N\omega_1$, and ω is a flat section of type (n,n) everywhere by [I, Lemma 16.2]. ∎

Theorem 2.2 (Buchweitz and Flenner [BF03]; Pridham [Pri24])**.** *Let X be a smooth projective complex variety, E a perfect complex on X, and $X \hookrightarrow \mathcal{X} \to \operatorname{Spec} R$ a deformation of X over an Artin local $\mathbb{C}$-algebra along which $\mathrm{ch}(E)$ stays of Hodge type. If the semiregularity map*

$$\sigma_E\colon \operatorname{Ext}^2_X(E,E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(X, \Omega^q_X), \qquad \sigma_E(\zeta) = \sum_q \tfrac{1}{q!}\operatorname{tr}\big(\zeta \circ \mathrm{At}(E)^q\big),$$

is injective, then E extends to a perfect complex on $\mathcal{X}$.

In this generality, for perfect complexes and for every obstruction annihilated by σ_E rather than only the curvilinear ones, the theorem is proved by Pridham [Pri24], who identifies σ_E with the tangent map of a generalised Abel–Jacobi map on the derived moduli of perfect complexes; for coherent sheaves see also [BLM23]. Perry [Per26] extends it to noncommutative varieties with group actions, which covers twisted derived categories and answers a question of Markman.

Theorem 2.3 (Semiregularity at one point closes the family)**.** *Let $n \geq 2$, let s_0 and E be as in Proposition 2.1, and suppose the semiregularity map σ_E of Theorem 2.2 is injective. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$ in that family.*

Proof. Let $\mathfrak{M}$ be the stack of perfect complexes with vanishing negative self-extensions on the fibres of $f\colon \mathcal{A} \to \mathcal{S}$. It is algebraic and locally of finite presentation over $\mathcal{S}$ [Lie06], and at a point $[E]$ over s_0 its deformation theory relative to $\mathcal{S}$ has tangent space $\operatorname{Ext}^1(E,E)$ and obstruction space $\operatorname{Ext}^2(E,E)$. If $\operatorname{Ext}^{<0}(E,E) \neq 0$, which that stack excludes,[4] use instead the analytic Kuranishi germ of pairs from the proof of Theorem 3.5, which has the same tangent and obstruction spaces; the argument below then gives a nonempty analytic open set in place of a Zariski open one, which is all that [I, Corollary 16.5] needs.

By Proposition 2.1 the Chern character of E stays of Hodge type along every direction in $\mathcal{S}$, because it is a flat section of Hodge classes on the whole base. So Theorem 2.2 applies to every infinitesimal deformation of A_0 inside the family, and injectivity of σ_E

[4] Lieblich's stack parametrises the complexes he calls universally gluable, those with $\operatorname{Ext}^{<0}(E,E) = 0$; without that condition a complex can have automorphisms of negative degree, and the natural moduli object is the derived stack of Toën and Vaquié.

kills every obstruction. The morphism $\mathfrak{M} \to \mathcal{S}$ is therefore formally smooth at $[E]$, and being locally of finite presentation it is smooth on an open neighbourhood of $[E]$, which we take connected. Smooth morphisms are open, so the image of that neighbourhood is a nonempty Zariski open subset $U \subseteq \mathcal{S}$.

For $s \in U$ pick a point of the neighbourhood over s, that is, a perfect complex E_s on A_s. Chern characters are locally constant in families, so $\mathrm{ch}(E_s)$ is the parallel transport of $\mathrm{ch}(E)$: it is $N\omega_1$ in degree n and zero elsewhere. Chern characters of perfect complexes are algebraic, so $N\omega_1$, and hence ω_1, is algebraic on A_s; the rest of the Weil line follows by [I, Lemma 16.3]. So $p^{-1}(U) \subseteq \Sigma$, which therefore contains a nonempty open set, and [I, Corollary 16.5] gives $\Sigma = \mathcal{D}_{n,n}$. Apply [I, Theorem 15.14(a)]. ∎

Remark 2.4 (The hypothesis is not met by the object above). Injectivity of σ_E is automatic for some objects and not for others, and its degree-zero component already shows both. For a line bundle L one has $\mathrm{Ext}^2(L,L) = H^2(\mathcal{O}_X)$ and $\sigma_0 = \mathrm{tr}$ is the identity on it, so σ_L is injective for every line bundle on every variety, and Theorem 2.2 returns the classical fact that a divisor class deforms wherever it stays of type $(1,1)$. The object of Proposition 2.1 has rank zero, so its trace component is not an isomorphism, and in general not injective, the source having no reason to be as small as $h^2(\mathcal{O}_{A_0}) = \binom{2n}{2}$. The hypothesis of Theorem 2.3 therefore rests on the components with $q \geq 1$, which is where semiregularity stops being formal and becomes a computation.

For the object of Proposition 2.1 the hypothesis fails: by [I, Proposition 9.10] it has $\dim \ker \sigma_E \geq n^2$ at every base point constructed in this paper (see also [I, Remark 9.13]). The hypothesis concerns a choice of perfect complex representing a class in K_0, and [I, Proposition 9.10] does not reach every representative. Theorem 2.5 and the results after it describe what a representative satisfying the hypothesis would have to look like, and Theorem 3.5 shows that there is none. The proof of Theorem 2.3 applies word for word to the polarised criterion (P2′) of Remark 3.6, and Remark 3.13 records which of the results below carry over to it.

2.2. **Consequences of the rank condition.** The hypothesis of Theorem 2.3 is a condition on a rank, which cannot be inspected until the object is written down. It nevertheless has consequences that hold for every representative at once, because the semiregularity map composed with the evaluation map of Hochschild cohomology is contraction into the Chern character (Setup 4.54), and the Chern character here is a Weil class. The annihilator of a Weil class in HT^2 is large, and injectivity forces the evaluation map to vanish on the whole of it. We compute the annihilator and draw the consequences. They are the exact analogue for (P2) of what Theorem 4.61 and Theorem 4.89 do for the secant route in Section 4, and they rest on the same local products.

Theorem 2.5 (Consequences of injectivity)**.** *Let A be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $n \geq 2$, and write $H^1(A,\mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of K, $V_\pm^{1,0}$, $V_\pm^{0,1}$ for their Hodge components and $T_\pm \subset H^0(T_A) = T_0A$ for the eigenspaces of K on the tangent space, each of dimension n. Let $\omega = a\alpha_+ + b\alpha_-$ be a class on the Weil line with $ab \neq 0$, for instance any nonzero rational one, and let E be a nonzero perfect complex on A with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

(i) *The annihilator of $\mathrm{ch}(E)$ in $HT^2(A)$ is*

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\,E) = (V_+^{0,1} \otimes V_-^{0,1}) \;\oplus\; \mathrm{Hom}_K(H^{1,0}, H^{0,1}) \;\oplus\; (T_+ \otimes T_-),$$

of dimension $n^2+2n^2+n^2=4n^2$: the mixed block of $H^{0,2}$, which is the annihilator of [I, Lemma 9.6]*; the K-linear first order deformations of A, which contain the tangent space of the polarised Weil family; and the mixed bivectors.*

(ii) *If σ_E is injective, then ev_E vanishes on this space. Explicitly: $z\cdot\mathrm{id}_E=0$ for every $z\in V_+^{0,1}\otimes V_-^{0,1}$; E extends to first order along every K-linear deformation of A; and the translation classes satisfy $A_\theta A_{\theta'}=0$ in $\mathrm{Ext}^2(E,E)$ for every $\theta\in T_+$ and $\theta'\in T_-$.*

(iii) *Under the same hypothesis the algebra map $H^*(\mathcal{O}_A)\to\mathrm{Ext}^*(E,E)$, $z\mapsto z\cdot\mathrm{id}_E$, kills the ideal generated by the mixed block, which is $\bigoplus_{a,b\geq 1}\bigwedge^a V_+^{0,1}\otimes\bigwedge^b V_-^{0,1}$; so the image of $H^{0,k}(A)$ has dimension at most $2\binom{n}{k}$ for $1\leq k\leq n$ and is zero for $k>n$, and every endomorphism f of E has $\mathrm{tr}(f)=0$ in $H^0(\mathcal{O}_A)=\mathbb{C}$.*

(iv) *Under the same hypothesis, if at some point p of its support E is locally isomorphic to i_*F with F locally free of positive rank on a smooth closed subvariety $Z\ni p$ of codimension at least two, then $T_+\subseteq T_pZ$ or $T_-\subseteq T_pZ$.*

Proof. (i) Write $H^*(A,\mathbb{C})=\bigwedge^*(H^{1,0}\oplus H^{0,1})$ and $\alpha_\pm$ for a generator of $\bigwedge^{2n}V_\pm=\bigwedge^n V_\pm^{1,0}\otimes\bigwedge^n V_\pm^{0,1}$. The three summands of HT^2 act as follows: $z\in H^{0,2}=\bigwedge^2 H^{0,1}$ by wedge product; $v\in H^1(T_A)=\mathrm{Hom}(H^{1,0},H^{0,1})$ as the derivation that replaces one $(1,0)$ factor by its image; $\pi\in H^0(\bigwedge^2 T_A)=\bigwedge^2(H^{1,0})^\vee$ by double contraction of $(1,0)$ factors. For z, the computation is [I, Lemma 9.6]. For v, decompose $\mathrm{Hom}(H^{1,0},H^{0,1})$ into its four blocks by eigenspaces. An element of the block $\mathrm{Hom}(V_+^{1,0},V_+^{0,1})$ replaces a factor of $\bigwedge^n V_+^{1,0}$ in α_+ by an element of $V_+^{0,1}$, which is already exhausted in α_+, so it kills α_+; it acts trivially on α_-, which has no $V_+^{1,0}$ factor. Likewise $\mathrm{Hom}(V_-^{1,0},V_-^{0,1})$ kills ω. A nonzero element of the block $\mathrm{Hom}(V_+^{1,0},V_-^{0,1})$ carries α_+ to a nonzero element of $\bigwedge^{n-1}V_+^{1,0}\otimes V_-^{0,1}\otimes\bigwedge^n V_+^{0,1}$ and kills α_-, and a nonzero element of $\mathrm{Hom}(V_-^{1,0},V_+^{0,1})$ carries α_- to a nonzero element of a different summand of $H^{n-1,n+1}$ and kills α_+. Since $ab\neq 0$ and the two summands are independent, v kills ω if and only if its two mixed blocks vanish, that is, if and only if v is K-linear. The space of K-linear maps has dimension $2n^2$, and it contains the n^2-dimensional tangent space of the polarised family, which is its symmetric part. For π, a mixed bivector $\theta\wedge\theta'$ with $\theta\in T_+=(V_+^{1,0})^\vee$ and $\theta'\in T_-=(V_-^{1,0})^\vee$ kills α_+ because θ' kills every factor of $\bigwedge^n V_+^{1,0}$, and kills α_- symmetrically; a nonzero bivector in $\bigwedge^2 T_+$ carries α_+ to a nonzero element of $\bigwedge^{n-2}V_+^{1,0}\otimes\bigwedge^n V_+^{0,1}$ and kills α_-, and symmetrically for $\bigwedge^2 T_-$, the two images lying in different summands; so the annihilator in $H^0(\bigwedge^2 T_A)$ is $T_+\otimes T_-$. Since the images of the three summands of HT^2 lie in the three different Hodge components $H^{n,n+2}$, $H^{n-1,n+1}$ and $H^{n-2,n}$ of $H^*(A,\mathbb{C})$, the annihilator is the direct sum of the three annihilators computed, of dimensions n^2, $2n^2$ and n^2.

(ii) By the commutative square $\sigma_E\circ\mathrm{ev}_E=(\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 4.54, σ_E vanishes on $\mathrm{ev}_E(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,E))$; if σ_E is injective, ev_E vanishes there. The three explicit statements are (15) on the three summands of (i): on $H^{0,2}$ the evaluation map is $z\mapsto z\cdot\mathrm{id}_E$; on $H^1(T_A)$ it is the obstruction map $v\mapsto v\lrcorner\mathrm{At}(E)$, whose vanishing on v is the extension of E to first order along v [Tod09]; and on a bivector $\theta\wedge\theta'$ it is $\frac{1}{2}(\theta\wedge\theta')\lrcorner\mathrm{At}(E)^2=A_\theta A_{\theta'}$, as in the proof of Theorem 4.61.

(iii) The map $z\mapsto z\cdot\mathrm{id}_E$ is a homomorphism of graded algebras from $H^*(\mathcal{O}_A)=\bigwedge^* H^{0,1}$ to the graded centre of the Yoneda algebra, so its kernel is an ideal, and by (ii) it contains the mixed block $V_+^{0,1}\otimes V_-^{0,1}$. The ideal generated by the mixed block in $\bigwedge^*(V_+^{0,1}\oplus V_-^{0,1})=\bigwedge^* V_+^{0,1}\otimes\bigwedge^* V_-^{0,1}$ is the sum of the summands $\bigwedge^a V_+^{0,1}\otimes\bigwedge^b V_-^{0,1}$ with $a,b\geq 1$, since each

such summand consists of multiples of the mixed block, while the remaining summands, with $a = 0$ or $b = 0$, meet the ideal in zero; the quotient in degree $k \geq 1$ is $\bigwedge^k V_+^{0,1} \oplus \bigwedge^k V_-^{0,1}$, of dimension $2\binom{n}{k}$, and zero for $k > n$. In particular the top class of $H^{2n}(\mathcal{O}_A)$, which is mixed for $n \geq 1$, acts by zero. Serre duality on A, whose canonical bundle is trivial, is the perfect pairing $\mathrm{Ext}^{2n}(E,E) \times \mathrm{Hom}(E,E) \to H^{2n}(\mathcal{O}_A) \cong \mathbb{C}$, $(\xi, f) \mapsto \mathrm{tr}(\xi \circ f)$; for $\xi = w \cdot \mathrm{id}_E$ with w the top class this is $w \cdot \mathrm{tr}(f)$, where $\mathrm{tr}(f) \in H^0(\mathcal{O}_A) = \mathbb{C}$, so $w \cdot \mathrm{id}_E = 0$ forces $\mathrm{tr}(f) = 0$ for every f.

(iv) Let $\mathcal{O}$ be the local ring of A at p and choose coordinates with $Z = \{x_1 = \cdots = x_c = 0\}$, $c \geq 2$. Locally $E \cong (\mathcal{O}/J)^{\oplus r}$ with $J = (x_1, \ldots, x_c)$, a regular sequence, so $\mathrm{Ext}^*_{\mathcal{O}}(\mathcal{O}/J, \mathcal{O}/J) = \bigwedge^*_{\mathcal{O}/J} N$ as a graded algebra, where $N = \mathrm{Hom}(J/J^2, \mathcal{O}/J)$ is the normal module with basis the classes $\partial_1, \ldots, \partial_c$ dual to $x_1, \ldots, x_c$; the Ext^1 is computed on the Koszul resolution exactly as in the proof of Lemma 4.60, and the product is the wedge because the Yoneda algebra of a Koszul complex is the exterior algebra on the dual of the generators. For E of rank r the local Yoneda algebra is $\bigwedge^* N \otimes M_r(\mathcal{O}/J)$. By Lemma 4.59 the germ of A_θ at p is the jet class along θ; on the Koszul complex the jet class of a derivation D is the chain map $-D(d)$, and for $D = \theta = \sum_i c^i \partial_i$ its component in $\mathrm{Ext}^1 = N \otimes M_r(\mathcal{O}/J)$ is $\sum_{i \leq c} (c^i \bmod J)\, \partial_i \otimes \mathrm{id}$, the class of θ modulo T_Z. Multiplicativity of the germ map then gives the germ of $A_\theta A_{\theta'}$ as $(\theta \bmod T_Z) \wedge (\theta' \bmod T_Z) \otimes \mathrm{id}$ in $\bigwedge^2 N \otimes M_r(\mathcal{O}/J)$, whose value at p is $\bar\theta(p) \wedge \bar\theta'(p) \in \bigwedge^2 (T_pA/T_pZ)$. If $A_\theta A_{\theta'} = 0$ for all $\theta \in T_+$, $\theta' \in T_-$, then $\bar a \wedge \bar b = 0$ for all $\bar a$ in the image I_+ of T_+ and $\bar b$ in the image I_- of T_- in $N_p = T_pA/T_pZ$, and $I_+ + I_- = N_p$ since $T_+ \oplus T_- = T_pA$. If both $I_\pm$ were nonzero, every nonzero $\bar b \in I_-$ would be proportional to every nonzero $\bar a \in I_+$, so $I_+ = I_-$ would be a line and $N_p = I_+ + I_-$ of dimension one, against $c \geq 2$. Hence $I_+ = 0$ or $I_- = 0$, which is the claim. ∎

Corollary 2.6 (Locally free components are excluded)**.** *Let A be as in Theorem 2.5 and suppose that no proper abelian subvariety $B \subset A$ has $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$; equivalently, that A has no nonzero K-stable abelian subvariety on which K acts through a single embedding. This holds whenever A is simple, at every split member $X \times \widehat{X}$ with X simple of endomorphism algebra $\mathbb{Q}$, and at a general point of the quaternionic locus of* [I, Setup 15.24]*. Let E be a perfect complex on A with $\mathrm{ch}(E) = N\omega$ as above and σ_E injective. Then no irreducible component Z_0 of the support of E of codimension at least two has a dense open subset along which E is a locally free sheaf of positive rank on Z_0. In particular E is not a sheaf, and along every component of its support of codimension at least two it has, generically, either at least two nonvanishing cohomology sheaves or a cohomology sheaf that is not locally free on the reduced support.*

Proof. Suppose that Z_0 is such a component, with $U \subseteq Z_0^{\mathrm{sm}}$ dense open and $E|_U$ a locally free sheaf of positive rank on U. By Theorem 2.5(iv), $U = U_+ \cup U_-$ with $U_\pm = \{p \in U : T_\pm \subseteq T_pZ_0\}$, both closed in U; as U is irreducible, $U = U_+$ say. Then every constant vector field $\theta \in T_+$ is tangent to Z_0 along U, hence along Z_0^{sm}. The set $S = \{(p,t) \in Z_0 \times \mathbb{C} : p + \exp(t\theta) \in Z_0\}$ is a closed analytic subset of the irreducible space $Z_0 \times \mathbb{C}$, and it contains an open neighbourhood of $Z_0^{\mathrm{sm}} \times \{0\}$, because the local flow of the restriction of θ to the submanifold Z_0^{sm} exists on such a neighbourhood; so $S = Z_0 \times \mathbb{C}$, and $Z_0 + \exp(t\theta) \subseteq Z_0$, hence $= Z_0$, for every t. Let $G = \{x \in A : Z_0 + x = Z_0\}$, a closed algebraic subgroup of A, and B its identity component, an abelian subvariety. The connected set $\exp(T_+)$ contains 0 and lies in G, hence in B, so $T_+ \subseteq T_0B$. By hypothesis $B = A$, so $Z_0 = A$, against the codimension. The last sentence of the statement is the negation of the condition just excluded. If E were a sheaf, its support would have codimension at least n, because the Chern character of a nonzero sheaf begins in the

codimension of its support, with the support cycle, whose multiplicities are positive; this is excluded by Theorem 2.13(iii) below, whose proof uses nothing from the present one beyond the translation argument above.

For the equivalence in the hypothesis, let $\sqrt{-d}$ act on A through an order of K. If an abelian subvariety $B \neq A$ has $T_0B \supseteq T_+$, then $H_1(B,\mathbb{Q}) \cap \sqrt{-d}\,H_1(B,\mathbb{Q})$ is a K-stable sub-Hodge structure, hence $H_1(C,\mathbb{Q})$ for a K-stable abelian subvariety $C \subseteq B$, and $T_0C = T_0B \cap \sqrt{-d}\,T_0B$ still contains T_+, on which $\sqrt{-d}$ acts by a scalar. The K-stable complement C' of C for the polarisation is nonzero, since $C \subseteq B \neq A$, and T_0C' is a K-stable complement of $T_0C \supseteq T_+$, so $T_0C' \subseteq T_-$ and K acts on C' through a single embedding. Conversely, if a nonzero K-stable $C' \neq A$ has $T_0C' \subseteq T_-$, its K-stable complement B is proper with $T_0B \supseteq T_+$. For the three cases: a simple A has no nonzero proper abelian subvariety; for $X \times \widehat{X}$ with X simple and $\mathrm{End}^0(X) = \mathbb{Q}$ the sub-Hodge structures of $H_1(X) \otimes \mathbb{Q}^2$ are $H_1(X) \otimes W$ with $W \subseteq \mathbb{Q}^2$, and since K acts on $\mathbb{Q}^2$ through a quadratic field, W is K-stable only for $W = 0$ or $\mathbb{Q}^2$; and at a general quaternionic member A is either simple or, when the quaternion algebra of [I, Setup 15.24] splits, isogenous to $C \times C$ with C a general principally polarised abelian n-fold, whose sub-Hodge structures are again $H_1(C) \otimes W$ with the same conclusion. ∎

2.3. **The Hochschild action and the two branches.** Theorem 2.5 computes the annihilator summand by summand and reads off what the vanishing of the evaluation map on it means for E. That description hides a structure which is easily brought out and gives more: the annihilator is the mixed part of a splitting of $HH^1(A)$ into two halves, one for each of the two conjugate pieces α_+ and α_- of the Weil class, and the same splitting describes the annihilator in every degree. The results of this subsection concern the Hochschild cohomology of A alone, and their consequences for E follow formally.

Setup 2.7 (Hochschild cohomology in these terms)**.** For an abelian variety A the Hochschild–Kostant–Rosenberg isomorphism identifies

$$HH^*(A) \;\cong\; HT^*(A) = \bigoplus_{p,q} H^p(\textstyle\bigwedge^q T_A) = \bigwedge^*(H^{0,1}(A) \oplus H^0(T_A)),$$

as graded rings, the grading being $p+q$: the Todd class of A is trivial, so the twist by $\mathrm{td}^{1/2}$ in [Cal05], [CRV12] is the identity and the plain identification is already multiplicative. Write $HH^1(A) = H^{0,1}(A) \oplus H^0(T_A)$, of dimension $4n$, so that $HH^*(A)$ is the exterior algebra on it, of dimension 2^{4n}. It acts on $H^*(A,\mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$ by wedging with the classes of $H^{0,1}$ and contracting the classes of $H^{1,0}$ against $H^0(T_A)$; the two families of operators involve disjoint generators, so they pairwise anticommute and the action is an action of the exterior algebra. In degree two this is the operator $x\lrcorner$ of Setup 4.54, and $\mathrm{ev}_E \colon HH^*(A) \to \mathrm{Ext}^*(E,E)$ is a homomorphism of graded rings into the graded centre, with $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ [BF08, Section 6], [Cal05], [CRV12].

Theorem 2.8 (The annihilator is the mixed part of a splitting)**.** *Let A be of $(K,-1,n)$-Weil type with $n \geq 2$, and let $\omega = a\alpha_+ + b\alpha_-$ lie on the Weil line with $ab \neq 0$. Put*

$$P = \mathrm{Ann}_{HH^1}(\alpha_+) = V_+^{0,1} \oplus T_-, \qquad Q = \mathrm{Ann}_{HH^1}(\alpha_-) = V_-^{0,1} \oplus T_+.$$

(i) $\dim P = \dim Q = 2n$ *and* $HH^1(A) = P \oplus Q$*; in particular* $\mathrm{Ann}_{HH^1}(\omega) = 0$.

(ii) $\mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$*, of dimension* $4n^2$*. This is the space computed in Theorem 2.5(i), written as the mixed part of the splitting.*

(iii) *Contraction with α_- is injective on $\bigwedge^* P$, and contraction with α_+ is injective on $\bigwedge^* Q$; the two images meet in exactly one line, in degree $2n$.*

(iv) *For every $k \geq 1$,*

$$\operatorname{Ann}_{HH^k}(\omega) = \Big(\bigoplus_{\substack{i+j=k \\ i,j\geq 1}} \textstyle\bigwedge^i P \wedge \bigwedge^j Q \Big) \;\oplus\; \epsilon_k,$$

where $\epsilon_k = 0$ for $k \neq 2n$ and ϵ_{2n} is the line of (iii). *So* $\dim \operatorname{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$ *for $k \geq 1$, while* $\operatorname{Ann}_{HH^0}(\omega) = 0$, *and the annihilator is the ideal generated by its degree-two part together with that one line.*

Proof. Choose bases $x_1, \dots, x_n$ of $V_+^{1,0}$ and $y_1, \dots, y_n$ of $V_-^{1,0}$, and bases $\overline{x}_i$ of $V_+^{0,1}$ and $\overline{y}_j$ of $V_-^{0,1}$ (the bar is notation only: complex conjugation carries $V_+^{1,0}$ to $V_-^{0,1}$), so that $\alpha_+ = x_1 \cdots x_n \overline{x}_1 \cdots \overline{x}_n$ and $\alpha_- = y_1 \cdots y_n \overline{y}_1 \cdots \overline{y}_n$, and write $\partial_{x_i}, \partial_{y_j}$ for the dual basis of $H^0(T_A) = T_+ \oplus T_-$.

(i) Of the $4n$ basis operators, $\overline{x}_i \wedge$ kills α_+ because $\overline{x}_i$ already occurs in it, and $\partial_{y_j} \lrcorner$ kills α_+ because no y occurs in it; so $P \supseteq V_+^{0,1} \oplus T_-$. Conversely, the classes $\overline{y}_j \wedge \alpha_+$ are linearly independent, as are the classes $\partial_{x_i} \lrcorner \alpha_+$, and the two families lie in different summands of H^*, so no nonzero combination of the operators $\overline{y}_j \wedge$ and $\partial_{x_i} \lrcorner$ kills α_+: $P = V_+^{0,1} \oplus T_-$, of dimension $2n$. The computation for Q is the same with the roles of the two eigenspaces exchanged, and $P \cap Q = 0$ with $\dim P + \dim Q = 4n$, so $HH^1 = P \oplus Q$. A class of degree one that kills ω kills α_+ and α_- separately, the two images lying in different summands and $ab \neq 0$, so it lies in $P \cap Q = 0$.

(iii) A monomial $\overline{x}_I \partial_{y_J} \in \bigwedge^* P$ sends α_- to $\pm\, \overline{x}_I\, y_{[n]\setminus J}\, \overline{y}_{[n]}$, and these are linearly independent for distinct pairs (I, J), which is the injectivity; the statement for Q is symmetric. For the intersection, the image of $\bigwedge^* P$ is spanned by monomials with no factor x and all n factors $\overline{y}$, and the image of $\bigwedge^* Q$, by the same computation, by monomials with no factor y and all n factors $\overline{x}$. Both images are spanned by monomials, so their intersection is spanned by the monomials common to both, and the only such monomial is $\overline{x}_{[n]}\overline{y}_{[n]}$. It is the image of $\pi_P = \overline{x}_{[n]}\partial_{y_{[n]}}$ and of $\pi_Q = \overline{y}_{[n]}\partial_{x_{[n]}}$, both of degree $2n$. So the intersection is the single line spanned by that class.

(ii) and (iv) Decompose $\bigwedge^k HH^1 = \bigoplus_{i+j=k} \bigwedge^i P \wedge \bigwedge^j Q$. A summand with $i, j \geq 1$ annihilates α_+, because it has a factor in P and the action is an action of the algebra, and annihilates α_- for the same reason with Q; so it lies in the annihilator of ω. The sum of these summands over all k is the ideal generated by $P \wedge Q$, since each of their monomials has, after reordering, a factor $p \wedge q$ with $p \in P$ and $q \in Q$. On the remaining two summands, $\xi_P \in \bigwedge^k P$ has $\xi_P \lrcorner \omega = b\, \xi_P \lrcorner \alpha_-$ and $\xi_Q \in \bigwedge^k Q$ has $\xi_Q \lrcorner \omega = a\, \xi_Q \lrcorner \alpha_+$, so a combination $\xi_P + \xi_Q$ annihilates ω if and only if $b\, \xi_P \lrcorner \alpha_- = -a\, \xi_Q \lrcorner \alpha_+$. By (iii) both contractions are injective and their images meet only in degree $2n$, in one line; so this happens only for $k = 2n$, and then only on the line spanned by $b^{-1}\pi_P - a^{-1}\pi_Q$, with π_P and π_Q scaled so that $\pi_P \lrcorner \alpha_- = \pi_Q \lrcorner \alpha_+$. In degree zero the annihilator is zero because $\omega \neq 0$. Taking $k = 2$ and $n \geq 2$, so that $2 \neq 2n$, gives (ii), the dimension being $\dim P \cdot \dim Q = 4n^2$; the general count is $\binom{4n}{k} - \binom{2n}{k} - \binom{2n}{k}$ plus the exceptional line. ∎

Corollary 2.9 (Consequences for the Hochschild action)**.** *Let A and ω be as above and let E be a nonzero perfect complex on A with $\operatorname{ch}(E) = N\omega$, $N \neq 0$.*

(i) *Unconditionally,* $\dim \operatorname{Ext}^2(E, E) \geq 2n(2n-1)$.

(ii) *If σ_E is injective then ev_E vanishes on $P \wedge Q$, hence on the ideal it generates, and the ring homomorphism $\operatorname{ev}_E \colon HH^*(A) \to \operatorname{Ext}^*(E, E)$ factors through*

$$R \;=\; \textstyle\bigwedge^* P \;\oplus_{\mathbb{C}}\; \bigwedge^* Q, \qquad \dim_{\mathbb{C}} R = 2^{2n+1} - 1,$$

the two exterior algebras glued along their scalars, inside $HH^*(A)$ of dimension 2^{4n}. At $n = 2$ that is 31 out of 256.

(iii) *Granting the compatibility $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ in degree k [Cal05], [CRV12], the induced map on R is injective away from the line of Theorem 2.8(iii), so*

$$\dim \mathrm{Ext}^k(E,E) \;\geq\; 2\binom{2n}{k} \quad (1 \leq k < 2n), \qquad \sum_{k\geq 1} \dim \mathrm{Ext}^k(E,E) \;\geq\; 2^{2n+1} - 3.$$

Proof. (i) By Setup 2.7, $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^2}(\mathrm{ch}\,E)$ in degree two, since $\sigma_E(\mathrm{ev}_E(\xi)) = \xi\lrcorner\,\mathrm{ch}(E)$ and the right side is nonzero off the annihilator. Hence $\dim \mathrm{Ext}^2(E,E) \geq \dim HH^2(A) - \dim \mathrm{Ann}_{HH^2}(\omega) = \binom{4n}{2} - 4n^2 = 2n(2n-1)$ by Theorem 2.8(ii). No hypothesis on σ_E is used.

(ii) ev_E vanishes on the annihilator by Theorem 2.5(ii), its kernel is an ideal because it is a ring map, and the ideal generated by $P \wedge Q$ in $\bigwedge^*(P \oplus Q) = \bigwedge^* P \otimes \bigwedge^* Q$ is the sum of the summands with $i, j \geq 1$, whose complement is $\bigwedge^* P \oplus_{\mathbb{C}} \bigwedge^* Q$ of dimension $2 \cdot 2^{2n} - 1$.

(iii) With the compatibility in degree k, the argument of (i) gives $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^k}(\omega)$, and by Theorem 2.8(iv) that annihilator meets $\bigwedge^k P \oplus \bigwedge^k Q$ only in the exceptional line. So the map induced on R has kernel contained in that one line and its image has dimension at least $\dim R - 1 = 2^{2n+1} - 2$. One of those dimensions sits in degree zero, where the image is $\mathbb{C} \cdot \mathrm{id}_E$, nonzero because E is; the remaining $2^{2n+1} - 3$ lie in positive degrees, which is the second inequality. For $1 \leq k < 2n$ the exceptional line has no component in degree k, so $\bigwedge^k P \oplus \bigwedge^k Q$, of dimension $2\binom{2n}{k}$, maps injectively, which is the first. ∎

Remark 2.10 (The two branches). The ring R of Corollary 2.9(ii) is the ring of functions on two odd affine spaces of dimension $2n$ crossing at the origin. In these terms, if E satisfies the criterion, the deformations of $D^b(A)$ along which its Hochschild action survives form two branches, P and Q, which meet only at the trivial deformation and are the annihilators of the two conjugate pieces α_+ and α_- of the Weil class. The criterion says that E sees the two halves of the class separately and their interaction not at all.

The splitting $HH^1 = P \oplus Q$ is not the splitting into eigenspaces of K, which are $V_+^{0,1} \oplus T_+$ and $V_-^{0,1} \oplus T_-$; P and Q cut across them. It is the splitting into the annihilators of α_+ and of α_-, and it exists because each of the two pieces of the Weil class is a top form on an eigenspace. Nor is the conclusion a contradiction in itself. R is a perfectly good ring, and the bound of (iii) is consistent with the value of $\chi(E,E)$ computed in [I, Proposition 9.11], since $\sum_k (-1)^k 2\binom{2n}{k} = 0$. The corollary says how large such an E must be and how little of the Hochschild cohomology it may see, but nothing in the Hochschild computation forbids an E of the required shape. The exclusion comes from a different direction, deformation to non-algebraic Weil tori, in Theorem 3.5.

2.4. **The numerical form of the criterion and the support.** The same square gives two further results. The first converts the rank of σ_E into a single dimension, the form in which a candidate can be tested. The second locates the support of E, and it excludes the first place one would look: a cycle representing the Weil class.

Theorem 2.11 (The criterion as a dimension)**.** *Let A, ω and E be as in Corollary 2.9, so that $\mathrm{ch}(E) = N\omega$ with $N \neq 0$ and no other component.*

(i) *If $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$, the least value that Corollary 2.9(i) allows, then $\mathrm{ev}_E \colon HH^2(A) \to \mathrm{Ext}^2(E,E)$ is surjective with kernel exactly $P \wedge Q$, the map σ_E is injective, and every class of $\mathrm{Ext}^2(E,E)$ is a sum of products $\mathrm{ev}_E(p)\,\mathrm{ev}_E(p') +$*

$\mathrm{ev}_E(q)\,\mathrm{ev}_E(q')$ *with* $p,p' \in P$ *and* $q,q' \in Q$*, hence lies in the graded centre of* $\mathrm{Ext}^*(E,E)$.

(ii) *Conversely, if* σ_E *is injective, then* $\mathrm{ev}_E(HH^2(A))$ *has dimension exactly* $2n(2n-1)$*, and* σ_E *carries it isomorphically onto* $HH^2(A)\lrcorner\omega$.

(iii) *Consequently, if at one base point* s_0 *of a family some perfect complex* E *with* $\mathrm{ch}(E) = N\omega_1$ *has* $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$*, then* $\Sigma = \mathcal{D}_{n,n}$*, and the Hodge conjecture holds in every degree for every member of that family with* $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$.

Proof. (i) The square of Setup 2.7 gives $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$ in degree two, the equality being Theorem 2.8(ii). Hence $\operatorname{rank} \mathrm{ev}_E \geq \binom{4n}{2} - 4n^2 = 2n(2n-1) = \dim \mathrm{Ext}^2(E,E)$, so ev_E is onto and its kernel, of dimension $4n^2$ and contained in $P \wedge Q$, is all of $P \wedge Q$. Every class of $\mathrm{Ext}^2(E,E)$ is then $\mathrm{ev}_E(\xi)$ for some ξ, and $\sigma_E(\mathrm{ev}_E(\xi)) = N\,\xi\lrcorner\omega$ vanishes only for $\xi \in P \wedge Q = \ker \mathrm{ev}_E$; so σ_E is injective. Since $HH^2(A) = \bigwedge^2 P \oplus (P \wedge Q) \oplus \bigwedge^2 Q$ and ev_E is a ring map into the graded centre, the image is $\mathrm{ev}_E(\bigwedge^2 P) + \mathrm{ev}_E(\bigwedge^2 Q)$, which is the description of the classes.

(ii) By Theorem 2.5(ii) the kernel of ev_E in degree two contains $P \wedge Q$, and by the argument of (i) it is contained in it, so it equals $P \wedge Q$ and the image has dimension $2n(2n-1)$. The composite $\sigma_E \circ \mathrm{ev}_E$ is contraction into $\mathrm{ch}(E) = N\omega$, whose kernel is exactly $P \wedge Q = \ker \mathrm{ev}_E$, so σ_E maps the image isomorphically onto $HH^2(A)\lrcorner\omega$.

(iii) By (i) the map σ_E is injective, and Theorem 2.3 applies. ∎

Theorem 2.11(iii) is a sufficient condition for $\mathbf{W}(K,n,\delta)$ that mentions no map at all: one object, at one point, and one number. At $n=2$ the number is 12 and at $n=3$ it is 30. By Theorem 3.5 no complex attains it, but the first two parts of the theorem hold for the polarised criterion (P2′) with the number $r(\gamma)$ of Theorem 3.9(iv) in place of $2n(2n-1)$, and that is the form in which the polarised criterion can be tested.

The dimension of $\mathrm{Ext}^2(E,E)$ is computable in a way the rank of σ_E is not. For a complex built from sheaves whose cohomology is known, the local-to-global spectral sequence $E_2^{p,q} = H^p(A, \mathcal{E}xt^q(E,E)) \Rightarrow \mathrm{Ext}^{p+q}(E,E)$ reduces the count to the cohomology of the local $\mathcal{E}xt$ sheaves, and on an abelian variety the cohomology of a sheaf supported on an abelian subvariety, or of a homogeneous bundle, is a question of linear algebra. The rank of σ_E, by contrast, requires the trace of the exponential of the Atiyah class on every class of Ext^2, a computation with the object and not only with its invariants. Part (ii) says that, on the central part of Ext^2, nothing is lost by reading the criterion this way. Whether an injective σ_E forces the whole of $\mathrm{Ext}^2(E,E)$ to be central is, by Serre duality, a statement about Ext^{2n-2}, which the square in degree two does not reach; nothing below uses it.

The second result rests on a local fact about complexes supported at a point, which we state and prove first.

Lemma 2.12 (Two jet classes on a complex of finite length)**.** *Let* $R = \mathbb{C}[[x_1,\dots,x_c]]$ *with* $c \geq 2$*, let* M *be a bounded complex of finitely generated free* R*-modules whose cohomology has finite length, and put* $\chi(M) = \sum_i (-1)^i \ell(H^i(M))$*. For* $a \in \mathbb{C}^c$ *let* $J(a) \in \mathrm{Ext}^1_R(M,M)$ *be the jet class of* M *along the constant vector field* $\sum_i a_i\partial_i$*, represented by the chain map* $-\sum_i a_i\partial_i(d)$ *as in the proof of Theorem 2.5(iv). If* $\chi(M) \neq 0$ *and* a,b *are linearly independent, then* $J(a)J(b) \neq 0$ *in* $\mathrm{Ext}^2_R(M,M)$.

Proof. Let T be an abelian variety of dimension c and $t \in T$ a point. Linear coordinates on the universal cover identify the completed local ring $\widehat{\mathcal{O}}_{T,t}$ with R in such a way that the translation invariant vector fields become the constant fields $\sum_i a_i\partial_i$: the exponential

map from $T_tT = \mathbb{C}^c$ is a local biholomorphism carrying the constant vector fields to the translation invariant ones, and its germ at the origin identifies $\widehat{\mathcal{O}}_{T,t}$ with $\mathbb{C}[[x_1, \dots, x_c]]$ compatibly with derivations. A complex of R-modules with finite length cohomology is the same thing as a complex of $\mathcal{O}_{T,t}$-modules with finite length cohomology, since modules of finite length over either ring are modules over $\mathcal{O}_{T,t}/\mathfrak{m}^N$ for N large, so M defines a perfect complex M_T on T supported at t. Its local $\mathcal{E}xt$ sheaves are skyscrapers at t, so the local-to-global spectral sequence gives $\operatorname{Ext}^*_T(M_T, M_T) = \operatorname{Ext}^*_R(M, M)$, and by Lemma 4.59 the translation classes A_a, A_b of M_T restrict to $J(a)$, $J(b)$. Apply the square of Setup 4.54 on T to the bivector $a \wedge b$:

$$\sigma_{M_T}(A_a A_b) = (a \wedge b) \lrcorner \operatorname{ch}(M_T).$$

A sheaf of length ℓ supported at a point has Chern character $\ell \cdot [\mathrm{pt}]$, and Chern characters are additive over the cohomology of a complex, so $\operatorname{ch}(M_T) = \chi(M)\,[\mathrm{pt}]$. The class of a point is a nonzero multiple of $dz_1 \cdots dz_c\, d\bar{z}_1 \cdots d\bar{z}_c$, and contracting it with $a \wedge b$ removes two holomorphic factors and leaves a nonzero class of type $(c-2, c)$, because a and b are independent. So the right side is nonzero when $\chi(M) \neq 0$, hence so is $A_a A_b$, and so is its restriction $J(a)J(b)$, the restriction map being an isomorphism. ∎

The hypothesis on $\chi(M)$ cannot be dropped. The Euler characteristic is the only additive invariant here: the Grothendieck group of complexes with finite length cohomology over a regular local ring is $\mathbb{Z}$, generated by the residue field, since every module of finite length has a composition series whose factors are all isomorphic to it. The lemma says that the product of two independent jet classes detects this generator, and on a complex of Euler characteristic zero the product can vanish. Let K be the Koszul resolution of the residue field k of $\mathbb{Q}[x_1, x_2]$, on which the jet class of a derivation D is the chain map $-D(d_K)$, and let $u = \partial_2(d_K)\partial_1(d_K) \colon K \to K[2]$ represent $J(\partial_2)J(\partial_1)$. The entries of d_K are $\pm x_1, \pm x_2$, so u has constant entries. Let C be the cone of u, with $C^m = K^{m+1} \oplus K^{m+2}$ and $d_C(a, b) = (-d_K a,\ u(a) + d_K b)$. As u induces zero on cohomology, C has two cohomology modules, each isomorphic to k, so $\chi(C) = 0$. Since $\partial_i(u) = 0$, the matrix $\partial_i(d_C)$ is block diagonal, and the product $J(\partial_2)J(\partial_1)$ on C is $(a, b) \mapsto (u(a), u(b))$, which equals $d_C H + H d_C$ for $H(a, b) = (b, 0)$: it is null homotopic, the gluing has killed it. On the split complex $k \oplus k[1]$, whose Euler characteristic is also zero, the same product is $u \oplus u[1]$, which is not null homotopic because $u \neq 0$ in $\operatorname{Ext}^2(k, k)$ by the lemma applied to k.

Theorem 2.13 (The support of an object meeting the criterion)**.** *Let A be an abelian $2n$-fold of $(K, -1, n)$-Weil type with $n \geq 2$, let $\omega = a\alpha_+ + b\alpha_-$ with $ab \neq 0$, and let E be a perfect complex on A with $\operatorname{ch}(E) = N\omega$, $N \neq 0$, no other component, and σ_E injective. For an irreducible component Z of the support of E write*

$$m_Z(E) = \sum_i (-1)^i\, \ell_{\mathcal{O}_{A,Z}}(\mathcal{H}^i(E)_Z)$$

for the multiplicity of Z in the cycle of E.

(i) *If Z has codimension at least two and $m_Z(E) \neq 0$, then Z is a union of cosets of an abelian subvariety $B \subsetneq A$ with $T_0 B \supseteq T_+$ or $T_0 B \supseteq T_-$.*

(ii) *If A has no proper abelian subvariety B with $T_0 B \supseteq T_+$ or $T_0 B \supseteq T_-$, which by Corollary 2.6 holds when A is simple, at every split member $X \times \widehat{X}$ with X simple and $\operatorname{End}^0(X) = \mathbb{Q}$, and at a general point of the quaternionic locus, then $m_Z(E) = 0$ for every component Z of codimension at least two, and the cycle of E is a divisor.*

(iii) *At every member of the family, whatever its endomorphisms, the support of E has an irreducible component of codimension less than n.*

In particular no perfect complex whose support has codimension at least n, at any member of any Weil family, satisfies the criterion of Theorem 2.3. This includes every complex assembled from the structure sheaves of the components of a cycle representing ω, the base cycle of [I, Theorem 15.27] *among them.*

Proof. (i) Let $c \geq 2$ be the codimension of Z and let $U \subseteq Z$ be the dense open set of smooth points of Z_{red} that lie on no other component of the support. Fix $\theta \in T_+$ and $\theta' \in T_-$; by Theorem 2.5(ii), $A_\theta A_{\theta'} = 0$ in $\mathrm{Ext}^2(E,E)$. Suppose that at some $p \in U$ the images $\bar\theta, \bar\theta'$ of θ, θ' in $N_p = T_pA/T_pZ$ are linearly independent, and complete them by images of constant fields $\nu_3, \dots, \nu_c$ to a basis of N_p. In linear coordinates on the universal cover the constant fields are coordinate derivations, and the germ S at p of the translate of the span of $\theta, \theta', \nu_3, \dots, \nu_c$ is a smooth germ of dimension c, transverse to Z at p, to which θ and θ' are tangent. Let $j\colon S \to A$ be the inclusion and $M = Lj^*E$ on $\widehat{\mathcal{O}}_{S,p} \cong \mathbb{C}[[x_1, \dots, x_c]]$. Derived restriction is a ring map $\mathrm{Ext}^*(E,E) \to \mathrm{Ext}^*(M,M)$, and it carries the germ of A_θ to the jet class of M along $\theta|_S$, because the Atiyah class is functorial for pullback and θ is tangent to S [BF03, Section 3]. So $J(\theta)J(\theta') = 0$ on M. Near p the support of E is Z, and $Z \cap S = \{p\}$, so M has finite length cohomology; its Euler characteristic is the intersection multiplicity of the cycle of E with S at p, which, S being transverse to Z at a smooth point lying on no other component, is $m_Z(E)$, by Serre's formula for intersection multiplicities applied to each cohomology sheaf and summed with signs [Ser00, Chapter V].[5] Since θ and θ' are independent directions on S, Lemma 2.12 applied to M gives $m_Z(E) = 0$.

So if $m_Z(E) \neq 0$, then at every $p \in U$ and for all $\theta \in T_+$, $\theta' \in T_-$ the images satisfy $\bar\theta \wedge \bar\theta' = 0$ in $\bigwedge^2 N_p$. As in the proof of Theorem 2.5(iv), the images $I_\pm$ of $T_\pm$ in N_p span N_p, which has dimension $c \geq 2$, so one of them is zero: $T_+ \subseteq T_pZ$ or $T_- \subseteq T_pZ$. The two conditions cut out closed subsets of the irreducible set U whose union is U, so one of them, say the first, holds on all of U, and the argument of Corollary 2.6 shows that Z is stable under translation by the identity component B of its stabiliser, an abelian subvariety with $T_0B \supseteq T_+$. Since $Z \neq A$, $B \neq A$.

(ii) follows from (i), since no such B exists; if A itself is a component of the support, its multiplicity is the rank of E, which is zero, so the cycle of E is a divisor.

(iii) Suppose every component of the support has codimension at least n. Then $\mathrm{ch}_k(E) = 0$ for $k < n$ and $\mathrm{ch}_n(E) = \sum_Z m_Z(E)[Z]$, the sum over the components of codimension exactly n; this is Riemann–Roch without denominators in its simplest case [Ful98, Chapter 15]. By (i), each Z with $m_Z(E) \neq 0$ is $\pi^{-1}(\pi(Z))$ for the quotient $\pi\colon A \to A/B$ by an abelian subvariety B with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$, so $[Z] = \pi^*[\pi(Z)]$ lies in the subring of $H^*(A,\mathbb{C})$ generated by $\pi^*H^1(A/B,\mathbb{C})$, the space of classes of degree one that vanish on T_0B and on its conjugate. When $T_0B \supseteq T_+$, a $(1,0)$-form vanishing on T_+ lies in $V_-^{1,0}$, and a $(0,1)$-form vanishing on the conjugate of T_+ lies in $V_+^{0,1}$, because complex conjugation exchanges the two eigenvalues of K;[6] so the subring lies in $\bigwedge^*(V_-^{1,0} \oplus V_+^{0,1})$. When $T_0B \supseteq T_-$ it lies in $\bigwedge^*(V_+^{1,0} \oplus V_-^{0,1})$. Choose bases of the four pieces. Both exterior

[5] Serre's formula expresses $\sum_i (-1)^i \ell(\mathrm{Tor}_i(F, \mathcal{O}_S))$, for a module F and a smooth germ S meeting its support properly at p, as a sum over the components Z' of the support of $\ell(F_{Z'})$ times the intersection multiplicity of Z' and S at p. For S transverse to a component smooth at p the intersection multiplicity is one. For a complex the Tor spectral sequence gives the same formula with alternating sums over the cohomology, which define $m_Z(E)$.

[6] The field K acts on $H_1(A,\mathbb{Q})$ by rational endomorphisms, so on the complexification its action commutes with complex conjugation, which carries the eigenspace of $\sqrt{-d}$ for one embedding of K to that for the other. On the tangent space the conjugate of T_+ is the eigenspace in $T^{0,1}$ for the second embedding; it pairs with $V_-^{0,1}$, not with $V_+^{0,1}$.

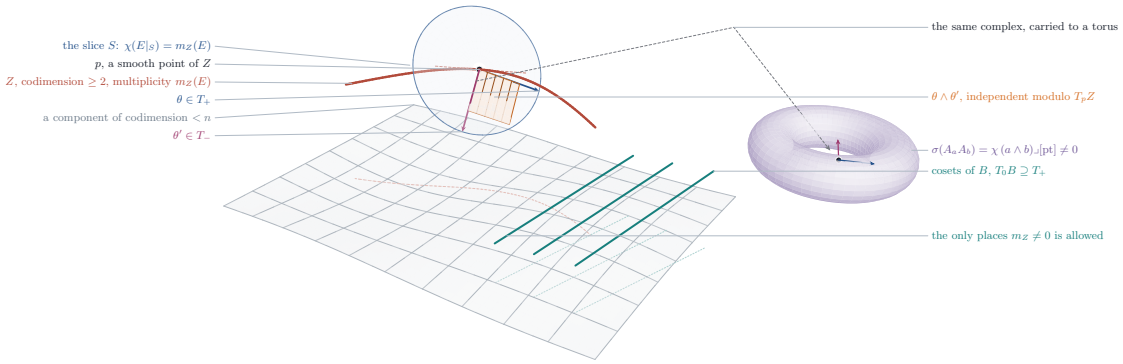


FIGURE 1. The proof of Theorem 2.13. The grey sheet stands for a component of the support of codimension less than n, which the theorem forces to exist; the clay curve Z for a component of codimension at least two. At a smooth point p of Z the directions $\theta \in T_+$ and $\theta' \in T_-$ span the slice S, transverse to Z, on which E restricts to a complex of finite length with Euler characteristic $m_Z(E)$; the amber parallelogram is the bivector $\theta \wedge \theta'$, independent modulo T_pZ. Carried to a torus of the same dimension, the product of the two jet classes is seen by the semiregularity map as $m_Z(E)$ times the class of a point contracted with the bivector, which is not zero, so $m_Z(E) = 0$ or the bivector degenerates. The teal lines are cosets of an abelian subvariety B tangent to T_+, the only components of codimension at least two on which a nonzero multiplicity survives. The scene is schematic in the dimensions and exact in the incidences.

algebras are spanned by monomials in these bases, so their sum is too, and α_+, a single monomial in n generators of $V_+^{1,0}$ and n of $V_+^{0,1}$, is not among the monomials spanning it; nor is α_-. As $ab \neq 0$, the class $N\omega$ has nonzero coordinates on both and is not in the sum, which contradicts $\mathrm{ch}_n(E) = N\omega$. So some component has codimension less than n. The last assertion follows, since a complex assembled from structure sheaves of the components of a cycle of codimension n is supported on that cycle. ∎

Figure 1 draws the slice and the bivector of the proof.

2.5. **Indecomposability and endomorphisms.** The numerical criterion of Theorem 2.11 fixes one number, $\dim \mathrm{Ext}^2(E,E)$. Together with Serre duality, the lower bounds of Corollary 2.9 in the other degrees and the Hodge–Riemann relations for the Weil class, it fixes a good deal more: the object that carries the class can be taken indecomposable, and it has at least three linearly independent endomorphisms at $n = 2$ and, granting the compatibility of Corollary 2.9(iii), at $n = 3$, the first case in which the criterion is needed at a nontrivial discriminant.

Theorem 2.14 (The shape of an object meeting the criterion)**.** *Let A, ω and E be as in Theorem 2.11, so that $\mathrm{ch}(E) = N\omega$ with $N \neq 0$ and no other component, and suppose that $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$. Write $e_k = \dim \mathrm{Ext}^k(E,E)$.*

(i) *$\chi(E,E) = \sum_k (-1)^k e_k = (-1)^n N^2 \int_A \omega^2$, and this number is positive.*

(ii) *$E \cong E_0 \oplus E'$ with E_0 indecomposable, $\mathrm{ch}(E_0) = \mathrm{ch}(E)$, $\dim \mathrm{Ext}^2(E_0,E_0) = 2n(2n-1)$, $\mathrm{ch}(E') = 0$ and $\mathrm{Ext}^2(E_0,E') = \mathrm{Ext}^2(E',E_0) = \mathrm{Ext}^2(E',E') = 0$. So E_0 meets the criterion as well.*

(iii) *If E is indecomposable, then* $\mathrm{End}(E) = \mathbb{C}\cdot\mathrm{id}\oplus J$ *with J the nilpotent radical, every $f \in J$ annihilates* $\mathrm{Ext}^2(E,E)$ *and* $\mathrm{Ext}^{2n-2}(E,E)$ *under the Yoneda product, and* $\mathrm{ev}_E(\bigwedge^{2n} P) = \mathrm{ev}_E(\bigwedge^{2n} Q)$ *is a single line in* $\mathrm{Ext}^{2n}(E,E)$*. If moreover $e_1 = 4n$, then J annihilates* $\mathrm{Ext}^1(E,E)$ *as well.*

(iv) *If $n = 2$ or $n = 3$, then*

$$\dim \mathrm{End}(E) \;\geq\; 2 + \tfrac{1}{2}\,\chi(E,E) \;>\; 2,$$

so E has at least three linearly independent endomorphisms, and at $n = 3$ also $e_1 \leq e_0 + 9$. In particular no simple object meets the criterion: no stable sheaf, no complex with $\mathrm{End} = \mathbb{C}$*, and no image of one under an autoequivalence of $D^b(A)$; and an indecomposable object that meets it has a nonzero nilpotent endomorphism. The smallest profiles $(e_0, \ldots, e_{2n})$ that the constraints allow are $(3,8,12,8,3)$ at $n = 2$, with $\chi = 2$, and $(3,12,30,40,30,12,3)$ and $(3,12,30,41,30,12,3)$ at $n = 3$, with $\chi = 2$ and $\chi = 1$.*

Parts (i) and (ii), and part (iv) at $n = 2$, use the compatibility $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 2.7 in degree two only, as Theorem 2.11 does. Part (iii), and part (iv) at $n = 3$, use it in the further degrees in which Corollary 2.9(iii) grants it: $2n-2$ and $2n$ for (iii), and three for (iv).

Proof. (i) The Todd class of A is trivial, so the Riemann–Roch theorem gives $\chi(E,E) = \int_A \mathrm{ch}(E)^\vee\,\mathrm{ch}(E)$, and $\mathrm{ch}(E)^\vee = (-1)^n N\omega$ since ω has degree $2n$. The polarisation pairs V_+ only with V_- (see [I, Figure 9]), so its class θ lies in $V_+ \otimes V_-$ and $\theta \wedge \alpha_\pm = 0$, because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So ω is a nonzero rational primitive class of type (n,n) in the middle degree, and the Hodge–Riemann relations give $(-1)^{n(2n-1)}\int_A \omega^2 > 0$, where $(-1)^{n(2n-1)} = (-1)^n$.

(ii) The category $D^b(A)$ has finite dimensional morphism spaces and split idempotents, so $E = \bigoplus_i E_i$ with each E_i indecomposable. The evaluation map and the semiregularity map are compatible with direct sums, and by Theorem 2.11(i) the kernel of ev_E in degree two is $P \wedge Q$; so $\mathrm{ev}_{E_i}(P \wedge Q) = 0$ and, composing with σ_{E_i}, $(P \wedge Q)\lrcorner\,\mathrm{ch}(E_i) = 0$ for every i. We claim that the only classes of Hodge type killed by $P \wedge Q$ are those of $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$. Write $H^*(A,\mathbb{C}) = \bigwedge^*(A_+ \oplus B_+ \oplus A_- \oplus B_-)$ with $A_\pm = V_\pm^{1,0}$ and $B_\pm = V_\pm^{0,1}$, and grade it by the four degrees (i,j,k,l). As in Setup 2.7, P acts by wedging with B_+ and contracting A_-, and Q by wedging with B_- and contracting A_+, so $P \wedge Q$ is spanned by four kinds of operators, each of which shifts the four degrees by a fixed amount: wedging with $b_+ \wedge b_-$ kills a component of degree (i,j,k,l) for all choices exactly when $j = n$ or $l = n$; wedging with b_+ after contracting a_+, exactly when $i = 0$ or $j = n$; contracting a_- and wedging with b_-, exactly when $k = 0$ or $l = n$; and contracting a_- and a_+, exactly when $i = 0$ or $k = 0$. Each of these is a statement about a tensor product of maps that are injective unless the degree in question is extreme. A class of type (p,p) has $i+k = j+l = p$ in every component. If $j = n$, then $l = n$ would force $i = k = n$, against the fourth condition, so $l < n$ and the third condition gives $k = 0$; hence $i = n + l$, so $i = n$ and $l = 0$: the component lies in $\bigwedge^n A_+ \otimes \bigwedge^n B_+ = \mathbb{C}\alpha_+$. If $j < n$, the first two conditions give $l = n$ and $i = 0$, hence $k = n$ and $j = 0$: the component lies in $\mathbb{C}\alpha_-$. This proves the claim. Since $\mathrm{ch}(E_i)$ is a rational class of Hodge type killed by $P \wedge Q$, it lies on the rational Weil plane, and when it is nonzero Corollary 2.9(i) gives $\dim \mathrm{Ext}^2(E_i,E_i) \geq 2n(2n-1)$, a nonzero rational Weil class having both coefficients a and b nonzero. As $\mathrm{Ext}^2(E,E) = \bigoplus_{i,j}\mathrm{Ext}^2(E_i,E_j)$ has dimension $2n(2n-1)$ and $\sum_i \mathrm{ch}(E_i) = N\omega \neq 0$, exactly one summand E_0 has nonzero Chern character and every other block of Ext^2 vanishes.

(iii) An indecomposable object of a category with finite dimensional morphism spaces and split idempotents has a local endomorphism ring, so $\mathrm{End}(E) = \mathbb{C}\cdot\mathrm{id}\oplus J$ with J nilpotent. By Theorem 2.11(i), ev_E maps $\bigwedge^2 P \oplus \bigwedge^2 Q$ isomorphically onto $\mathrm{Ext}^2(E,E)$. By Serre duality $e_{2n-2} = e_2$, and contraction with ω is injective on $\bigwedge^{2n-2} P \oplus \bigwedge^{2n-2} Q$ by Theorem 2.8(iv), since $2n-2 \neq 2n$; so, with the compatibility in degree $2n-2$, ev_E maps that space isomorphically onto $\mathrm{Ext}^{2n-2}(E,E)$. In degree $2n$ the semiregularity map is the trace, since $H^{q+2n}(\Omega_A^q) = 0$ for $q > 0$, so $\operatorname{tr}\mathrm{ev}_E(\xi)$ is the $H^{0,2n}$ component of $\xi \lrcorner \mathrm{ch}(E)$; for the generators π_P, π_Q of $\bigwedge^{2n} P$ and $\bigwedge^{2n} Q$ it is Nc_P and Nc_Q with $c_P c_Q \neq 0$. Indeed π_P wedges with a basis of B_+ and contracts a basis of A_-, so it kills α_+, whose factor $\bigwedge^n B_+$ is already full, and carries α_- to $\pm$ the generator $\bigwedge^n B_+ \wedge \bigwedge^n B_-$ of $H^{0,2n}$; so $c_P = \pm b$ and likewise $c_Q = \pm a$, both nonzero. For $f \in \mathrm{End}(E)$ and $x \in \bigwedge^2 P \oplus \bigwedge^2 Q$ write $f\cdot \mathrm{ev}_E(x) = \mathrm{ev}_E(\varphi_f(x))$; then $f \mapsto \varphi_f$ is a homomorphism of algebras. Let $x \in \bigwedge^2 P$ and $y \in \bigwedge^{2n-2} P$, and write $x \wedge y = \beta(x,y)\pi_P$. Since ev_E is a ring map that kills the ideal generated by $P \wedge Q$,

$$\beta(x,y)\operatorname{tr}(f\,\mathrm{ev}_E(\pi_P)) = \operatorname{tr}(f\,\mathrm{ev}_E(x)\,\mathrm{ev}_E(y)) = \operatorname{tr}\mathrm{ev}_E(\varphi_f(x)\wedge y) = Nc_P\,\beta(\varphi_f(x)_P, y),$$

and the pairing β is perfect, so $\varphi_f(x)_P = \mu_f x$ with $\mu_f = \operatorname{tr}(f\,\mathrm{ev}_E(\pi_P))/Nc_P$. The same computation with $y \in \bigwedge^{2n-2} Q$, where $x \wedge y$ lies in the ideal generated by $P \wedge Q$, gives $\varphi_f(x)_Q = 0$. So f acts on $\mathrm{ev}_E(\bigwedge^2 P)$ by a scalar μ_f, and in the same way on $\mathrm{ev}_E(\bigwedge^2 Q)$ by a scalar ν_f; μ and ν are characters of $\mathrm{End}(E)$ and vanish on J. So J kills $\mathrm{Ext}^2(E,E)$, on both sides because ev_E lands in the graded centre, and the same argument with the roles of the two degrees exchanged shows that it kills $\mathrm{Ext}^{2n-2}(E,E)$. In particular $\operatorname{tr}(f\,\mathrm{ev}_E(\pi_P)) = \mu_f Nc_P = 0$ for $f \in J$, so $\mathrm{ev}_E(\pi_P)$ and, likewise, $\mathrm{ev}_E(\pi_Q)$ lie in the annihilator of J under the perfect pairing of $\mathrm{End}(E)$ with $\mathrm{Ext}^{2n}(E,E)$. That annihilator is a line, J having codimension one, and both elements are nonzero, their traces being Nc_P and Nc_Q; so they span the same line. Finally, if $e_1 = 4n$ then $\mathrm{Ext}^1(E,E) = \mathrm{ev}_E(HH^1)$, because ev_E is injective on HH^1, as shown in (iv). For $f \in J$ and $h \in HH^1$ write $f\,\mathrm{ev}_E(h) = \mathrm{ev}_E(h')$; then $\mathrm{ev}_E(h' \wedge h'') = f\,\mathrm{ev}_E(h \wedge h'') = 0$ for every $h'' \in HH^1$, so $h' \wedge HH^1 \subseteq P \wedge Q$, which forces $h' = 0$ since $\dim P = \dim Q \geq 2$.

(iv) By Serre duality $e_k = e_{2n-k}$. The map ev_E is injective on HH^1 using degree two alone: if $\mathrm{ev}_E(h) = 0$ then $\mathrm{ev}_E(h \wedge h') = 0$ for every h', so $h \wedge HH^1 \subseteq \mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$, and $h = 0$ as in (iii). So $e_1 \geq 4n$, and Corollary 2.9(iii) gives $e_3 \geq 2\binom{6}{3} = 40$ at $n = 3$. At $n = 2$ the Euler characteristic is $2e_0 - 2e_1 + 12$, and at $n = 3$ it is $2e_0 - 2e_1 + 60 - e_3$. So $2e_0 = \chi + 2e_1 - 12 \geq \chi + 4$ at $n = 2$, and $2e_0 = \chi + 2e_1 + e_3 - 60 \geq \chi + 4$ at $n = 3$, where also $e_1 = e_0 + 30 - \frac{1}{2}(\chi + e_3) \leq e_0 + 9$. With (i) this gives the bound. An autoequivalence preserves End, and a stable sheaf is simple. An indecomposable object with $\dim \mathrm{End} \geq 3$ has $J \neq 0$. For the smallest profiles take $e_0 = 3$. At $n = 2$, χ is even and positive, so $2e_1 = 18 - \chi \leq 16$, and $e_1 \geq 8$ forces $e_1 = 8$ and $\chi = 2$. At $n = 3$, $2e_1 + e_3 = 66 - \chi \leq 65$ with $e_1 \geq 12$ and $e_3 \geq 40$ forces $e_1 = 12$ and $e_3 \in \{40, 41\}$, with $\chi = 2$ and $\chi = 1$. ∎

Remark 2.15 (Objects excluded by the shape). Theorem 2.14 gives necessary conditions only, and their effect is to exclude the objects one would construct first. Line bundles, semi-homogeneous bundles, skyscraper sheaves, stable sheaves and their Fourier–Mukai transforms are simple, and direct sums of them are excluded by (ii) together with [III, Theorem 2.32]. At $n = 3$, granting the compatibility of Corollary 2.9(iii), the shape allowed is an indecomposable object with a local endomorphism algebra of dimension at least three whose radical kills Ext^2 and Ext^4 and pairs to zero with the line $\mathrm{ev}_E(\bigwedge^6 P)$. The smallest conceivable one has $\mathrm{Hom}(E,E)$ of dimension three, $\mathrm{Ext}^1(E,E)$ equal to the image

of $HH^1(A)$, of dimension twelve and killed by the radical, and $\mathrm{Ext}^3(E,E)$ of dimension forty or forty-one. Iterated self-extensions and thickenings are the natural objects with nilpotent endomorphisms, but an extension of a simple object F by itself has $\mathrm{ch} = 2\,\mathrm{ch}(F)$, so F would already have to carry the class, and the question returns one level down. For $n \geq 4$ write $e_k = 2\binom{2n}{k} + \delta_k$ for $1 \leq k \leq 2n-1$, with $\delta_k \geq 0$ and $\delta_2 = \delta_{2n-2} = 0$. Since $\sum_{k=1}^{2n-1}(-1)^k\binom{2n}{k} = -2$, the Euler characteristic is $2e_0 - 4 + \sum_k (-1)^k\delta_k$, and an excess δ_k in an even degree $4 \leq k \leq 2n-4$ can make it positive with $e_0 = 1$; so the bound on End disappears. By Theorem 3.5 none of these shapes is realised.

3. The semiregularity criterion in its pure and polarised forms

3.1. **Weil tori and the pure form of the criterion.** The object E of [I, Section 16] has Chern character exactly $N\omega$. For such an object, injectivity of σ_E would make E extend along every K-linear deformation of A (Theorem 2.5(ii)). These deformations reach non-algebraic tori, on which the Weil class is a Hodge class but not a Chern class of any coherent sheaf. This settles the pure form of the criterion, negatively, in every dimension.

Setup 3.1 (Weil tori)**.** Let A be an abelian $2n$-fold of $(K, -1, n)$-Weil type, $V = H^1(A, \mathbb{Q})$ with its action of K, and $V \otimes \mathbb{C} = V_+ \oplus V_-$ the two eigenspaces, each of dimension $2n$. For complementary n-dimensional subspaces $X, Y \subset V_+$ the subspace $X \oplus \overline{Y}$ of $V \otimes \mathbb{C}$ is the space of holomorphic 1-forms of a complex structure on the real torus underlying A; call the resulting complex torus $T_{X,Y}$. On $T_{X,Y}$ the field K acts with signature (n,n), so $T_{X,Y}$ is a Weil torus in the sense of [Voi02b], and every complex structure on that real torus commuting with K and of signature (n,n) arises in this way. The pairs (X,Y) form a connected complex manifold S of dimension $2n^2$, containing the point of A; the polarised Weil family is the submanifold of dimension n^2 on which the polarisation of A is of type $(1,1)$, and the tangent space of S is the space $\mathrm{Hom}_K(H^{1,0}, H^{0,1})$ of K-linear first order deformations in Theorem 2.5(i). Since α_+ spans $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$ and X is holomorphic while Y is antiholomorphic, the Weil classes are of type (n,n) on every $T_{X,Y}$. We say that a property holds for a very general member of S if it holds outside a countable union of proper analytic subsets.

Lemma 3.2 (The Hodge classes of a very general Weil torus)**.** *For a very general member T of S, $\mathrm{Hdg}^k(T) = 0$ for $1 \leq k \leq 2n-1$, $k \neq n$, and $\mathrm{Hdg}^n(T)$ is the rational Weil plane.*

Proof. There are countably many rational classes, and the Hodge locus of each is an analytic subset of the connected manifold S; so a rational class that is of Hodge type at a very general member is of Hodge type on all of S. Let $G = \mathrm{Res}_{K/\mathbb{Q}}\,\mathrm{GL}_{2n}$ be the group of K-linear automorphisms of V. Its real points act transitively on S, since $\mathrm{GL}(V_+)$ acts transitively on pairs of complementary subspaces. The Mumford–Tate group M of a very general member lies in G, since K acts by endomorphisms of every member, and it contains the Mumford–Tate group of every member, whose Hodge classes include those of the very general one. So the real Lie algebra of M contains the complex structure J_s of every member, which is the derivative of its Hodge cocharacter, and therefore contains the $\mathrm{Ad}\,G(\mathbb{R})$-span of one J_s. This span is an ideal of $\mathfrak{g}_{\mathbb{R}} = \mathfrak{gl}_{2n}(\mathbb{C})$ containing the non-central element J_s, which acts on V_+ by i on X and by $-i$ on Y; hence it contains $\mathfrak{sl}_{2n}(\mathbb{C})$, and M contains $G^{\mathrm{der}} = \mathrm{Res}_{K/\mathbb{Q}}\,\mathrm{SL}_{2n}$. Over $\mathbb{C}$ this is $\mathrm{SL}(V_+) \times \mathrm{SL}(V_-)$ acting on $\bigwedge^*(V_+ \oplus V_-) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$, whose invariants are spanned by 1, α_+, α_- and $\alpha_+\alpha_-$. Hodge classes are invariant under M, which gives the statement. ∎

Lemma 3.3 (A very general Weil torus has no subvarieties)**.** *A very general member T of S has no analytic subsets other than finite sets and T itself.*

Proof. Let $Z \subset T$ be an irreducible analytic subset of codimension c with $0 < c < 2n$. For every Kähler class κ the class $[Z]$ satisfies $\int_T [Z]\,\kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$. By Lemma 3.2 this forces $c = n$ and $[Z]$ into the Weil plane. Take for κ_0 the class of a positive definite hermitian form that is block diagonal for the splitting $H^{1,0} = X \oplus \overline{Y}$. Its blocks lie in $X \otimes \overline{X}$ and $\overline{Y} \otimes Y$, both inside $V_+ \otimes V_-$, so every power of κ_0 has a factor from V_+ and a factor from V_- in each term, and $\alpha_\pm \wedge \kappa_0^n = 0$ because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So $\int_T [Z]\,\kappa_0^n = 0$, which is impossible. ∎

Proposition 3.4 (Coherent sheaves on a very general Weil torus)**.** *Let $n \geq 2$ and let T be a very general member of S. Every coherent analytic sheaf F on T has $\mathrm{ch}_k(F) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \leq k \leq 2n-1$. For $n = 2$ this is Voisin's theorem that every coherent sheaf on a general Weil torus of dimension four has $c_2(F) = 0$* [Voi02b, Theorem 10]*, whose proof we follow.*

Proof. Chern classes of coherent sheaves on T exist in rational Deligne cohomology, extend those of bundles and are additive [Gri10], so they are Hodge classes. By Lemma 3.3 the torsion of F, the locus where the torsion-free quotient is not locally free, and the cokernel of the map from that quotient to its reflexive hull are all supported on finite sets, and a sheaf supported on a finite set has Chern character in the top degree only. So it suffices to treat a reflexive sheaf G. Every coherent sheaf on T has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$ (Lemma 3.2); so every torsion-free sheaf has slope zero for every Kähler class, and a torsion-free sheaf is stable for one Kähler class exactly when it has no saturated subsheaf of intermediate rank, which does not depend on the class. A Jordan–Hölder filtration of G therefore has stable torsion-free factors Q_j, stable for every Kähler class, and $\mathrm{ch}(G) = \sum_j \mathrm{ch}(Q_j)$. By the theorem of Bando and Siu [BS94] the stable reflexive sheaf Q_j^{**} carries an admissible Hermite–Einstein metric for every Kähler class κ and satisfies $\int_T (2rc_2 - (r-1)c_1^2)(Q_j^{**})\,\kappa^{2n-2} \geq 0$, with equality exactly when Q_j^{**} is a projectively flat vector bundle. Here $c_1 = 0$ and $c_2(Q_j^{**}) \in \mathrm{Hdg}^2(T)$. For $n \geq 3$ that group is zero; for $n = 2$ it is the Weil plane, and then $\int_T c_2(Q_j^{**})\,\kappa_0^2 = 0$ for the class κ_0 of the proof of Lemma 3.3. In either case equality holds for some κ, so Q_j^{**} is a projectively flat bundle with $c_1 = 0$, hence flat, and its rational Chern classes of positive degree vanish. Since Q_j and Q_j^{**} differ only on a finite set, $\mathrm{ch}_k(Q_j) = 0$ for $1 \leq k \leq 2n-1$, and the same holds for G and for F. ∎

Theorem 3.5 (The pure form of the criterion is never met)**.** *Let $n \geq 2$, let A be an abelian $2n$-fold of $(K, -1, n)$-Weil type, ω a nonzero rational Weil class, and E a perfect complex on A with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component. Then σ_E is not injective. In particular $\dim \mathrm{Ext}^2(E, E) > 2n(2n-1)$, the hypotheses of Theorem 2.3 and Theorem 2.11(iii) are satisfied by no object at any member of any Weil family of an imaginary quadratic field, and the propagation statement* (P2) *of Remark 3.7, in the form stated there, is false.*

Proof. Suppose that σ_E is injective. Since A is projective, E is isomorphic to a bounded complex $E^\bullet$ of holomorphic vector bundles. The class $N\omega$ is of Hodge type on every member of S (Setup 3.1), so by the theorem of Buchweitz and Flenner in Pridham's form (Theorem 2.2, [BF03, Pri24]), every obstruction to extending E along an Artinian deformation of A inside S vanishes, exactly as in the proof of Theorem 2.3. The deformations of the complex structure inside S together with those of the holomorphic structures and

differentials of $E^\bullet$ are governed by a differential graded Lie algebra of Dolbeault type with finite dimensional cohomology, and Kuranishi's method [Kur65], in the form of [GM88, Sections 2–3], gives an analytic germ (M, m_0) with a morphism $\pi\colon (M, m_0) \to (S, s_0)$ and a complex of holomorphic vector bundles on the pulled back family of tori deforming $E^\bullet$, whose completion is a hull of the formal deformation functor. The vanishing of the obstructions says that π is formally smooth, hence smooth, so its image contains a neighbourhood U of s_0. Thus every member T_s with $s \in U$ carries a bounded complex of holomorphic vector bundles $E_s^\bullet$ whose Chern character is the parallel transport of $N\omega$. Choose $s \in U$ very general. By Proposition 3.4, $\mathrm{ch}_n(E_s) = \sum_i (-1)^i \mathrm{ch}_n(E_s^i) = 0$, while it is the transport of $N\omega$, which is not zero. This contradiction proves that σ_E is not injective. The argument uses only that $\mathrm{ch}(E)$ is of Hodge type on every member of S, so it applies verbatim when $\mathrm{ch}(E) - N\omega$ has components in degrees 0 and $4n$ as well. The remaining statements follow, since $\dim \mathrm{Ext}^2(E, E) \geq 2n(2n-1)$ by Corollary 2.9(i), and equality forces σ_E to be injective by Theorem 2.11(i). ∎

Remark 3.6 (The polarised criterion). Theorem 3.5 turns Theorem 2.5(ii) from a constraint into a contradiction: an object that extends along every K-linear deformation of A reaches non-algebraic tori, where the Weil class is a Hodge class but not a Chern class. The results that describe such an object, Theorem 2.5, Corollary 2.9(ii)–(iii), Theorem 2.11, Theorem 2.13, [III, Theorem 2.32] and Theorem 2.14, remain true as implications, but they are vacuous: no object satisfies their hypotheses. They record what the pure form asks of an object, and Remark 3.13 says which of them extend to the polarised form defined next.

The obstruction does not reach a Chern character with a component that is of Hodge type on the polarised family and not beyond it, unless a twist removes that component. Call (P2′) the statement that at one base point of each Weil family some perfect complex E with

$$\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\, \theta^k, \qquad N \neq 0,\ c_k \in \mathbb{Q},$$

θ the polarisation, has injective semiregularity map. The proof of Theorem 2.3 applies to it word for word, with the deformation taken over the polarised family, along which $\mathrm{ch}(E)$ stays of Hodge type: at a member A_s it produces $\mathrm{ch}_n(E_s) = N\omega_1 + c_n\theta^n$ algebraic, and θ^n is algebraic, so ω_1 is. The form above is already the most general one allowed by the subring generated by θ and the Weil classes: a Weil class is primitive, so $\theta\omega_1 = 0$ and that subring is spanned by the powers of θ and the Weil plane. Theorem 3.5 excludes directly a Chern character that remains of Hodge type on all of S, which by Lemma 3.2 is a multiple of the Weil class plus terms in degrees 0 and $4n$, and after a line bundle twist the characters $c\, e^{t\theta} + N\omega_1 + c'\theta^{2n}$ with $t\theta$ integral (Theorem 3.9(v)). For every other polarised character with some $c_k \neq 0$, $1 \leq k \leq 2n-1$, the component of the Hodge locus of $\mathrm{ch}(E)$ through A is the polarised family itself, whose members are abelian varieties, so nothing forces E onto non-algebraic tori and the argument cannot be run on E (Proposition 3.11); the one exception is a ratio at $n = 2$ at which the locus is five-dimensional (Remark 3.12). The semiregular objects in Markman's proofs for abelian fourfolds and for the sixfolds of trivial discriminant [Mar25a, Mar25b] are not of the pure form: their Chern characters are not multiples of a Weil class, and they are deformed only along families on which those characters stay of Hodge type, and nothing below depends on whether they have exactly the form above. From here on, in [III, Corollary 2.71], [III, Theorem 2.70] and the frontier of [III, Section 2], (P2) is read in this polarised form. The annihilator computations of [I, Section 16] assume that $\mathrm{ch}(E)$ is a pure Weil class; the next subsection carries them out for

a Chern character with components in every even degree (Theorem 3.9, Proposition 3.10, Proposition 3.11).

The same obstruction does not reach the Mumford target of [III, Section 3.2]. The exceptional classes there are rigid: their Hodge locus among all complex tori is the Mumford curve ([III, Theorem 3.21], [III, Proposition 3.24]), so a semiregular complex with Chern character $N\omega$ on the square of a Mumford fourfold would only have to deform along the curve, whose members are all algebraic.

Remark 3.7 (Two finite statements). After [I, Theorem 16.9] and Theorem 2.3 the Hodge conjecture for abelian varieties of Weil type follows from either of the following, each of which is a statement about explicit finite data and neither of which mentions a general point. The first is settled by [I, Theorem 16.17], and we record it because it is the form in which the question first presents itself. The second is false as stated here (Theorem 3.5), and its polarised form, (P2′) of Remark 3.6, is computed in closed form below.

(P1) *A bound.* At every point of the orbit $G(\mathbb{Q}) \cdot s_0$ the class ω is represented by a cycle whose Hilbert data comes from one finite set. [I, Theorem 16.15] shows that the transport of the base cycle does not satisfy this, for any cycle whose components have finite stabiliser, and [I, Theorem 16.17] shows that, read on an arbitrary representative, the statement is equivalent to $\mathbf{W}(K, n, \delta)$ itself. It is therefore a reformulation of the conjecture for the family, and we record it as such rather than as a criterion.

(P2) *A rank.* The semiregularity map of a perfect complex with the Chern character of Proposition 2.1 at one base point is injective. In this form the statement is false (Theorem 3.5); its polarised form, in Remark 3.6, allows the Chern character to carry powers of the polarisation, and what follows records what the pure form asks. The source of the map is $\mathrm{Ext}^2(E, E)$ for a single object, and its target has dimension $\sum_q h^{q,q+2}(A_0) = \binom{4n}{2n-2}$, computed in [I, Proposition 9.2]. So the target is large, which is the direction that helps, and the question is the rank of the map. Full rank would force the following (Theorem 2.5): the evaluation map of Hochschild cohomology must vanish on the $4n^2$-dimensional annihilator of the Weil class, so E must be killed by the mixed block of $H^{0,2}$, must extend along every K-linear deformation, must have all products of transverse translation classes zero, and must have no endomorphism of nonzero trace; and at a base point with no abelian subvariety tangent to an eigenspace it can be a locally free sheaf on no smooth germ of its support of codimension at least two (Corollary 2.6). The representative that the construction supplies, a virtual sum of line bundles, fails at the first of these ([I, Proposition 9.10]), and any representative that is a sheaf fails at the last. Theorem 2.8 identifies that annihilator as the mixed part of a splitting $HH^1 = P \oplus Q$ into the annihilators of the two conjugate pieces of the class, and Corollary 2.9 draws the consequence: the whole Hochschild action of E factors through a ring of dimension $2^{2n+1} - 1$ inside one of dimension 2^{4n}, while $\dim \mathrm{Ext}^2(E, E) \geq 2n(2n-1)$ for every representative, semiregular or not. Theorem 2.11 turns the rank condition into that number: equality suffices. Theorem 2.13 locates the support: at every member it has a component of codimension less than n, so no complex carried by a cycle representing ω can serve, and at a general member every component of codimension at least two has multiplicity zero. Theorem 2.14 fixes its shape: an indecomposable summand carries the class and meets the criterion alone, and at $n = 2$, and at $n = 3$ granting the compatibility of Corollary 2.9(iii), it has at least three independent endomorphisms, so no simple object serves.

Neither statement is the variational Hodge conjecture: (P1) says nothing about deformations, and (P2) is an infinitesimal statement at one point, the global propagation being supplied by [I, Theorem 16.4], which is what [I, Remark B.11] identifies as missing. Nor is either implied by the results of [I, Section 15.5]: the base points and the cycles are constructed there, and (P1) and (P2) are questions about those same cycles that the construction does not answer. In the forms stated here both are settled: (P1) is equivalent to its conclusion, and (P2) is false. Its polarised form in Remark 3.6 is the criterion that the rest of this section computes.

3.2. **The polarised criterion in closed form.** The annihilator computations of [I, Section 16] assume that $\mathrm{ch}(E)$ is a pure Weil class, as Remark 3.6 notes. This subsection carries them out for a Chern character with components in every even degree. In degree two the answer is complete and holds for every n; in the other degrees it is a closed formula, with one correction term in the middle degree. The same computation gives the Hodge locus of the character, and with it the list of the obstructions of this paper that apply to polarised characters (Proposition 3.11, Remark 3.13). None of this constructs an object or decides whether one exists. It gives, for each shape of Chern character, the number that $\dim \mathrm{Ext}^2(E,E)$ has to reach and the exact kernel that ev_E must then have, which is the form in which a candidate can be tested.

Setup 3.8 (Polarised characters)**.** Let A be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $n \geq 2$, with polarisation θ, and let $\omega = u\alpha_+ + \overline{u}\,\alpha_-$ be a nonzero real class on the Weil line, in the notation of Theorem 2.5. For complex numbers $c_0, \dots, c_{2n}$ put

$$p(\theta) = \sum_{k=0}^{2n} c_k \theta^k, \qquad \gamma = \omega + p(\theta), \qquad \mu_m = m!\, c_m,$$

and let $\rho \in \{0,1,2,3\}$ be the rank of the $(2n-1) \times 3$ Hankel matrix $H = [\mu_{j+i}]$, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$. Write $r(\gamma)$ for the rank of contraction $HT^2(A) \to H^*(A,\mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. For $a < n \leq b$, in the basis of Theorem 2.8 adapted to an eigenbasis e_j of V_+ and f_j of V_- with e_j, f_j of type $(1,0)$ for $j < n$ and $j \geq n$ respectively, write $z_{ab} = w(e_b)w(f_a) \in V_+^{0,1} \otimes V_-^{0,1}$, $v^1_{ab} = w(e_b)c(e_a)$ and $v^2_{ab} = w(f_a)c(f_b)$ in $H^1(T_A)$, and $\pi_{ab} = c(e_a)c(f_b) \in T_+ \otimes T_-$, where w is wedge product with a $(0,1)$ class and c contraction of a $(1,0)$ class.

The matrix H is a catalecticant of the moment sequence $\mu_m = m!\, c_m$. A nonzero vector (x,λ,w) in its kernel is a linear recurrence $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ of order at most two, so $\rho \leq 2$ says that the moments satisfy one, and the shapes of Theorem 3.9(ii) are its solutions, read off from the roots of $wT^2 + \lambda T + x$. For a Chern character, $\gamma = N\omega_1 + \sum_k c_k\theta^k$ with rational c_k, and u is determined by $N\omega_1$. Since $\alpha_\pm^2 = 0$, $\int_A \omega^2 = 2|u|^2 \int_A \alpha_+\alpha_-$; scaling the eigenbases so that $\int_A \alpha_+\alpha_- = (-1)^n \int_A \theta^{2n}/(2n)!$, which is possible because the sign of $\int_A \alpha_+\alpha_-$ is $(-1)^n$ by the Hodge–Riemann relations, one has $|u|^2 = (-1)^n \frac{(2n)!}{2} \int_A \omega^2 / \int_A \theta^{2n}$. The rank ρ does not change when γ is multiplied by $e^{t\theta}$, which acts on the moments by a binomial transform, and $\omega e^{t\theta} = \omega$ because a Weil class is primitive; so ρ is an invariant of γ up to twist.

Theorem 3.9 (The polarised criterion as a number)**.** *In Setup 3.8:*

(i) *For $n \geq 3$,*

$$\mathrm{Ann}_{HT^2}(\gamma) = \bigoplus_{a<n\leq b} \{x\, z_{ab} + y_1 v^1_{ab} + y_2 v^2_{ab} + w\, \pi_{ab} \ : \ (x,\ i(y_1-y_2),\ w) \in \ker H\},$$

so that $\dim \mathrm{Ann}_{HT^2}(\gamma) = n^2(4-\rho)$ *and* $r(\gamma) = (4+\rho)n^2 - 2n$*, whatever* $u \neq 0$ *is.*

(ii) $\rho = 3$ *for a general* p*. If* $\rho = 3$*, and either* $n \geq 3$ *or* $n = 2$ *and* $e = 0$ *in the notation of (iii), then* $\mathrm{Ann}_{HT^2}(\gamma)$ *is exactly the tangent space* $T_{[A]}\mathcal{D}_{n,n}$ *of the polarised Weil family, spanned by the* n^2 *vectors* $v^1_{ab} + v^2_{ab}$*, inside* $H^1(T_A)$*, with no component in* $H^{0,2}$ *or in* $H^0(\bigwedge^2 T_A)$*; and* $r(\gamma) = \binom{4n}{2} - n^2 = 7n^2 - 2n$*. Moreover* $\rho \leq 2$ *exactly when* p *is one of* $Ae^{s\theta} + Be^{t\theta}$*,* $(A + B\theta)e^{t\theta}$*,* $Ae^{t\theta} + B\theta^{2n}$*,* $A\theta^{2n-1} + B\theta^{2n}$*, and* $\rho \leq 1$ *exactly when* $p \in \mathbb{C}e^{t\theta} \cup \mathbb{C}\theta^{2n}$*; the pure case* $\rho = 0$ *gives back* $r = 2n(2n-1)$ *of Corollary 2.9(i).*

(iii) *For* $n = 2$*,* $\dim \mathrm{Ann}_{HT^2}(\gamma) = 4(4-\rho) + e$ *with* $e = \dim \ker((HK)^2 - |u|^2) \leq 2$ *and* K *the antidiagonal matrix with entries* $1, -2, 1$*; so* $r(\gamma) \in \{12, 16, 18, 20, 22, 23, 24\}$*, and* $e > 0$ *exactly when* $X(X-I)^2 = 4J^2$*, where* $X = |u|^2$ *and* I*,* J *are the two invariants of the binary quartic* $\sum_m \binom{4}{m}\mu_m x^{4-m}y^m$*.*

(iv) *Let* E *be a perfect complex on* A *with* $\mathrm{ch}(E) = \gamma$ *and* $N \neq 0$*. Then* $\dim \mathrm{Ext}^2(E,E) \geq r(\gamma)$ *with no hypothesis on* σ_E*. If equality holds, then* ev_E *is onto with kernel* $\mathrm{Ann}_{HT^2}(\gamma)$ *and* σ_E *is injective, so that* (P2′) *holds at that point and, by Remark 3.6 and Theorem 2.3, the Hodge conjecture holds for every member of the family with full Hodge group. Conversely, if* σ_E *is injective then* $\ker \mathrm{ev}_E = \mathrm{Ann}_{HT^2}(\gamma)$*.*

(v) *If* $p \in \mathbb{C}e^{t\theta} + \mathbb{C}\theta^{2n}$ *with* $t\theta$ *the class of a line bundle* L*, a set of shapes with* $\rho \leq 2$ *that contains* $\mathbb{C}\theta^{2n}$*, then no perfect complex* E *with* $\mathrm{ch}(E) = \gamma$ *has injective* σ_E*.*

Proof. (i) Let the torus $(\mathbb{C}^\times)^{2n}$ act by $e_j \mapsto \tau_j e_j$, $f_j \mapsto \tau_j^{-1} f_j$. It preserves the Hodge decomposition and θ, and scales $\alpha_\pm$ by $(\prod_j \tau_j)^{\pm 1}$; contraction is equivariant. A weight of HT^2 has at most two nonzero coordinates, so for $n \geq 3$ two such weights never differ by a nonzero multiple of $(1, \dots, 1)$, and the images of one weight component in the θ-part and in the two α-parts land in distinct weight spaces. Hence $\mathrm{Ann}(\gamma) = \mathrm{Ann}(p(\theta)) \cap \mathrm{Ann}(\alpha_+) \cap \mathrm{Ann}(\alpha_-)$, and the last two meet in $P \wedge Q$ by Theorem 2.8(ii). The space $P \wedge Q$ is the sum over $a < n \leq b$ of the four-dimensional weight spaces spanned by $z_{ab}, v^1_{ab}, v^2_{ab}, \pi_{ab}$, and $\mathrm{Ann}(p(\theta))$ is torus-stable, so the computation splits pair by pair. Put $m = e_b \wedge f_a$ and $\theta' = \theta - t_a - t_b$, where $t_j = is_j e_j \wedge f_j$ with $s_j = \pm 1$ the sign that makes $\theta = \sum_j t_j$ positive. Since the t_j are even with $t_j^2 = 0$, direct evaluation gives

$$z \lrcorner \theta^k = m\,\theta'^k, \quad v^1 \lrcorner \theta^k = ik\,m\,\theta'^{k-1}, \quad v^2 \lrcorner \theta^k = -ik\,m\,\theta'^{k-1}, \quad \pi \lrcorner \theta^k = k(k-1)\,m\,\theta'^{k-2},$$

so for $\xi = xz + y_1 v^1 + y_2 v^2 + w\pi$ and $\lambda = i(y_1 - y_2)$,

$$\xi \lrcorner p(\theta) = m \wedge \sum_{j=0}^{2n-2} \frac{\theta'^j}{j!}(x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2}).$$

The classes $m \wedge \theta'^j$, $0 \leq j \leq 2n-2$, are nonzero and of distinct degrees, so ξ kills $p(\theta)$ exactly when $(x, \lambda, w) \in \ker H$. The map $(x, y_1, y_2, w) \mapsto (x, \lambda, w)$ is onto with kernel spanned by $v^1 + v^2$, which gives $1 + (3 - \rho)$ dimensions for each of the n^2 pairs. (ii) The vector $v^1_{ab} + v^2_{ab}$ is K-linear and kills θ, and these n^2 vectors span the tangent space of the polarised family by [I, Proposition 9.14]; when $\rho = 3$ they are all of the annihilator. If $\rho \leq 2$, a nonzero $(x, \lambda, w) \in \ker H$ turns the equations $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ into a linear recurrence of full rank on $\mu_0, \dots, \mu_{2n}$, whose two-dimensional space of solutions is spanned by the listed shapes according to the roots of $wT^2 + \lambda T + x$; $\rho \leq 1$ is the intersection of two such spaces. (iii) At $n = 2$ the argument of (i) fails for exactly one pair of weights, $\epsilon_2 + \epsilon_3$ in $\bigwedge^2 P$ and $-\epsilon_0 - \epsilon_1$ in $\bigwedge^2 Q$, whose difference is $(1, 1, 1, 1)$; these give one coupled block of size eight, whose kernel is isomorphic to $\ker(X - T)$ with T

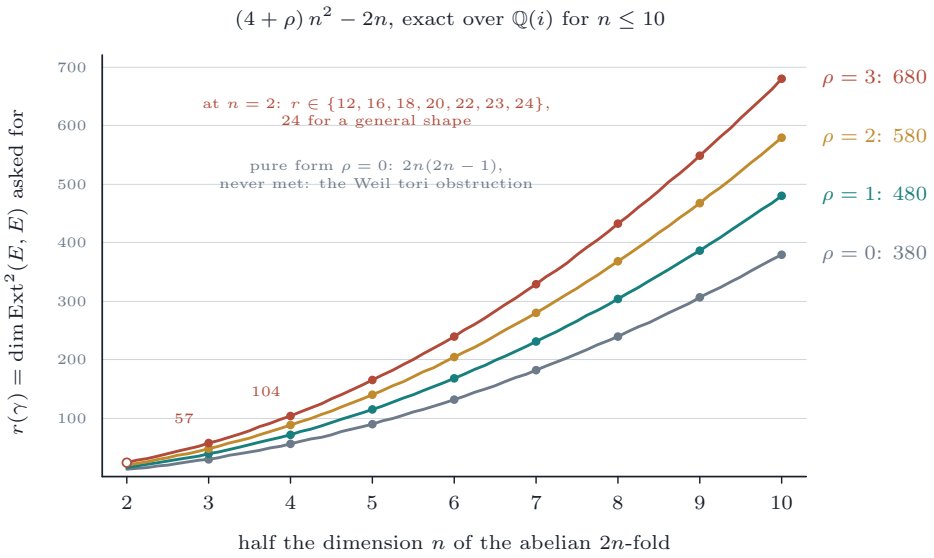


FIGURE 2. The dimension $r(\gamma) = (4+\rho)n^2 - 2n$ that $\mathrm{Ext}^2(E,E)$ must have for a complex with Chern character $\gamma = N\omega_1 + \sum_k c_k\theta^k$ to meet the polarised criterion, for the four Hankel ranks ρ of Theorem 3.9; the dots mark the values at $n = 2, \dots, 10$, and at $n = 10$ the space HT^2 has dimension 780. The pure form, $\rho = 0$, is the bottom curve $2n(2n-1)$, which no complex attains (Theorem 3.5); a general shape lies on the top curve $7n^2 - 2n$. At $n = 2$ the weight argument fails for one pair of weights and seven values occur.

conjugate on its image to $(HK)^2$, and the characteristic polynomial of HK is $y^3 - Iy - 2J$, which is traceless, so $e \le 2$. (iv) is the argument of Theorem 2.11(i)–(ii) with $\mathrm{Ann}_{HT^2}(\gamma)$ in place of $P \wedge Q$: $\sigma_E \circ \mathrm{ev}_E$ is contraction into γ, so $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\gamma)$. (v) If $\gamma - \omega$ has components in degrees 0 and $4n$ only, the proof of Theorem 3.5 applies verbatim. If $p = c\,e^{t\theta} + c'\theta^{2n}$ with $t\theta = c_1(L)$, then $\mathrm{ch}(E \otimes L^{-1}) = \gamma e^{-t\theta} = c + \omega + c'\theta^{2n}$, because $\omega\theta = 0$ and $\theta^{2n+1} = 0$, and $\sigma_{E\otimes L^{-1}} = e^{-c_1(L)} \cdot \sigma_E$ is injective exactly when σ_E is, so the first case applies to $E \otimes L^{-1}$. ∎

For a general shape, then, the polarised criterion asks of one complex that $\dim \mathrm{Ext}^2(E,E)$ be 24 at $n = 2$, 57 at $n = 3$ and 104 at $n = 4$, against 12, 30 and 56 for the pure form, and injectivity then forces nothing beyond first-order extension along the polarised family, which is what the criterion is for. The larger number comes from the components $c_k\theta^k$: every direction of HT^2 other than the n^2 directions of the family moves γ off the Hodge classes, so an object meeting the bound has ev_E injective on a complement of $T_{[A]}\mathcal{D}_{n,n}$. Figure 2 plots the number against n for the four Hankel ranks.

Proposition 3.10 (The Hochschild profile of a polarised character)**.** *In Setup 3.8, let ρ_k be the rank of contraction $HT^k(A) \to H^*(A,\mathbb{C})$ into γ, put $C_l = l!\,c_l$, let r' be the rank of the $(n+1)\times(n+1)$ Hankel matrix $H' = [C_{p+q}]$, and $d = \dim\ker\big((K'H')^2 - |u|^2\big)$ with $K' = \mathrm{antidiag}\big((-1)^a\binom{n}{a}\big)$.*

(i) $\rho_0 = \rho_{2n} = 1$, $\rho_k = 0$ *for* $k > 2n$, *and for* $1 \le k \le 2n-1$, $k \ne n$,

$$\rho_k = 2\binom{2n}{k} + M_k \min(k+1,\, 2n+1-k,\, r'), \qquad M_k = \sum_{\substack{i+j=k \\ 1\le i,j\le n-1}} \binom{n}{i}\binom{n}{j};$$

in the middle degree the same formula holds with d *subtracted. In particular* $\rho_k = \rho_{2n-k}$, *and the general profiles are* $(1,8,24,8,1)$ *at* $n=2$ *and* $(1,12,57,112,57,12,1)$ *at* $n=3$.

(ii) $\chi(E,E) = V\,Q(c) + W$ *for a perfect complex with* $\mathrm{ch}(E) = \gamma$, *where* $V = \int_A \theta^{2n} > 0$, $W = (-1)^n \int_A \omega^2 > 0$ *and* $Q(c) = \sum_k (-1)^k c_k c_{2n-k}$; *and* $\chi(E,E)$ *is even.*

(iii) *Granting the compatibility of Corollary 2.9(iii) in every degree,* $\dim \mathrm{Ext}^k(E,E) \ge \rho_k$. *At* $n=2$, *at the points with* $d=1$ *one has* $\rho_2 = 23$, *which is odd, so* $\dim \mathrm{Ext}^2(E,E) = r(\gamma)$ *cannot hold there, since* $\chi(E,E)$ *is even and Serre duality gives* $\chi(E,E) \equiv \dim \mathrm{Ext}^2(E,E) \bmod 2$.

Proof. (i) Split $HT^1 = Q \oplus P$ as in Theorem 2.8, so that $HT^k = \bigoplus_{i+j=k} \bigwedge^i Q \otimes \bigwedge^j P$ and Q, P act on the two tensor factors of $H^*(A) = H_+ \otimes H_-$ separately. A mixed monomial kills ω, and on $p(\theta)$ the block of mixed monomials with fixed index sets I, J has entries $C_{s+t+m+m'}$ depending only on $s+t$ and $m+m'$, so its rank is that of a catalecticant of the binary form $\sum_k c_k x^k y^{2n-k}/(2n-k)!$, which is $\min(k+1, 2n+1-k, r')$ by the classical Hilbert function of an apolar ideal. The pure sources $\bigwedge^k Q$ and $\bigwedge^k P$ kill α_- and α_+ respectively and contract the other class injectively (Theorem 2.8(iii)). To see where their images go, use the torus of the proof of Theorem 3.9(i) and put $I_Q = \{j < n\}$, $I_P = \{j \ge n\}$. The basis vectors $w(f_j), c(e_j)$ of Q have weight $-\epsilon_j$ for $j \in I_Q$, the vectors $w(e_j), c(f_j)$ of P have weight ϵ_j for $j \in I_P$, θ has weight 0 and $\alpha_\pm$ has weight $\pm\sum_j \epsilon_j$. If $\xi \in \bigwedge^k Q$ is a monomial, $k < 2n$, and another monomial η of HT^k sends $p(\theta)$ or α_- to the weight of $\xi \lrcorner \alpha_+$, which is $+1$ at every $j \in I_P$, then it is not α_-, which would need two factors from P at every $j \in I_P$; so η has one factor from P for each $j \in I_P$, ξ has weight at most -1 at every $j \in I_Q$, and $k \ge n$. For $k = n$ this forces ξ and η to lie in the spans $D_Q \subset \bigwedge^n Q$ and $D_P \subset \bigwedge^n P$ of the 2^n monomials with one factor for each index. The same holds with P and Q exchanged. A mixed monomial kills ω, so for $k < n$ the matrix of contraction is block triangular, the pure blocks injective onto targets of their own, and ρ_k is $2\binom{2n}{k}$ plus the rank of the mixed blocks; for $k > n$ the formula follows from [III, Lemma 3.33], which also gives $\rho_k = \rho_{2n-k}$.

For $k = n$, a pure monomial outside $D_Q \cup D_P$ repeats an index, so it kills 1 and t_j at that index and hence $p(\theta)$, and keeps a target of its own; no mixed monomial reaches the weights $\sum_{j\in I_P} \epsilon_j$ and $-\sum_{j\in I_Q} \epsilon_j$ of the images of $D_Q \oplus D_P$, since it would need n factors from one side. So ρ_n is the value of the formula minus the dimension of the kernel of the block $\beta\colon D_Q \oplus D_P \to T_1 \oplus T_2$, T_1 and T_2 being the targets of those two weights. For $R \subseteq I_Q$ and $J \subseteq I_P$ let $X_R = \prod_{j\in I_Q} z_j$ with $z_j = w(f_j)$ for $j \in R$ and $c(e_j)$ otherwise, and $Y_J = \prod_{j\in I_P} z_j$ with $z_j = c(f_j)$ for $j \in J$ and $w(e_j)$ otherwise, the factor of largest index acting first, and write $\alpha_+ = a\, e_0 \cdots e_{2n-1}$, $\alpha_- = b\, f_0 \cdots f_{2n-1}$. Since $\theta^{2n}/(2n)! = \prod_j t_j = \prod_j s_j\, e_0 \cdots e_{2n-1} f_0 \cdots f_{2n-1}$, the normalisation of Setup 3.8 gives $ab = (-1)^n \prod_j s_j$. As the t_j are even with $t_j^2 = 0$, $p(\theta) = \sum_S \mu_{|S|} \prod_{j\in S} t_j$. Each operator acts on one factor of $H^*(A) = \bigotimes_j \bigwedge^*\langle e_j, f_j\rangle$ with the Koszul sign of the factors before it, and in each of the four products below the parities of those factors do not depend on R or J; the signs are $+1$ on $p(\theta)$, $\epsilon_1 = (-1)^{n(n-1)/2}$ for X_R on α_+ and $\epsilon_2 = (-1)^{n(3n-1)/2}$ for Y_J on α_-. With $\tau_R = e_n \cdots e_{2n-1} \prod_{j\in R} t_j$ and $\tau'_{R'} = f_0 \cdots f_{n-1} \prod_{j\in R'} t_j$, and using

$c(f_j)t_j = -is_je_j$, $c(e_j)t_j = is_jf_j$ and $e_jf_j = -is_jt_j$, one finds

$$X_R \lrcorner \alpha_+ = \epsilon_1 a \prod_{j\in R}(is_j)\,\tau_R, \qquad Y_J \lrcorner p(\theta) = \sum_R \mu_{|J|+|R|} \prod_{j\in J}(-is_j)\,\tau_R,$$

$$Y_J \lrcorner \alpha_- = \epsilon_2 b \prod_{j\in I_P\setminus J}(-is_j)\,\tau'_{I_P\setminus J}, \qquad X_R \lrcorner p(\theta) = \sum_{R'} \mu_{n-|R|+|R'|} \prod_{j\in I_Q\setminus R}(is_j)\,\tau'_{R'}.$$

For $x = \sum_R x_R X_R$ and $y = \sum_J y_J Y_J$ put $\tilde{x}_R = \prod_{j\in R}(is_j)\,x_R$ and $\tilde{y}_J = \prod_{j\in J}(-is_j)\,y_J$, and let $\Pi_P = \prod_{j\in I_P}(-is_j)$, $\Pi_Q = \prod_{j\in I_Q}(is_j)$. Then $\beta(x,y) = 0$ says

$$\epsilon_1 a\,u\,\tilde{x}_R = -\sum_J \mu_{|J|+|R|}\,\tilde{y}_J, \qquad \epsilon_2 b\,\overline{u}\,\Pi_P(-1)^{|J|}\tilde{y}_J = -\Pi_Q \sum_R \mu_{2n-|R|-|J|}(-1)^{|R|}\tilde{x}_R$$

for all R and J. So $\tilde{x}_R = g_{|R|}$ and $(-1)^{|J|}\tilde{y}_J = h_{|J|}$ depend only on $|R|$ and $|J|$, and with $D = \operatorname{diag}((-1)^c\binom{n}{c})$ and the exchange matrix J_0, so that $DJ_0 = K'$ and $J_0DJ_0 = (-1)^nD$, the conditions read $\epsilon_1 a\,u\,g = -H'Dh$ and $\epsilon_2 b\,\overline{u}\,\Pi_P h = -\Pi_Q J_0 H' J_0 Dg$. Eliminating h gives $(H'K')^2 g = \lambda g$ with $\lambda = (-1)^n\epsilon_1\epsilon_2\,ab\,|u|^2\,\Pi_P/\Pi_Q$, and $\epsilon_1\epsilon_2 = (-1)^n$, $ab = (-1)^n\prod_j s_j$, $\Pi_P/\Pi_Q = (-1)^n\prod_j s_j$, so $\lambda = |u|^2$. Conversely an eigenvector g determines h, x and y. Hence $\dim\ker\beta = \dim\ker\big((H'K')^2 - |u|^2\big) = d$, as $H'K'$ and $K'H'$ are conjugate.

(ii) $\operatorname{ch}(E)^\vee$ changes the sign of the components of odd codegree, $\omega\theta = 0$ kills the cross terms, and Hodge–Riemann gives $W > 0$; parity holds because the Chern character of a perfect complex on a complex torus is integral, and the intersection form on $\bigwedge^{2n} H^1(A,\mathbb{Z})$ pairs each monomial only with its complement, so $\int x^2$ is even. (iii) is the argument of Corollary 2.9(iii). ∎

The profile also shows what the polarised criterion leaves unconstrained. For the pure form, Theorem 2.14(iv) derives $\dim\operatorname{End}(E) \geq 3$ at $n = 2$ and $n = 3$ from $\chi(E,E) > 0$. For a polarised character, $\chi(E,E) = VQ(c) + W$ can have either sign, and when it is small enough the Euler characteristic forces nothing. At $n = 2$, in the model of [I, Setup 4.1] with $d = 1$, where $\int\eta^4 = 24$ and $\int\omega_1^2 = 8$ (see the proof of Proposition 3.11), the character $\omega_1 + p(\eta)$ with $c = (2,1,0,1,-1)$ has $Q(c) = 2c_0c_4 - 2c_1c_3 + c_2^2 = -6$, so $\chi(E,E) = -144 + 8 = -136$, and a simple object is not excluded.

Proposition 3.11 (The Hodge locus of a polarised character)**.** (i) *Let T be a complex torus, let x be a real class of type (k,k) on T, and for $v \in \operatorname{Hom}(H^{1,0}, H^{0,1}) = H^1(T_T)$ let J_v be the complex structure on the underlying real torus with $H^{1,0}_v = (1+v)H^{1,0}$. Then x is of type (k,k) for J_v if and only if $v \lrcorner x = 0$. So in this chart the Hodge locus of x is the linear space $\operatorname{Ann}_{H^1(T_T)}(x)$, and it coincides with its first-order locus.*

(ii) *Let J be any complex structure on the real torus underlying A. If θ^k is of type (k,k) for J for some $1 \leq k \leq 2n-1$, then θ is of type $(1,1)$ for J.*

(iii) *In Setup 3.8 with $n \geq 3$, the germ at A of the Hodge locus of γ among all complex structures is the polarised Weil family $\mathcal{D}_{n,n}$, of dimension n^2, if $c_k \neq 0$ for some $1 \leq k \leq 2n-1$, and the family of all K-linear deformations, of dimension $2n^2$, otherwise. In the first case the connected component of the Hodge locus through A is $\mathcal{D}_{n,n}$, and all its members are abelian varieties polarised by θ.*

(iv) *For $n = 2$ the same holds, with one exception: if $c_1 = c_3 = 0$ and $c_2^2\int_A\theta^4 = \frac{3}{4}\int_A\omega^2$, the germ has dimension 5. In the model of [I, Setup 4.1] this is the character $N\omega_1 \pm \frac{d}{2}N\eta^2 + c_0 + c_4\eta^4$; with rational data the ratio is attained exactly in the families of trivial discriminant.*

Proof. (i) Write D_v for the derivation of $\bigwedge^* H^1$ extending v. Since $v^2=0$ on H^1, $\bigwedge^*(1+v)=\exp(D_v)$, so the Hodge filtration of J_v is $F_v^p=\exp(D_v)F^p$. Now $\exp(-D_v)x=\sum_m(-1)^m D_v^m x/m!$ with $D_v^m x$ of type $(k-m,k+m)$, so $x\in F_v^k$ exactly when $D_v x=v\lrcorner x=0$; and a real class lies in $F_v^k\cap\overline{F_v^k}=H_v^{k,k}$ as soon as it lies in F_v^k. (ii) Let $t\mapsto t_J$ be the circle acting on $H^{p,q}$ by t^{p-q}. It acts on $\bigwedge^* H^1(A,\mathbb{R})$ by algebra automorphisms and fixes θ^k, so the path $t\mapsto t_J\theta$ lies in $\{y\in\bigwedge^2 H^1(A,\mathbb{R}) : y^k=\theta^k\}$. At a nondegenerate y the differential $h\mapsto k\,y^{k-1}h$ of $y\mapsto y^k$ is injective on $\bigwedge^2$ when $2+(k-1)\le 2n$, by the linear hard Lefschetz theorem for the symplectic form y on a space of dimension $4n$: for a nondegenerate $y\in\bigwedge^2 U^\vee$ on a real space U of dimension $2m$, multiplication by y^j is injective from $\bigwedge^p$ to $\bigwedge^{p+2j}$ whenever $p+j\le m$, because y generates an $\mathfrak{sl}_2$-action on $\bigwedge^* U^\vee$ in which $\bigwedge^p$ has weight $p-m$; here $m=2n$, $p=2$ and $j=k-1$. So near θ that set is discrete, the path is constant, and θ is of type $(1,1)$. The bound $k\le 2n-1$ is sharp, since θ^{2n} is a multiple of the volume form for every complex structure. (iii) By (i), applied to each homogeneous component, the Hodge locus of γ in the chart is $\mathrm{Ann}_{H^1(T_A)}(\gamma)$, the part of $\mathrm{Ann}_{HT^2}(\gamma)$ in $H^1(T_A)$. In the description of Theorem 3.9(i) that part is $x=w=0$ with $(0,\lambda,0)\in\ker H$; the vector $(0,1,0)$ lies in $\ker H$ exactly when $\mu_1=\cdots=\mu_{2n-1}=0$, so the part is spanned by the n^2 vectors $v^1_{ab}+v^2_{ab}$, which span $T_{[A]}\mathcal{D}_{n,n}$, when some c_k with $1\le k\le 2n-1$ is nonzero, and by all v^1_{ab},v^2_{ab}, the K-linear deformations, otherwise. A germ of submanifold of dimension n^2 inside a linear space of the same dimension is that space. On $\mathcal{D}_{n,n}$ the class θ stays of type $(1,1)$ and nondegenerate, so the signature of the hermitian form it defines is constant and every member is polarised by θ. The same argument at every member shows that $\mathcal{D}_{n,n}$ is open in the Hodge locus; it is closed in the space of complex structures, since K-linearity is a closed condition and a limit of positive hermitian forms that stays nondegenerate stays positive; and it is connected. (iv) At $n=2$ the weight argument of Theorem 3.9(i) fails for one pair of weights. The other weights contribute as in (iii): the K-linear deformations, of dimension 8, when $c_1=c_2=c_3=0$, and the tangent space of the polarised family, of dimension 4, otherwise. The remaining pair of weights couples $\xi'\in\mathrm{Hom}(V_+^{1,0},V_-^{0,1})$ with $\xi''\in\mathrm{Hom}(V_-^{1,0},V_+^{0,1})$. Choose bases $x_{\pm,k}$ of $V_\pm^{1,0}$ and $y_{\pm,k}$ of $V_\pm^{0,1}$, $k=1,2$, with $\theta=\sum_k(x_{+,k}y_{-,k}+x_{-,k}y_{+,k})$, put $\beta_\pm=x_{\pm,1}x_{\pm,2}y_{\pm,1}y_{\pm,2}$, a generator of $\bigwedge^4 V_\pm$, and write $\omega=u_+\beta_+ + u_-\beta_-$. The component of $(\xi'+\xi'')\lrcorner\gamma$ of degree 3 in V_- and 1 in V_+ is $u_-\,\xi''\lrcorner\beta_- + c_2\,\xi'\lrcorner\theta^2$, because $c_1=c_3=0$ and $\xi'\lrcorner\theta^4=4\theta^3(\xi'\lrcorner\theta)=0$; the component of degree 3 in V_+ is the same with $+$ and $-$ exchanged. Here $\xi'\lrcorner\theta$ lies in the line spanned by $y_{-,1}y_{-,2}$, so $\xi'\lrcorner\theta^2=2\theta\,(\xi'\lrcorner\theta)$ is a multiple of $\theta\,y_{-,1}y_{-,2}$. The map $\xi''\mapsto\xi''\lrcorner\beta_-$ is injective, and $\theta\,y_{-,1}y_{-,2}=\xi''_0\lrcorner\beta_-$ for $\xi''_0(x_{-,1})=-y_{+,2}$, $\xi''_0(x_{-,2})=y_{+,1}$, which has $\xi''_0\lrcorner\theta=2y_{+,1}y_{+,2}$. With ξ'_0 defined in the same way, a solution has $\xi'=a\xi'_0$ and $\xi''=b\xi''_0$ with $u_+a+4c_2b=0$ and $4c_2a+u_-b=0$. So the annihilator meets this pair in a line when $16c_2^2=u_+u_-$ and in zero otherwise; since $\theta^4=24\,\beta_+\beta_-$ and $\omega^2=2u_+u_-\,\beta_+\beta_-$, the condition is $c_2^2\int\theta^4=\frac{3}{4}\int\omega^2$. In the model, where $\int\theta^4$ and $\int\omega^2$ are 24 and $2|u|^2$ times the same volume, the ratio reads $|u|=4|c_2|$. In the model of [I, Setup 4.1], $\eta^4=4!\prod_j\epsilon_j x_j y_j$ gives $\int\eta^4=24$; in ω_1^2 only the products $w^{(T)}w^{(T^c)}$ survive, and each of the eight with $|T|$ even contributes d^2, so $\int\omega_1^2=8d^2$; likewise $\int\omega_2^2=8d$, and $\omega_1\omega_2=0$ because T and T^c have the same parity. This gives $c_2=\pm dN/2$. For a general rational Weil class $\omega=x\omega_1+y\omega_2$ and hermitian form $\mathrm{diag}(-a_1,-a_2,a_3,a_4)$ one has $\int\omega^2=8d\,\mathrm{N}_{K/\mathbb{Q}}(y+x\sqrt{-d})$ and $\int\theta^4=24\,a_1a_2a_3a_4$, so the ratio asks $(c_2/N)^2=d\,\mathrm{N}_{K/\mathbb{Q}}(y+x\sqrt{-d})/(4a_1a_2a_3a_4)$, a rational square exactly when $a_1a_2a_3a_4$ is a norm from K. ∎

The argument of Theorem 3.5 therefore does not extend to the polarised form. A semiregular complex deforms along the Hodge locus of its Chern character. For the pure form that locus is the whole $2n^2$-dimensional family of K-linear tori, whose very general member is not algebraic; for a polarised character with some $c_k \neq 0$, $1 \leq k \leq 2n-1$, it is the polarised family itself, all of whose members are abelian varieties, and the argument through Weil tori has nothing to act on. This concerns the argument run on E itself: an autoequivalence can still remove a component, and a line bundle twist is how the argument reaches the polarised characters of Theorem 3.9(v). The only other case it can reach is the exceptional ratio of (iv), where the Hodge locus leaves the polarised family.

Remark 3.12 (The exceptional ratio at $n = 2$). At $n = 2$, with $c_1 = c_3 = 0$ and $c_2^2 \int \theta^4 = \frac{3}{4} \int \omega^2$, put $\gamma_2 = \omega + c_2\theta^2$, and let $\mathfrak{g}$ be its stabiliser in $\mathfrak{gl}(H^1(A, \mathbb{R}))$. In the notation of the proof of Proposition 3.11(iv), put $(e_1, \dots, e_4) = (x_{+,1}, x_{+,2}, y_{+,1}, y_{+,2})$ and $(f_1, \dots, f_4) = (y_{-,1}, y_{-,2}, -x_{-,1}, -x_{-,2})$; then $\theta = \sum_j e_j f_j$, $\omega = u_+ e_{1234} + u_- f_{1234}$ with $e_{1234} = e_1 \wedge \dots \wedge e_4$, and the ratio is $16c_2^2 = u_+ u_-$. Passing to the basis se_j, $s^{-1}f_j$ with $s^8 = u_+/u_-$ fixes θ and makes the two coefficients of ω equal, and passing from f_j to $\sqrt{-1}f_j$, if necessary, changes the sign of θ^2 and fixes f_{1234}. So over $\mathbb{C}$ the class γ_2 is a nonzero multiple of

$$\tfrac{1}{2}(e_{1234} + f_{1234}) - \tfrac{1}{8}\Big(\textstyle\sum_j e_j f_j\Big)^2,$$

which is the Cayley form $\frac{1}{2}\omega_0^2 + \Re\Omega_0$ of $\mathbb{C}^4$, with $\omega_0 = \frac{\sqrt{-1}}{2} \sum dz_j \wedge d\bar{z}_j$ and $\Omega_0 = dz_1 \wedge \dots \wedge dz_4$, written in the basis $e_j = dz_j$, $f_j = d\bar{z}_j$ [HL82]. The stabiliser of the Cayley form in $\mathrm{GL}(8, \mathbb{R})$ is $\mathrm{Spin}(7)$, acting on $\mathbb{R}^8$ by its spin representation, and its invariants in $\bigwedge^* \mathbb{R}^8$ are spanned by 1, the form and the volume form [Bry87]. A stabiliser in $\mathfrak{gl}$ is cut out by linear equations, so it commutes with extension of scalars, and $\mathfrak{g} \otimes \mathbb{C} \cong \mathfrak{so}(7, \mathbb{C})$ acts on $H^1(A, \mathbb{C})$ by the spin representation, irreducibly, with invariants in $\bigwedge^* H^1$ spanned by 1, γ_2 and the class of a point. At an ordinary ratio the stabiliser is the Lie algebra $\mathfrak{su}(2, 2)$ of the Hodge group, of dimension 15, which also fixes γ_2 here.

The real form is $\mathfrak{so}(4, 3)$. The spin representation of $\mathfrak{so}(7, \mathbb{C})$ carries an invariant symmetric form, unique up to a scalar by irreducibility, so $\mathfrak{g}$ preserves a real symmetric form B on $H^1(A, \mathbb{R})$. On $H^1(A, \mathbb{C}) = V_+ \oplus V_-$ the subalgebra $\mathfrak{su}(2, 2) \otimes \mathbb{C} = \mathfrak{sl}(V_+)$ acts on V_+ and dually on V_-, and its only invariant bilinear forms pair V_+ with V_-; so B is a multiple of the real part of the hermitian form of θ, of signature $(4, 4)$, and $\mathfrak{g} \subseteq \mathfrak{so}(4, 4)$. The real forms of $\mathfrak{so}(7, \mathbb{C})$ are the $\mathfrak{so}(p, q)$ with $p + q = 7$, and the maximal compact subalgebra $\mathfrak{so}(p) \oplus \mathfrak{so}(q)$ lies in a maximal compact subalgebra of $\mathfrak{so}(4, 4)$, which is $\mathfrak{so}(4) \oplus \mathfrak{so}(4) \cong \mathfrak{su}(2)^4$. A simple subalgebra of $\mathfrak{su}(2)^4$ projects injectively to a factor, so it has dimension 3; hence $p, q \leq 4$, and $\{p, q\} = \{4, 3\}$. So γ_2 is the invariant 4-form of $\mathrm{Spin}(4, 3)$, a split form of the Cayley form.

The Hodge decomposition of $\mathfrak{g}$ has dimensions 5, 11, 5. Indeed, the complex structure J of A fixes the class γ_2 of type $(2, 2)$ and lies in the image of the Hodge group, which is connected and contained in the group of $\mathfrak{g}$; as $J^2 = -1$ is the nontrivial central element of $\mathrm{Spin}(4, 3)$, the image of J in $\mathrm{SO}(7, \mathbb{C})$ is an involution with $2k$ eigenvalues -1 and k odd, since a lift of such an involution squares to $(-1)^k$. The eigenvalue -1 of $\mathrm{Ad}\, J$ on $\mathfrak{g} \otimes \mathbb{C}$ then has multiplicity $2k(7 - 2k)$, which is 10 or 6, and it contains the (-1)-eigenspace of $\mathfrak{sl}(V_+)$, of dimension 8; so $k = 1$, and $\mathfrak{g}^{-1,1} \oplus \mathfrak{g}^{1,-1}$ has dimension 10.

The Hodge locus of γ near A is the orbit of A under the connected real group G of $\mathfrak{g}$, since that orbit has dimension $\dim \mathfrak{g}^{-1,1} = 5$ and lies in the locus, which near A is smooth of dimension 5 (Proposition 3.11(i) and (iv)). The Hodge group of a very general member T has Lie algebra an ideal of $\mathfrak{g}$, hence all of it, so the Hodge classes of T are 1,

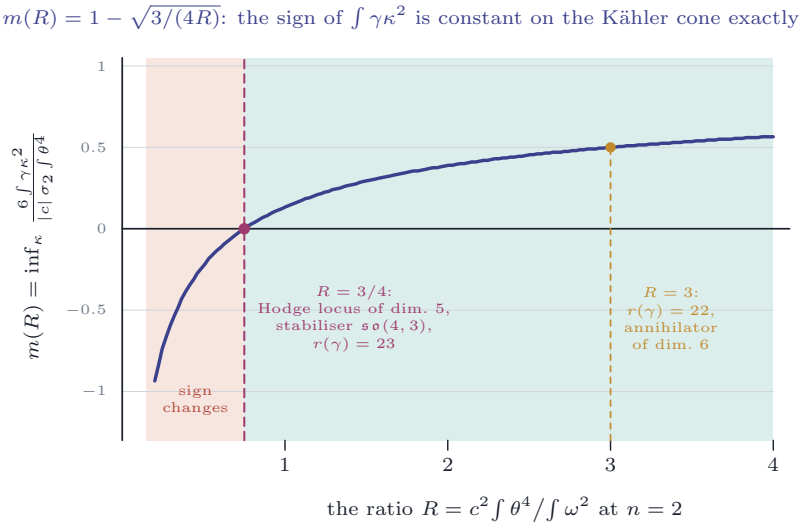


FIGURE 3. The exceptional ratio of Remark 3.12. By the identity displayed there, which is linear in γ, the infimum over the Kähler cone of $6\int\gamma\kappa^2/(|c|\,\sigma_2(h)\int\theta^4)$ is $m(R) = 1 - \sqrt{3/(4R)}$, approached at the boundary of the cone by a positive hermitian form of rank two whose Plücker coordinates satisfy $|\xi_{01}\xi_{23}| = \frac{1}{4}$. It is negative exactly for $R < 3/4$, and $R = 3/4$ is also where the Hodge locus of γ jumps from dimension 4 to 5 and its stabiliser becomes $\mathfrak{so}(4,3)$.

γ_2 and the class of a point, and $\mathrm{NS}(T) = 0$. The locus therefore contains tori that are not algebraic, and the proof of Theorem 3.5 can be started there. It breaks down at its last step: Proposition 3.4 needs a Kähler class on which the degree-four Chern character integrates to zero, and here

$$\int_A \gamma_2\,\kappa_h^2 = \frac{|c_2|}{6}\Big(\operatorname{sign}(c_2)\,\sigma_2(h) + 4\operatorname{Re}\big(\phi\det h_{01,23}\big)\Big)\int_A \theta^4, \qquad |\phi| = 1,$$

for the Kähler class $\kappa_h = i\sum h_{jk}z_j\wedge\overline{z}_k$ of a positive hermitian h in a unitary frame for θ, and the right-hand side has the sign of c_2. Indeed, writing $h = \sum_j \lambda_j w_j w_j^*$ with (w_j) orthonormal, Cauchy–Binet gives $\sigma_2(h)\pm 4\operatorname{Re}(\phi\det h_{01,23}) = \sum_{j<k}\lambda_j\lambda_k\big(1\pm 4\operatorname{Re}(\phi\,\xi_{01}\overline{\xi}_{23})\big)$ with $\xi = w_j\wedge w_k$ of norm one. The Plücker relation and the inequality of the means give $|\xi_{01}\xi_{23}| \leq \frac{1}{4}$, so each term is nonnegative, and by Parseval the brackets sum to 6 over $j<k$, so the sum is at least $6\min_j\lambda_j^2 > 0$. Since G fixes γ_2 and carries Kähler cones to Kähler cones, the same holds on every member of the locus. So at this ratio the argument excludes at most a torsion-free sheaf with $c_2 > 0$, which Bogomolov's inequality already forbids to be θ-semistable, and it excludes no complex. With $R = c_2^2\int\theta^4/\int\omega^2$, the ratio that equals $\frac{3}{4}$ here, Figure 3 draws the infimum of the ratio as a function of R. Since Markman's theorem settles every Weil fourfold, this remark concerns the shape of (P2′), not the conjecture.

Remark 3.13 (The obstructions for a polarised character). The obstruction theorems proved from [I, Section 9] on for a Chern character on the Weil line divide as follows for a polarised character. The following hold unchanged, with $r(\gamma)$ in place of $2n(2n-1)$

where a number occurs: [I, Theorem 9.20] and Corollary 4.21, whose only hypothesis is that $\mathrm{ch}(E)$ stay of Hodge type to first order along $\mathcal{D}_{n,n}$; Theorem 4.5 and Corollary 4.6, since a polarised character is a flat Hodge class in every degree; Lemma 2.12; and the first two parts of Theorem 2.11, which become Theorem 3.9(iv). The following hold only when $c_0 = \cdots = c_{2n-2} = 0$, so that $H^{0,2}$ still kills γ: [I, Theorem 9.7], [I, Proposition 9.10], [I, Corollary 9.18], [I, Theorem 9.19] and Theorem 2.5(iii). The following hold only when $\ker H$ contains a vector with $w \neq 0$, which needs $\rho \leq 2$, the vector giving a compensated class of $P \wedge Q$ with a component in $T_+ \otimes T_-$: Theorem 2.5(iv), Corollary 2.6 and Theorem 2.13. The following do not extend: Theorem 3.5, outside Theorem 3.9(v); the positivity of $\chi(E,E)$ in [I, Proposition 9.11] and with it Theorem 2.14(i)–(iii), by Proposition 3.10(ii); and [III, Theorem 2.32]. So for a general shape, $\rho = 3$, what these results exclude is a line bundle summand off the polarisation line and a Chern character effective in the exponential basis, and nothing else; Proposition 3.14 below extends the first to summands with a scalar Atiyah class. An object with no such summand, whose Chern character is not effective, and with $\dim \mathrm{Ext}^2(E,E) = 7n^2 - 2n$, is not excluded by anything proved here, and none is known; by Lemma 3.16 below it may be taken indecomposable.

Proposition 3.14 (Summands with a scalar Atiyah class)**.** *Let A be of $(K,-1,n)$-Weil type with $n \geq 2$, let $T = T_{[A]}\mathcal{D}_{n,n}$, and let E be a perfect complex on A with $v \lrcorner \mathrm{ch}(E) = 0$ for every $v \in T$. Suppose E has a direct summand $F[p]$ with $\mathrm{At}(F) = \delta \cdot \mathrm{id}_F$, where δ is a real class of type $(1,1)$ that is not a real multiple of θ, and $\mathrm{rk}\, F \neq 0$. Then there is $v \in T$ with $\mathrm{ev}_E(v) \neq 0$ and $\sigma_E(\mathrm{ev}_E(v)) = 0$. In particular no finite direct sum of shifted bundles $\pi_* L$, π an isogeny, among them the simple semi-homogeneous bundles, has a polarised character γ with $N \neq 0$ and injective semiregularity map.*

Proof. As in [I, Theorem 9.20], $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$ on $H^1(T_A)$ and the Atiyah class of a direct sum is block diagonal, so the component of $\mathrm{ev}_E(v)$ in $\mathrm{Ext}^2(F,F)$ is $(v \lrcorner \delta) \cdot \mathrm{id}_F$; the map $z \mapsto z \cdot \mathrm{id}_F$ on $H^{0,2}$ is injective because $\mathrm{tr}(z \cdot \mathrm{id}_F) = \mathrm{rk}(F)\, z$, and some $v \in T$ has $v \lrcorner \delta \neq 0$ by [I, Proposition 9.15(i)]. On the other hand $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E) = 0$. For the last sentence, the proof of Lemma 4.91(i) gives $\mathrm{At}(\pi_* L) = \delta \cdot \mathrm{id}$ with $\pi^* \delta = c_1(L)$ for any line bundle L, and $\mathrm{ch}(\pi_* L) = \mathrm{rk}(\pi_* L)\, e^{\delta}$; if every δ were a multiple of θ the Chern character of the sum would lie in $\mathbb{Q}[\theta]$, which does not contain ω because $\omega\theta = 0$ and $\theta^{n+1} \neq 0$. ∎

Theorem 2.3 asks that σ_E be injective, but its proof uses only that the relative deformations of E dominate the family, and Pridham's form of Theorem 2.2, in which σ_E annihilates every obstruction, gives this under a weaker hypothesis. That hypothesis allows an object that is obstructed to first order, which is exactly the way in which [I, Theorem 9.20], Corollary 4.21 and Proposition 3.14 exclude an object.

Proposition 3.15 (A flatness form of the criterion)**.** *Let $n \geq 2$, $s_0 \in \mathcal{D}_{n,n}$, and let E be a bounded complex of vector bundles on A_{s_0} with $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k \theta^k$ and $N \neq 0$. Let K_E be the base of a miniversal deformation of E on the fixed variety A_{s_0}, put $e = \dim \mathrm{Ext}^1(E,E)$ and $k = \dim \ker \sigma_E$. By Theorem 2.2 in Pridham's form, σ_E annihilates every obstruction, both on A_{s_0} and along the Artinian deformations of A_{s_0} in $\mathcal{D}_{n,n}$.*

(i) $\dim K_E \geq e - k$.

(ii) *If $\dim K_E = e - k$, then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of the family with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$.*

(iii) $\dim K_E = e - k$ *as soon as the cone $\{\xi \in \mathrm{Ext}^1(E,E) : \xi \circ \xi = 0\}$ has codimension k.*

For $k = 0$ this is Theorem 2.3.

Proof. Let R be the completed local ring of $\mathcal{D}_{n,n}$ at s_0, regular of dimension n^2. By hypothesis $\ker \sigma_E$ is a complete obstruction space for the deformations of the pair (A_s, E) over Artinian R-algebras, whose relative tangent space is $\mathrm{Ext}^1(E,E)$; so this functor has a hull P/J with $P = R[[x_1, \dots, x_e]]$ and J generated by at most k elements [FM98], and its fibre over the closed point of R is a hull of the absolute functor, of dimension $\dim K_E$. Krull's height theorem gives $\dim P/J \geq n^2 + e - k$, and the same count on the fibre gives (i). If $\dim K_E = e - k$, the dimension inequality for a local homomorphism [Mat86, Theorem 15.1] gives $\dim P/J \leq n^2 + e - k$, so equality holds, J has height k and is generated by a regular sequence, P/J is Cohen–Macaulay, and it is flat over R by miracle flatness [Mat86, Theorem 23.1]: a local homomorphism from a regular local ring to a Cohen–Macaulay local ring is flat exactly when the dimension of the target is the dimension of the source plus that of the fibre, and here both sides equal $n^2 + e - k$. The analytic Kuranishi germ of pairs, as in the proof of Theorem 3.5, has P/J as its completion, so it is flat over the germ of $\mathcal{D}_{n,n}$ at s_0 and therefore open onto a neighbourhood U of s_0. Every A_s with $s \in U$ then carries a bounded complex of vector bundles, algebraic by GAGA, whose Chern character is the parallel transport of $\mathrm{ch}(E)$, so $N\omega_{1,s}$ differs from an algebraic class by a polynomial in θ and is algebraic; [I, Corollary 16.5] gives (ii). For (iii), the quadratic part of the Kuranishi map of E is $\xi \mapsto \xi \circ \xi$, so the tangent cone of K_E lies in the cone of the hypothesis, and $\dim K_E \leq e - k$. Every step uses only deformations along $\mathcal{D}_{n,n}$, which are projective, whereas the proof of Theorem 3.5 needs non-projective ones. ∎

No object is known that meets (iii) with $k > 0$. Among direct sums the flatness form fails for a large class (Remark 3.29 below); for indecomposable objects nothing in this paper excludes it, and the proposition says only that injectivity of σ_E is more than propagation needs.

3.3. **Candidate objects for the polarised criterion.** The polarised criterion asks for one perfect complex. The results below say what such a complex can be made of, for K imaginary quadratic: a direct sum meets the criterion only through one indecomposable summand (Lemma 3.16); no sum of objects whose Ext algebra is an exterior algebra meets it (Corollary 3.17); the Orlov products of secant objects meet its twisted form, with $\dim \mathrm{Ext}^2$ equal to the number of Theorem 3.9, at $n = 2$ and $n = 3$ (Theorem 3.19), and cannot meet it at $n = 4$ when the factors have no negative self-extensions (Proposition 3.20); line bundles on smooth divisors, which give the certificates, give no factor from $n = 4$ on, nor do bundles restricted from the abelian variety at $n = 5$ (Proposition 3.23); and at $n = 4$ a combination of natural objects needs at least eight of them, on one conic (Theorem 3.27). None of this gives a new case of the Hodge conjecture: the certificates are for twisted objects, in families where Markman's theorem already holds.

Lemma 3.16 (The summands of an object meeting the criterion)**.** *Let A be a member of a $(K, -1, n)$-Weil family, K imaginary quadratic and $n \geq 2$, let $T = T_{[A]}\mathcal{D}_{n,n} \subseteq H^1(T_A)$, and let $E = \bigoplus_i F_i[k_i]$ be a finite direct sum of perfect complexes with $v \lrcorner \mathrm{ch}(E) = 0$ for every $v \in T$, as holds for every polarised character $N\omega_1 + \sum_k c_k \theta^k$. Suppose that σ_E is injective on $\mathrm{ev}_E(T)$; this holds if σ_E is injective, and under the weakened criterion of* [Mar25b, Question 11.4].

(i) *For every i and every $v \in T$, $\mathrm{ev}_{F_i}(v) = 0$ and $v \lrcorner \mathrm{ch}(F_i) = 0$, and every $\mathrm{ch}(F_i)$ lies in $\bigoplus_k \mathbb{Q}\theta^k \oplus W(A)$.*

(ii) *If* $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ *with* $N \neq 0$, *some* F_i *has* $\mathrm{ch}(F_i) = \omega' + \sum_k c'_k\theta^k$ *with* ω' *a nonzero rational Weil class, and then* $\dim \mathrm{Ext}^2(F_i, F_i) \geq r(\mathrm{ch}\, F_i) \geq 2n(2n-1)$. *If* σ_E *is injective, so is every* σ_{F_i}, *and that* F_i *meets the polarised criterion on its own.*

So the polarised criterion is a statement about indecomposable objects.

Proof. For $v \in T$, $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E) = 0$ by the commutative square of Setup 4.54, so $\mathrm{ev}_E(v) = 0$. The Atiyah class is natural, so $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$ is block diagonal with the blocks $\pm\,\mathrm{ev}_{F_i}(v)$, which therefore vanish, and $v \lrcorner \mathrm{ch}(F_i) = \sigma_{F_i}(\mathrm{ev}_{F_i}(v)) = 0$. By Proposition 3.11(i) every homogeneous component of $\mathrm{ch}(F_i)$, a rational class of type (k,k), stays of Hodge type on the germ of $\mathcal{D}_{n,n}$ at A, which in the chart of that proposition is the linear space T; so it is a Hodge class at a member with Hodge group $\mathrm{SU}(V,H)$, and [I, Theorem 4.14] puts it in $\mathbb{Q}\theta^k$, or in $\mathbb{Q}\theta^n \oplus W$ for $k = n$. That is (i). For (ii), the Weil parts of the $\pm\,\mathrm{ch}(F_i)$ add up to $N\omega_1 \neq 0$, so one of them is nonzero; Theorem 3.9 gives $\dim \mathrm{Ext}^2(F_i, F_i) \geq r(\mathrm{ch}\, F_i)$ with no hypothesis on σ, and $r \geq (4+0)n^2 - 2n = 2n(2n-1)$ for $n \geq 3$, while at $n = 2$ the seven values of Figure 2 are at least 12. Injectivity passes to the summands by [III, Lemma 3.38]. By the Krull–Schmidt property of $D^b(A)$ the F_i may be taken indecomposable. ∎

The Weil part ω' of the summand need not be a multiple of ω_1; propagating any nonzero rational Weil class propagates the whole Weil plane, through the algebraic action of K. The lemma is the Weil-family analogue of [III, Corollary 3.41(ii)].

Corollary 3.17 (Sums of objects with exterior Ext algebras)**.** *In Lemma 3.16, let every* F_i *have* $\dim \mathrm{Ext}^2(F_i, F_i) < 2n(2n-1)$. *This holds when* $\mathrm{Ext}^*(F_i, F_i) \cong \bigwedge^* \mathbb{C}^{2n}$, *and so for line bundles, for the sheaves* $\mathcal{O}_B \otimes L$ *with* $B \subseteq A$ *an abelian subvariety, for simple semi-homogeneous bundles, for skyscraper sheaves, and for the images of all of these under autoequivalences of* $D^b(A)$. *Then no such sum with* $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ *and* $N \neq 0$ *has* σ_E *injective on* $\mathrm{ev}_E(T)$*: it fails the polarised criterion and the weakened criterion, some* $v \in T$ *obstructs it to first order, and* $\dim \mathrm{Ext}^2(E, E) > r(\mathrm{ch}\, E)$.

Proof. The first assertion is Lemma 3.16(ii), since $\binom{2n}{2} = n(2n-1) < 2n(2n-1)$, and $\dim \mathrm{Ext}^2(E, E) \geq r(\mathrm{ch}\, E)$ with equality only if σ_E is injective (Theorem 3.9). For $\mathcal{O}_B$ choose a complementary abelian subvariety B'; the isogeny $\mu\colon B \times B' \to A$ is étale with group $G = B \cap B'$, and $\mu^*\mathcal{O}_B = \bigoplus_{g \in G} \mathcal{O}_{B \times \{g\}}$ with G permuting the summands simply transitively, so $\mathrm{Ext}_A(\mathcal{O}_B, \mathcal{O}_B) = \mathrm{Ext}_{B \times B'}(\mathcal{O}_{B \times 0}, \mathcal{O}_{B \times 0}) = \bigwedge^*(\mathbb{C}^{\dim B} \oplus \mathbb{C}^{\dim B'})$. For a simple semi-homogeneous bundle V, $V \otimes V^\vee$ is a sum of line bundles in Pic^0 of which $\mathcal{O}$ occurs once [Muk78], so $\mathrm{Ext}^*(V, V) = H^*(\mathcal{O}_A)$. Equivalences preserve Ext algebras. ∎

By Lemma 3.16 the objects that Remark 3.13 allows may be taken indecomposable. The corollary recovers, for polarised characters, the part of [III, Theorem 2.32] that concerns objects with exterior Ext algebras, and it extends Proposition 3.14, whose summand has a scalar Atiyah class, to the subtori, the points and the Fourier–Mukai images. In the language of pure spinors, a summand whose character is a rational multiple of the pure spinor of a rational maximal isotropic subspace is excluded unless that character lies in $\mathbb{Q}[\theta]$, because the rational maximal isotropic subspaces whose pure spinors are flat over $\mathcal{D}_{n,n}$ are those with pure spinor 1, the class of a point or $e^{c\theta}$.

The secant objects of [I, Section 14] have their own Hochschild profile, which Theorem 4.56(i) and (ii) compute in degrees one and two.

Proposition 3.18 (The Hochschild profile of a secant class)**.** *Let* X *be a principally polarised abelian* n*-fold,* $n \geq 2$, *with polarisation* θ, *let* $d \geq 1$, *and let* $v = a\,u_\theta + b\,v_\theta$ *with*

$(a,b)\in\mathbb{Q}^2\setminus\{0\}$, *in the notation of* [I, (30)]. *The rank of contraction* $HT^k(X)\to H^*(X,\mathbb{C})$, $\xi\mapsto\xi\lrcorner v$, *is* 1 *for* $k=0$ *and* $k=n$, $2\binom{n}{k}$ *for* $1\le k\le n-1$, *and* 0 *for* $k>n$. *A perfect complex* F *on* X *with* $\operatorname{ch}(F)=v$ *has* $\dim\operatorname{Ext}^k(F,F)\ge 2\binom{n}{k}$ *for* $k=1,2$, *and*

$$\chi(F,F)=\begin{cases}(-1)^{n/2}\,2^{n-1}d^{\,n/2-1}(a^2d+b^2) & n \text{ even},\\ 0 & n \text{ odd}.\end{cases}$$

Proof. Put $s=\sqrt{-d}$ and $w=\frac12(a-bs/d)$. By [I, Theorem 14.5(i)], $v=w\,e^{s\theta}+\overline{w}\,e^{-s\theta}$, and $w\neq 0$. By Lemma 4.55(i), $\xi\lrcorner e^{\pm s\theta}=y_\pm e^{\pm s\theta}$, where $y_\pm\in H^{0,k}(X)$ is the component of $e^{\pm s\theta}\xi$ in $\bigwedge^k H^{0,1}$. The map $\xi\mapsto(y_+,y_-)$ is onto $H^{0,k}\oplus H^{0,k}$ for $k\ge1$: $\xi=z\in\bigwedge^k H^{0,1}$ gives (z,z), and $\xi=z'\otimes\partial$ with $z'\in\bigwedge^{k-1}H^{0,1}$ gives $\pm s\,z'\wedge(\partial\lrcorner\theta)$, where $\partial\lrcorner\theta$ runs over $H^{0,1}$. If $w\,y_+e^{s\theta}+\overline{w}\,y_-e^{-s\theta}=0$, the components of degrees k and $k+2$ give $wy_+ +\overline{w}y_-=0$ and $(wy_+-\overline{w}y_-)\wedge\theta=0$, and for $k<n$ wedge product with $\theta=\sum_a p_a\wedge q_a$ is injective on $\bigwedge^k H^{0,1}$, so $y_\pm=0$. So the rank is $2\binom{n}{k}$ for $1\le k<n$. For $k=n$, $y\wedge\theta=0$ for $y\in H^{0,n}$ and the image is the line spanned by $wy_+ +\overline{w}y_-$; for $k>n$, $H^{0,k}=0$; and for $k=0$ the rank is one. The bounds on Ext^1 and Ext^2 follow, as in Theorem 4.56(i) and (ii), from the compatibility of the characteristic morphism with contraction into the Chern character. Finally the Todd class is trivial, so $\chi(F,F)=\int_X\operatorname{ch}(F)^\vee\operatorname{ch}(F)$ with $\operatorname{ch}(F)^\vee=w\,e^{-s\theta}+\overline{w}\,e^{s\theta}$, and $\int_X(e^{2s\theta}+e^{-2s\theta})=(2s)^n(1+(-1)^n)$ because $\int_X\theta^n=n!$; with $|w|^2=(a^2d+b^2)/(4d)$ this is the formula. ∎

At $n=4$ and $(a,b)=(1,3)$, the class of Proposition 4.52, this gives $\chi(\mathcal{F},\mathcal{F})=8d(d+9)$, the rank recorded there.

Theorem 3.19 (Twisted certificates at $n=2$ and $n=3$)**.** *Let* $d\ge1$, $K=\mathbb{Q}(\sqrt{-d})$, *let* Φ *be Orlov's equivalence of Proposition 4.52(v), and for a perfect complex* F *on a principally polarised* X *put* $E=\Phi(F\boxtimes F^\vee)$ *on* $A=X\times\widehat{X}$, *with the* K*-action of* [I, (29)], *the Poincaré class* ℓ *and the polarisation* $\eta=d\beta+\widehat{\beta}$ *of the split member (*[I, Notation 15.1]*).*

(i) *Let* X *be the Jacobian of a curve* C *of genus two,* $C\subset X$ *by an Abel–Jacobi map, and* $F=\mathcal{O}_C(p)$ *for a point* $p\in C$. *Then* $\operatorname{ch}(F)=v_\theta$, $\dim\operatorname{Ext}^k(F,F)=1,4,1$, *and* $\dim\operatorname{Ext}^k(E,E)=1,8,18,8,1$ *for* $k=0,\dots,4$, *which are the ranks of contraction from* $HT^k(A)$ *into* $\operatorname{ch}(E)$. *So* $\dim\operatorname{Ext}^2(E,E)=18=r(\operatorname{ch}E)$ *and* σ_E *is injective.*

(ii) *Let* X *be the Jacobian of a non-hyperelliptic curve* C *of genus three,* $i\colon C^{(2)}\to X$ *the embedding of the theta divisor,* x *the class of* $\{p+q:q\in C\}$ *and* θ *the restriction of the polarisation, and let* $d=k(k+1)$ *with* $k\ge1$ *and* $F=i_*\mathcal{O}_{C^{(2)}}((2k+1)x-k\theta)$. *Then* $\operatorname{ch}(F)=v_\theta$, $\dim\operatorname{Ext}^k(F,F)=1,6,6,1$, *and* $\dim\operatorname{Ext}^k(E,E)=1,12,48,74,48,12,1$. *So* $\dim\operatorname{Ext}^2(E,E)=48=r(\operatorname{ch}E)$ *and* σ_E *is injective.*

(iii) *In both cases* $\operatorname{ch}(E)$ *is not killed by* $T=T_{[A]}\mathcal{D}_{n,n}$, *and no line bundle twist of* E *has a character killed by* T; *the twisted character* $\kappa=\operatorname{ch}(E)e^{\ell/2}$ *is killed by* T, *has the form* $N'\omega'+\sum_k c_k\eta^k$ *with* ω' *a nonzero rational Weil class, and has the same contraction ranks as* $\operatorname{ch}(E)$. *A class* $B\in\operatorname{NS}(A)_\mathbb{Q}$ *with* $\operatorname{ch}(E)e^B$ *killed by* T *is congruent to* $\ell/2$ *modulo* $\mathbb{Q}\eta$. *At* $n=2$, κ *has* $c_1=c_3=0$ *and lies at the exceptional ratio of Remark 3.12, and its annihilator in* $H^1(T_A)$ *has dimension* 5; *at* $n=3$ *that annihilator has dimension* $9=n^2$.

Proof. (i) By Grothendieck–Riemann–Roch $\operatorname{ch}(i_*\mathcal{O}_C(p))=[C]+(1+1-2)[\mathrm{pt}]=\theta=v_\theta$, since $v_\theta=\theta$ when $n=2$. For the curve, $\operatorname{Ext}^1(\mathcal{O}_C,\mathcal{O}_C)=H^1(\mathcal{O}_C)\oplus H^0(N_C)$ with $N_C=K_C$ by adjunction, of dimension $2+2$, and $\operatorname{Ext}^0=\operatorname{Ext}^2=\mathbb{C}$ by Serre duality; twisting by $\mathcal{O}_C(p)$ changes nothing. The Ext algebra of $F^\vee$ is that of F, Φ is an equivalence, and by

the Künneth formula $\mathrm{Ext}^*(E,E) = \mathrm{Ext}^*(F,F) \otimes \mathrm{Ext}^*(F^\vee,F^\vee)$, which gives $1,8,18,8,1$. By Proposition 3.18 and the Künneth decomposition $HT^2(X\times X) = HT^2\otimes 1 \oplus HT^1 \otimes HT^1 \oplus 1\otimes HT^2$ the rank of contraction into $\mathrm{ch}(F)\boxtimes \mathrm{ch}(F^\vee)$ is $1+4\cdot 4+1 = 18$, and it is transported by Φ as in the proof of Theorem 4.56(iv). The other degrees are the same count. Equality in degree two forces σ_E injective, as in Theorem 3.9. (ii) On $C^{(2)}$ one has $x^2 = 1$, $x\theta = 3$, $\theta^2 = 6$ and $K_{C^{(2)}} = \theta$, $i_*x = \frac{1}{2}\Theta^2$ and $i_*\theta = \Theta^2$, so Riemann–Roch gives $\mathrm{ch}_2(F) = 0$ and $\mathrm{ch}_3(F) = -k(k+1)[\mathrm{pt}]$, that is $\mathrm{ch}(F) = \Theta - d[\mathrm{pt}] = v_\theta$. The local-to-global spectral sequence, with $h^p(\mathcal{O}_{C^{(2)}}) = 1,3,3$ and $h^p(K_{C^{(2)}}) = 3,3,1$, gives $\dim \mathrm{Ext}^1, \dim \mathrm{Ext}^2 \le 6$, and Proposition 3.18 gives ≥ 6; $\mathrm{Ext}^0 = \mathbb{C}$ and Serre duality gives $\mathrm{Ext}^3 = \mathbb{C}$. The rest is as in (i), with $r(\mathrm{ch}\,E) = 6n^2 - 2n = 48$ by Theorem 4.56(iv).

(iii) By Proposition 3.21 below, whose proof does not use this theorem, κ lies in $\mathbb{Q}[\eta]\oplus W(A)$ and is killed by T; both summands are defined over $\mathbb{Q}$, so its Weil part $N'\omega'$ is rational, and $r(\kappa) = r(\mathrm{ch}\,E)$ is Lemma 4.55(i). The proof of Proposition 3.21 also gives κ explicitly. At $n=2$, $\mathrm{ch}(F) = \theta = \frac{1}{2s}(e^{s\theta} - e^{-s\theta})$ and $\mathrm{ch}(F^\vee) = -\theta$, so, with $t = s/2d$,

$$\kappa = -(e^{t\eta} + e^{-t\eta}) - \tfrac{1}{4d}(\gamma^2 - d\ell^2) = -2 + \tfrac{1}{4d}\eta^2 - \tfrac{1}{192d^2}\eta^4 - \tfrac{1}{4d}(\gamma^2 - d\ell^2).$$

Hence $c_1 = c_3 = 0$ and $c_2 = 1/4d$. Write $\gamma - s\ell = a_1 + a_2$ and $\gamma + s\ell = b_1 + b_2$ for the decompositions over the two factors of $X = E_1\times E_2$, as in the proof of Proposition 3.21. Then $a_i^2 = b_i^2 = 0$, so the Weil part is $\omega = -\frac{1}{4d}(a_1a_2 + b_1b_2)$, and $a_ib_i = \gamma_i^2 + d\ell_i^2 = -4d\,\beta_i\widehat{\beta}_i$; so $\int\omega^2 = \frac{2}{16d^2}\int a_1b_1a_2b_2 = 2$, while $\int\eta^4 = 6\int\eta_1^2\eta_2^2 = 24d^2$. Thus $c_2^2\int\eta^4 = \frac{3}{2} = \frac{3}{4}\int\omega^2$, the exceptional ratio of Proposition 3.11(iv), where the annihilator has dimension 5. At $n=3$, $\mathrm{ch}(F) = v_\theta = \frac{1}{2s}(e^{s\theta} - e^{-s\theta})$ and $\mathrm{ch}(F^\vee) = -v_\theta$, and the same computation gives the η-part $-2s(e^{t\eta} - e^{-t\eta}) = 2\eta + \cdots$; so $c_1 = 2 \neq 0$ and the annihilator is $T_{[A]}\mathcal{D}_{3,3}$, of dimension 9, by Proposition 3.11(iii).

For B, put $B' = B - \frac{1}{2}\ell$, so that $\mathrm{ch}(E)e^B = \kappa e^{B'}$. Contraction with $v\in T$ is a derivation and kills κ, so $v\lrcorner(\kappa e^{B'}) = \kappa\,(v\lrcorner B')\,e^{B'}$. Multiplication by κ is injective on $H^2(A,\mathbb{C})$: at $n=2$ the degree zero part of κ is -2, and at $n=3$ its lowest part is 2η, and multiplication by η is injective from H^2 to H^4 on the sixfold A. So T kills $\kappa e^{B'}$ exactly when $v\lrcorner B' = 0$ for all $v\in T$. By Proposition 3.11(i) the real class B' of type $(1,1)$ then stays of type $(1,1)$ on the germ of $\mathcal{D}_{n,n}$ at A, hence on all of $\mathcal{D}_{n,n}$, and at a very general member $\mathrm{NS}\otimes\mathbb{Q} = \mathbb{Q}\eta$ ([I, Lemma 7.5]); so $B'\in\mathbb{Q}\eta$. The classes $x_j\xi_j$ appear in $\frac{1}{2}\ell + c\eta$ with coefficient $\frac{1}{2}$ for every c, so no integral class is congruent to $\frac{1}{2}\ell$ modulo $\mathbb{Q}\eta$, on any X: no line bundle twist of E, and in particular not E itself, has a character killed by T. ∎

In both cases the numerical form of the criterion holds with equality, so the injectivity of σ_E comes out of a dimension count and does not need [Mar25b, Question 11.4]. The certificates concern the twisted object only: the character κ stays of Hodge type along $\mathcal{D}_{n,n}$, while $\mathrm{ch}(E)$ does so only on a locus of dimension 3 at $n=2$ and 6 at $n=3$, so propagation needs the extension of Theorem 2.2 to twisted objects, which Perry proves [Per26], and an argument extracting untwisted cycles, as in [Mar25b, Section 2]. The families concerned, of trivial discriminant with $n=2$ and $n=3$, are Markman's.

Proposition 3.20 (No Orlov product at $n=4$)**.** *Let X be a principally polarised abelian fourfold, $d\ge 1$, and F_1, F_2 perfect complexes on X with characters in the secant plane $P_\theta\setminus\{0\}$ and $\mathrm{Ext}^{<0}(F_i,F_i) = 0$, for instance sheaves. Then $E = \Phi(F_1\boxtimes F_2^\vee)$ has $\dim\mathrm{Ext}^2(E,E) > 88 = r(\mathrm{ch}\,E)$.*

Proof. By the Künneth formula $\dim\mathrm{Ext}^2(E,E) = e_0e_2' + e_1e_1' + e_2e_0'$ with $e_k = \dim\mathrm{Ext}^k(F_1,F_1)$ and e_k' likewise for F_2, and by Proposition 3.18 $e_0\ge 1$, $e_1\ge 8$, $e_2\ge 12$;

so the sum is at least $12+64+12=88$, which is $r(\operatorname{ch} E)$ by Theorem 4.56(iv). Equality would force $e_0,e_1,e_2=1,8,12$. Since $e_k=0$ for $k<0$ and Serre duality gives $e_k=e_{4-k}$, this would give $\chi(F_1,F_1)=1-8+12-8+1=-2$, while Proposition 3.18 gives $\chi(F_1,F_1)=8d(a^2d+b^2)>0$. ∎

Without $\operatorname{Ext}^{<0}=0$ the count changes, because negative Ext groups of the F_i contribute products $e_{-j}e'_{2+j}$ to $\operatorname{Ext}^2(E,E)$ and change the Euler characteristic. The next two results treat the template in every dimension: the twisted character is always flat, and the count says exactly which secant objects would give equality.

Proposition 3.21 (Flatness in every dimension)**.** *Let (X,θ) be a principally polarised abelian n-fold, $n\geq 1$, let $d\geq 1$ and $s=\sqrt{-d}$, and put $\Psi(Z)=\operatorname{ch}(\Phi(Z))e^{\ell/2}$ for classes Z on $X\times X$, extended linearly, with $\gamma=d\beta-\widehat{\beta}$ as in* [I, Theorem 15.2] *and $\eta=d\beta+\widehat{\beta}$. For $\sigma=\pm1$, $\Psi(e^{\sigma s\theta}\boxtimes e^{\sigma s\theta})$ is a nonzero multiple of $e^{\sigma(s/2d)\eta}$, and $\Psi(e^{\sigma s\theta}\boxtimes e^{-\sigma s\theta})$ is a nonzero multiple of the Weil class $(\gamma-\sigma s\ell)^n$. Consequently, for perfect complexes F_1,F_2 with $\operatorname{ch}(F_i)\in P_\theta$, the class $\kappa=\operatorname{ch}(\Phi(F_1\boxtimes F_2^\vee))e^{\ell/2}$ lies in $\mathbb{Q}[\eta]\oplus W(A)$, is killed by $T=T_{[A]}\mathcal{D}_{n,n}$, and has Hankel rank $\rho\leq 2$; for $F_1=F_2\neq 0$ its Weil part is nonzero.*

Proof. A symplectic basis of $H^1(X,\mathbb{Z})$ for θ identifies $H^*(X,\mathbb{Q})$, with θ, with the tensor product of n copies of the cohomology of an elliptic curve with its class θ_i, $\theta=\sum_i\theta_i$. The action of Φ on cohomology is the correspondence $(1\times\Phi_P)\circ\mu_*$ with $\mu(a,b)=(a-b,b)$, and μ, the Poincaré class $\ell=\sum_i\ell_i$, the classes η and γ, and the exponentials $e^{\pm s\theta}=\bigotimes_i e^{\pm s\theta_i}$ all decompose as tensor products over i of classes of even degree, so no signs arise. Each identity is therefore the tensor product of n copies of the identity for $n=1$, up to a nonzero scalar, where for the Weil class one uses $(\gamma-\sigma s\ell)^n=n!\prod_i(\gamma_i-\sigma s\ell_i)$, valid because the factors commute and $(\gamma_i-\sigma s\ell_i)^2=0$: by [I, Theorem 15.2(i)] at $n=1$ the class $\gamma_i-\sigma s\ell_i$ spans the top exterior power of a two-dimensional eigenspace.

It remains to treat $n=1$. Let y_1,y_2 be a symplectic basis of $H^1(X)$, z_1,z_2 its copy on the second factor of $X\times X$ and ξ_1,ξ_2 the dual basis of $H^1(\widehat{X})$, so that $\theta=\beta=y_1y_2$, $\widehat{\beta}=\xi_1\xi_2$, $\ell=y_1\xi_1+y_2\xi_2$ and $\ell^2=-2\beta\widehat{\beta}$. On cohomology Φ sends w to $\int_z\mu^*w\cdot e^{z_1\xi_1+z_2\xi_2}$, the integral being over the second factor; with the opposite sign of the Poincaré class, ℓ is replaced by $-\ell$ in this proposition and in Theorem 3.19. For $a,b\in\{\pm1\}$,

$$\mu^*(e^{as\theta}\boxtimes e^{bs\theta})=(1+as(y_1+z_1)(y_2+z_2))(1+bs\,z_1z_2),$$
$$e^{z_1\xi_1+z_2\xi_2}=1+z_1\xi_1+z_2\xi_2-z_1z_2\xi_1\xi_2,$$

and collecting the coefficient of z_1z_2 in the product gives

$$\operatorname{ch}\Phi(e^{as\theta}\boxtimes e^{bs\theta})=(a+b)s-ab\,d\,\beta-\widehat{\beta}-as\,\ell-as\,\beta\widehat{\beta}.$$

Multiplying by $e^{\ell/2}=1+\frac12\ell-\frac14\beta\widehat{\beta}$, and using $\beta\ell=\widehat{\beta}\ell=0$, gives $\Psi=2as-d\beta-\widehat{\beta}-\frac{as}{2}\beta\widehat{\beta}=2as\,e^{(as/2d)\eta}$ when $a=b$, and $\Psi=d\beta-\widehat{\beta}-as\ell=\gamma-as\ell$ when $a=-b$. So for general n the multiples are $(2\sigma s)^n$ and $1/n!$.

For the consequence, write $\operatorname{ch}(F_1)=w_1e^{s\theta}+\overline{w}_1e^{-s\theta}$ and $\operatorname{ch}(F_2^\vee)=w_2e^{-s\theta}+\overline{w}_2e^{s\theta}$ as in the proof of Proposition 3.18; the terms $e^{\pm s\theta}\boxtimes e^{\pm s\theta}$ give multiples of $e^{\pm(s/2d)\eta}$ and the mixed terms give the Weil part $w_1w_2c_+\alpha_+ + \overline{w}_1\overline{w}_2c_-\alpha_-$, with $c_\pm\neq 0$. The classes of $\mathbb{Q}[\eta]\oplus W(A)$ are Hodge classes on every member of the family, so T kills κ, and the η-part lies in the span of the two exponentials $e^{\pm(s/2d)\eta}$, which gives $\rho\leq 2$. ∎

Proposition 3.22 (The equality criterion for the Orlov template)**.** *Let (X,t) be a polarised abelian n-fold, $n\geq 2$, $d\geq 1$, and F_1, F_2 perfect complexes with characters in $P_t\setminus\{0\}$; put $e_k=\dim\operatorname{Ext}^k(F_1,F_1)$ and $e'_k=\dim\operatorname{Ext}^k(F_2,F_2)$ for all $k\in\mathbb{Z}$, and $E=\Phi(F_1\boxtimes F_2^\vee)$.*

(i) $\dim \mathrm{Ext}^2(E,E) = \sum_{k\in\mathbb{Z}} e_k e'_{2-k} \geq R_n$, *where* $R_n = 6n^2 - 2n$ *for* $n \geq 3$ *and* $R_2 = 18$, *and* $R_n = r(\mathrm{ch}\, E) = r(\kappa)$.

(ii) *Equality holds if and only if* $(e_0, e_1, e_2) = (e'_0, e'_1, e'_2) = (1, 2n, r_2)$, *with* $r_2 = n(n-1)$ *for* $n \geq 3$ *and* $r_2 = 1$ *for* $n = 2$, *and* $e_{-j}e'_{2+j} = e_{2+j}e'_{-j} = 0$ *for every* $j \geq 1$.

(iii) *For* $F_1 = F_2 = F$ *and* $n \geq 3$ *the last condition says that the set* $S = \{j \geq 1 : \mathrm{Ext}^{-j}(F,F) \neq 0\}$ *lies in* $[n-1, \infty)$ *and that* $j \in S$ *implies* $j - (n-2) \notin S$; *for* $n = 2$ *it says* $\mathrm{Ext}^{<0}(F,F) = 0$.

(iv) *Equality implies that* σ_E *is injective.*

Proof. (i) Φ is an equivalence and the Künneth formula holds in all degrees, negative ones included, and dualising is an anti-equivalence, so $\mathrm{Ext}^*(E,E) = \mathrm{Ext}^*(F_1,F_1) \otimes \mathrm{Ext}^*(F_2,F_2)$. By Proposition 3.18, $e_0 \geq 1$, $e_1 \geq 2n$ and $e_2 \geq r_2$, so the three terms with $k = 0, 1, 2$ already give R_n; the equality $R_n = r(\mathrm{ch}\, E)$ is the Künneth count of Theorem 4.56(iv) and Theorem 3.19, and $r(\kappa) = r(\mathrm{ch}\, E)$ by Lemma 4.55(i). (ii) Every term is nonnegative, and $e_0 e'_2 \geq e'_2 \geq r_2 > 0$ with equality only if $e_0 = 1$ and $e'_2 = r_2$, and likewise for the other two terms; every other product must vanish. (iii) Serre duality gives $e_k = e_{n-k}$. For $1 \leq j \leq n-2$, e_{2+j} is at least the rank of contraction from $HT^{2+j}(X)$ into $\mathrm{ch}(F)$, which is positive by Proposition 3.18, because the characteristic morphism is compatible with contraction into the Chern character in every degree [CRV12]; so $e_{-j} = 0$. For $j \geq n-1$, $e_{2+j} = e_{-(j-(n-2))}$. (iv) The composite $\sigma_E \circ \mathrm{ev}_E$ is contraction into $\mathrm{ch}(E)$, of rank $R_n \leq \mathrm{rk}\, \sigma_E \leq \dim \mathrm{Ext}^2(E,E)$, so equality forces σ_E to be injective. ∎

For $n = 2$ and $n = 3$ the certificates of Theorem 3.19 attain the equality; for $n = 4$ it is excluded by Proposition 3.20 when $\mathrm{Ext}^{<0} = 0$, with negative Ext groups in degrees at most $-(n-1)$ the Euler characteristic leaves room in every dimension, and from $n = 5$ on it leaves room without them (Proposition 3.23(i)). If a secant object F attaining the equality existed for one d and unboundedly many n, then, since κ is flat in every dimension by Proposition 3.21, a twisted form of Theorem 2.3 together with [III, Proposition 2.13] would make the Weil classes of every family of $\mathbb{Q}(\sqrt{-d})$ algebraic. That is the secant route of (C1) and (C2) in [III, Section 2.5], with the question of [Mar25b] replaced by a dimension count, and it lies off every minimal route ([III, Theorem 2.70(iv)]).

Both certificates are sheaves on theta divisors, a curve of genus two and the surface $C^{(2)}$. The next result explains why this construction stops at $n = 3$ and what a continuation would need.

Proposition 3.23 (Sheaves on divisors in the Orlov template)**.** *Let* (X, θ) *be a principally polarised abelian* n*-fold and* $d \geq 1$; *put* $S(x) = \sum_{j\geq 0} (-d)^j x^{2j}/(2j+1)!$, *so that* $v_\theta = \theta S(\theta)$, *and* $T_b(x) = bx/(1 - e^{-bx})$ *for an integer* $b \geq 1$. *Write* $\beta_0 = \beta_n = 1$, $\beta_k = 2\binom{n}{k}$ *for* $0 < k < n$ *and* $\beta_k = 0$ *otherwise; these are the Betti numbers of the connected sum of two real* n*-tori.*

(i) *Let* $n \geq 5$ *and let* F *be a perfect complex with* $\mathrm{ch}(F) \in P_\theta \setminus \{0\}$. *There are integers* e_k, *zero for* $k \notin [0,n]$, *with* $e_k = e_{n-k}$, $(e_0, e_1, e_2) = (1, 2n, n(n-1))$, $e_k \geq \beta_k$ *for all* k, *and* $\sum_k (-1)^k e_k = \chi(F,F)$. *So Serre duality, the contraction bounds* $e_k \geq \beta_k$ *(Proposition 3.18 and* [CRV12]*) and the Euler characteristic allow equality in Proposition 3.22 with* $\mathrm{Ext}^{<0}(F,F) = 0$. *For odd* n *one may take* $e_k = \beta_k$, *and at* $n = 5$ *this is the only choice; for even* n *the sequence* (β_k) *has Euler characteristic* $-2 \neq \chi(F,F)$, *so some* e_k *with* $3 \leq k \leq n-3$ *must exceed* β_k.

(ii) *Let* $n \geq 3$, *let* $i\colon D \hookrightarrow X$ *be a smooth effective divisor, and let* G *be a vector bundle of rank* $r \geq 1$ *on* D *with* $\mathrm{ch}(G) \in i^*H^*(X, \mathbb{Q})$. *If* $\mathrm{ch}(i_*G) \in P_\theta$, *then* $[D] = b\theta$

for an integer $b \geq 1$ and $\operatorname{ch}(G) = r\, i^*(S(\theta)\, T_b(\theta))$; in particular $c_1(G) = \frac{rb}{2}\,\theta|_D$, $\operatorname{ch}_2(G) = r(\frac{b^2}{12} - \frac{d}{6})\theta^2|_D$ and $r \geq 2$. If $n \geq 4$, every line bundle on D satisfies the hypothesis, so no line bundle L on a smooth divisor has $\operatorname{ch}(i_*L) \in P_\theta$, and rb is even. If $n \geq 5$, $b = 1$ and $r = 2$, then $c_2(G) = \frac{d+1}{3}\,\theta^2|_D$, which is integral only for $d \equiv 2 \pmod 3$.

(iii) *Let $n \geq 2$, let $i\colon D \hookrightarrow X$ be a smooth divisor with $[D] = b\theta$, V a vector bundle on X, $G = V|_D$ and $F = i_*G$. Then*

$$\dim \operatorname{Ext}^k(F,F) = h^k(\operatorname{End} G) + h^{k-1}\big(\operatorname{End} G \otimes \mathcal{O}_D(D)\big)$$

for every k. For $b = 1$ this is $\beta_k + h^k(\operatorname{End}_0 G) + h^{k-1}(\operatorname{End}_0 G \otimes \mathcal{O}_D(D))$, so for a line bundle V the sheaf F has exactly the profile required in Proposition 3.22(ii). For $b \geq 2$, $\dim \operatorname{Ext}^1(F,F) \geq b^n + n - 1 > 2n$.

(iv) *At $n = 5$, for no smooth divisor D and no vector bundle V on X is $i_*(V|_D)$ a factor of an Orlov product attaining the equality of Proposition 3.22.*

Proof. (i) The rank a of F is an integer, and $c_1(F) = b\theta$ with θ primitive, so $b \in \mathbb{Z}$; for even n the number $c = 2^{n-1}d^{n/2-1}(a^2d + b^2)$ is therefore a positive integer divisible by 2^{n-1}, and $\chi(F,F) = (-1)^{n/2}c$ by Proposition 3.18, while $\chi(F,F) = 0$ for odd n. The conditions fix e_0, e_1, e_2 and, by symmetry, e_{n-2}, e_{n-1}, e_n, with the values β_k; for $n = 5$ these are all the e_k. Since $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$, the sequence (β_k) has Euler characteristic 0 for odd n and -2 for even n. For odd n take $e_k = \beta_k$. For even $n \geq 6$ take $e_k = \beta_k$ for $k \neq n/2$ and $e_{n/2} = \beta_{n/2} + c + 2(-1)^{n/2} \geq \beta_{n/2}$, which is possible because $c \geq 2^{n-1} > 2$ and $3 \leq n/2 \leq n - 3$; its Euler characteristic is $-2 + (-1)^{n/2}(c + 2(-1)^{n/2}) = \chi(F,F)$.

(ii) The rank of i_*G is zero, so $\operatorname{ch}(i_*G) = b' v_\theta$ for some $b' \in \mathbb{Q}$, and comparing degree two gives $r[D] = b'\theta$. As $[D]$ is integral and nonzero and θ is primitive, $[D] = b\theta$ with b a positive integer, and D is ample. The tangent bundle of X is trivial and the normal bundle of D is $\mathcal{O}_D(D)$, so $\operatorname{td}(D) = i^*T_b(\theta)^{-1}$; writing $\operatorname{ch}(G) = i^*y$, Grothendieck–Riemann–Roch and the projection formula give

$$\operatorname{ch}(i_*G) = i_*\big(i^*(y\, T_b(\theta)^{-1})\big) = y\, T_b(\theta)^{-1}\, b\theta = y\,(1 - e^{-b\theta}).$$

So $\operatorname{ch}(i_*G) = rb\,\theta S(\theta)$ says $b\theta\, z\, T_b(\theta)^{-1} = 0$ for $z = y - r\, S(\theta)T_b(\theta)$, that is $\theta z = 0$. Let z_k be the component of z in $H^{2k}(X,\mathbb{Q})$. If $2k \leq n - 1$, multiplication by θ is injective on $H^{2k}(X,\mathbb{Q})$ by the hard Lefschetz theorem, so $z_k = 0$. If $2k \geq n$, the Gysin map i_* is injective on $H^{2k}(D,\mathbb{Q})$, being dual to the restriction $H^{2n-2-2k}(X,\mathbb{Q}) \to H^{2n-2-2k}(D,\mathbb{Q})$, which is onto by the Lefschetz hyperplane theorem, and $i_*i^*z_k = b\theta z_k = 0$; so $i^*z_k = 0$. Hence $\operatorname{ch}(G) = r\, i^*(S(\theta)T_b(\theta))$, and the expansion $S(x)T_b(x) = 1 + \frac{b}{2}x + (\frac{b^2}{12} - \frac{d}{6})x^2 + O(x^3)$ gives c_1 and ch_2. If $r = 1$ then $\operatorname{ch}_2(G) = \frac{1}{2}c_1(G)^2$, so $(\frac{b^2}{24} + \frac{d}{6})\theta^2|_D = 0$, which is false: $\int_D (\theta|_D)^{n-1} = b\,n! \neq 0$ and $n - 1 \geq 2$. If $n \geq 4$, the Lefschetz hyperplane theorem gives $H^2(X,\mathbb{Z}) \cong H^2(D,\mathbb{Z})$, so every line bundle L on D has $\operatorname{ch}(L) = i^*e^\lambda$ for some $\lambda \in H^2(X,\mathbb{Z})$, and $\theta|_D$ is primitive, so the integrality of $c_1(G)$ makes rb even. If $n \geq 5$, $b = 1$ and $r = 2$, then $c_2(G) = \frac{1}{2}c_1^2 - \operatorname{ch}_2 = \frac{d+1}{3}\theta^2|_D = \frac{2(d+1)}{3}\, i^*(\frac{1}{2}\theta^2)$. The class $\frac{1}{2}\theta^2$ is integral and primitive. The affine variety $X \setminus D$ has the homotopy type of a CW complex of dimension n, so by Lefschetz duality $H^4(X,D;\mathbb{Z}) \cong H_{2n-4}(X \setminus D;\mathbb{Z}) = 0$ and $H^5(X,D;\mathbb{Z}) \cong H_{2n-5}(X \setminus D;\mathbb{Z})$ is torsion free; hence $i^*\colon H^4(X,\mathbb{Z}) \to H^4(D,\mathbb{Z})$ is injective with torsion-free cokernel, so $i^*(\frac{1}{2}\theta^2)$ is primitive modulo torsion, and $c_2(G)$ is integral only if 3 divides $2(d+1)$.

(iii) Let s be the section of $\mathcal{O}_X(D)$ vanishing on D. The sheaf F has the locally free resolution $0 \to V(-D) \xrightarrow{s} V \to F \to 0$, and applying $\operatorname{Hom}(-, F)$ gives a long exact sequence with $\operatorname{Ext}^k(V, F) = H^k(\operatorname{End} G)$ and $\operatorname{Ext}^k(V(-D), F) = H^k(\operatorname{End} G \otimes \mathcal{O}_D(D))$, in which the map between these two groups is multiplication by $s|_D = 0$. So it splits into short exact sequences $0 \to H^{k-1}(\operatorname{End} G \otimes \mathcal{O}_D(D)) \to \operatorname{Ext}^k(F, F) \to H^k(\operatorname{End} G) \to 0$. Write $\operatorname{End} G = \mathcal{O}_D \oplus \operatorname{End}_0 G$. On an abelian variety $H^k(\mathcal{O}_X(D)) = 0$ for $k > 0$ and $H^k(\mathcal{O}_X(-D)) = 0$ for $k < n$, and $h^0(\mathcal{O}_X(D)) = b^n$ by Riemann–Roch. From $0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$ one gets $h^k(\mathcal{O}_D) = \binom{n}{k}$ for $k \le n-2$, and also for $k = n-1$ when $b = 1$, because $H^n(\mathcal{O}_D) = 0$ makes the map $H^n(\mathcal{O}_X(-D)) \to H^n(\mathcal{O}_X)$ between lines onto. From $0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to \mathcal{O}_D(D) \to 0$ one gets $h^0(\mathcal{O}_D(D)) = b^n - 1 + n$ and $h^k(\mathcal{O}_D(D)) = \binom{n}{k+1}$ for $k \ge 1$. For $b = 1$ the two contributions of $\mathcal{O}_D$ add up to β_k in every degree, which gives the formula; for a line bundle V, $\operatorname{End}_0 G = 0$, and F is a sheaf, so $\operatorname{Ext}^{<0}(F, F) = 0$. For $b \ge 2$, $\dim \operatorname{Ext}^1(F, F) \ge h^0(\mathcal{O}_D(D)) = b^n + n - 1$, and $b^n > n + 1$.

(iv) Such a product needs $\operatorname{ch}(F) \in P_\theta$ for $F = i_*(V|_D)$, so $[D] = b\theta$ by (ii), and its factors have $(e_0, e_1, e_2) = (1, 10, 20)$ by Proposition 3.22(ii); by (iii) $b = 1$. Then Serre duality gives $e_k = \beta_k$ for every k, so by (iii) $H^*(\operatorname{End}_0 G) = 0$ and $\chi(D, \operatorname{End}_0 G) = 0$. On the other hand (ii) gives $\operatorname{ch}(G) = r\, i^*(S(\theta)T_1(\theta))$, so $\operatorname{ch}(\operatorname{End} G) = \operatorname{ch}(G)\operatorname{ch}(G)^\vee = r^2 i^*\big(S(\theta)^2 T_1(\theta) T_1(-\theta)\big)$, and Hirzebruch–Riemann–Roch on D gives

$$\chi(D, \operatorname{End} G) = r^2 \int_X S(\theta)^2 \, \frac{\theta^2}{e^\theta - 1} = \frac{r^2(32d^2 - 20d - 1)}{6},$$

while $\chi(D, \mathcal{O}_D) = \int_X (1 - e^{-\theta}) = 1$. So $6\chi(D, \operatorname{End}_0 G) = r^2(32d^2 - 20d - 1) - 6 \ge 5$, because $32d^2 - 20d - 1 \ge 11$ for $d \ge 1$. ∎

So from $n = 5$ on the numbers allow a secant object attaining the equality, the line bundles on theta divisors have the Ext groups it requires in every dimension, and what fails is their character. The line bundle of Theorem 3.19(ii) has $c_1 = (2k+1)x - k\theta$, which by Proposition 3.23(ii), valid at $n = 3$, is not restricted from the Jacobian; from $n = 4$ on the Lefschetz theorem forbids such classes. At $n = 5$, by (ii) and (iv), a vector bundle G on a smooth divisor with i_*G a factor attaining the equality is neither a line bundle nor restricted from X; if G lives on a theta divisor and has Chern character restricted from X, its rank is even, and rank two needs $d \equiv 2 \pmod 3$. Bundles of this kind that are not restricted from X, sheaves on singular divisors, sheaves that are not locally free on their support, and objects of positive rank remain; we know no object of these kinds with a secant character and a small Ext algebra.

The singular divisors closest to the two certificates are the theta divisors of Jacobians: both certificates are line bundles on the symmetric product $C^{(g-1)}$, pushed forward along the Abel–Jacobi map, for $g = 2, 3$. From $g = 4$ on the theta divisor is singular and the Lefschetz argument of Proposition 3.23(ii) no longer applies, since $C^{(g-1)}$ carries the class x, which does not come from the Jacobian. The character excludes this continuation all the same.

Proposition 3.24 (The Jacobian continuation stops at genus three)**.** *Let C be a smooth projective curve of genus $g \ge 2$, $X = J(C)$ with its principal polarisation Θ, and $u\colon C^{(g-1)} \to X$ the Abel–Jacobi map, whose image is a translate of the theta divisor. Let x be the class of the divisor $p + C^{(g-2)}$ and $\theta = u^*\Theta$, let L be a line bundle on $C^{(g-1)}$ with $c_1(L) = \alpha x + \beta\theta$, and put $\gamma = \beta - \frac{1}{2}$. Then the perfect complex Ru_*L has*

$$\operatorname{ch}(Ru_*L) = \Theta + \Big(\frac{\alpha}{2} + \gamma\Big)\Theta^2 + c_3\Theta^3 + c_4\Theta^4 + \cdots,$$

and when $\alpha = -2\gamma$, that is $\alpha + 2\beta = 1$,

$$c_3 = -\frac{4\gamma^2 - 1}{24}, \qquad c_4 = -\frac{\gamma(4\gamma^2 - 1)}{72}.$$

Consequently $\operatorname{ch}(Ru_*L) \in P_\Theta$ for some $d \geq 1$ exactly when $g = 2$ and $\alpha + 2\beta = 1$, or $g = 3$, $\alpha + 2\beta = 1$ and $d = \beta(\beta - 1)$; these are the sheaves of Theorem 3.19(i) and (ii). For $g \geq 4$ no line bundle on $C^{(g-1)}$ with c_1 in the span of x and θ has a secant character.

Proof. The Todd class of X is 1, so by Grothendieck–Riemann–Roch $\operatorname{ch}(Ru_*L) = u_*(e^{\alpha x + \beta\theta} \operatorname{td}(C^{(g-1)}))$. By Macdonald [Mac62], [ACGH85, Chapter VII, Section 5], the total Chern class of $C^{(k)}$ is $(1+x)^{k-g+1}e^{-\theta/(1+x)}$; for $k = g-1$ the first factor is 1. In Macdonald's description of the cohomology ring $\theta = \sum_{i=1}^g \sigma_i$ with $\sigma_i^2 = 0$, so $e^{-\theta/(1+x)} = \prod_i (1 + x - \sigma_i)/(1+x)$ and, with $\operatorname{Td}(y) = y/(1-e^{-y})$ and $q = \operatorname{Td}'/\operatorname{Td}$,

$$\operatorname{td}(C^{(g-1)}) = \prod_i \frac{\operatorname{Td}(x - \sigma_i)}{\operatorname{Td}(x)} = \prod_i (1 - \sigma_i q(x)) = e^{-\theta q(x)}.$$

Since $\log \operatorname{Td}(y) = \frac{y}{2} - \frac{y^2}{24} + O(y^4)$, we have $-q(x) = -\frac{1}{2} + \frac{x}{12} + O(x^3)$. The class x^a is that of the image of $C^{(g-1-a)}$ under $D \mapsto D + p_1 + \cdots + p_a$, which u maps birationally onto a translate of W_{g-1-a}, of class $\Theta^{a+1}/(a+1)!$ by Poincaré's formula [ACGH85, Chapter I, Section 5]; with $\theta = u^*\Theta$ and the projection formula, $u_*(x^a\theta^b) = \Theta^{a+b+1}/(a+1)!$. So the coefficient c_k of Θ^k is $\sum_{a+b=k-1} m_{ab}/(a+1)!$, where m_{ab} is the coefficient of $x^a\theta^b$ in $e^{\alpha x}e^{\theta(\gamma + x/12)}$ for $k \leq 4$ (the term $O(x^3)$ first contributes to Θ^5), namely $m_{ab} = \sum_j \frac{\alpha^{a-j}}{(a-j)!}\frac{\gamma^{b-j}}{(b-j)!}\frac{1}{12^j j!}$. This gives $c_1 = 1$, $c_2 = \frac{\alpha}{2} + \gamma$,

$$c_3 = \frac{\alpha^2}{12} + \frac{\alpha\gamma}{2} + \frac{1}{24} + \frac{\gamma^2}{2}, \qquad c_4 = \frac{\alpha^3}{144} + \frac{\alpha^2\gamma}{12} + \frac{\alpha}{72} + \frac{\alpha\gamma^2}{4} + \frac{\gamma}{24} + \frac{\gamma^3}{6},$$

and at $\alpha = -2\gamma$ these are the stated values.

The rank of Ru_*L is 0, so $\operatorname{ch}(Ru_*L) \in P_\Theta$ means $\operatorname{ch}(Ru_*L) = v_\Theta = \Theta - \frac{d}{6}\Theta^3 + \frac{d^2}{120}\Theta^5 - \cdots$, the coefficient of Θ being 1. For $g = 2$ this is $c_2 = 0$. For $g \geq 3$ it also asks $c_3 = -d/6$, that is $d = \gamma^2 - \frac{1}{4} = \beta(\beta - 1)$, which settles $g = 3$. For $g \geq 4$ it asks in addition $c_4 = 0$; as $4\gamma^2 - 1 = 4d$, this is $\gamma d = 0$, so $d = 0$ or $\gamma = 0$, where $d = -\frac{1}{4}$. At $g = 3$, $L = \mathcal{O}((2k+1)x - k\theta)$ has $\beta = -k$ and $d = k(k+1)$, and at $g = 2$, $C^{(1)} = C$ and $L = \mathcal{O}_C(p)$ has $\alpha = 1$, $\beta = 0$. ∎

So the continuation of the two certificates must leave the line bundles on $C^{(g-1)}$: a bundle of higher rank, a sheaf on the theta divisor of a principally polarised abelian variety that is not a Jacobian, or a complex of positive rank.

Remark 3.25 (Untwisting). The template $E = \Phi(F_1 \boxtimes F_2^\vee)$ does not give an untwisted object meeting the polarised criterion by the obvious operations. No line bundle twist makes $\operatorname{ch}(E)$ flat (Theorem 3.19(iii)). At $n = 2$ the object $S^2(E[1]) \otimes P$, for P a line bundle with $c_1(P) = \ell$, is untwisted with flat character, of rank 3 and at the exceptional ratio after a twist by a multiple of η, but its number is $r = 23$, which is odd, while Serre duality and the evenness of the Mukai pairing on an abelian variety make $\dim \operatorname{Ext}^2 \equiv \chi(E,E) \equiv 0 \pmod 2$; so equality is impossible. The object $\bigwedge^2(E[1]) \otimes P$ has no Weil part. For $E \otimes V$ with V a simple semi-homogeneous bundle of slope $\ell/2$ [Muk78], the character $\operatorname{rk}(V)\kappa$ is flat, but $\sigma_{E\otimes V}$ kills the summands $\operatorname{Ext}^2(E, E \otimes P_\xi)$, $\xi \neq 0$, of $\operatorname{Ext}^2(E \otimes V, E \otimes V)$ coming from the decomposition of $V \otimes V^\vee$, and at $n = 2$ these are nonzero; this last point uses the decomposition of $\operatorname{End} V$ and the way Φ intertwines $\otimes P_\xi$ with translations, which we recall

without proof. None of this shows that the polarised criterion must be met in twisted form; it shows that this template meets it only there.

At $n = 4$ the natural objects of the split member can be described completely, and the description turns the criterion into a statement about points on a quadric.

Setup 3.26 (The quadric of Lagrangians)**.** Let X be a very general principally polarised abelian fourfold, $U = H^1(X, \mathbb{Q})$, and $A = X \times \widehat{X}$ with the K-action of [I, (29)], so that $H^1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^2$ and the rational Hodge ring $\mathcal{H}_A$ of A is the ring of $\mathrm{Sp}(U)$-invariants in $\bigwedge^*(U \otimes \mathbb{Q}^2)$, of dimension 55, with $1, 3, 6, 10, 15, 10, 6, 3, 1$ in degrees $0, 2, \dots, 16$. The symmetric pairing on $H^1(A, \mathbb{Q}) \oplus H_1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^4$ is the product of the symplectic form of U and a symplectic form on $\mathbb{Q}^4$, so the maximal isotropic subspaces stable under $\mathrm{Sp}(U)$ are the $U \otimes L$ with L a Lagrangian plane in $\mathbb{Q}^4$. Let $Q \subset \mathbb{P}^4$ be the Lagrangian Grassmannian $\mathrm{LG}(2, 4)$, a smooth quadric threefold in its Plücker embedding. Call a perfect complex on A *natural* if it is a line bundle, a sheaf $\mathcal{O}_B \otimes L$ on an abelian subvariety, a point, or the image of one of these under an autoequivalence. Then the character of a natural F is a multiple of the pure spinor of $U \otimes L_F$ for a rational point $L_F \in Q$, and there is a linear isomorphism $\Phi_4 \colon H^0(Q, \mathcal{O}(4))^\vee \to \mathcal{H}_A$ with $\mathrm{ch}(F) = \Phi_4(\nu_4(L_F))$ up to a rational factor, ν_4 the fourth Veronese map. Under it the powers of η span the image of a conic $C_\eta \subset Q$, and the Weil plane is the rational part of the line spanned by $\nu_4(L_+)$ and $\nu_4(L_-)$ for a conjugate pair of K-points $L_\pm$ of Q; the line $l_W = L_+L_-$ meets Q only in $L_\pm$, and the plane of C_η is its polar plane.

The pure spinor description of the characters of natural objects is classical, and we recall it without proof. The abelian subvarieties $B_{(p,q)} = \{(px, q\varphi x)\}$, with $\varphi \colon X \to \widehat{X}$ the polarisation, form a conic of natural objects, and on it the two points $L_\pm$ have coordinates $(p : q) = (1 : \mp\sqrt{-d})$: by [I, Theorem 14.2(iv)] the eigenspaces $W_\pm$ are the graphs of $\mp\sqrt{-d}\,\beta$. So the Weil classes are the values of the octic $(p, q) \mapsto [B_{(p,q)}]$ at those points.

Theorem 3.27 (Minimal support at $n = 4$)**.** *In Setup 3.26, let $F_1, \dots, F_s$ be natural objects whose points L_{F_i} are pairwise distinct and not on C_η, and suppose that $\sum_i m_i \,\mathrm{ch}(F_i) + P = N\omega_1 + \sum_k c_k \eta^k$ with $m_i \in \mathbb{Q}$, $P \in \mathbb{Q}[\eta]$ and $N \neq 0$. Then $s \geq 8$. If $s = 8$, the eight points lie on one smooth conic $\Pi \cap Q$ with Π a plane containing l_W, the conic is disjoint from C_η, and $\sum_i m_i \,\mathrm{ch}(F_i) = N\omega_1$ exactly. Conversely any eight rational points of such a conic carry such a relation with every $m_i \neq 0$.*

Proof. Through Φ_4 the relation reads $\sum_i m_i \nu_4(L_{F_i}) + P' = c_+\nu_4(L_+) + c_-\nu_4(L_-)$ in $H^0(Q, \mathcal{O}(4))^\vee$, with P' in the span of $\nu_4(C_\eta)$ and $c_\pm$ conjugate and nonzero because $N \neq 0$. A quartic f vanishing on C_η, at L_- and at every L_{F_i} gives $c_+ f(L_+) = 0$; so it suffices to produce such an f with $f(L_+) \neq 0$ unless the configuration is as stated. Let h_- be the linear form $B(L_-, \cdot)$, with B the bilinear form of the quadric Q; it vanishes on the polar plane of l_W, hence on C_η, and at L_-, but not at L_+, since $l_W \not\subset Q$. If the points split into at most three groups G_k with L_+ outside the span of each, linear forms l_k vanishing on G_k and not at L_+ give the quartic $h_- l_1 l_2 l_3$, which kills everything but c_+, so $c_+ = 0$. A rational line through L_+ also passes through L_-, so it is l_W, which has no rational point on Q; hence groups of one or two points never contain L_+ in their span, and a triple does only if it lies in a plane through l_W. For $s \leq 6$ split into pairs. For $s = 7$, either some triple is not in a plane through l_W, or all seven lie in one such plane Π, hence on $C'' = \Pi \cap Q$. That conic is smooth, since a degenerate one is a pair of conjugate lines with at most one rational point, and it misses C_η, since Π meets the plane of C_η in one point z and $z \in Q$ would put the lines $zL_\pm$ in Q. The quartics vanishing on C_η restrict onto all binary octics

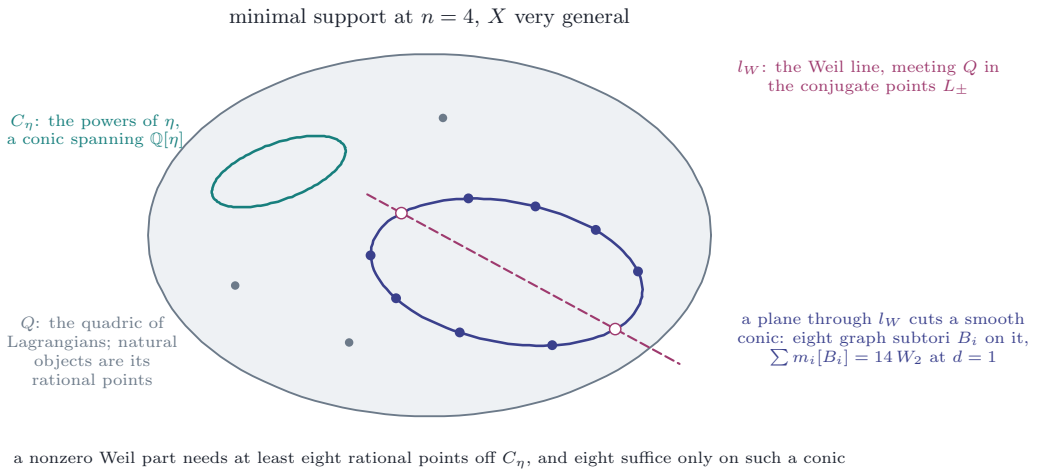


FIGURE 4. Theorem 3.27. The quadric Q of Lagrangians; the rational points are the Chern characters of the natural objects on the split member at $n = 4$, up to the Veronese map ν_4. The powers of the polarisation fill the conic C_η (teal), and the Weil line is spanned by the conjugate points $L_\pm$ (open circles) in which the line l_W (dashed) meets Q. A combination of natural characters with a nonzero Weil part needs at least eight rational points off C_η, and with eight they lie on one conic cut out by a plane through l_W (indigo), such as the eight graph subvarieties B_i of Example 3.28; the grey points are natural objects off any such conic, which no relation of this size can use.

on $C'' \cong \mathbb{P}^1$, and nine distinct points impose independent conditions on octics, so $L_\pm$ and the seven points are independent. For $s = 8$, if no plane through l_W holds seven of the points, choose two triples not in such planes and a pair; if exactly seven lie in Π, modify the quartic by a multiple of one vanishing on Π and at the eighth point; if all eight lie on C'', the ten points $L_\pm, L_{F_1}, \dots, L_{F_8}$ of the rational normal curve $\nu_4(C'')$ of degree eight satisfy exactly one relation, with every coefficient nonzero, and it is rational by Galois symmetry. Its part along the span of $\nu_4(C_\eta)$ vanishes, because quartics restrict onto the sum of the octics on the two disjoint conics, so the relation is pure. ∎

Figure 4 draws the quadric of Lagrangians, the conic C_η and a conic through eight natural characters.

Example 3.28 (Fourteen times a Weil class)**.** Let $d = 1$ and write $(\gamma - \sqrt{-1}\,\ell)^4 = W_1 + \sqrt{-1}\,W_2$ with W_1, W_2 rational, as in [I, Theorem 15.2(iii)]. The eight subvarieties $B_i = B_{(p_i,q_i)}$ with

$$(p_i, q_i) = (1,-3),\ (1,-2),\ (1,2),\ (1,3),\ (2,-1),\ (2,1),\ (3,-1),\ (3,1)$$

and $m = (-1, 16, -16, 1, -16, 16, 1, -1)$ satisfy

$$\textstyle\sum_i m_i[B_i] = 14\,W_2,$$

with the sign of φ for which $B_{(p,q)}$ has conormal forms $p\xi_j - qx_{n+j}$ and $p\xi_{n+j} + qx_j$; with the opposite sign the right side is $-14W_2$. Among the 12870 sets of eight subvarieties with $|p|, |q| \le 3$ this one has the least $\sum_i m_i^2$, namely 1028, and the least $\sum_i |m_i|$, namely 68. In general the coefficients of the relation of Theorem 3.27 for eight subvarieties are proportional

to $\big((q_i^2+dp_i^2)\prod_{j\neq i}(p_iq_j-p_jq_i)\big)^{-1}$. The sum $E=\bigoplus_i \mathcal{O}_{B_i}^{\oplus|m_i|}[k_i]\oplus\mathcal{O}(\eta)\oplus\mathcal{O}(2\eta)\oplus\mathcal{O}(3\eta)$, with $k_i=0$ for $m_i>0$ and 1 otherwise, has $\mathrm{ch}(E)=14W_2+\sum_k c_k\eta^k$ with $k!\,c_k=1+2^k+3^k$, so $\rho=3$ and $r=104$, while $\dim\mathrm{Ext}^2(E,E)=28868$; by Corollary 3.17 it is far from the criterion, and every direct sum of natural objects with $N\neq0$ has $\dim\mathrm{Ext}^2\geq 8\cdot 28=224$.

Remark 3.29 (The flatness form for direct sums). Lemma 3.16 does not apply to the flatness form of Proposition 3.15, but a direct sum fails that form too when its cross-extensions cannot interact. Let $E=\bigoplus_i(\mathcal{O}_{B_i}\otimes L_i)^{\oplus m_i}[k_i]$ with the B_i pairwise distinct abelian subvarieties of dimension n meeting pairwise in finite sets, such as the $B_{(p,q)}$. The extensions between different summands are concentrated in degree n, so an arrow of Ext^1 between summands raises the shift by $n-1$, and a chain of $a\geq1$ arrows lands in $\mathrm{Ext}^{2+a(n-1)}\neq\mathrm{Ext}^2$. Every term of the Maurer–Cartan equation that involves a cross-extension therefore lands in a zero group, the base K_E is the product of the varieties of commuting $2n$-tuples of $m_i\times m_i$ matrices with the space of cross-extensions, and a count of dimensions, using the simultaneously diagonalisable component, gives $\dim K_E>e-k$; so (ii) and (iii) of Proposition 3.15 fail. We give this in outline only. Indecomposable objects, among them iterated extensions of at least eight natural blocks, and sums whose cross-extensions compose to nonzero classes are not covered.

4. Secant objects on the split member

4.1. **Sums of line bundles and the positivity obstruction.** Theorem 2.3 asks for one perfect complex at one base point. The cheapest candidate is a sum of line bundles on a split member, because there the Weil classes are already polynomials in β, $\widehat{\beta}$ and ℓ by [I, Theorem 15.2], so the Chern character one wants is manifestly in reach. This subsection settles that candidate. It does not exist, and the obstruction is not semiregularity: it is the Chern character condition, and it fails because the norm form of K is positive definite.

The norm rules out three different constructions in this paper, by three different mechanisms, and they should be kept apart. [I, Theorem 5.6] uses the norm *character* of $K^\times$ and an equivariance hypothesis. [I, Proposition 15.18] uses the fact that the part R of H^2 on which $K^\times$ acts through τ^2 and $\overline{\tau}^2$ has no Hodge classes at a general point, which is a statement about the Hodge group. Theorem 4.5 below uses neither: it uses that $\mathrm{N}_{K/\mathbb{Q}}$ is a positive definite quadratic form on K and that the multiplicities of a filtration are positive integers. The three are independent, and only the third is available at a special point, which is exactly where the object of Proposition 2.1 has to live.

Lemma 4.1 (The semiregularity map of a split object)**.** *Let Y be a smooth projective complex variety and let*

$$E=\bigoplus_{i=1}^{s} L_i[p_i]^{\oplus m_i},\qquad \lambda_i=c_1(L_i),$$

with the L_i line bundles, the λ_i pairwise distinct and $p_i\in\mathbb{Z}$. Then for $\zeta=(\zeta_{ij})\in\mathrm{Ext}^2(E,E)$,

$$\sigma_E(\zeta)_q=\frac{1}{q!}\sum_{i=1}^{s}(-1)^{p_i}\,\mathrm{tr}(\zeta_{ii})\,\lambda_i^q, \tag{1}$$

so σ_E annihilates $\mathrm{Ext}^2(L_i[p_i],L_j[p_j])$ for $i\neq j$ and the traceless part of each diagonal block. Consequently σ_E is injective if and only if all three of the following hold:

(i) *$m_i=1$ for every i;*
(ii) *$H^{2+p_i-p_j}(Y,L_jL_i^{-1})=0$ for all $i\neq j$;*

(iii) *the map*

$$\Phi\colon H^2(\mathcal{O}_Y)^{\oplus s} \longrightarrow H^*(Y,\mathbb{C}), \qquad (\zeta_i) \longmapsto \sum_i e^{\lambda_i}\zeta_i,$$

is injective.

Proof. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of a line bundle is its first Chern class, so $\mathrm{At}(E)$ is diagonal with entries λ_i and $\mathrm{At}(E)^q$ is diagonal with entries λ_i^q. The matrix product $\zeta \circ \mathrm{At}(E)^q$ therefore has (i,j) entry $\zeta_{ij}\lambda_i^q$, and the trace keeps only the diagonal; a shift by p multiplies the trace by $(-1)^p$. That is (1), and it exhibits the two spaces the map kills.

For the equivalence, the off-diagonal part of $\mathrm{Ext}^2(E,E)$ is $\bigoplus_{i\neq j}\mathrm{Ext}^2(L_i[p_i],L_j[p_j]) = \bigoplus_{i\neq j} H^{2+p_i-p_j}(L_jL_i^{-1})$, which must vanish, giving (ii); the diagonal block at i is $H^2(\mathcal{O}_Y)\otimes \mathrm{End}(\mathbb{C}^{m_i})$ and only its trace survives, so $m_i = 1$, giving (i). What is left is $\bigoplus_i H^2(\mathcal{O}_Y)$, and by (1) an element (ζ_i) of it dies exactly when $\sum_i \lambda_i^q\zeta_i = 0$ for every $q \geq 0$, which is (iii) after absorbing the signs $(-1)^{p_i}$ into the ζ_i and the factorials into the grading. ∎

Condition (ii) is cheap, because the shifts p_i are at our disposal: only their parities are fixed by the Chern character, and a line bundle whose first Chern class is nondegenerate of index k has cohomology in degree k alone, so one may choose the p_i to avoid every index that occurs. Condition (iii) is a rank condition, and condition (i) is free. What is not free is the Chern character itself, and that is the subject of the rest of this subsection.

Notation 4.2 (Coordinates on the divisor classes). Let $A_0 = X \times \widehat{X}$ be a split member, $n \geq 2$, and keep $\gamma = d\beta - \widehat{\beta}$ from [I, Notation 15.1]. Write

$$\theta^{\pm} = \gamma \mp \sqrt{-d}\,\ell \in \textstyle\bigwedge^2 V_{\pm}, \qquad R_0 = \mathbb{Q}\gamma \oplus \mathbb{Q}\ell \subset R.$$

Every $\theta \in R_0$ is $\theta = u\theta^+ + \overline{u}\,\theta^-$ for a unique $u \in K$, and in coordinates $u = g + h\sqrt{-d}$ corresponds to $\theta = 2g\gamma + 2dh\,\ell$. By [I, Proposition 15.19(iv)] the group $\mathrm{NS}(A_0)_{\mathbb{Q}}$ at a split member with X general is $\mathbb{Q}\eta \oplus R_0$, so every divisor class λ is $e\eta + \theta$ with $e \in \mathbb{Q}$ and $\theta \in R_0$, and we write $u(\lambda) \in K$ for the coordinate of its R_0-part.

Lemma 4.3 (A basis in degree four)**.** *For $n \geq 2$ the six products $\beta^2, \widehat{\beta}^2, \ell^2, \beta\widehat{\beta}, \beta\ell, \widehat{\beta}\ell$ are linearly independent in $H^4(A_0,\mathbb{Q})$. Equivalently the six classes η^2, $\eta\theta^+$, $\eta\theta^-$, $(\theta^+)^2$, $(\theta^-)^2$, $\theta^+\theta^-$ are a basis of the degree four part of the subring generated by the divisor classes.*

Proof. Grade $H^*(A_0,\mathbb{Q}) = \bigwedge^* U \otimes \bigwedge^* U^\vee$ by bidegree. Then β has bidegree $(2,0)$, $\widehat{\beta}$ has $(0,2)$ and ℓ has $(1,1)$, so the six products have bidegrees $(4,0)$, $(0,4)$, $(2,2)$, $(2,2)$, $(3,1)$, $(1,3)$. All are distinct except for the pair $\beta\widehat{\beta}$ and ℓ^2, and those two are independent in $\bigwedge^2 U \otimes \bigwedge^2 U^\vee$ because $\beta\widehat{\beta} = \beta \otimes \widehat{\beta}$ is decomposable there while $\ell^2 = -2\sum_{i<j}(x_ix_j)\otimes(\xi_i\xi_j)$ is diagonal in the induced bases, of tensor rank $\binom{2n}{2} > 1$. The second description follows because the change of basis between the two sets of six classes is invertible over K: $\eta = d\beta + \widehat{\beta}$ and $\gamma = d\beta - \widehat{\beta}$ span the same plane as β and $\widehat{\beta}$ since $\det\left(\begin{smallmatrix} d & 1 \\ d & -1 \end{smallmatrix}\right) = -2d \neq 0$, and $\theta^{\pm}$ span the same plane as γ and ℓ over $\mathbb{C}$. ∎

The next lemma says the same thing at every abelian variety of Weil type, not only at a split member, and it says it without coordinates. It is the fact on which the positivity argument turns.

Lemma 4.4 ($\theta^+\theta^-$ is never proportional to η^2)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type with $n \geq 2$, and let $0 \neq \theta^{\pm} \in \bigwedge^2 V_{\pm}$. Then $\theta^+\theta^-$ is not a scalar multiple of η^2 in $H^4(A,\mathbb{C})$.*

Proof. Both classes lie in the summand $\bigwedge^2 V_+ \otimes \bigwedge^2 V_-$ of $H^4(A,\mathbb{C})$, the isotypic component for the character N^2, so the question is one of tensor rank there. The polarisation class η lies in $V_+ \otimes V_-$ and is the hermitian form of [I, Proposition 3.7], hence nondegenerate: in dual bases $\eta = \sum_{a=1}^{2n} v_a \wedge w_a$ with v_a a basis of V_+ and w_a a basis of V_-. Squaring,

$$\eta^2 = \sum_{a,b} (v_a \wedge w_a)(v_b \wedge w_b) = -2 \sum_{a<b} (v_a v_b) \otimes (w_a w_b),$$

which is diagonal in the induced bases of $\bigwedge^2 V_+$ and $\bigwedge^2 V_-$ and therefore has tensor rank exactly $\binom{2n}{2}$. The class $\theta^+\theta^-$ is, up to sign, $\theta^+ \otimes \theta^-$, of tensor rank one. For $n \geq 2$ one has $\binom{2n}{2} \geq 6 > 1$, so the two cannot be proportional. ∎

Theorem 4.5 (The positivity obstruction)**.** *Let (A, ι, η) be of $(K, -1, n)$-Weil type with $n \geq 2$, and suppose that the only divisor classes on which $K^\times$ acts through the norm character are the multiples of η, that is*

$$\mathrm{NS}(A)_{\mathbb{Q}} \cap (V_+ \otimes V_-) = \mathbb{Q}\,\eta. \tag{2}$$

Let E be a perfect complex on A whose Chern character is effective in the exponential basis, meaning

$$\mathrm{ch}(E) = \sum_{i=1}^{s} m_i\, e^{\lambda_i}, \qquad m_i \in \mathbb{Q}_{>0}, \quad \lambda_i \in \mathrm{NS}(A)_{\mathbb{Q}}, \tag{3}$$

and write $\lambda_i = e_i\eta + \theta_i$ with θ_i in the part R of $H^2(A,\mathbb{Q})$ carrying the characters τ^2 and $\overline{\tau}^2$, and $u_i \in K$ for the θ^+-coordinate of θ_i. If $\mathrm{ch}_2(E)$ is a flat Hodge class, then

$$\sum_{i=1}^{s} m_i\, \mathrm{N}_{K/\mathbb{Q}}(u_i) = 0, \tag{4}$$

and since $\mathrm{N}_{K/\mathbb{Q}}$ is positive definite and every m_i is positive, $u_i = 0$ for every i. Hence every λ_i is a rational multiple of η, and $\mathrm{ch}_k(E) \in \mathbb{Q}\,\eta^k$ for every k. In particular $\mathrm{ch}_n(E)$ has no component on the Weil line.

Proof. From (3), $\mathrm{ch}_2(E) = \frac{1}{2}\sum_i m_i \lambda_i^2$, and hypothesis (2) is what makes the decomposition $\lambda_i = e_i\eta + \theta_i$ available with e_i a scalar. Decompose $H^4(A,\mathbb{C})$ into $K^\times$-isotypic pieces and keep the one for N^2, namely $\bigwedge^2 V_+ \otimes \bigwedge^2 V_-$. Writing $\theta_i = u_i\theta^+ + \overline{u}_i\theta^-$ for a fixed choice of generators $\theta^\pm$ of $\bigwedge^2 V_\pm$, one has

$$\lambda_i^2 = e_i^2\eta^2 + 2e_i\eta\theta_i + u_i^2(\theta^+)^2 + \overline{u}_i^2(\theta^-)^2 + 2\mathrm{N}_{K/\mathbb{Q}}(u_i)\,\theta^+\theta^-,$$

and of these five terms only the first and the last are in the N^2 piece: $\eta\theta_i$ carries $\mathrm{N}\tau^2$ and $\mathrm{N}\overline{\tau}^2$, and $(\theta^\pm)^2$ carry τ^4 and $\overline{\tau}^4$, all four distinct from N^2. So the N^2-part of $\mathrm{ch}_2(E)$ is

$$\tfrac{1}{2}\Big(\sum_i m_i e_i^2\Big)\eta^2 + \Big(\sum_i m_i \mathrm{N}_{K/\mathbb{Q}}(u_i)\Big)\theta^+\theta^-.$$

Now compare with the flat Hodge classes of degree four. For $n \geq 3$ they are $\mathbb{Q}\eta^2$ by [I, Theorem 4.14]. For $n = 2$ they are $\mathbb{Q}\eta^2 \oplus W(A)$, and $W(A)$ spans the summands for τ^4 and $\overline{\tau}^4$, which are not N^2. In both cases the N^2-part of a flat Hodge class of degree four is a multiple of η^2. By Lemma 4.4 the class $\theta^+\theta^-$ is not such a multiple, so its coefficient must vanish, which is (4).

Finally $\mathrm{N}_{K/\mathbb{Q}}(g + h\sqrt{-d}) = g^2 + dh^2$ with $d > 0$, so $\mathrm{N}_{K/\mathbb{Q}}(u) \geq 0$ with equality only at $u = 0$, and a sum of nonnegative terms with positive weights vanishes only when every term does. Then $\mathrm{ch}_k(E) = \frac{1}{k!}(\sum_i m_i e_i^k)\eta^k$, and η^k is not on the Weil line by [I, Theorem 4.14]. ∎

Corollary 4.6 (Objects filtered by line bundles)**.** *Let A satisfy* (2)*, which holds at every split member of* [I, Theorem 15.2] *and at every quaternionic base point of* [I, Theorem 15.25]*. Then Theorem 2.3 cannot be applied at A with any of the following:*

(a) *a coherent sheaf admitting a filtration by line bundles, in particular any direct sum of line bundles;*
(b) *a simple semi-homogeneous vector bundle in the sense of Mukai* [Muk78]*, since such a bundle of rank r has $\mathrm{ch}(E) = r\,e^{c_1(E)/r}$;*
(c) *any iterated extension of the objects in (a) and (b).*

In each case the failure is not of the semiregularity hypothesis but of the Chern character hypothesis of Proposition 2.1: no such E has a Chern character which is a flat section of Hodge classes with a nonzero Weil component.

Proof. First, (2) holds at the points named. At a split member $\mathrm{NS}(A)_{\mathbb{Q}}$ has rank three with basis $\beta, \widehat{\beta}, \ell$ by [I, Proposition 15.19(iv)], and of these η spans the norm-character part while γ and ℓ span R. At a quaternionic point $\mathrm{NS}(A)_{\mathbb{Q}}$ is the space of Rosati-symmetric elements of $B = (-d, b)_{\mathbb{Q}}$, which is spanned by 1, j and ij; the first gives η and the other two are semilinear over K, hence lie in R. In both cases the norm-character part of $\mathrm{NS}(A)_{\mathbb{Q}}$ is exactly $\mathbb{Q}\eta$.

Each of (a), (b) and (c) has a Chern character of the shape (3) with positive multiplicities, by additivity of ch along a filtration and by Mukai's computation of the Chern character of a simple semi-homogeneous bundle. The conclusion of Theorem 4.5 then contradicts the requirement that $\mathrm{ch}_n(E)$ have a nonzero component on the Weil line. ∎

Remark 4.7 (Scope of the theorem, and the objects of Markman)*.* At a very general member of the family $\mathrm{NS}(A)_{\mathbb{Q}} = \mathbb{Q}\eta$, so Theorem 4.5 is true for the empty reason that there are no θ_i to begin with. Its content is at the special members, which is exactly where Theorem 2.3 needs its object to live and where the base points of [I, Theorem 15.27] are. That is the opposite of the usual situation in this paper, where the special members are the easy ones.

The theorem does not touch the objects of Markman [Mar25a, Mar25b]. Those are sheaves whose class in K_0 is not a positive combination of exponentials of divisor classes, so (3) fails for them at the first step; and they live at very general points, where, as just noted, the statement is vacuous. What the theorem does is remove the class of objects one would reach for first, and it removes it for a reason which is visible before any cohomology is computed.

Remark 4.8 (Signed sums of line bundles)*.* Theorem 4.5 uses positivity of the multiplicities and nothing else, so it says exactly where to go next: the object must be a complex whose class in K_0 is not effective, that is one in which the line bundles appear with multiplicities of both signs. Such objects exist. Writing $E = F \oplus G[1]$ with F and G sums of line bundles, the class in K_0 is $[F] - [G]$ and (4) becomes $\sum_{L_i \subset F} \mathrm{N}(u_i) = \sum_{L_j \subset G} \mathrm{N}(u_j)$, which is satisfiable; and Lemma 4.1 still applies verbatim, with the shifts absorbing condition (ii).

The object of Theorem 2.3, if it exists at a split member, is not a sum of line bundles in any arrangement of signs and shifts: Corollary 4.21 below proves this in every dimension. It must have Ext^2 that the trace does not see, that is, it must be indecomposable in the derived category. This description of the remaining construction is sharper than the one in Remark 3.7.

4.2. **An explicit object and the Prouhet–Tarry–Escott problem.** Proposition 2.1 produces the object of Theorem 2.3 by an existence argument: the Chern character is an isomorphism onto the Chow ring with rational coefficients, so a K-theory class with the

right character exists. That is enough for the theorem and tells one nothing about the object. This subsection writes it down. The answer turns the question into a Diophantine one, and the Diophantine problem has a classical name.

Setup 4.9 (Objects supported on R)**.** Let A be of $(K,-1,n)$-Weil type, $n\geq 2$, satisfying (2), and keep $\theta^{\pm}$ from Notation 4.2. For $u\in K$ with $\theta(u)=u\theta^{+}+\overline{u}\,\theta^{-}\in \mathrm{NS}(A)_{\mathbb{Q}}$, and for multiplicities $m_i\in\mathbb{Z}$, consider

$$v=\sum_{i=1}^{s} m_i\, e^{\theta(u_i)}, \qquad T_{q,r}=\sum_{i=1}^{s} m_i\, u_i^{q}\,\overline{u}_i^{\,r} \quad (q,r\geq 0).$$

Proposition 4.10 (The Chern character in terms of the u_i)**.** *In Setup 4.9,*

$$v_k=\sum_{\substack{q+r=k\\ 0\leq q,r\leq n}} \frac{T_{q,r}}{q!\,r!}\,(\theta^{+})^{q}(\theta^{-})^{r}, \tag{5}$$

because $(\theta^{\pm})^{q}=0$ for $q>n$. Consequently v is a flat section of Hodge classes with a nonzero Weil component if

$$T_{q,r}=0 \ \text{ for all } 0\leq q,r\leq n \text{ with } (q,r)\notin\{(0,0),(n,0),(0,n)\}, \qquad T_{n,0}\neq 0, \tag{6}$$

and then $v_0=T_{0,0}=\sum_i m_i$, $v_k=0$ for $0<k<2n$ with $k\neq n$, and

$$v_n=\frac{1}{n!}\Big(T_{n,0}(\theta^{+})^{n}+\overline{T}_{n,0}(\theta^{-})^{n}\Big),$$

a nonzero rational multiple of a generator of the Weil line, by [I, Theorem 15.2].

Proof. Expand $\theta(u_i)^k$ by the binomial theorem, divide by $k!$ and collect; the vanishing of $(\theta^{\pm})^{q}$ for $q>n$ is because $\theta^{\pm}\in\bigwedge^{2}V_{\pm}$ and $\dim_{\mathbb{C}}V_{\pm}=2n$. Under (6) the only surviving terms of (5) are the ones listed, and $(\theta^{+})^{n}$ spans $\bigwedge^{2n}V_{+}$, which is the Weil line over $\mathbb{C}$. ∎

So the search for the object of Theorem 2.3 is the search for a finitely supported integer-valued function on $\mathcal{O}_K$ that annihilates every monomial $u^{q}\overline{u}^{\,r}$ with $q,r\leq n$ except three of them. Two things follow at once, and they pull in opposite directions.

Lemma 4.11 (Equal norms force the Prouhet–Tarry–Escott condition)**.** *Suppose all the u_i in Setup 4.9 have the same norm $N=\mathrm{N}_{K/\mathbb{Q}}(u_i)>0$. Then (6) holds if and only if*

$$\sum_i m_i u_i^{k}=0 \ \text{ for } k=0,1,\dots,n-1, \qquad \sum_i m_i u_i^{n}\neq 0. \tag{7}$$

With $m_i=\pm 1$ this says that the u_i split into two multisets on the circle $\mathrm{N}=N$ whose power sums agree through degree $n-1$ and differ at degree n: a Prouhet–Tarry–Escott pair of degree $n-1$, constrained to a norm circle of $\mathcal{O}_K$.

Proof. For $q\geq r$ one has $u^{q}\overline{u}^{\,r}=N^{r}u^{q-r}$, so $T_{q,r}=N^{r}\sum_i m_i u_i^{q-r}$, and the pairs (q,r) with $q,r\leq n$ realise every difference $k=q-r$ from 0 to n. The excluded pairs $(0,0)$ and $(n,0)$ are the ones with $r=0$ and $k=0$ or $k=n$; but $k=0$ also occurs at $(1,1),\dots,(n,n)$, which are not excluded, so $\sum_i m_i=0$ is still required. The remaining differences $k=1,\dots,n-1$ each occur at some non-excluded pair, and $k=n$ occurs only at $(n,0)$. ∎

Theorem 4.12 (Equal norms cannot be semiregular)**.** *Let $E=\bigoplus_i L_i[p_i]$ realise (7) with all norms equal. Then $\mathrm{ch}_0(E)=0$, and the semiregularity map σ_E has a kernel of dimension at least n^2.*

Proof. By Lemma 4.11 the case $k=0$ of (7) gives $\sum_i m_i = 0$, which is $\mathrm{ch}_0(E)$. For the kernel, take $\zeta_i = m_i\zeta$ for a single $\zeta \in H^2(\mathcal{O}_A)$; then by Lemma 4.1(iii)

$$\Phi\big((m_i\zeta)_i\big) = \Big(\sum_i m_i e^{\lambda_i}\Big)\zeta = \mathrm{ch}(E)\,\zeta,$$

which by Proposition 4.10 is a multiple of $\omega\,\zeta$ with ω a generator of the Weil line. Now $W(A)_{\mathbb{C}}$ is spanned by $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, and the Weil type condition says $\dim V_\pm^{0,1} = n$; writing $H^2(\mathcal{O}_A) = \bigwedge^2(V_+^{0,1} \oplus V_-^{0,1})$, the product of a generator of $\bigwedge^{2n} V_+$ with ζ kills the components of ζ in $\bigwedge^2 V_+^{0,1}$ and in $V_+^{0,1} \otimes V_-^{0,1}$, and likewise with the roles exchanged. So $\omega\,\zeta = 0$ for every ζ in $V_+^{0,1} \otimes V_-^{0,1}$, a space of dimension n^2, and all of those lie in the kernel of Φ. ∎

Corollary 4.13 (At least $n+1$ norms)**.** *If E as in Setup 4.9 satisfies (6) and $\mathrm{ch}_0(E) \neq 0$, then the u_i take at least $n+1$ distinct norms.*

Proof. Let $N_1, \dots, N_\nu$ be the distinct norms and M_j the sum of the m_i over the level $\mathrm{N}(u_i) = N_j$. The conditions $T_{r,r} = 0$ for $r = 1, \dots, n$ read $\sum_j M_j N_j^r = 0$ for $r = 1, \dots, n$. The matrix $(N_j^r)_{1\le r\le n,\, 1\le j\le \nu}$ has rank $\min(n,\nu)$ because the N_j are distinct and nonzero, so $\nu \le n$ forces every $M_j = 0$ and hence $\mathrm{ch}_0(E) = \sum_j M_j = 0$. ∎

The smallest solution. At $n=3$ and $K = \mathbb{Q}(\sqrt{-3})$ the Prouhet–Tarry–Escott condition of Lemma 4.11 has a particularly simple solution: the norm circle $\mathrm{N} = 1$ carries the six roots of unity of K, and the two cosets of the cube roots of -1 and of $+1$ are a Prouhet–Tarry–Escott pair of degree two. Explicitly, with ζ_6 a primitive sixth root of unity, the cube roots of -1 are $\zeta_6, \zeta_6^3, \zeta_6^5$ and those of $+1$ are $1, \zeta_6^2, \zeta_6^4$; each triple sums to zero and has zero sum of squares, while the sums of cubes are -3 and $+3$.

Proposition 4.14 (An explicit object at $n=3$)**.** *Let $K = \mathbb{Q}(\sqrt{-3})$, $n=3$, and let $A_0 = X \times \widehat{X}$ be a split member. In the basis $\beta, \widehat{\beta}, \ell$ of $\mathrm{NS}(A_0)_{\mathbb{Q}}$ put*

$$\lambda_1 = (3,-1,3), \quad \lambda_2 = (-6,2,0), \quad \lambda_3 = (3,-1,-3),$$

and $\lambda_{3+i} = -\lambda_i$ for $i = 1,2,3$, with $m_i = +1$ for $i \le 3$ and $m_i = -1$ for $i \ge 4$. Then the six classes are distinct and integral, the perfect complex

$$E = (L_1 \oplus L_2 \oplus L_3) \;\oplus\; (L_4 \oplus L_5 \oplus L_6)[3]$$

has

$$\mathrm{ch}_k(E) = 0 \;\; (k \neq 3), \qquad \mathrm{ch}_3(E) = 12\,\omega_1,$$

and every difference $\lambda_j - \lambda_i$ with $i \neq j$ is nondegenerate of index exactly 3, so that conditions (i) and (ii) of Lemma 4.1 hold for this shift. Condition (iii) fails: the map Φ has rank 75 instead of 90, with kernel of dimension $15 = \dim H^2(\mathcal{O}_{A_0})$.

Proof. With $d=3$, the class $\theta(u) = 2g\gamma + 6h\ell$ of $u = g + h\sqrt{-3}$ has coordinates $(6g, -2g, 6h)$, so $\lambda_1, \lambda_2, \lambda_3$ are the images of the cube roots $\zeta_6, \zeta_6^3, \zeta_6^5$ of -1 and $\lambda_4, \lambda_5, \lambda_6$ those of the cube roots of $+1$; the six classes are distinct and integral. All six have norm 1, and $\sum_i m_i u_i^k$ is 0 for $k = 0, 1, 2$ and $-3 - 3 = -6$ for $k = 3$, so (7) holds at $n = 3$. By Lemma 4.11 and Proposition 4.10, $\mathrm{ch}_k(E) = 0$ for $k \neq 3$ and $\mathrm{ch}_3(E) = -\big((\theta^+)^3 + (\theta^-)^3\big)$, and by [I, Example 15.3] at $n=3$, $\omega_1 \pm \sqrt{-3}\,\omega_2 = -\frac{1}{6}(\theta^{\mp})^3$; so $\mathrm{ch}_3(E) = 12\,\omega_1$.

Every difference is $\lambda_j - \lambda_i = \theta(w)$ with $w = u_j - u_i \neq 0$, and $\theta(w)$ is nondegenerate of index n for every $w \in K^\times$. Indeed $\theta(1) = 2\gamma$ and $\theta(\sqrt{-d}) = 2d\ell$ are of type $(1,1)$, hence so are $\theta^\pm$, so $\theta^+ \in V_+^{1,0} \otimes V_+^{0,1}$, and $V_+^{0,1}$ is the conjugate of $V_-^{1,0}$. In the splitting

of the tangent space dual to $H^{1,0} = V_+^{1,0} \oplus V_-^{1,0}$, the hermitian form of $\theta(w)$ therefore has the block form $\left(\begin{smallmatrix} 0 & wP \\ \bar{w}P^* & 0 \end{smallmatrix}\right)$, with P the block of 2γ, which is invertible because γ is nondegenerate. Its eigenvalues are plus and minus the singular values of wP, so it has signature (n, n), and a line bundle with this class has cohomology only in degree n [BL04, Chapter 3]. For the shift, condition (ii) asks that $2 + p_i - p_j \neq 3$ whenever $i \neq j$, that is $p_i - p_j \neq 1$; within each of the two blocks the difference is 0, and between them it is ± 3.

For the rank, the u_i are the six sixth roots of unity, each once, and $\overline{u}_i = u_i^{-1}$. Absorb the signs into the ζ_i as in Lemma 4.1(iii) and expand $e^{\lambda_i} = e^{u_i\theta^+} e^{\overline{u}_i\theta^-}$:

$$\Phi(\zeta) = \sum_{a,b=0}^{3} \frac{(\theta^+)^a (\theta^-)^b}{a!\, b!}\, Z_{a-b}, \qquad Z_k = \sum_i u_i^k \zeta_i,$$

with k read modulo 6. The $Z_0, \dots, Z_5$ determine the ζ_i, by a Vandermonde matrix. Split $H^{0,2}(A_0) = \bigwedge^2 V_+^{0,1} \oplus (V_+^{0,1} \otimes V_-^{0,1}) \oplus \bigwedge^2 V_-^{0,1}$, write $Z_k = Z_k^+ + Z_k^0 + Z_k^-$ accordingly, and grade $H^*(A_0) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$ by the Hodge types of the two factors. The term $(\theta^+)^a(\theta^-)^b Z_k^0$ has types $(a, a+1)$ and $(b, b+1)$, the term with Z_k^+ has types $(a, a+2)$ and (b, b), and the term with Z_k^- has types (a, a) and $(b, b+2)$. These are different for different (a, b) and different pieces, so every term vanishes separately. Since $\theta^\pm$ pairs $V_\pm^{1,0}$ nondegenerately with $V_\pm^{0,1}$, both of dimension 3, multiplication by $(\theta^\pm)^a$ is injective on $\bigwedge^j V_\pm^{0,1}$ when $a + j \leq 3$ and zero otherwise; in bases with $\theta^+ = \sum_i e_i f_i$ it sends f_J to a signed sum of the $e_I f_{I \cup J}$ with $I \cap J = \emptyset$, $|I| = a$, and different J give disjoint supports. Hence $Z_k^0 = 0$ unless no $a, b \leq 2$ have $a - b \equiv k$, which leaves $k = 3$; $Z_k^+ = 0$ unless no $a \leq 1$, $b \leq 3$ have $a - b \equiv k$, which leaves $k = 2$; and likewise $Z_k^- = 0$ unless $k = 4$. So $\ker \Phi$ is $V_+^{0,1} \otimes V_-^{0,1}$ at $k = 3$ together with $\bigwedge^2 V_+^{0,1}$ at $k = 2$ and $\bigwedge^2 V_-^{0,1}$ at $k = 4$, of dimension $9 + 3 + 3 = 15 = \dim H^2(\mathcal{O}_{A_0})$, and Φ has rank $90 - 15 = 75$. The first summand is the kernel of Theorem 4.12. ∎

Remark 4.15 (What the construction settles). Three things are settled. The object of Proposition 2.1 is not an abstraction: at $n = 3$ over $\mathbb{Q}(\sqrt{-3})$ it is six line bundles and a shift, and its Chern character is exactly 12 times a Weil class with every other graded piece zero. The construction is completely general, and Lemma 4.11 identifies the constraint as a Prouhet–Tarry–Escott problem on a norm circle, a question of a kind that can be attacked directly. And Corollary 4.13 says what any semiregular solution must look like: the u_i must take at least $n + 1$ distinct norms, because otherwise the rank is zero and the argument of Theorem 4.12 produces a kernel of dimension n^2.

This leaves the existence of a solution with $n + 1$ norms. The level sums are then forced up to scale (Proposition 4.20), and Corollary 4.21 shows that no solution could serve, because a line bundle summand off the polarisation line is obstructed along the family whatever the norms.

The obstruction classes of the explicit object. [I, Corollary 9.18] and [I, Remark 9.21] say that the kernel of σ_E contains a copy of the tangent space of the Weil family, and that this is a statement about where the semiregularity map is blind, not a statement that the object is obstructed. For the explicit object the obstruction classes can be written down, and they are not zero. The computation is elementary and we record it in general.

Recall the conventions of [I, Section 9.3]: a first-order deformation of A is $v \in \mathrm{Hom}(H^{1,0}, H^{0,1}) = H^1(A, T_A)$, extended by zero on $H^{0,1}$ and acting on $H^*(A, \mathbb{C})$ as a derivation $v \lrcorner$; a class ξ of type (p, p) stays of type (p, p) to first order along v exactly when $v \lrcorner \xi = 0$. For a line bundle L with $\lambda = c_1(L)$ the Atiyah class is $\lambda \in H^1(\Omega_A^1)$, its

contraction with v is $v \lrcorner \lambda \in H^2(\mathcal{O}_A) = H^{0,2}(A)$, and this is the obstruction to deforming L along v; a shift does not change it. For $E = \bigoplus_i L_i[p_i]$ the Atiyah class is diagonal with entries λ_i, as in the proof of Lemma 4.1, so the obstruction to deforming E along v is the diagonal element

$$\mathrm{ob}_E(v) = \big(v \lrcorner \lambda_i\big)_i \in \bigoplus_i H^{0,2}(A) \subset \mathrm{Ext}^2(E,E). \tag{8}$$

Proposition 4.16 (The obstruction map of a split object)**.** *Let $A_0 = X \times \widehat{X}$ be a split member with β principal and X general, so that $\mathrm{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \widehat{\beta}, \ell \rangle$ by* [I, Proposition 15.19(iv)]*. Let $T = T_{[A_0]}\mathcal{D}_{n,n}$ be the space of polarised first-order deformations commuting with K, of dimension n^2 by* [I, Proposition 9.14]*, and put*

$$T_{\mathrm{spl}} = \big\{\, v \in \mathrm{Hom}(H^{1,0}, H^{0,1}) \;:\; v \lrcorner \beta = v \lrcorner \widehat{\beta} = v \lrcorner \ell = 0 \,\big\}.$$

Let $E = \bigoplus_{i=1}^s L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and let $\mathrm{ob}_E \colon T \to \bigoplus_i H^{0,2}(A_0)$ be the map (8) *restricted to T.*

(i) *T_{spl} is the tangent space to the split locus: it has dimension $n(n+1)/2$, it is contained in T, and it consists of the deformations of $X \times \widehat{X}$ induced by the polarised deformations of (X, β).*

(ii) *If the classes λ_i together with η span $\mathrm{NS}(A_0)_{\mathbb{Q}}$, then $\ker \mathrm{ob}_E = T_{\mathrm{spl}}$ and $\mathrm{rank}\, \mathrm{ob}_E = n(n-1)/2$, the codimension of the split locus in $\mathcal{D}_{n,n}$.*

(iii) *If $\mathrm{ch}_k(E) \in \mathbb{Q}\eta^k$ for $k \neq n$ and $\mathrm{ch}_n(E) \in \mathbb{Q}\eta^n \oplus W(A_0)$, then $\sigma_E \circ \mathrm{ob}_E = 0$ on T: for every $v \in T$,*

$$\sigma_E\big(\mathrm{ob}_E(v)\big) = v \lrcorner \mathrm{ch}(E) = 0. \tag{9}$$

(iv) *For the object of Proposition 4.14, at $n = 3$ and $K = \mathbb{Q}(\sqrt{-3})$: $\ker \mathrm{ob}_E = T_{\mathrm{spl}}$ has dimension 6 and ob_E has rank 3; the image of ob_E lies in $\ker \sigma_E$; the subspace $\Delta = \{z \cdot \mathrm{id}_E : z \in \mathrm{Ann}_{H^{0,2}}(\omega_1)\}$ of* [I, Corollary 9.18] *has dimension 9 and meets the image of ob_E only in zero, so that $\Delta \oplus \mathrm{im}\, \mathrm{ob}_E$ is a twelve-dimensional subspace of the fifteen-dimensional kernel of σ_E; and $\mathrm{ob}_E(v)$ has components $v \lrcorner \lambda_i$ on L_i and $-v \lrcorner \lambda_i$ on $L_i^{-1}[3]$, so it lies on the line of vectors of the shape $(z, \dots, z)$ only when it vanishes. In particular E has a nonzero first-order obstruction along every $v \in T$ not tangent to the split locus, and no element $z \cdot \mathrm{id}_E$ of $\mathrm{Ext}^2(E,E)$ cancels it.*

Proof. (i) Let $P = H^{1,0}(X) \subset H^1(X, \mathbb{C}) = U_{\mathbb{C}}^\vee$ with basis $p_1, \dots, p_n$, so that $p_1, \dots, p_n, \overline{p}_1, \dots, \overline{p}_n$ is a basis of $U_{\mathbb{C}}^\vee$, and let $q_a, \overline{q}_a$ be the dual basis of $U_{\mathbb{C}} = H^1(\widehat{X}, \mathbb{C})$; the dual of a conjugate basis is the conjugate of the dual basis, so the notation is consistent. The Poincaré class is the identity element of $U^\vee \otimes U$,

$$\ell = \sum_{a=1}^n (p_a \wedge q_a + \overline{p}_a \wedge \overline{q}_a),$$

and it is of type $(1,1)$, which forces $H^{1,0}(\widehat{X}) = \langle \overline{q}_a \rangle$ and $H^{0,1}(\widehat{X}) = \langle q_a \rangle$: the Hodge structure of $\widehat{X}$ is the dual one. Since β is of type $(1,1)$ and nondegenerate, $\beta = \sum_{a,b} h_{ab}\, p_a \wedge \overline{p}_b$ with $h \in \mathrm{GL}_n(\mathbb{C})$; and since β is principal with Gram matrix B in a symplectic basis of U and $\widehat{\beta}$ has Gram matrix $-B^{-1} = B$ in the dual basis of $U^\vee$, the Gram matrix of $\widehat{\beta}$ in the basis $p_a, \overline{p}_a$ is minus the inverse of the Gram matrix $\left(\begin{smallmatrix} 0 & h \\ -h^\top & 0 \end{smallmatrix}\right)$ of β in the basis $q_a, \overline{q}_a$, which gives $\widehat{\beta} = -\sum_{a,b} (h^{-1})_{ab}\, \overline{q}_a \wedge q_b$.

Now $H^{1,0}(A_0) = \langle p_a\rangle \oplus \langle \overline{q}_a\rangle$ and $H^{0,1}(A_0) = \langle \overline{p}_a\rangle \oplus \langle q_a\rangle$, so a deformation v is four $n\times n$ matrices,

$$v(p_a) = \sum_c (A_{ca}\,\overline{p}_c + C_{ca}\,q_c), \qquad v(\overline{q}_a) = \sum_c (B_{ca}\,\overline{p}_c + D_{ca}\,q_c),$$

and v kills $\overline{p}_a$ and q_a. Contracting,

$$v\lrcorner\,\ell = \sum_{a,c}\Big(A_{ca}\,\overline{p}_c\wedge q_a + C_{ca}\,q_c\wedge q_a + B_{ca}\,\overline{p}_a\wedge\overline{p}_c + D_{ca}\,\overline{p}_a\wedge q_c\Big),$$

and reading off the three kinds of monomials, $v\lrcorner\,\ell = 0$ if and only if $D = -A^{\top}$ and B, C are symmetric. Next

$$v\lrcorner\,\beta = \sum_{a,b,c} h_{ab}\Big(A_{ca}\,\overline{p}_c\wedge\overline{p}_b + C_{ca}\,q_c\wedge\overline{p}_b\Big),$$

so $v\lrcorner\,\beta = 0$ if and only if $Ch = 0$, that is $C = 0$, and Ah is symmetric. Likewise $v\lrcorner\,\widehat{\beta} = 0$ if and only if $B = 0$ and Dh^{-1} is symmetric. Given $D = -A^{\top}$, the last condition reads $A^{\top}h^{-1} = h^{-\top}A$, which after multiplication by $h^{\top}$ on the left and by h on the right is $h^{\top}A^{\top} = Ah$, the symmetry of Ah again. Hence

$$T_{\mathrm{spl}} = \{(A,B,C,D) = (A,0,0,-A^{\top}) \;:\; Ah \text{ symmetric}\},$$

of dimension $n(n+1)/2$. It is the tangent space to the split locus: the same computation on X alone shows that the polarised first-order deformations of (X,β) are the matrices A with Ah symmetric, the space $T_{[X]}\mathcal{A}_n$ of dimension $n(n+1)/2$; the deformation of $\widehat{X}$ induced by A is the dual one, $-A^{\top}$; and the map $A\mapsto(A,0,0,-A^{\top})$ is injective. Finally $T_{\mathrm{spl}}\subseteq T$: for $v\in T_{\mathrm{spl}}$, $v\lrcorner\,\eta = 0$ because $\eta\in\langle\beta,\widehat{\beta}\rangle$, so v is polarised, and $v\lrcorner\,\omega_1 = v\lrcorner\,\omega_2 = 0$ because ω_1 and ω_2 are polynomials in $\beta,\widehat{\beta},\ell$ by [I, Theorem 15.2] and $v\lrcorner$ is a derivation; by [I, Proposition 9.14] such a v commutes with K.

(ii) Every $v\in T$ has $v\lrcorner\,\eta = 0$. If the λ_i and η span $\mathrm{NS}(A_0)_{\mathbb{Q}} = \langle\beta,\widehat{\beta},\ell\rangle$, then $v\lrcorner\,\lambda_i = 0$ for all i is equivalent, for $v\in T$, to $v\lrcorner\,\beta = v\lrcorner\,\widehat{\beta} = v\lrcorner\,\ell = 0$, so $\ker\mathrm{ob}_E = T\cap T_{\mathrm{spl}} = T_{\mathrm{spl}}$ by (i), and the rank is $n^2 - n(n+1)/2 = n(n-1)/2$, which is the codimension of [I, Proposition 15.19(iii)].

(iii) By (1), $\sigma_E(\zeta) = \sum_i (-1)^{p_i}\zeta_{ii}\,e^{\lambda_i}$ for a diagonal ζ. Since $v\lrcorner$ is a derivation and $v\lrcorner\,\lambda_i$ has even degree, $v\lrcorner\,\lambda_i^{q} = q\,\lambda_i^{q-1}(v\lrcorner\,\lambda_i)$ and so $v\lrcorner\,e^{\lambda_i} = (v\lrcorner\,\lambda_i)\,e^{\lambda_i}$. Therefore

$$\sigma_E(\mathrm{ob}_E(v)) = \sum_i (-1)^{p_i}(v\lrcorner\,\lambda_i)\,e^{\lambda_i} = v\lrcorner\sum_i (-1)^{p_i}e^{\lambda_i} = v\lrcorner\,\mathrm{ch}(E).$$

For $v\in T$ one has $v\lrcorner\,\eta^k = k\eta^{k-1}(v\lrcorner\,\eta) = 0$, and $v\lrcorner\,\omega = 0$ for every $\omega\in W(A_0)$ by [I, Proposition 9.14]; under the hypothesis on $\mathrm{ch}(E)$ this gives (9).

(iv) The three classes $\lambda_1 = \gamma + 3\ell$, $\lambda_2 = -2\gamma$, $\lambda_3 = \gamma - 3\ell$ span $\langle\gamma,\ell\rangle$, and $\langle\eta,\gamma\rangle = \langle\beta,\widehat{\beta}\rangle$ by the determinant in the proof of Lemma 4.3, so (ii) applies and gives the kernel and the rank; Proposition 4.14 gives the hypothesis of (iii), so the image lies in $\ker\sigma_E$. The subspace Δ has dimension $\dim\mathrm{Ann}_{H^{0,2}}(\omega_1) = n^2 = 9$ by [I, Lemma 9.6]. The components of $\mathrm{ob}_E(v)$ on L_i and on $L_i^{-1}[3]$ are $v\lrcorner\,\lambda_i$ and $v\lrcorner\,(-\lambda_i) = -v\lrcorner\,\lambda_i$, so an element of the image of the shape $(z,\dots,z)$ has $z = -z$, hence is zero; this is $\Delta\cap\mathrm{im}\,\mathrm{ob}_E = 0$, and with $\dim\ker\sigma_E = 15$ from Proposition 4.14 the dimension count follows. ∎

Remark 4.17 (Meaning of the obstruction classes). Proposition 4.16 sharpens [I, Remark 9.21] in one respect and exposes a limit of the criterion in another. First, for the explicit object the obstruction classes along the family are nonzero, and they are nonzero exactly in the $n(n-1)/2$ directions normal to the split locus, which are the directions in

which the Weil class has to be transported. Second, those classes lie in the kernel of σ_E by (9), and (9) holds for every object whose Chern character is a flat section of Hodge classes along the family, so the semiregularity map cannot detect the obstruction of any such object. The obstruction is therefore nonzero, and the criterion cannot see it. For the explicit object the situation is the worst possible: the twelve dimensions $\Delta \oplus \operatorname{im} \mathrm{ob}_E$ of the kernel are accounted for by the two mechanisms of [I, Corollary 9.18] and (8), the remaining three lie on the even side of [I, Corollary 9.8] as well, and none of the fifteen is an artefact. The classes of the shape $z \cdot \mathrm{id}_E$ are the ones a change of the twisting class would contribute, one and the same z to every rank one summand; the last assertion of (iv) says that no such change removes the obstruction. So the explicit object does not move off the split locus to first order, either as a complex or as a twisted complex, and the question of [I, Question 15.13] for it is not one of verifying a hypothesis but of replacing the object.

The positive content is a recipe. By (ii), a split object whose classes span $\mathrm{NS}(A_0)_{\mathbb{Q}}$ with η has obstruction map of rank $n(n-1)/2$ against the family, and the spanning hypothesis is more than is needed: a single class $\theta \in \mathrm{NS}(A_0)_{\mathbb{Q}}$ not proportional to η already has $\{v \in T : v \lrcorner \theta = 0\} = T_{\mathrm{spl}}$. Indeed, every $v \in T$ kills η and commutes with K, so it is a pair (v_+, v_-) with $v_\pm \in \mathrm{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$. Since $\theta^\pm$ is of type $(1,1)$ in $\bigwedge^2 V_\pm$, the R_0-part $u\theta^+ + \overline{u}\,\theta^-$ of θ, with $u \neq 0$, gives $v \lrcorner \theta = u\, v_+ \lrcorner \theta^+ + \overline{u}\, v_- \lrcorner \theta^-$, whose two terms lie in the different summands $\bigwedge^2 V_+^{0,1}$ and $\bigwedge^2 V_-^{0,1}$ of $H^{0,2}(A_0)$. So the kernel of $v \mapsto v \lrcorner \theta$ on T does not depend on $u \neq 0$; it is therefore the common kernel for γ and ℓ, the cases $u = 1$ and $u = \sqrt{-d}$, which is T_{spl} by (ii). So every split object built from line bundles other than powers of η is obstructed in exactly the normal directions to the split locus, and Theorem 4.5 says that an object built from powers of η carries no Weil class. What survives is the statement of [I, Remark 9.9]: the object must have positive rank.

The evaluation map of the explicit object. [I, Theorem 9.19] places the copy Δ of the tangent space inside the image of the evaluation map. For a split object the whole of ev_E can be written down, and for the explicit object the entire kernel of σ_E turns out to lie in its image. We identify $HH^2(A_0)$ with

$$HT^2(A_0) = H^{0,2}(A_0) \;\oplus\; \mathrm{Hom}\big(H^{1,0}, H^{0,1}\big) \;\oplus\; \textstyle\bigwedge^2 \big(H^{1,0}\big)^\vee \tag{10}$$

by the Hochschild–Kostant–Rosenberg isomorphism, the three summands being $H^2(\mathcal{O})$, $H^1(T)$ and $H^0(\bigwedge^2 T)$ for the trivial tangent bundle. An element $x = (z, v, \pi)$ acts on $H^*(A_0, \mathbb{C})$ by $z\wedge$, by the derivation $v\lrcorner$ of [I, Section 9.3], and by the double contraction $\pi\lrcorner$, which removes two $(1,0)$ slots; we write $x \lrcorner \xi$ for the sum of the three.

The double contraction is a second order operator. For a class λ of type $(1,1)$, which has a single $(1,0)$ slot, $\pi \lrcorner \lambda = 0$ and $\pi \lrcorner \lambda^k = \binom{k}{2} \lambda^{k-2}\, (\pi \lrcorner \lambda^2)$, because $\pi\lrcorner$ is a composite of two interior products, each of which is an antiderivation, and the only nonzero terms are those in which the two interior products hit two different factors. Summing over k with the factorials gives $\pi \lrcorner e^\lambda = \frac{1}{2}(\pi \lrcorner \lambda^2)\, e^\lambda$, exactly parallel to $v \lrcorner e^\lambda = (v \lrcorner \lambda)\, e^\lambda$ for the first order operator $v\lrcorner$. This is the whole content of the compatibility (12) below: the semiregularity map of a line bundle is multiplication by e^λ, and every summand of HT^2 acts on e^λ as multiplication by a class in $H^{0,2}$ which is precisely the value of ev_L. The same identity, for arbitrary E, is the commutative square of Buchweitz and Flenner [BF03, Corollary 4.3] and its Hochschild form in [Mar25b, Lemma 11.3]; on an abelian variety the Todd class is trivial and no correction to the HKR map is needed [Cal05, BF08].

Proposition 4.18 (The evaluation map of a split object)**.** *Let A_0 be a split member with X general and let $E = \bigoplus_{i=1}^{s} L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and all $\operatorname{Ext}^2(L_i[p_i], L_j[p_j])$, $i \neq j$, zero, so that $\operatorname{Ext}^2(E,E) = \bigoplus_i H^{0,2}(A_0)$.*

(i) *Under (10), ev_E is diagonal:*

$$\mathrm{ev}_E(z, v, \pi)_i = z + v \lrcorner \lambda_i + \tfrac{1}{2}\, \pi \lrcorner \lambda_i^2 \;\in\; H^{0,2}(A_0) = \operatorname{Ext}^2(L_i[p_i], L_i[p_i]). \tag{11}$$

(ii) *For every $x \in HT^2(A_0)$,*

$$\sigma_E\big(\mathrm{ev}_E(x)\big) = x \lrcorner \operatorname{ch}(E), \tag{12}$$

and consequently $\ker \sigma_E \cap \operatorname{im} \mathrm{ev}_E = \mathrm{ev}_E\big(\{x : x \lrcorner \operatorname{ch}(E) = 0\}\big)$. The weakened hypothesis of [Mar25b, Question 11.4] *holds for E if and only if ev_E kills every x that preserves $\operatorname{ch}(E)$ to first order.*

(iii) *$\dim HT^2(A_0) = 2\binom{2n}{2} + 4n^2$ and $\dim \operatorname{Ext}^2(E,E) = s\binom{2n}{2}$; for $s \geq 5$ and every $n \geq 2$ the second exceeds the first, so ev_E is not surjective and* [Mar25b, Lemma 11.3] *cannot apply to a sum of five or more line bundles.*

(iv) *For the object of Proposition 4.14, with $\lambda_i = u_i\theta^+ + \overline{u}_i\theta^-$ as in Notation 4.2: the space $\{x : x \lrcorner \operatorname{ch}(E) = 0\}$ is $\operatorname{Ann}_{H^{0,2}}(\omega_1) \oplus \{v : v \lrcorner \omega_1 = 0\} \oplus \{\pi : \pi \lrcorner \omega_1 = 0\}$, of dimensions 9, 18 and 9; the first and the third summands are both carried by ev_E onto Δ; the second is carried onto*

$$\Sigma = \Big\{ (u_i a + \overline{u}_i b)_{i=1}^{6} \;:\; a \in \textstyle\bigwedge^2 V_+^{0,1},\; b \in \bigwedge^2 V_-^{0,1} \Big\}, \qquad \dim \Sigma = n(n-1) = 6,$$

which contains the image of the obstruction map of Proposition 4.16; $\Delta \cap \Sigma = 0$; and

$$\ker \sigma_E = \Delta \oplus \Sigma \subseteq \operatorname{im} \mathrm{ev}_E\,.$$

So every direction in which σ_E is blind is the value of ev_E on a Hochschild class that preserves $\operatorname{ch}(E)$ to first order, and the weakened hypothesis fails for E by all fifteen dimensions. The rank of ev_E is 45.

Proof. (i) The evaluation map is the action of $HH^*(A_0)$ on $\operatorname{Ext}^*(E,E)$, which under HKR is contraction of a polyvector in $H^q(\bigwedge^p T)$ with $\frac{1}{p!}\operatorname{At}(E)^p$ [BF08]; for $p = 0$ this is $z \cdot \mathrm{id}_E$ as in the proof of [I, Theorem 9.19], for $p = 1$ it is $v \lrcorner \operatorname{At}(E)$, the map (8), and for $p = 2$ it is $\frac{1}{2}\pi \lrcorner \operatorname{At}(E)^2$. The Atiyah class of E is diagonal with entries λ_i, so all three are diagonal with the entries displayed.

(ii) By (1), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_i e^{\lambda_i}$ for diagonal ζ. The three terms of (11) are exactly the factors by which $z\wedge$, $v\lrcorner$ and $\pi\lrcorner$ multiply e^{λ_i}: the first trivially, the second because $v\lrcorner$ is a derivation and $v \lrcorner \lambda_i$ has even degree, the third by the formula for $\pi \lrcorner e^{\lambda}$ above. Hence $\sigma_E(\mathrm{ev}_E(x)) = \sum_i (-1)^{p_i}\, x \lrcorner e^{\lambda_i} = x \lrcorner \operatorname{ch}(E)$. The description of $\ker \sigma_E \cap \operatorname{im} \mathrm{ev}_E$ follows: $\mathrm{ev}_E(x)$ lies in $\ker \sigma_E$ if and only if $x \lrcorner \operatorname{ch}(E) = 0$.

(iii) The three summands of (10) have dimensions $\binom{2n}{2}$, $4n^2$ and $\binom{2n}{2}$. The inequality $s\binom{2n}{2} > 2\binom{2n}{2} + 4n^2$ is $(s-2)(2n-1) > 4n$, and $4n/(2n-1) \leq 8/3$ for $n \geq 2$, so it holds for $s \geq 5$.

(iv) Since $\operatorname{ch}(E) = 12\omega_1$ is of type $(3,3)$, the three summands of $x \lrcorner \operatorname{ch}(E)$ have types $(3,5)$, $(2,4)$ and $(1,3)$, so the kernel is the direct sum of the three kernels. The first is $\operatorname{Ann}_{H^{0,2}}(\omega_1)$, of dimension $n^2 = 9$ by [I, Lemma 9.6]. The second is $\operatorname{Hom}(V_+^{1,0}, V_+^{0,1}) \oplus \operatorname{Hom}(V_-^{1,0}, V_-^{0,1})$, of dimension $2n^2 = 18$, by the proof of [I, Proposition 9.14], whose argument for (i)⇔(iii) does not use the polarisation. For the third, decompose $\bigwedge^2 (H^{1,0})^\vee$ as $\bigwedge^2 (V_+^{1,0})^\vee \oplus (V_+^{1,0})^\vee \otimes (V_-^{1,0})^\vee \oplus \bigwedge^2 (V_-^{1,0})^\vee$; here ω_1 is a combination of $(\theta^+)^3$ and $(\theta^-)^3$ with both coefficients nonzero, $(\theta^\pm)^3$ spans $\bigwedge^3 V_\pm^{1,0} \otimes \bigwedge^3 V_\pm^{0,1}$, the outer blocks of π act

injectively on the respective factor $\bigwedge^3 V_\pm^{1,0}$ and land in different summands of $H^{1,3}$, and the middle block kills both; so the third kernel is the middle block, of dimension $n^2 = 9$.

At a split point $\theta^\pm$ are of type $(1,1)$, so $\theta^\pm \in V_\pm^{1,0} \otimes V_\pm^{0,1}$, and they are nondegenerate pairings because $(\theta^\pm)^3 \neq 0$. Write $\lambda_i^2 = u_i^2(\theta^+)^2 + \overline{u}_i^2(\theta^-)^2 + 2\mathrm{N}(u_i)\,\theta^+\theta^-$. A middle block π kills $(\theta^\pm)^2$, which have no $V_\mp^{1,0}$ slot, so $\frac{1}{2}\pi \lrcorner \lambda_i^2 = \mathrm{N}(u_i)\,\pi\lrcorner(\theta^+\theta^-)$; the six u_i of Proposition 4.14 are $\pm\zeta_6, \pm 1, \pm\zeta_6^{-1}$, all of norm one, so this value is independent of i and $\mathrm{ev}_E(\pi) = z_\pi \cdot \mathrm{id}_E$ with $z_\pi = \pi\lrcorner(\theta^+\theta^-)$. The map $\pi \mapsto z_\pi$ from the middle block to $V_+^{0,1} \otimes V_-^{0,1}$ is the tensor product of the two nondegenerate pairings, hence an isomorphism onto the middle block of $H^{0,2}$, which is $\mathrm{Ann}_{H^{0,2}}(\omega_1)$ by [I, Lemma 9.6]. So ev_E carries the third kernel onto Δ, as it carries the first by [I, Theorem 9.19].

For $v = (v_+, v_-)$ in the second kernel, $v\lrcorner\lambda_i = u_i\, v_+\lrcorner\theta^+ + \overline{u}_i\, v_-\lrcorner\theta^-$ with $a = v_+\lrcorner\theta^+ \in \bigwedge^2 V_+^{0,1}$ and $b = v_-\lrcorner\theta^- \in \bigwedge^2 V_-^{0,1}$; as $v_\pm$ ranges over $\mathrm{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$, a and b range over the whole of $\bigwedge^2 V_\pm^{0,1}$ because the pairings $\theta^\pm$ are nondegenerate, which is the surjectivity of $\mathrm{alt}(Vc)$ in [I, (26)]. So the image is Σ. The map $(a,b) \mapsto (u_i a + \overline{u}_i b)_i$ is injective, because $(u_1, \overline{u}_1)$ and $(u_2, \overline{u}_2)$ are linearly independent in $\mathbb{C}^2$, $u_1/\overline{u}_1 = \zeta_6^2$ differing from $u_2/\overline{u}_2 = 1$; hence $\dim\Sigma = 2\binom{n}{2} = n(n-1) = 6$. The obstruction map of Proposition 4.16 is the restriction to the polarised K-commuting v, so its image lies in Σ. An element of Σ of the shape $(z, \dots, z)$ has $u_i a + \overline{u}_i b = z = -z$ on comparing i with $3+i$, so $z = 0$ and then $a = b = 0$: $\Delta \cap \Sigma = 0$. By (ii), $\Delta \oplus \Sigma \subseteq \ker\sigma_E \cap \mathrm{im}\,\mathrm{ev}_E$, and its dimension $9 + 6 = 15$ equals $\dim\ker\sigma_E$ by Proposition 4.14, so the three spaces coincide.

For the rank, each term of (11) is a scalar depending on i times a class independent of i. Write $\mathbf{u}^k = (u_i^k)_i \in \mathbb{C}^6$ and $M = V_+^{0,1} \otimes V_-^{0,1}$. The class z contributes $\mathbf{u}^0 \otimes z$. Split v into its four blocks $v_{\varepsilon\varepsilon'} \colon V_\varepsilon^{1,0} \to V_{\varepsilon'}^{0,1}$; then $v\lrcorner\theta^+$ is the sum of $v_{++}\lrcorner\theta^+ \in \bigwedge^2 V_+^{0,1}$ and $v_{+-}\lrcorner\theta^+ \in M$, the first ranging over $\bigwedge^2 V_+^{0,1}$ as above and the second over M because θ^+ is a nondegenerate pairing, and symmetrically for $v\lrcorner\theta^-$ with the blocks v_{--} and v_{-+}. As $\overline{u}_i = u_i^{-1}$, the values on the second summand of (10) are therefore $\mathbf{u}^1 \otimes (a+c) + \mathbf{u}^{-1} \otimes (b+c')$ with a, b, c, c' ranging independently over $\bigwedge^2 V_+^{0,1}$, $\bigwedge^2 V_-^{0,1}$, M, M. On the third summand, the outer block π_{++} kills $(\theta^-)^2$ and $\theta^+\theta^-$, and $\pi_{++} \mapsto \frac{1}{2}\pi_{++}\lrcorner(\theta^+)^2$ is an isomorphism onto $\bigwedge^2 V_+^{0,1}$ (in bases with $\theta^+ = \sum_j e_j f_j$ it sends $e_j^\vee \wedge e_k^\vee$ to $\pm f_j f_k$); likewise for π_{--}, and the middle block contributes $\mathbf{u}^0 \otimes z_\pi$. So

$$\mathrm{im}\,\mathrm{ev}_E = \mathbf{u}^0 \otimes H^{0,2} + \mathbf{u}^{\pm 1} \otimes \left(\textstyle\bigwedge^2 V_\pm^{0,1} \oplus M\right) + \mathbf{u}^{\pm 2} \otimes \textstyle\bigwedge^2 V_\pm^{0,1}.$$

The u_i are the six sixth roots of unity, so the vectors $\mathbf{u}^k$ for $k = 0, \pm 1, \pm 2$ are distinct characters of the cyclic group they form and are linearly independent. The sum is direct, and the rank is $15 + 2 \cdot 12 + 2 \cdot 3 = 45$. ∎

Figure 5 draws the three kernels and their images in $\mathrm{Ext}^2(E,E)$.

Remark 4.19 (Question 11.4 for split objects). For a split object the two criteria stand or fall together, and for the explicit object both fall completely. [I, Theorem 9.19] makes the first n^2 dimensions of the failure uniform over every concentrated object in every dimension; Proposition 4.18(iv) accounts for the remaining $n(n-1)$ at $n = 3$ by the deformations of the two halves $V_\pm$ of the Hodge structure separately, and the equality $\ker\sigma_E = \Delta \oplus \Sigma$ says that the two mechanisms exhaust the kernel: nothing in it is accidental. None of this touches the case for which the question was asked. A sheaf of positive rank has $z \cdot \mathrm{ch}(E) \neq 0$ for every $z \neq 0$, so its Δ is zero, and whether ev_E kills the deformations that preserve $\kappa(E)$ is a statement about the sheaf and not about its Chern character. Markman's candidate in [Mar25b, Section 12], a rank one complex on a principally polarised abelian fourfold

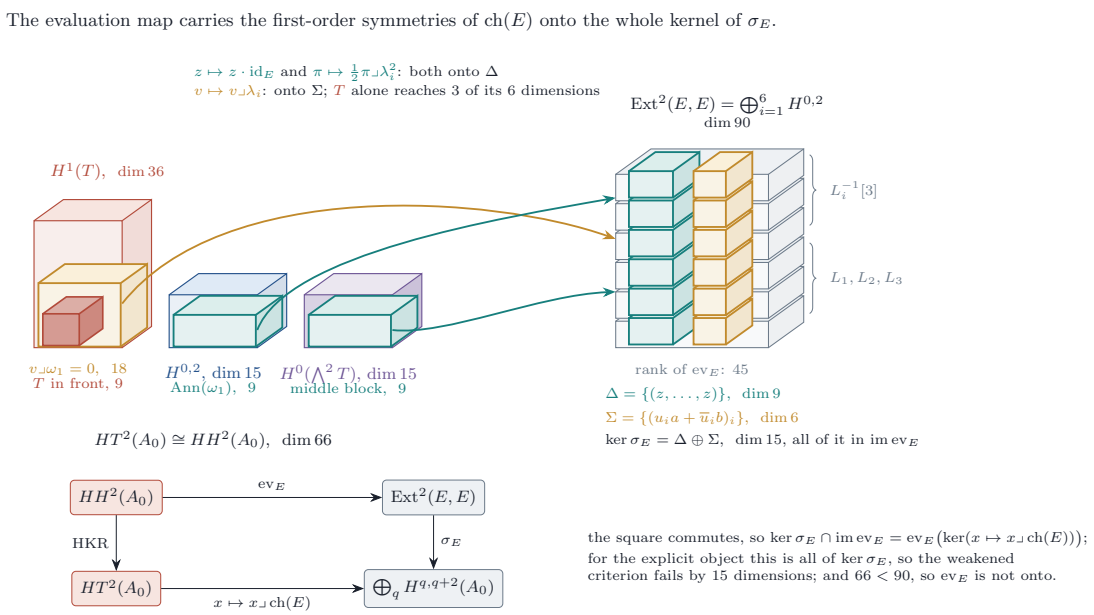


FIGURE 5. Proposition 4.18 for the explicit object at $n = 3$. Left: the three summands of $HT^2(A_0)$, drawn with heights proportional to their dimensions, and inside each the part that preserves $\mathrm{ch}(E)$ to first order: the annihilator of ω_1, the deformations killing ω_1 with the polarised K-commuting tangent space T in front, and the middle block of bivectors. Right: $\mathrm{Ext}^2(E,E)$ as six stacked copies of $H^{0,2}$, one per summand of E, and inside it the kernel of σ_E: the column Δ, one and the same class in every layer, and the column Σ, whose entries depend on the summand through u_i. The three arrows are the evaluation map on the three kernels; the two teal ones both land on Δ, the ochre one on Σ, and together they cover the kernel. Below, the commutative square that makes this a statement about the weakened criterion of [Mar25b, Question 11.4].

with Chern character on the secant $\langle e^{\pm\sqrt{-d}\Theta}\rangle$, is exactly such a sheaf, and he records that the verification remains to be done. Theorem 4.56 reduces that verification to one identity on the fourfold, and Theorem 4.61 shows that the identity fails: the candidate does not satisfy the weakened criterion either.

Solutions of nonzero rank. We return to the Diophantine problem of Setup 4.9. By Corollary 4.13 a solution with $\mathrm{ch}_0(E) \neq 0$ needs at least $n+1$ distinct norms, and multiplicity one bounds each level sum by the number of elements of $\mathcal{O}_K$ of that norm.

Proposition 4.20 (The level sums)**.** *Keep Setup 4.9 and suppose E satisfies (6) with $\mathrm{ch}_0(E) \neq 0$, so that the u_i take $\nu \geq n+1$ distinct norms by Corollary 4.13. Write $N_1 < \cdots < N_\nu$ for those norms, M_j for the sum of the m_i at level N_j, and $r_K(N) = \#\{u \in \mathcal{O}_K : \mathrm{N}_{K/\mathbb{Q}}(u) = N\}$. If $\nu = n+1$ then, with $P = \prod_k N_k$, $S = \mathrm{ch}_0(E)$ and $D_j = N_j \prod_{k\neq j}(N_j - N_k)$,*

$$M_j = (-1)^n \frac{S\,P}{D_j}, \tag{13}$$

the least $S > 0$ making every M_j an integer is $S_{\min} = \operatorname{lcm}_j(|D_j|/\gcd(|D_j|, P))$, and multiplicity one, Lemma 4.1(i), forces $|M_j| \le r_K(N_j)$ for every j.

Proof. The conditions $T_{r,r} = 0$ for $r = 1, \dots, n$ say that the linear functional $f \mapsto \sum_j M_j f(N_j)$ kills $x, x^2, \dots, x^n$, hence sends every polynomial f of degree at most n to $S\, f(0)$. Applying it to $f(x) = \prod_{k \ne j}(x - N_k)$, which has degree n, gives

$$M_j \prod_{k \ne j} (N_j - N_k) = S\,(-1)^n \prod_{k \ne j} N_k,$$

and multiplying by N_j turns this into (13). The value of $S_{\min}$ is the definition of a least common multiple, and the bound on $|M_j|$ holds because each u of norm N_j contributes at most one term $m_i = \pm 1$ to M_j. ∎

So the level sums are determined by the norms up to the scale S, and they compete with the supply $r_K(N_j)$ of elements at each level. That competition is arithmetic. A second mechanism, in the cohomology of A, closes the split route for every n and every set of norms at once.

Corollary 4.21 (Sums of line bundles fail both criteria)**.** *Let A be of $(K, -1, n)$-Weil type with $n \ge 2$, and let $E = \bigoplus_i L_i[p_i]^{\oplus m_i}$ be a sum of shifted line bundles, twisted by a common class or not, whose (twisted) Chern character remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component. Then σ_E is not injective on the image of ev_E: E is neither semiregular nor semiregular in the weakened sense of* [Mar25b, Question 11.4]*, and it is obstructed to first order along some direction of the Weil family. In particular every object of Setup 4.9 satisfying (6), for every $n \ge 2$, every field K, every rank and every set of norms, fails both criteria.*

Proof. Write λ_i for the twisted first Chern classes of the summands. If every λ_i were a real multiple of η, it would be a rational multiple, both classes being rational, and the twisted Chern character $\sum_i m_i e^{\lambda_i}$ would be a polynomial in η; its component in degree $2n$ would then lie in $\mathbb{Q}\eta^n$, which meets the Weil line in zero by [I, Proposition 5.1], against the hypothesis. So some λ_i is not a real multiple of η, and [I, Theorem 9.20] applies to the summand $L_i[p_i]$. For an object of Setup 4.9 the classes are $\lambda_i = \theta(u_i) \in R_0$, and $R_0 \cap \mathbb{Q}\eta = 0$; no twist is needed because $c_1(E) = \theta(T_{1,0}) = 0$ by (6), as $(1, 0)$ is not an excluded pair for $n \ge 2$; and (6) gives a flat Chern character with a nonzero Weil component by Proposition 4.10. ∎

Remark 4.22 (What the corollary asks of an object)*.* By Corollary 4.21 there is no sum of line bundles, at any member of any Weil family in any dimension, that could run the semiregularity argument in either form. The reason is [I, Proposition 9.15(i)]: the only divisor class that stays of type $(1, 1)$ across the whole family is the polarisation, so every line bundle summand off that line is obstructed in some direction of the family, and the semiregularity map, whose composite with the obstruction is contraction with the flat class $\mathrm{ch}(E)$, cannot see it. The object the argument needs is one with no rank one summand off the polarisation line, of positive rank, indecomposable enough that its Atiyah class contracted with every direction of T vanishes. Markman's reflexive sheaves at $n = 3$ are of that kind, and nothing here says that their analogues at $n \ge 4$ do not exist.

4.3. **Convolutions of line bundles.** After Corollary 4.21 the next objects built from line bundles are their convolutions, the iterated cones of shifted line bundles. Their Chern character is the same signed sum of exponentials as for a direct sum, but their Ext groups are cut down by the maps between the pieces, so the argument of [I, Theorem 9.20]

no longer applies as it stands. This subsection shows that fewer than $2n$ pieces with nondegenerate differences never serve (Theorem 4.26); writes down, at a CM member for $K = \mathbb{Q}(i)$, a signed sum of $2 \cdot 4^{n-1}$ line bundles that is exactly a real Weil class (Proposition 4.27); and determines, for the convolutions of those pieces with three multiples of the polarisation at $n = 3$, which products in the twisted differential can act on the 525 diagonal classes of degree two, of which at least 468 must disappear before the polarised criterion can hold (Theorem 4.31). Nothing enters the diagonal, cup products remove at most 276 of these classes, and everything else would have to be removed by fourfold products along a path through the three multiples and one of the Weil pieces: one family of them for some placements of the multiples, and two families that exclude each other for the others (Proposition 4.32); one order of the pieces gives a convolution that solves the Maurer–Cartan equation (Proposition 4.33). Off the diagonal, the classes between two pieces that differ in every coordinate are reached by no product (Lemma 4.34), and no convolution of these pieces meets the polarised criterion (Theorem 4.35). At $n = 4$ the same classes, now between pieces that differ in three coordinates, are reached by no product in any convolution, and they exclude every convolution whose 128 pieces L_ζ occupy two shifts (Theorem 4.37) or three consecutive ones (Theorem 4.38); with the groups between pieces whose shifts differ by two or three (Lemma 4.39) they exclude every convolution whose pieces have their shifts within five consecutive values (Corollary 4.44); the diagonal classes at a lonely shift (Lemma 4.45) and the interleaved shifts (Proposition 4.47) extend this to six (Corollary 4.48).

Setup 4.23 (Convolutions)**.** Let A be an abelian variety of dimension g with a flat Kähler metric, and let $L_1, \dots, L_m$ be line bundles on A with hermitian metrics of constant curvature, hermitian forms $H_1, \dots, H_m$ and pairwise distinct first Chern classes. For integers $d_1, \dots, d_m$ put $P_a = L_a[-d_a]$. A *convolution* of $P_1, \dots, P_m$ is an object E given by a one-sided twisted complex[7]

$$\Big(\bigoplus_a \mathcal{A}^{0,*}(L_a)[-d_a],\ \bar\partial + \delta\Big), \qquad \delta = (\delta_{ab})_{a<b}, \quad \delta_{ab} \in \mathcal{A}^{0,1+d_a-d_b}(L_a^{-1} \otimes L_b),$$

with $\bar\partial\delta + \delta \circ \delta = 0$, so that $\mathrm{ch}(E) = \sum_a (-1)^{d_a} e^{c_1(L_a)}$. Write $\mathcal{A}(a,b) = \mathcal{A}^{0,*}(L_a^{-1} \otimes L_b)$ and $\mathcal{H}(a,b)$ for its harmonic forms, so that $\mathcal{H}^q(a,b) \cong H^q(A, L_a^{-1} \otimes L_b)$, and write π, G and $h = \bar\partial^* G$ for the harmonic projection, the Green operator and the homotopy, with $1 - \pi = \bar\partial h + h\bar\partial$ and $\pi h = h\pi = h^2 = 0$. A form of degree q in $\mathcal{A}(a,b)$ is a morphism $P_a \to P_b$ of degree $w = q - d_a + d_b$; we call q its Ext degree and, when $w = 1$, call $q - 1$ its *excess*. When $H_b - H_a$ is nondegenerate, $\mathcal{H}(a,b)$ is concentrated in the single degree $n(H_b - H_a)$, the number of negative eigenvalues [BL04, Chapter 3].

Lemma 4.24 (The harmonic model)**.** *In Setup 4.23:*

(i) *E is isomorphic to a convolution of the same pieces with $h\delta = 0$, and then $\delta = \delta^H - h(\delta\circ\delta)$ with $\delta^H = \pi\delta$ harmonic. Substituting repeatedly, every δ_{ab} is a finite sum of planar binary trees whose leaves are components δ^H_{cd} with $c < d$, whose internal vertices compose, and whose internal edges carry $-h$.*

(ii) *$\mathrm{Ext}^*(E,E)$ is the cohomology of $\mathcal{H}_E = \bigoplus_{a,b} \mathcal{H}(a,b)$, graded by w, with the differential $d_E = \sum_{k\ge 0} \pi\, Q(-hQ)^k$, where $Q = [\delta, -]$.*

[7] Every object obtained from $P_1, \dots, P_m$ by iterated cones in this order is of this form, because the Dolbeault complexes form a differential graded enhancement of the full subcategory of line bundles. All the objects of this subsection are given in this way.

(iii) *For $x \in \mathcal{H}(a,b)$, the component of $d_E(x)$ in $\mathcal{H}(c,e)$ is a sum of terms $\pm\pi\, T$, one for each planar binary tree T with $k \geq 2$ leaves, internal vertices composing, $-h$ on the internal edges other than the root, whose leaves read in order are the components along a path $c = a_0 \to a_1 \to \cdots \to a_k = e$ in which one step is $x\colon a \to b$ and the others are components $\delta^H_{a_j a_{j+1}}$ with $a_j < a_{j+1}$. The* Ext *degree of such a term is the sum of the* Ext *degrees of its leaves minus $k-2$.*

The terms with k leaves in (iii) are the Massey products of length k of the pieces;[8] (iii) is all that is used of them below.

Proof. (i) Suppose $h\delta_{cd} = 0$ for $d - c < r$. Conjugating by $1 + g$, with g of degree 0 supported at distance r, replaces δ_{ab} for $b - a = r$ by $X_{ab} + \bar\partial g_{ab}$, where X_{ab} depends only on δ, and leaves the components at smaller distance unchanged. With $g_{ab} = -hX_{ab}$ the new component is $X_{ab} - \bar\partial h X_{ab}$, and $h\bar\partial h = h(1 - \pi - h\bar\partial) = h$ gives $h(X_{ab} - \bar\partial h X_{ab}) = 0$. Induction on r gives $h\delta = 0$, and then $\delta = \pi\delta + \bar\partial h\delta + h\bar\partial\delta = \delta^H - h(\delta \circ \delta)$ by the Maurer–Cartan equation. The component $(\delta \circ \delta)_{ab}$ involves only components at distance less than $b - a$, so the substitution ends. (ii) The endomorphism complex of E is $\bigoplus_{a,b} \mathcal{A}(a,b)$ with the differential $\bar\partial + Q$. Blockwise, the inclusion of harmonic forms, π and $-h$ form a strong deformation retraction of $(\bigoplus \mathcal{A}(a,b), \bar\partial)$ onto $(\mathcal{H}_E, 0)$, and Q raises $b-a$, so hQ is nilpotent. The basic perturbation lemma gives a homotopy equivalence of $(\bigoplus \mathcal{A}(a,b), \bar\partial + Q)$ with $(\mathcal{H}_E, d_E)$, with the chain maps $\iota_E = \sum_k (-hQ)^k$ on $\mathcal{H}_E$ and $\pi_E = \pi \sum_k (-Qh)^k$. (iii) Each Q in d_E composes with one component of δ on the left or on the right; writing that component as in (i) gives the trees, the edges from (i) and those from d_E being exactly the internal edges other than the root. Composition adds degrees, h lowers them by one, and a binary tree with k leaves has $k-2$ internal edges. ∎

Lemma 4.25 (The index of a sum)**.** *For hermitian forms X, Y on $\mathbb{C}^g$, $n(X+Y) \leq n(X) + n(Y)$. If $H_1 + \cdots + H_k = 0$, then $\sum_j n(H_j) \geq g - \dim\ker H_i$ for every i; in particular $\sum_j n(H_j) \geq g$ when one H_i is nondegenerate.*

Proof. A subspace on which $X+Y$ is negative definite meets the subspaces on which $X \geq 0$ and $Y \geq 0$, of codimensions $n(X)$ and $n(Y)$, only in zero, so its dimension is at most $n(X) + n(Y)$. For the second statement, $-H_i = \sum_{j\neq i} H_j$ gives $g - n(H_i) - \dim\ker H_i = n(-H_i) \leq \sum_{j\neq i} n(H_j)$. ∎

Theorem 4.26 (Fewer than $2n$ pieces)**.** *Let A be of $(K, -1, n)$-Weil type with $n \geq 2$, and let E be a convolution of $m < 2n$ shifted line bundles whose differences $H_b - H_a$ are all nondegenerate. Suppose that* $\mathrm{ch}(E)$ *remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component, as every polarised character $N\omega_1 + p(\theta)$ with $N \neq 0$ does. Then there is $v \in T_{[A]}\mathcal{D}_{n,n}$ with*

$$\mathrm{ev}_E(v) \neq 0 \qquad \text{and} \qquad \sigma_E(\mathrm{ev}_E(v)) = 0,$$

so σ_E is not injective, and for a polarised character $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\, E)$ *by Theorem 3.9(iv).*

Proof. First, no element of degree one of $\mathcal{H}_E$ has a differential with a nonzero component in the diagonal classes $\mathcal{H}^2(c,c) = H^2(A, \mathcal{O}_A)$. Such a component is a sum of terms of Lemma 4.24(iii) whose k leaves have degree one and run around a closed path $c \to$

[8] In the language of A_∞-categories they are the higher products m_k of the minimal model of the line bundles, evaluated on the Maurer–Cartan element δ^H of the twisted complex; for $k = 2$ they are cup products.

$\cdots \to c$; the pieces before the step x increase from c and those after it increase to c, so the path passes through k distinct pieces and $k \le m$. Its leaves have Ext degrees $q_j = n(H_{a_{j+1}} - H_{a_j})$, and the output has Ext degree $\sum_j q_j - (k-2) = 2$, so $\sum_j q_j = k < 2n$. The k differences are nondegenerate and add up to zero, so Lemma 4.25 gives $\sum_j q_j \ge g = 2n$, a contradiction. Hence taking diagonal components gives a well defined map $\pi_D\colon \operatorname{Ext}^2(E,E) \to \bigoplus_a H^2(A, \mathcal{O}_A)$.

Next, $\operatorname{ev}_E(v) = v \lrcorner \operatorname{At}(E)$, as in the proof of [I, Theorem 9.20]. In the twisted complex the Atiyah class is represented by the commutator of the Chern $(1,0)$-connection $\bigoplus_a \nabla_a$ with $\bar\partial + \delta$; its diagonal entries are the curvatures F_a, constant forms representing $c_1(L_a)$, and its other entries $\nabla \delta_{ab}$ lie above the diagonal. So $\operatorname{ev}_E(v)$ is represented by an upper triangular z with $z_{aa} = v \lrcorner F_a$, and in $\mathcal{H}_E$ by $\pi_E(z)$. The operators Q and h preserve upper triangular forms and Q makes them strictly upper triangular, so the diagonal of $\pi_E(z)$ is that of $\pi(z)$, and $\pi_D(\operatorname{ev}_E(v)) = (v \lrcorner c_1(L_a))_a$.

If every $c_1(L_a)$ were a real multiple of the polarisation, it would be a rational one and $\operatorname{ch}(E)$ a polynomial in it, whose component in degree $2n$ meets the Weil line in zero by [I, Proposition 5.1]. So some $c_1(L_a)$ is not, and by [I, Proposition 9.15(i)] some $v \in T_{[A]}\mathcal{D}_{n,n}$ has $v \lrcorner c_1(L_a) \ne 0$; then $\pi_D(\operatorname{ev}_E(v)) \ne 0$, while $\sigma_E(\operatorname{ev}_E(v)) = v \lrcorner \operatorname{ch}(E) = 0$. ∎

[I, Theorem 9.20] is the case $\delta = 0$, with any number of pieces. A convolution reaches its diagonal classes once it has $2n$ pieces, when a cycle through all of them can have total index $2n$, the equality case of Lemma 4.25. A convolution meeting the polarised criterion must therefore have at least $2n$ pieces, or pieces with degenerate differences. The next construction has both.

Proposition 4.27 (A pure Weil class from line bundles)**.** *Let $E_0 = \mathbb{C}/\mathbb{Z}[i]$, $A = E_0^{2n}$ with coordinates $z_1, \dots, z_{2n}$ and $n \ge 2$, let $K = \mathbb{Q}(i)$ act by i on $z_1, \dots, z_n$ and by $-i$ on $z_{n+1}, \dots, z_{2n}$, and let θ be the product principal polarisation; then A is of $(\mathbb{Q}(i), -1, n)$-Weil type, a CM member as in* [I, Remark 17.6]*. Write B_j for the surface with coordinates (z_j, z_{n+j}), so that $A = B_1 \times \cdots \times B_n$. For $p \in \mathbb{Z}$ and $\zeta \in \mu_4^n$ let L_ζ be a line bundle whose hermitian form is $\left(\begin{smallmatrix} p & \zeta_j \\ \bar\zeta_j & p \end{smallmatrix}\right)$ on B_j and zero between different surfaces, and put $\alpha = \prod_j \frac{i}{2}\, dz_j \wedge d\bar z_{n+j}$.*

(i) *α and $\overline{\alpha}$ span the Weil classes of A, and*

$$\sum_{\prod \zeta = 1} e^{c_1(L_\zeta)} - \sum_{\prod \zeta = -1} e^{c_1(L_\zeta)} = 2 \cdot 4^{n-1}\,(\alpha + \overline{\alpha}).$$

(ii) *For distinct integers t_1, t_2, t_3 and signs σ_i, the character $\gamma = 2\cdot 4^{n-1}(\alpha+\overline{\alpha}) + \sum_i \sigma_i e^{t_i\theta}$ has Hankel rank 3, so $r(\gamma) = 7n^2 - 2n$. A convolution with this character of the $2\cdot 4^{n-1}$ bundles L_ζ, with $(-1)^d = \prod \zeta$, and of three line bundles M_i with $c_1(M_i) = t_i\theta$ has $2\cdot 4^{n-1} + 3$ pieces and $(2\cdot 4^{n-1}+3)\, n(2n-1)$ diagonal classes of degree two:* 525 *against $r = 57$ at $n = 3$, and* 3668 *against* 104 *at $n = 4$.*[9]

Proof. The hermitian forms have imaginary parts integral on $\mathbb{Z}[i]^{2n}$, so the line bundles exist. On B_j write c_1 of the restriction as $p(a_j + b_j) + \zeta_j c_j + \overline{\zeta}_j c_j'$, where $a_j = \frac{i}{2} dz_j \wedge d\bar z_j$, $b_j = \frac{i}{2} dz_{n+j} \wedge d\bar z_{n+j}$, $c_j = \frac{i}{2} dz_j \wedge d\bar z_{n+j}$ and $c_j' = \frac{i}{2} dz_{n+j} \wedge d\bar z_j$. Since $c_j^2 = c_j'^2 = 0$,

$$e^{c_1(L_\zeta)} = \prod_j f_j(\zeta_j), \qquad f_j(z) = e^{p(a_j+b_j)}(1 + z c_j + \overline{z} c_j' + c_j c_j').$$

[9] At $n = 2$ the same construction has $8 + 3 = 11$ pieces and 66 diagonal classes against $r = 24$.

For $u \in \mu_4$, $\frac{1}{2}(u+\overline{u})$ is u when $u = \pm 1$ and 0 when $u = \pm i$, so the left side of (i) is $\frac{1}{2}(\prod_j \sum_z z f_j(z) + \prod_j \sum_z \overline{z} f_j(z))$. Now $\sum_{z\in\mu_4} z = \sum z^2 = 0$ and $\sum |z|^2 = 4$, so $\sum_z z f_j(z) = 4e^{p(a_j+b_j)}c'_j = 4c'_j$, because $a_j c'_j$ and $b_j c'_j$ repeat a differential, and likewise $\sum_z \overline{z} f_j(z) = 4c_j$. This gives $2\cdot 4^{n-1}(\prod c'_j + \prod c_j) = 2\cdot 4^{n-1}(\overline{\alpha}+\alpha)$. The differentials dz_j, $j \le n$, and $d\bar{z}_{n+j}$ span the i-eigenspace of K on $H^1(A,\mathbb{C})$, and α spans its top exterior power, which is a Weil class, and $\overline{\alpha}$ the conjugate one. (ii) The moments $\mu_m = \sum_i \sigma_i t_i^m$ make the Hankel matrix of Setup 3.8 a product of Vandermonde matrices of rank 3, and Theorem 3.9(ii) gives r. The counts are immediate. ∎

The pieces L_ζ and $L_{\zeta'}$ have a degenerate difference as soon as ζ and ζ' share a coordinate, so Theorem 4.26 does not apply to them, and there are more than $2n$ of them. The rest of this subsection computes what the Massey products can do to the diagonal classes of such a convolution. The pieces are external products over the surfaces B_j, and the first tool is a bound on how far a product can lower the degree on each surface.

Lemma 4.28 (Degree drop on a product)**.** *Let $A = B_1 \times \cdots \times B_n$ with a product metric, and let every L_a be an external product with a product metric. Decompose the leaves of a term of Lemma 4.24(iii) with k leaves into tensor products of harmonic forms on the factors. Then the output has degree $s_j - d_j$ on B_j, where s_j is the sum of the degrees of the leaves on B_j and $d_j \ge 0$, $\sum_j d_j = k-2$, and*

$$d_j \le \max(0,\, u_j - 2),$$

where u_j is the number of leaves whose component on B_j is not a multiple of the unit section of $\mathcal{O}_{B_j}$ (such a unit occurs only where source and target agree on B_j).

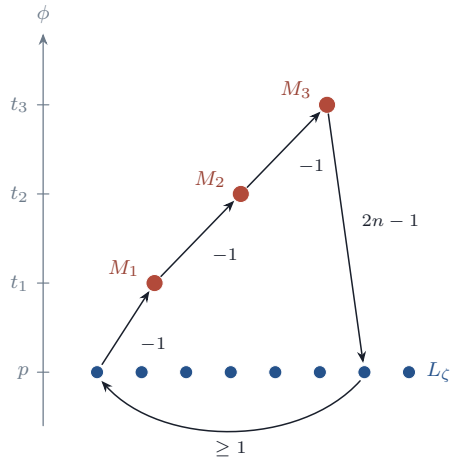
Proof. The Laplacian is $\Delta = \sum_j \Delta_j$, so $h = \sum_j h_j$ with $h_j = \bar{\partial}_j^* G = G\bar{\partial}_j^*$, which lowers the degree on B_j by one and keeps the others; expand every internal edge accordingly, and let d_j be the number of edges carrying h_j. Let V_j, U_j and W_j be the forms whose component on B_j is Δ_j-harmonic, a multiple of the unit, or in the image of $\bar{\partial}_j^*$. The operators G and h_i, $i \ne j$, preserve all three, h_j kills V_j and W_j, π kills W_j, and multiplication by an element of U_j preserves V_j, U_j and W_j. For an edge e carrying h_j let $N(e)$ be the set of leaves below e whose component on B_j is not a unit. If $|N(e)| \le 1$ the value below e lies in V_j and the term vanishes. If e lies below the next such edge e' with $N(e) = N(e')$, everything composed in between lies in U_j, the value at e' lies in W_j, and the term vanishes; if $N(e)$ is the set of all u_j such leaves for the highest such edge, the value at the root lies in W_j and π kills it. So in a nonzero term the sets $N(e)$ are distinct, of size at least two, different from the whole set, and pairwise nested or disjoint. Together with the whole set they form a family of that kind, which has at most $u_j - 1$ members, since its members are the internal vertices of a forest with u_j leaves in which every internal vertex has at least two children. Hence $d_j \le u_j - 2$. ∎

Proposition 4.29 (Nothing enters the diagonal)**.** *Keep Proposition 4.27 with $n \ge 3$, and let t_1, t_2, t_3 be distinct with $|t_i - p| \ge 2$. For every convolution E of the pieces L_ζ and M_i, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero, no element of degree one of $\mathcal{H}_E$ has a differential with a nonzero diagonal component. Consequently*

$$\dim \operatorname{Ext}^2(E,E) \;\ge\; \dim \ker\big(d_E|_{D^2}\big), \qquad D^2 = \bigoplus_a H^2(A,\mathcal{O}_A).$$

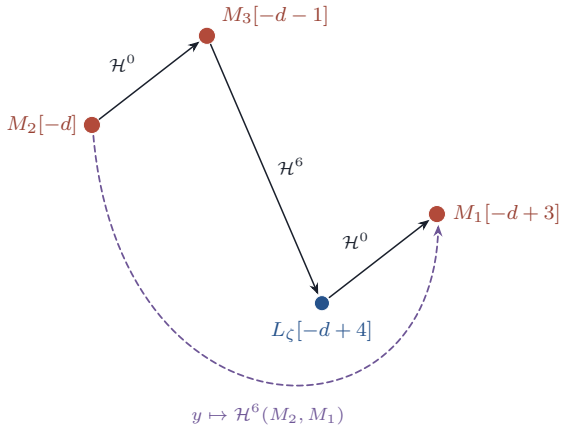
Proof. As in the proof of Theorem 4.26, a term that reaches the diagonal runs around a closed path of distinct pieces whose k steps have degree one and total excess 0. The step

(a) the excess of a cycle, $n \geq 3$



excess of a step of degree one: -1 up (Ext^0), $2n-1$ down (Ext^{2n}), at least 1 from L_ζ to $L_{\zeta'}$ this cycle: $3\cdot(-1)+(2n-1)+1 = 2n-3 > 0$; reaching $H^2(\mathcal{O}_A)$ needs a cycle of excess 0

(b) the one path that acts, $n = 3$



Ext degrees $0, 6, 0$, excesses $-1, 5, -1$: the product with $y \in H^2(\mathcal{O}_A)$ lands in degree 6, in a space of dimension $(t_2 - t_1)^6$; the shifts force $\sigma_1 = \sigma_3 = -\sigma_2$

FIGURE 6. The pieces of Proposition 4.27 at their values ϕ, with $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$. (a) A step of degree one between pieces with a definite difference has Ext degree 0 going up and $2n$ going down, so excess -1 or $2n-1$; a step between two L_ζ has excess at least 1. The cycle drawn runs up through all three M_i and down once, the worst case, and still has excess $2n-3 > 0$, so for $n \geq 3$ no term reaches the diagonal (Proposition 4.29). (b) The one path whose products act on the diagonal at $n = 3$ when the t_i lie above p (Theorem 4.31(i)), drawn at the same heights and in the order of the convolution, with the Ext degrees of its steps and the shifts they force; the dashed arrow is the value of the product with a diagonal class y.

from L_ζ to $L_{\zeta'}$ has Ext degree 1 on each surface where $\zeta_j \neq \zeta'_j$, the difference there having eigenvalues $\pm|\zeta'_j - \zeta_j|$, and at least 0 elsewhere. If $\prod\zeta = \prod\zeta'$ the two differ on at least two surfaces; otherwise the step has even Ext degree, the shifts having opposite parities. Either way its excess is at least 1. The difference between L_ζ and M_i has eigenvalues $t_i - p \pm 1$ on every surface, so it is definite: the step has Ext degree 0 and excess -1 if it goes up the value ϕ, where $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$, and Ext degree $2n$ and excess $2n-1$ if it goes down; the same holds between the M_i. A maximal run of the cycle through the M_i starts and ends at the value p, or is the whole cycle, so it has at least one step down and at most three steps up; its excess is at least $2n-1-3 \geq 2$. So every cycle has excess at least 1. Twisting by line bundles of degree zero leaves the forms unchanged and can only make some $\mathcal{H}(a,b)$ vanish. For the inequality, the diagonal components of cocycles of degree two vanish on coboundaries, so $\dim \mathrm{Ext}^2(E,E)$ is at least the dimension of their image, which contains the kernel of d_E on D^2. ∎

At $n = 2$ the bound on a run is 0 and a cycle through the three multiples can reach the diagonal; this is why the analysis below starts at $n = 3$.

Proposition 4.30 (Cup products on the diagonal)**.** *Keep Proposition 4.29 with $n = 3$, so that D^2 has dimension $35 \cdot 15 = 525$. The terms of d_E on D^2 with two leaves, the cup products, vanish on the 45 diagonal classes of the M_i, and on the 480 diagonal classes of the L_ζ they have a kernel of dimension at least 204. So their kernel on D^2 has dimension at least 249.*

Proof. For $y = (y_a) \in D^2$ the cup products in $\mathcal{H}(a, b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, since y has even degree. On a surface where the pieces differ, δ^H_{ab} has degree 1, the only degree of the cohomology there, and the product with a component of y of positive degree lands in a degree where the cohomology vanishes. The pieces M_i differ from every other piece on every surface, which gives the first statement. Write y_a by its type $\tau \in \{0, 1, 2\}^3$, $|\tau| = 2$, the degrees on the three surfaces: three types $(2, 0, 0)$ of dimension one and three types $(1, 1, 0)$ of dimension four. The degree count on each surface and the parity of Ext degrees show that L_ζ and $L_{\zeta'}$ can carry a nonzero cup product on the type τ only if either ζ' differs from ζ by a sign in one coordinate outside the support of τ, or ζ' is ζ multiplied by (i, i) or $(-i, -i)$ in two coordinates and τ is 2 on the third. Let Γ_τ be the graph on the 32 pieces with these edges. If each component y^τ is constant on the connected components of Γ_τ, every cup product vanishes. For $\tau = (2, 0, 0)$ the coordinate ζ_1 is constant along the edges, and the eight pieces with a given ζ_1 are connected, so Γ_τ has 4 components; for $\tau = (1, 1, 0)$ the edges change the sign of ζ_3 only, and Γ_τ has 16 components. The kernel therefore contains a space of dimension $3 \cdot 4 \cdot 1 + 3 \cdot 16 \cdot 4 = 12 + 192 = 204$. ∎

Theorem 4.31 (The diagonal on E_0^6)**.** *Let $n = 3$ and let E be a convolution of the 32 bundles L_ζ and of M_1, M_2, M_3 as in Proposition 4.27, with $p + 2 \leq t_1 < t_2 < t_3$ and signs σ_i, so that $\mathrm{ch}(E) = 32(\alpha + \overline{\alpha}) + \sum_i \sigma_i e^{t_i \theta}$ and the polarised criterion asks for $\dim \mathrm{Ext}^2(E, E) = 57$.*

(i) *The Massey products of length at least three vanish on D^2, except those of length four along a path $M_2 \to M_3 \to L_\zeta \to M_1$ taken in this order in the convolution, which occur only if $\sigma_1 = \sigma_3 = -\sigma_2$ and take values in $\mathcal{H}^6(M_2, M_1) \cong H^6(A, \mathcal{O}_A(-(t_2 - t_1)\theta))$, of dimension $(t_2 - t_1)^6$.*

(ii) *$\dim \mathrm{Ext}^2(E, E) \geq 249 - \min(249, (t_2 - t_1)^6)$, and $\dim \mathrm{Ext}^2(E, E) \geq 249$ unless the order and the signs are as in (i). In particular E does not meet the polarised criterion when $t_2 - t_1 \leq 2$, since $249 - 64 = 185 > 57$, nor when the signs or the order differ from (i).*

(iii) *If E meets the polarised criterion, then the length four products of (i) have rank at least $249 - 57 = 192$ on the kernel of the cup products, and $t_2 - t_1 \geq 3$.*

The same holds for $t_1 < t_2 < t_3 \leq p - 2$ with E replaced by its dual.

Proof. (i) By Lemma 4.24(iii) a term with $k \geq 3$ leaves acting on $y \in H^2(\mathcal{O}_A)$ at a piece c runs along a chain $s = a_0 \to \cdots \to a_{k-1} = e$ of distinct pieces, increasing in the order of the convolution, whose $k - 1$ steps are components of degree one, with y inserted at c. Its value lies in $\mathcal{H}(s, e)$, in Ext degree $2 + \sum_j q_j - (k - 2) = 3 + \varepsilon$, where ε is the sum of the excesses of the steps; so $-3 \leq \varepsilon \leq 3$, and $\mathcal{H}(s, e)$ must be nonzero in degree $3 + \varepsilon$. By the proof of Proposition 4.29 a step between two L_ζ has excess at least 1, with equality only between pieces of opposite parities, and a step between pieces at different values of ϕ has excess -1 and Ext degree 0 going up, excess 5 and Ext degree 6 going down. Let U and D be the numbers of steps up and down and λ the total excess of the steps between two L_ζ, so that $\varepsilon = 5D - U + \lambda$. For two distinct L_ζ, $L_{\zeta'}$ differing in $m \geq 1$ coordinates,

$\mathcal{H}(L_\zeta, L_{\zeta'})$ vanishes above degree $6-m$; for pieces at different values it sits in degree 0 if the target is higher and in degree 6 if it is lower.

Two observations remove chains. First, a chain of total excess -2 has no step down, and so consists of steps between L_ζ followed by at most three steps up. Second, let $a \to b \to c$ be two consecutive steps of Ext degree 0, sections x_{ab} and x_{bc} of line bundles. The Maurer–Cartan equation $\pi(\delta \circ \delta) = 0$ of the proof of Proposition 4.33(i) has, in $\mathcal{H}^0(a, c)$, one term for each chain from a to c of total excess -2. If $a \to b \to c$ is the only such chain, the equation reads $x_{ab}x_{bc} = 0$, a product of holomorphic sections being holomorphic and so harmonic; since A is connected, x_{ab} or x_{bc} vanishes, and every term along a chain through both steps is zero.

Here every step up ends at one of M_1, M_2, M_3, so $U \le 3$, and $D \le 1$. If $D = 0$, the chain is $r \ge 0$ pieces L_ζ followed by multiples in increasing order. When it ends at an L_ζ, $k \ge 3$ gives $\lambda \ge 2$, so s and e differ in one coordinate and $\lambda = 2$; then the two steps change the parity twice, whereas s and e have opposite parities. When it ends at a multiple, e lies above s, so $\varepsilon = -3$, which forces $U = 3$, $\lambda = 0$ and the chain $L_\zeta \to M_1 \to M_2 \to M_3$; by the first observation $L_\zeta \to M_1 \to M_2$ is the only chain of excess -2 from L_ζ to M_2, and the second removes it. If $D = 1$, then $U \ge 2 + \lambda$, so (U, λ) is $(2, 0)$, $(3, 0)$ or $(3, 1)$ and ε is 3, 2 or 3. For $\varepsilon = 3$ the piece s is a multiple and e lies below it. With $(3, 1)$ the three steps up would end at three multiples other than s, which do not exist. With $(2, 0)$ the chain has three steps, and a step down lands at an L_ζ, since a fourth multiple does not exist. It is $M_1 \to M_2 \to M_3 \to L_\zeta$, removed by the second observation at $M_1 \to M_2 \to M_3$; or $M_a \to M_b \to L_\zeta \to M_c$ with $t_a < t_b$ and $t_c < t_a$, that is $M_2 \to M_3 \to L_\zeta \to M_1$; or $M_a \to L_\zeta \to M_b \to M_c$ with $t_b < t_c < t_a$, that is $M_3 \to L_\zeta \to M_1 \to M_2$, removed at $L_\zeta \to M_1 \to M_2$. For $\varepsilon = 2$, with $(U, \lambda) = (3, 0)$, the pieces s and e are L_ζ differing in one coordinate; the last step is the step down, preceded by three steps up from s, so the chain is $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, removed at $L_\zeta \to M_1 \to M_2$.

Only $M_2 \to M_3 \to L_\zeta \to M_1$ remains. Its steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_L = d_{M_3} - 5$ and $d_{M_1} = d_L + 1$; this gives the signs, and the value lies in $\mathcal{H}^{3+d_{M_2}-d_{M_1}}(M_2, M_1) = \mathcal{H}^6(M_2, M_1)$, whose dimension is $|\chi(\mathcal{O}_A((t_1-t_2)\theta))| = (t_2-t_1)^6$. (ii) By Proposition 4.29 it suffices to bound the kernel of d_E on D^2. By Proposition 4.30 the cup products have a kernel Y of dimension at least 249, and on Y the differential is the sum of the products of (i), of rank at most $(t_2 - t_1)^6$. (iii) follows from (ii). For the dual, $E \mapsto E^\vee$ preserves Ext^2 and sends the configuration with p, t_i to the one with $-p$, $-t_i$, the set of the L_ζ being stable under $\zeta \mapsto -\zeta$. ∎

The placements of the t_i on both sides of p are not covered by Theorem 4.31. There two families of fourfold products occur instead of one, and no convolution carries both.

Proposition 4.32 (Placements on both sides of p)**.** *Keep Theorem 4.31, but let $t_1 \le p - 2$ and $p + 2 \le t_2 < t_3$.*

(i) *The Massey products of length at least three vanish on D^2 unless $\sigma_1 = \sigma_2 = -\sigma_3$. In that case the only ones that act have length four and run along one of the paths*

$$\text{(a)}\quad M_2 \to M_3 \to M_1 \to L_\zeta, \qquad \text{(b)}\quad M_3 \to M_1 \to L_\zeta \to M_2,$$

taken in this order in the convolution, with $\prod \zeta = \sigma_3$. The products along (a) take values in $\mathcal{H}^6(M_2, L_\zeta)$, of dimension $\left((t_2 - p)^2 - 1\right)^3$, and those along (b) in $\mathcal{H}^6(M_3, M_2)$, of dimension $(t_3 - t_2)^6$. A convolution carries at most one of the two kinds, since (a) puts M_2 before M_3 and (b) puts M_3 before M_2.

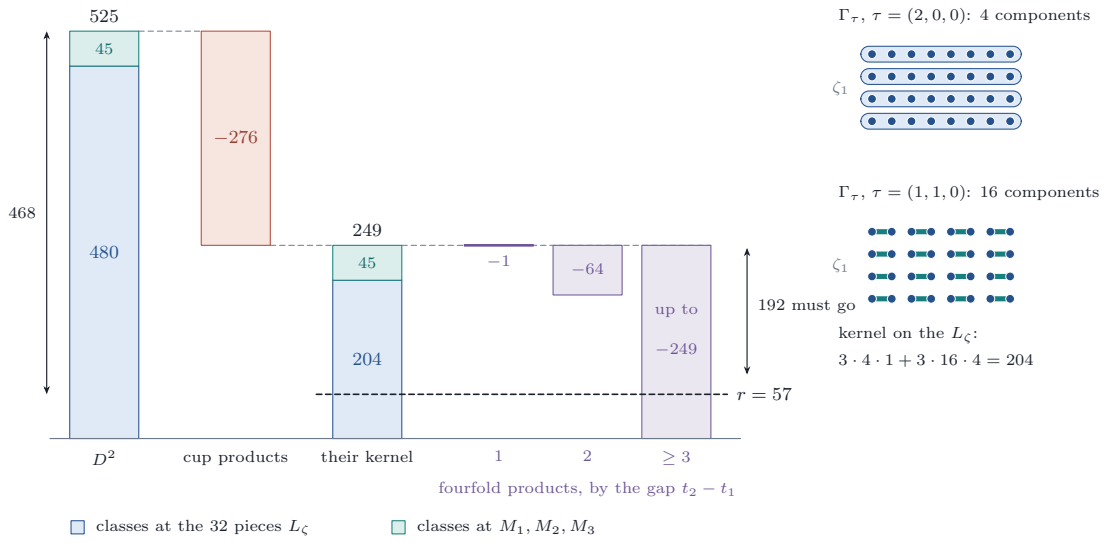


FIGURE 7. The 525 diagonal classes of degree two at $n = 3$ (Proposition 4.30 and Theorem 4.31). The cup products remove 276 of them for general components, and at most that many in any case, leaving 204 at the pieces L_ζ and the 45 at the M_i; when the t_i lie on one side of p, the fourfold products of the surviving path remove at most $(t_2 - t_1)^6$ more, which is 1 or 64 for the gaps 1 and 2 and no constraint from the third gap on. The criterion asks for 57, so 468 classes must disappear in all and 192 after the cup products. On the right, the graphs Γ_τ on the 32 pieces, one row for each value of ζ_1: for $\tau = (2,0,0)$ each row is a component, for $\tau = (1,1,0)$ the components are the pairs differing in the sign of ζ_3; a class constant on the components of every Γ_τ has vanishing cup products.

(ii) *If E meets the polarised criterion, then $\sigma_1 = \sigma_2 = -\sigma_3$, and the products along (a) or those along (b) have rank at least* 192 *on the kernel of the cup products. For (b) this needs $t_3 - t_2 \geq 3$. The sixteen targets of (a) have dimension $16\big((t_2 - p)^2 - 1\big)^3 \geq 432$, so for (a) the count gives no condition.*

For $t_1 < t_2 \leq p - 2$ and $t_3 \geq p + 2$ the same holds for $E^\vee[1]$, and the condition on the signs becomes $\sigma_2 = \sigma_3 = -\sigma_1$.

Proof. (i) We use the notation and the two observations of the proof of Theorem 4.31(i). Now $\phi(M_1) = t_1 < p < t_2 < t_3$; a step up ends at M_2, at M_3, or at an L_ζ coming from M_1, so again $U \leq 3$ and $D \leq 1$. If $D = 0$, the values do not decrease along the chain. When e lies above s, $\varepsilon = -3$ forces $U = 3$, $\lambda = 0$ and the chain $M_1 \to L_\zeta \to M_2 \to M_3$, removed at $L_\zeta \to M_2 \to M_3$, the only chain of excess -2 from L_ζ to M_3. When e and s have the same value, they are two L_ζ joined by steps between L_ζ, which is excluded as before. If $D = 1$, (U, λ) is $(2,0)$, $(3,0)$ or $(3,1)$. For $\varepsilon = 3$, e lies below s. With $(3,1)$ the steps up end at M_2, M_3 and an L_ζ following M_1; then s is an L_ζ, so $e = M_1$, which is followed by a step: impossible. With $(2,0)$ the chain has three steps and no two consecutive L_ζ. As M_1 is the lowest piece, it is never the end of a step up, and a step up into an L_ζ comes from M_1; so two consecutive steps up end at M_2 or M_3. The chain cannot start at M_1. If it starts at an L_ζ, it ends at M_1 with the step down, after two

steps up: $L_\zeta \to M_2 \to M_3 \to M_1$, removed at $L_\zeta \to M_2 \to M_3$. If it starts at M_3, the first step goes down and the last two go up, ending at M_2; the middle piece is then an L_ζ entered from M_1, and the chain is $M_3 \to M_1 \to L_\zeta \to M_2$, path (b). If it starts at M_2, then $e \in \{L_\zeta, M_1\}$, so the last two steps are not both up; the first step goes up to M_3, the second down, and the third up into $e = L_\zeta$ from M_1: the chain is $M_2 \to M_3 \to M_1 \to L_\zeta$, path (a). For $\varepsilon = 2$, $(U, \lambda) = (3, 0)$ and s, e are L_ζ differing in one coordinate. The last step enters e from M_1, since a step down into e would follow three steps up from s, one more than the values allow; M_1 is entered by the step down, after two steps up from s: $L_\zeta \to M_2 \to M_3 \to M_1 \to L_{\zeta'}$, removed at $L_\zeta \to M_2 \to M_3$. Along (a) the steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_{M_1} = d_{M_3} - 5$ and $d_L = d_{M_1} + 1$. Along (b) they have Ext degrees 6, 0 and 0, which forces $d_{M_1} = d_{M_3} - 5$, $d_L = d_{M_1} + 1$ and $d_{M_2} = d_L + 1$. This gives the signs and puts the values in $\mathcal{H}^{3+d_{M_2}-d_L}(M_2, L_\zeta) = \mathcal{H}^6(M_2, L_\zeta)$ and in $\mathcal{H}^{3+d_{M_3}-d_{M_2}}(M_3, M_2) = \mathcal{H}^6(M_3, M_2)$. On B_j the line bundle $L_\zeta \otimes M_2^{-1}$ has the hermitian form $\left(\begin{smallmatrix} p-t_2 & \zeta_j \\ \overline{\zeta}_j & p-t_2 \end{smallmatrix}\right)$, with eigenvalues $p - t_2 \pm 1 < 0$ and determinant $(t_2 - p)^2 - 1$, which gives the first dimension. The second is that of Theorem 4.31(i) with t_3, t_2 in place of t_2, t_1. (ii) Propositions 4.29 and 4.30 do not depend on the placement of the t_i, so as in Theorem 4.31(ii) $\dim \mathrm{Ext}^2(E, E) \geq 249 - \rho$, where ρ is the rank on the kernel of the cup products of the products of (i) that E carries, and $\rho = 0$ unless $\sigma_1 = \sigma_2 = -\sigma_3$. The criterion asks for 57, so $\rho \geq 192$, and $(t_3 - t_2)^6 \geq 192$ exactly when $t_3 - t_2 \geq 3$. For the last statement, $E \mapsto E^\vee[1]$ preserves Ext^2, sends p and the t_i to $-p$ and the $-t_i$, reverses the order of the pieces and changes every sign. ∎

The Maurer–Cartan equation can be solved for one order of the pieces.

Proposition 4.33 (An explicit convolution)**.** *Keep Theorem 4.31 with $p = 0$ and $t = (2, 5, 6)$, take the pieces to be external products $L_\zeta = N_{1,\zeta_1} \boxtimes N_{2,\zeta_2} \boxtimes N_{3,\zeta_3}$ and $M_i = M_{i,1} \boxtimes M_{i,2} \boxtimes M_{i,3}$ of line bundles on the B_j, and order them as M_2, M_3, the sixteen L_ζ with $\prod \zeta = 1$, M_1, the sixteen L_ζ with $\prod \zeta = -1$, with the shifts 0, 1, −4, −3, −5.*

(i) *The components that the degrees allow are $x_1 \in \mathcal{H}^0(M_2, M_3)$, $x_2^\zeta \in \mathcal{H}^6(M_3, L_\zeta)$, $x_3^\zeta \in \mathcal{H}^0(L_\zeta, M_1)$, components in $\mathcal{H}^2(L_\zeta, L_{\zeta'})$ for the $48 + 96$ pairs with $\prod \zeta = 1 = -\prod \zeta'$ that differ in one or two coordinates, and components in $\mathcal{H}^6(M_2, L_{\zeta'})$, $\prod \zeta' = -1$, which we take to be zero. For harmonic components the Maurer–Cartan equation is exactly*

$$x_1 x_2^\zeta = 0 \text{ in } H^6(A, M_2^{-1} \otimes L_\zeta) \text{ for every } \zeta, \qquad \sum_\zeta x_2^\zeta x_3^\zeta = 0 \text{ in } H^6(A, M_3^{-1} \otimes M_1),$$

and every solution is the harmonic part of a convolution E with $\mathrm{ch}(E) = 32(\alpha + \overline{\alpha}) + e^{5\theta} - e^{6\theta} - e^{2\theta}$.

(ii) *There are solutions with every $x_2^\zeta \neq 0$ and $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$.*

(iii) *For every solution $\dim \mathrm{Ext}^2(E, E) \geq 2560$, so E does not meet the polarised criterion.*

Proof. (i) A component from P_a to P_b has Ext degree $1 + d_a - d_b$, and the forms give the cohomology of each pair: $\mathcal{H}(M_2, M_3)$ and $\mathcal{H}(L_\zeta, M_1)$ sit in degree 0, $\mathcal{H}(M_3, L_\zeta)$, $\mathcal{H}(M_2, L_{\zeta'})$ and $\mathcal{H}(M_2, M_1)$ in degree 6, and $\mathcal{H}(L_\zeta, L_{\zeta'})$ in degree 2 exactly for the pairs listed; no other pair has cohomology in the degree a component needs. Put $\delta = \delta^H - h(\delta \circ \delta)$ recursively, as in Lemma 4.24(i), and $\Phi = \bar{\partial}\delta + \delta \circ \delta$. Then $\Phi = \pi(\delta \circ \delta) + h\bar{\partial}(\delta \circ \delta)$ and $\bar{\partial}(\delta \circ \delta) = \Phi \circ \delta - \delta \circ \Phi$, so by induction on $b - a$ the twisted complex exists exactly when $\pi(\delta \circ \delta) = 0$. The components of $\pi(\delta \circ \delta)$ come from paths of two or more components; those

that land in a nonzero group are $x_1 x_2^\zeta$ in $\mathcal{H}^6(M_2, L_\zeta)$ and $\sum_\zeta x_2^\zeta x_3^\zeta$ in $\mathcal{H}^6(M_3, M_1)$, while the product along $M_2 \to M_3 \to L_\zeta \to M_1$ would lie in $\mathcal{H}^5(M_2, M_1) = 0$ and x_2^ζ composed with a component in $\mathcal{H}^2(L_\zeta, L_{\zeta'})$ in degree eight. The signs $(-1)^d$ are $+, -, +, -, -$, which gives the Chern character by Proposition 4.27(i). (ii) Take x_1 and the x_3^ζ to be external products of nonzero sections on the B_j. On each B_j the products with their factors map $H^2(B_j, M_3^{-1} L_\zeta)$, of dimension 35, to spaces of dimensions 24 and 16, so they have kernels of dimensions at least 11 and 19. A sum of external products whose factor on one surface lies in the first kernel and whose factor on another lies in the second is a nonzero x_2^ζ with $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$, which solves (i).[10] (iii) The sixteen L_ζ with $\prod \zeta = 1$ have shift -4 and the others -5, so Lemma 4.34 below applies to every pair of them that differ in all three coordinates: there are 112 such pairs, and their groups $\mathcal{H}^3$ have dimensions adding up to $16 \cdot 160 = 2560$. ∎

The classes counted in (iii) are the ones no product can reach. Two pieces L_ζ, $L_{\zeta'}$ that differ in every coordinate have a difference that is indefinite on every surface, so $\mathcal{H}(L_\zeta, L_{\zeta'})$ is $\bigotimes_j H^1(B_j, \cdot)$, concentrated in degree three, of dimension $D(\zeta, \zeta') = \prod_j |\zeta_j - \zeta'_j|^2$; no component can join them, and if their shifts differ by one their classes have degree two.

Lemma 4.34 (Pieces that differ everywhere)**.** *Keep Proposition 4.29 with $n = 3$. Suppose that for some integer r either $p + 2 \le t_1 < t_2 < t_3$ and M_1, M_2, M_3 have the shifts $r + 1$, $r + 4$, $r + 5$, or $t_1 \le p - 2 < p + 2 \le t_2 < t_3$ and they have the shifts $r - 1$, $r + 3$, $r + 4$ or $r - 1$, $r + 1$, $r + 4$; these are the shifts forced by the paths of Theorem 4.31(i) and of Proposition 4.32(i), (a) and (b), with r the shift of the L_ζ on the path. Let L_ζ, $L_{\zeta'}$ differ in all three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then d_E vanishes on $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, and no element of degree one has a differential with a nonzero component there. Consequently*

$$\dim \operatorname{Ext}^2(E, E) \;\ge\; \dim \ker\bigl(d_E|_{D^2}\bigr) + \sum D(\zeta, \zeta'),$$

the sum running over the pairs of this kind.

Proof. Only the shifts are used, and the order of the pieces only removes components, so we ignore it. A component from P_a to P_b of Ext degree q has $d_b = d_a + 1 - q$. Between distinct L_ζ the degree q is at least 2: there is no $\mathcal{H}^0$, and $\mathcal{H}^1$ occurs only for pieces differing in one coordinate, which have opposite parities, while $q = 1$ would need equal shifts. So a component between two L_ζ lowers the shift by at least one, and by exactly one only in degree two. For a multiple M_i let μ_i and ν_i be the shifts of an L_ζ that a component from M_i reaches and from which a component reaches M_i: $\mu_i = d_{M_i} - 5$ and $\nu_i = d_{M_i} - 1$ when $t_i > p$, and $\mu_i = d_{M_i} + 1$, $\nu_i = d_{M_i} + 5$ when $t_i < p$. Between the multiples the degrees allow only $M_2 \to M_3$ in the first case, $M_2 \to M_3 \to M_1$ in the second, and $M_3 \to M_1$ in the third. Let λ_i be the largest μ_k and κ_i the least ν_k over the multiples M_k reachable from M_i, respectively reaching M_i, through these steps, M_i included. The values, in the order M_1, M_2, M_3 and relative to r, are

	μ			ν			λ			κ		
$r+1, r+4, r+5$	-4	-1	0	0	3	4	-4	0	0	0	3	3
$r-1, r+3, r+4$	0	-2	-1	4	2	3	0	0	0	2	2	2
$r-1, r+1, r+4$	0	-4	-1	4	0	3	0	-4	0	3	0	3

[10] Such solutions are far from unique: the equations of (i) are linear in each of x_1, x_2^ζ, x_3^ζ separately, and the components on the 144 pairs do not enter them at all.

and in all three cases, for all i and k,

$$\lambda_i \leq r \leq \kappa_k, \qquad \kappa_i \geq \lambda_i + 2, \qquad \lambda_i \leq \mu_i + 2, \qquad \nu_i \leq \kappa_i + 2, \tag{14}$$

while $\mu_i \leq \lambda_i$ and $\nu_i \geq \kappa_i$ by definition. A run $L \to M_k \to \cdots \to M_i \to L'$ through multiples goes from the shift $\nu_k \geq \kappa_k$ to $\mu_i \leq \lambda_k \leq \kappa_k - 2$, so every chain of components from one L to another lowers the shift, by at least two if it passes through a multiple and by exactly one only if it is a single component of degree two. In particular a chain from an L of shift x into M_i has $x \geq \kappa_i$, and a chain from M_i to an L of shift y has $y \leq \lambda_i$.

By Lemma 4.24(iii), a term of d_E on a class u of Ext degree q_u in $\mathcal{H}(a', b')$ runs along a chain $c \to \cdots \to a'$, the step u, and a chain $b' \to \cdots \to e$, and since the degrees of the components telescope, its value lies in $\mathcal{H}(c, e)$ in Ext degree

$$q_u + 1 + (d_c - d_{a'}) + (d_{b'} - d_e).$$

Write $a = L_\zeta$ and $b = L_{\zeta'}$, so that $d_a = d_b + 1$.

Nothing reaches $\mathcal{H}^3(a, b)$. Let u have degree one, so $q_u = 1 + d_{a'} - d_{b'}$, with chains from a to a' and from b' to b and $(a', b') \neq (a, b)$. If a' and b' are both pieces L, then $d_{a'} \leq d_a$ and $d_{b'} \geq d_b$, not both with equality, so $q_u \leq 1$; $q_u = 0$ is impossible, and $q_u = 1$ needs either $a' = b'$, and then a chain from a to b lowering the shift by one, a single component in $\mathcal{H}^2(a, b) = 0$, or a', b' differing in one coordinate, of opposite parities, against $d_{a'} = d_{b'}$. If $a' = M_i$ and b' is an L, then u lies in the degree of a component, so $d_{b'} = \mu_i$, while $d_{b'} \geq d_b = d_a - 1 \geq \kappa_i - 1 \geq \lambda_i + 1 > \mu_i$. If a' is an L and $b' = M_k$, then $d_{a'} = \nu_k$ and $d_a - d_b \geq \nu_k - \lambda_k \geq \kappa_k - \lambda_k \geq 2$. If $a' = M_i$ and $b' = M_k$, then $d_a - d_b \geq \kappa_i - \lambda_k$; for $i = k$ this is at least 2, and for $i \neq k$ the class u lies in the degree of a step $M_i \to M_k$, so $\lambda_k \leq \lambda_i$ and again $d_a - d_b \geq 2$. Each case contradicts $d_a - d_b = 1$.

d_E vanishes on $\mathcal{H}^3(a, b)$. Here u is the class itself and a term lands in $\mathcal{H}(c, e)$, $(c, e) \neq (a, b)$, in degree $3 + d_c - d_e$, with chains from c to a and from b to e. If c and e are both pieces L, then $d_c - d_e \geq 2$; $c = e$ is impossible, and otherwise a class of degree at least 5 needs c, e differing in one coordinate, in degree 5, so $d_c - d_e = 2$ against their opposite parities. If $c = M_i$ and e is an L, the degree is that of a component when $d_e = \mu_i + 2$, but $d_e \leq d_b < d_a \leq \lambda_i \leq \mu_i + 2$. If c is an L and $e = M_k$, it is that of a component when $d_c = \nu_k - 2$, but $d_c \geq d_a > d_b \geq \kappa_k \geq \nu_k - 2$. If $c = M_i$ and $e = M_k$, then $d_a \leq \lambda_i \leq r \leq \kappa_k \leq d_b$, against $d_a > d_b$.

For the inequality: a coboundary has no component on D^2 (Proposition 4.29) and none on these groups, so the cocycles of $\ker(d_E|_{D^2})$ and of these groups span a space that meets the coboundaries only in zero. Twisting the pieces by line bundles of degree zero does not change $D(\zeta, \zeta')$, the forms being nondegenerate on every surface, and can only remove components. ∎

Theorem 4.35 (No convolution of these pieces meets the criterion)**.** *Keep Proposition 4.29 with $n = 3$: let E be any convolution of the 32 bundles L_ζ and of M_1, M_2, M_3, with $t_1 < t_2 < t_3$ and $|t_i - p| \geq 2$, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero. Then* $\dim \mathrm{Ext}^2(E, E) \geq 69$. *In particular no such convolution meets the polarised criterion, which asks for* 57.

Proof. Replacing E by $E^\vee$ or $E^\vee[1]$, which preserves Ext^2, we may assume that the t_i are as in Theorem 4.31 or in Proposition 4.32. If no fourfold product of those results acts, $\dim \mathrm{Ext}^2(E, E) \geq 249$. Otherwise the path fixes the shifts of the M_i as in Lemma 4.34, with r the shift of every L_ζ on a path; let P be the sixteen pieces of the parity of r, Q the other sixteen, and m the number of pieces on paths.

The fourfold products act on D^2 only through the classes at the M_i and at the m pieces on paths, so their rank is at most $45 + 15m$, and by Proposition 4.30 $\dim\ker(d_E|_{D^2}) \geq 249 - 45 - 15m = 204 - 15m$. If $m \leq 9$ this is at least 69.

Let $m \geq 10$, so that at least ten pieces of P have the shift r. Every $W \in Q$ has 3, 6 and 7 partners in P differing from it in one, two and three coordinates, and for the seven $D = 16$ six times and $D = 64$ once.[11] If $d_W = r \pm 1$, at most six pieces of P lack the shift r, so one of the seven has it, and Lemma 4.34 gives at least 16 classes. If $d_W \neq r \pm 1$, consider for each type τ of the form $(1, 1, 0)$ the partner $W^\tau \in P$ obtained by changing the sign of the coordinate where τ vanishes. By the proof of Proposition 4.30, W^τ is the only piece with which W can carry a nonzero cup product on the type τ, and conversely; a cup product between them needs a component of degree two, hence shifts differing by one. If they do, the classes of type τ equal at W and W^τ and zero elsewhere have vanishing cup products, and W^τ, of shift $d_W \pm 1 \neq r$, lies on no path; if they do not, the classes of type τ at W alone have vanishing cup products. Either way these 4 classes lie in $\ker(d_E|_{D^2})$, 12 for each such W, and those of different W are independent, since $W \mapsto W^\tau$ is injective. Adding over Q, Lemma 4.34 gives $\dim \mathrm{Ext}^2(E, E) \geq 16 \cdot 12 = 192$. ■

At $n = 4$ there are 128 pieces L_ζ, and the classes that no product reaches are again those between pieces that differ in three coordinates and have adjacent shifts. The shifts along a chain are now controlled by counting alone.

Lemma 4.36 (Shifts along chains at $n = 4$)**.** *Keep Proposition 4.27 with $n = 4$, take the L_ζ to be external products $N_{1,\zeta_1} \boxtimes \cdots \boxtimes N_{4,\zeta_4}$ of line bundles on the B_j, let t_1, t_2, t_3 be distinct with $|t_i - p| \geq 2$, and let E be a convolution of the L_ζ and of M_1, M_2, M_3 with shifts of the given parities. Call a component of δ^H, or an element of degree one of $\mathcal{H}_E$ in one group $\mathcal{H}(a, b)$, a step from P_a to P_b; a step of* Ext *degree q has $d_b = d_a + 1 - q$. Put $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$.*

(i) *A step between distinct L_ζ has* Ext *degree at least two, so it lowers the shift by at least one, and by exactly one only in* Ext *degree two. A step in $\mathcal{H}^1(a, a) = H^1(A, \mathcal{O}_A)$ keeps the shift.*

(ii) *A step between an L_ζ and a multiple, or between two multiples, has* Ext *degree* 0 *if it goes up the value ϕ and* 8 *if it goes down, so it raises the shift by one or lowers it by seven.*

(iii) *A sequence of steps from an L_ζ to an $L_{\zeta'}$ through multiples, distinct before one marked step and distinct after it, where the marked step may lie in $\mathcal{H}^1(a, a)$ at a multiple, lowers the shift by at least four. Consequently a sequence of steps from one L_ζ to another, at most one of them not a component, lowers the shift by at least one, and by exactly one only if it consists of steps between pieces L_ζ, one of* Ext *degree two and the others in $\mathcal{H}^1(a, a)$.*

Proof. (i) For $\zeta \neq \zeta'$ the group $\mathcal{H}(L_\zeta, L_{\zeta'})$ is the tensor product over the surfaces of H^1 of a line bundle with a nondegenerate form, on the $k \geq 1$ surfaces where ζ and ζ' differ, and of $H^*(B_j, \mathcal{O}_{B_j})$ on the others, the factors of L_ζ and $L_{\zeta'}$ on those being equal. So its degrees are at least k, and degree one needs $k = 1$. Then the ratio $\zeta'_j/\zeta_j \neq 1$ in the one coordinate is the ratio of $\prod \zeta'$ and $\prod \zeta$, which is -1, and the shifts have opposite parities, while a step of Ext degree one keeps the shift. (ii) The forms of $M_i \otimes L_\zeta^{-1}$ have the eigenvalues $t_i - p \pm 1$ on every surface, which have the sign of $t_i - p$; the form of $M_k \otimes M_i^{-1}$ is $(t_k - t_i)\,\mathrm{Id}$. A

[11] The ratios ζ'_j/ζ_j of two pieces of opposite parities have product -1; when all three differ from 1 they are $(-1, -1, -1)$, with $D = 4^3$, or a permutation of $(i, -i, -1)$, with $D = 2 \cdot 2 \cdot 4$.

definite form has cohomology in degree 0 if it is positive and $2n = 8$ if it is negative [BL04, Chapter 3].[12] (iii) Let U and D be the numbers of steps up and down outside $\mathcal{H}^1(a,a)$; by (ii) the shift changes by $U - 7D$. The values start and end at p and pass through the values of multiples, so $D \geq 1$. If $D = 1$, the values increase strictly before the step down and strictly after it, so those before it lie above p and those after it below p, each multiple occurs once, and $U \leq 3$: the drop $7 - U$ is at least four.[13] If $D \geq 2$, at most six multiples occur, counted with repetition, so there are at most seven steps outside $\mathcal{H}^1(a,a)$, $U \leq 7 - D$, and the drop $7D - U \geq 8D - 7$ is at least nine. For the consequence, split the sequence at the pieces L_ζ it visits: each part through multiples lowers the shift by at least four, and by (i) each other part lowers it by at least one, unless it lies in $\mathcal{H}^1(a,a)$. ∎

Theorem 4.37 (Two shifts on E_0^8)**.** *Keep Lemma 4.36, and let L_ζ and $L_{\zeta'}$ differ in exactly three coordinates and have the shifts $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then $\mathcal{H}(L_\zeta, L_{\zeta'})$ has its classes of degree two in $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$, which is 16 or 64, and:*

(i) *for every convolution, no element of degree one has a differential with a nonzero component in $\mathcal{H}^3(L_\zeta, L_{\zeta'})$;*

(ii) *if the L_ζ occupy two consecutive shifts, those with $\prod \zeta = \epsilon$ the shift s and the others $s - 1$, for a sign ϵ and an integer s, then d_E vanishes on these groups, and for every order of the pieces, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \operatorname{Ext}^2(E,E) \;\geq\; 64 \cdot 640 = 40960.$$

In particular E does not meet the polarised criterion, which asks for $\dim \operatorname{Ext}^2(E,E) = 104$.

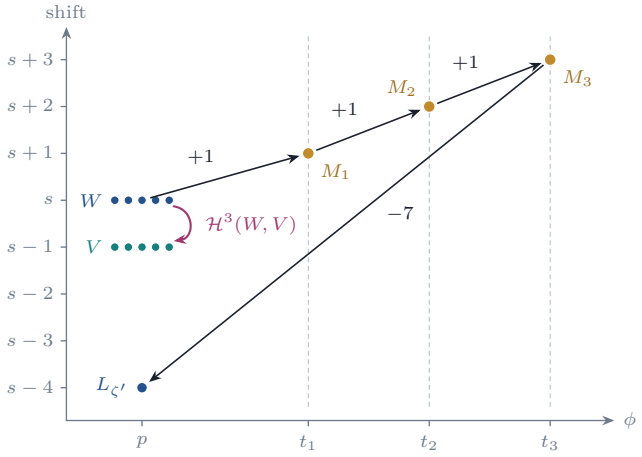
Proof. Write $W = L_\zeta$ and $V = L_{\zeta'}$. By the proof of Lemma 4.36(i), $\mathcal{H}(W,V)$ has degrees 3, 4 and 5, the last two from the surface where ζ and ζ' agree, and its classes of degree $w = 2$ have Ext degree $2 + d_W - d_V = 3$; the dimension of $\mathcal{H}^3(W,V)$ is the product over the three other surfaces of the determinants $|\zeta_j - \zeta'_j|^2$, as in Lemma 4.34.

(i) As in the proof of Lemma 4.34, a term of the differential of an element u of degree one in $\mathcal{H}(a',b')$ with a component in $\mathcal{H}(W,V)$ runs along a chain of components $W \to \cdots \to a'$, the step u, and a chain $b' \to \cdots \to V$, with $(a',b') \neq (W,V)$. This is a sequence of steps from W to V, one of them u, lowering the shift by one. By Lemma 4.36(iii) it consists of steps between pieces L_ζ, one of Ext degree two and at most one, u, in $\mathcal{H}^1(a,a)$. If u is not in $\mathcal{H}^1(a,a)$, it is the step of degree two and both chains are empty, against $(a',b') \neq (W,V)$. If u is in $\mathcal{H}^1(a,a)$, the rest is a single component $W \to V$ of Ext degree two, in $\mathcal{H}^2(W,V) = 0$.

(ii) A term of d_E on $x \in \mathcal{H}^3(W,V)$ runs along chains of components $c \to \cdots \to W$ and $V \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c,e)$ in Ext degree $3 + 1 + (d_c - d_W) + (d_V - d_e) = 3 + d_c - d_e$. No piece $L_{\zeta''}$ other than W lies on the first chain: by Lemma 4.36(iii) it would have the shift at least $d_W + 1 = s + 1$. Likewise only multiples follow V on the second chain, and each chain has at most three pieces besides W or V. Let U and D be the

[12] This is Mumford's index theorem: a line bundle on an abelian variety whose hermitian form is nondegenerate with i negative eigenvalues has cohomology only in degree i. On $B_j = E_0 \times E_0$ the form of $M_i \otimes L_\zeta^{-1}$ is $\left(\begin{smallmatrix} t_i - p & -\zeta_j \\ -\bar{\zeta}_j & t_i - p \end{smallmatrix}\right)$, with eigenvalues $t_i - p \pm 1$, and $|t_i - p| \geq 2$ makes both of the sign of $t_i - p$; over the four surfaces the index is 0 or 8. The hypothesis $|t_i - p| \geq 2$ is what keeps the steps between a piece and a multiple out of the middle degrees.

[13] The bound is attained: with p below the three values t_i, the run $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, up three times and down once, changes the shift by $3 - 7 = -4$. With p in the middle, $t_1 < p < t_2 < t_3$, the run up through M_2 and M_3, down to M_1 and up to $L_{\zeta'}$ changes it by $3 - 7 = -4$ as well.



Two shifts on E_0^8

W: the 64 pieces with $\prod\zeta=\epsilon$, at shift s;
V: the other 64, at shift $s-1$;
a step up in ϕ has Ext degree 0: shift $+1$;
a step down has Ext degree 8: shift -7;
a run through the multiples drops ≥ 4,
so no product reaches $\mathcal{H}^3(W,V)$,
and no term of d_E leaves it;
each top piece has 28 partners V, with
$D=16$ or 64, 640 classes in all:
$\dim \mathrm{Ext}^2(E,E)\geq 64\cdot 640=40960$,
against 104 for the polarised criterion;
three consecutive: ≥ 22528; more stay open

FIGURE 8. Lemma 4.36 and Theorem 4.37, in the placement $p<t_1<t_2<t_3$. Horizontal: the value ϕ of a piece; vertical: its shift. The pieces L_ζ with $\prod\zeta=\epsilon$ sit at the shift s and the others at $s-1$. A run through the multiples, up three times with Ext degree 0 and down once with Ext degree 8, ends four shifts lower, the least drop of Lemma 4.36(iii), so it never joins a top piece W to a partner V one shift lower; the group $\mathcal{H}^3(W,V)$ survives in $\mathrm{Ext}^2(E,E)$.

numbers of steps up and down along both chains, so that $U+D\leq 6$; by Lemma 4.36(ii), $d_W=d_c+U_1-7D_1$ and $d_e=d_V+U_2-7D_2$, and the Ext degree is

$$3+d_c-d_e=4-U+7D.$$

If $c=e$, this degree is 3, so $U=7D+1$, and $D\geq 1$ because the values return to $\phi(c)\neq p$; then $U\geq 8$, which is impossible. If $c\neq e$, the group $\mathcal{H}(c,e)$ sits in degree 0 if $\phi(e)>\phi(c)$ and in degree 8 if $\phi(e)<\phi(c)$, by Lemma 4.36(ii). Degree 0 needs $U=4+7D\geq 4$ with $D=0$, so the values increase strictly, the multiples are distinct and $U\leq 3$. Degree 8 needs $7D=U+4$, so $D=1$ and $U=3$; the values increase strictly before the step down, from $\phi(c)$, and after it, up to $\phi(e)<\phi(c)$, so the five values along the path, with W and V counted once, are distinct, and four of them are values of multiples, of which there are three. So no term reaches a nonzero group, and d_E vanishes on $\mathcal{H}^3(W,V)$.

By (i) and (ii) the classes of these groups are cocycles of degree two, and no nonzero combination of them is a coboundary, since every coboundary has zero components there. Each piece of the shift s has 28 partners of the other parity differing from it in exactly three coordinates,[14] all of the shift $s-1$, and the 28 groups have dimensions adding up to 640; there are 64 pieces of the shift s. ■

Figure 8 draws the shifts along a run. The hypothesis of (ii) is the arrangement of the explicit convolution of Proposition 4.33 at $n=3$, where the sixteen pieces of each parity share a shift. The pieces must not be twisted by line bundles of degree zero here, unlike

[14]The ratios ζ_j'/ζ_j in the three coordinates have product -1, so they are $(-1,-1,-1)$, with $D=64$, or a permutation of $(i,-i,-1)$, with $D=16$; with four places for the coordinate where they agree this gives $4\cdot 7=28$ partners and $4\cdot(64+6\cdot 16)=640$ classes.

in Theorem 4.35: a twist that is nontrivial on the surface where ζ and ζ' agree removes $\mathcal{H}(L_\zeta, L_{\zeta'})$.

With a third shift, a component of Ext degree two can follow or precede a class of $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, so the differential no longer vanishes on these classes. When the three shifts are consecutive, such terms reach only the groups between the highest and the lowest shift, which are too small to remove the classes.

Theorem 4.38 (Three consecutive shifts on E_0^8)**.** *Keep Lemma 4.36, and suppose that the shifts of the L_ζ lie in $\{s, s-1, s-2\}$ for an integer s. By their parities, for a sign ϵ, the pieces with $\prod \zeta = \epsilon$ have the shift $s-1$ and the others the shift s or $s-2$; let T and B be the sets of pieces of the shifts s and $s-2$. Then, for every order of the pieces, every choice of the shifts of the M_i and every placement of the t_i:*

(i) *every two pieces L_ζ, $L_{\zeta'}$ that differ in exactly three coordinates have shifts differing by one, and the groups $\mathcal{H}^3(L_\zeta, L_{\zeta'})$ of Theorem 4.37, taken from the piece of the larger shift, have dimensions adding up to* 40960*;*

(ii) *d_E maps the sum of these groups into $\bigoplus_{c\in T,\, e\in B} \mathcal{H}^5(c,e)$, and on them it consists of cup products with a single component of* Ext *degree two;*

(iii)
$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 40960 - \sum_{c\in T,\, e\in B} \dim \mathcal{H}^5(c,e) \;\geq\; 40960 - 18432 = 22528.$$

In particular E does not meet the polarised criterion. When T or B is empty this is Theorem 4.37(ii).

Proof. (i) The ratios ζ'_j/ζ_j in the three coordinates have product -1, so the two pieces have opposite parities: one has the shift $s-1$ and the other s or $s-2$. Each of the 64 pieces of the shift $s-1$ has 28 such partners, whose groups have dimensions adding up to 640, as in the proof of Theorem 4.37; this gives $64 \cdot 640 = 40960$.

(ii) Let $x \in \mathcal{H}^3(a,b)$ with $d_a = d_b + 1$. As in the proof of Theorem 4.37(ii), a term of $d_E(x)$ runs along chains of components $c \to \cdots \to a$ and $b \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c,e)$ in Ext degree $3 + d_c - d_e$. The shifts of the pieces L_ζ differ by at most two, so by Lemma 4.36(iii) no part of a chain between two pieces L_ζ passes through a multiple. Hence a chain $c \to \cdots \to a$ that starts at a multiple M_i is a run $M_i = m_0 \to \cdots \to m_r \to L_1$ of distinct multiples, $r \leq 2$, followed by pieces L_ζ only; let U_1 and D_1 be the numbers of its steps up and down the value ϕ before L_1, so that $d_{L_1} = d_{M_i} + U_1 - 7D_1$ by Lemma 4.36(ii) and $U_1 + D_1 = r + 1 \leq 3$. Likewise a chain $b \to \cdots \to e$ ending at a multiple M_k ends with a run from a piece L_2 with U_2 steps up and D_2 down, $U_2 + D_2 \leq 3$ and $d_{M_k} = d_{L_2} + U_2 - 7D_2$. Put $L_1 = c$ if c is a piece L_ζ and $L_2 = e$ if e is one, and $\delta = d_{L_1} - d_{L_2}$; then $1 = d_a - d_b \leq \delta \leq 2$.

If c and e are both pieces L_ζ, each nonempty chain lowers the shift by at least one, so $3 + d_c - d_e \geq 5$, while $\delta \leq 2$ gives at most 5. So the degree is 5, $c \in T$, $e \in B$, and the two chains together consist of one component, of Ext degree two by Lemma 4.36(i): the term is a cup product of x with that component.

If $c = M_i$ and e is a piece L_ζ, the degree is $3 + \delta - U_1 + 7D_1$, and $\mathcal{H}(c,e)$ sits in degree 0 or 8 by Lemma 4.36(ii). Degree 0 needs $U_1 \geq 3 + \delta \geq 4$, and degree 8 needs $7D_1 - U_1 = 5 - \delta \in \{3, 4\}$, so $D_1 \geq 1$ and $U_1 \geq 3$; both contradict $U_1 + D_1 \leq 3$. The case of a piece c and a multiple e is the same with the second run.

If $c = M_i$ and $e = M_k$, the degree is $3 + \delta - U + 7D$ with $U = U_1 + U_2 \leq 6$ and $D = D_1 + D_2$. For $i = k$ it is 3, so $U - 7D = \delta \geq 1$; $D = 0$ is impossible, since the values

would increase from t_i to p along the first run and from p to t_i along the second, and $D \geq 1$ gives $U \geq 8$. For $i \neq k$, $\mathcal{H}(c,e)$ sits in degree 0 if $t_k > t_i$ and 8 if $t_k < t_i$. Degree 0 needs $U = 3 + \delta + 7D$, so $D = 0$ and $U \geq 4$; but then the multiples of the first run have values below p and those of the second values above p, so they are distinct and $U \leq 3$. Degree 8 needs $7D - U = 5 - \delta$, so $D = 1$ and $U \geq 3$: the runs have at least four steps, so some multiple lies on both. One run increases strictly. If it is the second, its values lie in $(p, t_k]$, while those of the first lie at or above $t_i > t_k$ before its step down and below p after it. If it is the first, its values lie in $[t_i, p)$, while those of the second lie above p before its step down and at or below $t_k < t_i$ after it. Either way no multiple lies on both runs, a contradiction.

(iii) By Theorem 4.37(i), which holds for every convolution, no coboundary has a component in these groups, so the kernel of d_E on their sum injects into $\operatorname{Ext}^2(E,E)$, and by (ii) its dimension is at least 40960 minus the dimension of the targets. For pieces c, e of the same parity, $\mathcal{H}^5(c,e)$ is nonzero only when they differ in two or three coordinates, and each piece has such partners with groups $\mathcal{H}^5$ of dimensions adding up to 960.[15] So the sum in (iii) is at most $\min(|T|,|B|) \cdot 960 \leq 32 \cdot 960 = 30720$, which already gives $\dim \operatorname{Ext}^2(E,E) \geq 10240$. The dimension of $\mathcal{H}^5(c,e)$ depends only on the ratio e/c, so the sum is the weight of the cut (T,B) of a weighted Cayley graph on the group of ratios of two pieces of one parity, of order 64 and degree 960. For the vector $x = \pm 1$ of the cut, the weight is $\frac{1}{4}(64 \cdot 960 - x^{\mathsf{T}} A x) \leq 16\,(960 - \lambda)$, with A the weighted adjacency matrix and λ its least eigenvalue. Its eigenvalues are the character sums $\sum_g \dim \mathcal{H}^5(1,g)\,\chi(g)$. A character is $\chi(g) = \prod_j g_j^{k_j}$ with $k \in (\mathbb{Z}/4)^4$, up to adding $(1,1,1,1)$; put $\omega_j = i^{k_j}$, $W = \sum_j \omega_j$, $\varrho = \sum_j \omega_j^2$ and $\Omega = \prod_j \omega_j$. By the list in the footnote, the ratios that move the pair of coordinates $\{j,l\}$ contribute $64\,\omega_j^2\omega_l^2 + 32\,\Re(\omega_j\overline{\omega}_l)$, and those that fix only the coordinate o contribute $32\,\Re\big(\Omega\overline{\omega}_o(W - \omega_o)\big)$. Summing,

$$\chi\big(\textstyle\sum_g \dim \mathcal{H}^5(1,g)\,g\big) = 32\varrho^2 + 16|W|^2 - 192 + 32\,\Re(\Omega)\big(|W|^2 - 4\big).$$

If $\Re\Omega = 0$, the sum of the k_j is odd, so an odd number of them are odd, $|\varrho| = 2$, and the value is at least -64. If $\Re\Omega = 1$, an even number are odd: with none or four, $|\varrho| = 4$ and the value is at least 192; with two, $\varrho = 0$, and $W = 0$ would force the k_j to be $0,1,2,3$, whose sum is not divisible by 4, so $|W|^2 \geq 4$ and the value is at least -128. If $\Re\Omega = -1$, the value is $32\varrho^2 - 16|W|^2 - 64$, and $|W|^2 \leq 8 + 2\varrho^2$: when all k_j have the same parity, $|W| \leq 4 = |\varrho|$, and when two are odd, $\varrho = 0$ and W is a sum of two of ± 1 and two of $\pm i$, so $|W|^2 \leq 8$. Hence $\lambda \geq -192$,[16] and the sum is at most $16 \cdot 1152 = 18432$. ∎

Figure 9 draws the three shifts and the budget.

The groups of Theorem 4.37 join two pieces whose shifts differ by one. All groups between pieces whose shifts differ by three, and thirteen kinds between pieces whose shifts differ by two, are also left alone by the differential, and with them the argument covers every arrangement of the shifts within five consecutive values. For pieces a and b call $d_a - d_b$ the *spread* of $\mathcal{H}(a,b)$; its classes of degree two have Ext degree $2 + d_a - d_b$. For $\zeta, \zeta' \in \mu_4^4$ call $\zeta'/\zeta = (\zeta'_j/\zeta_j)_j$ their *ratio*; it *moves* the coordinates where it differs from 1,

[15] The ratios $e_j/c_j \neq 1$ have product 1. In two coordinates they are $(-1,-1)$, with $D = 16$, or $(i,-i)$, $(-i,i)$, with $D = 4$; then $\mathcal{H}^5 = \mathcal{H}^1 \otimes \mathcal{H}^1 \otimes \mathcal{H}^3(B_j \times B_k, \mathcal{O})$ has dimension $4D$, and the six pairs of coordinates give $6 \cdot 64 + 12 \cdot 16 = 576$. In three coordinates they are a permutation of $(i,i,-1)$ or of $(-i,-i,-1)$, with $D = 16$, and $\mathcal{H}^5 = (\mathcal{H}^1)^{\otimes 3} \otimes H^2(B_l, \mathcal{O})$ has dimension 16; the four triples give $4 \cdot 6 \cdot 16 = 384$. In four coordinates $\mathcal{H}(c,e)$ sits in degree 4.

[16] The eight values are 960, 384, 192, 96, -32, -64, -128 and -192. The bound 18432 is attained by the cut $\zeta_3\zeta_4 \in \{1,i\}$ against $\zeta_3\zeta_4 \in \{-1,-i\}$, so 22528 is the best bound that this count gives.

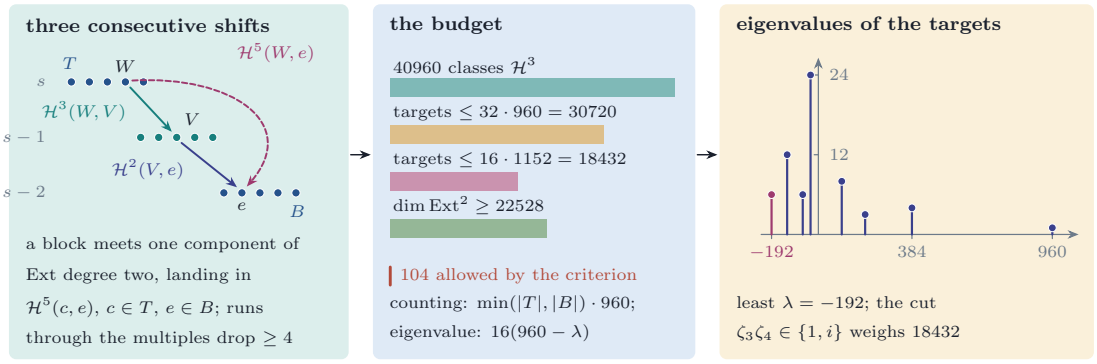


FIGURE 9. Theorem 4.38. Left: the pieces L_ζ at three consecutive shifts, those of one parity at $s-1$ and the others split between T at s and B at $s-2$; a group $\mathcal{H}^3$ between two pieces that differ in three coordinates (teal) can meet a component of Ext degree two (indigo) only in a group $\mathcal{H}^5(c,e)$ with $c \in T$ and $e \in B$ (magenta). Middle: the 40960 classes against the targets, at most 30720 by counting and 18432 by the least eigenvalue, against the 104 that the criterion allows. Right: the eigenvalues of the Cayley graph of the targets.

and it changes the parity, $\prod \zeta' = -\prod \zeta$, exactly when the product of its entries is -1. As before, $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$.

Lemma 4.39 (Groups of spread two and three)**.** *Keep Lemma 4.36, and let E be any convolution of the L_ζ and the M_i, in any order, with any shifts of the given parities and any placement of the t_i.*

(i) *For a class x of degree two in $\mathcal{H}(L_\zeta, L_{\zeta'})$ of positive spread, every term of $d_E(x)$ has all its leaves between pieces $L_{\zeta''}$.*

(ii) *Let $\prod \zeta = \prod \zeta'$, let ζ'/ζ move all four coordinates, one of them by -1, and let $d_{L_\zeta} = d_{L_{\zeta'}} + 2$. The classes of degree two of $\mathcal{H}(L_\zeta, L_{\zeta'})$ form $\mathcal{H}^4(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') \in \{64, 256\}$; no element of degree one has a differential with a nonzero component in it, and d_E vanishes on it.*

(iii) *Let $\prod \zeta = -\prod \zeta'$ and $d_{L_\zeta} = d_{L_{\zeta'}} + 3$. The classes of degree two of $\mathcal{H}(L_\zeta, L_{\zeta'})$ lie in $\mathcal{H}^5(L_\zeta, L_{\zeta'})$, and d_E vanishes on them.*

(iv) *Let ζ and ζ' differ in exactly three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Every term of d_E on $\mathcal{H}^3(L_\zeta, L_{\zeta'})$ runs along chains of components $c \to \cdots \to L_\zeta$ and $L_{\zeta'} \to \cdots \to e$ between pieces $L_{\zeta''}$ that together lower the shift by one or two.*

Proof. Write $a = L_\zeta$, $b = L_{\zeta'}$ and $\delta = d_a - d_b$. As in the proof of Theorem 4.37(ii), a term of d_E on a class of degree two in $\mathcal{H}(a,b)$ runs along chains of components $c \to \cdots \to a$ and $b \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c,e)$ in Ext degree $3 + d_c - d_e$, which is 3 if $c = e$.

(i) Let L_1 be the first piece $L_{\zeta''}$ of the first chain and L_2 the last of the second, and put $X = (d_{L_1} - d_a) + (d_b - d_{L_2})$. By Lemma 4.36(iii), $X \geq 0$, and $X \geq 4$ if a multiple lies between L_1 and a or between b and L_2. Before L_1 the first chain is a run of distinct multiples, with U_1 steps up and D_1 steps down the value ϕ and $U_1 + D_1 \leq 3$; after L_2 the second chain is a run with U_2 and D_2 steps, $U_2 + D_2 \leq 3$. By Lemma 4.36(ii) the term

has Ext degree $3+\delta+X-U+7D$, where $U=U_1+U_2$ and $D=D_1+D_2$. If c and e are pieces, this is $3+\delta+X$; a multiple on a chain makes it at least 8, but $\mathcal{H}^8(c,e)\neq 0$ only for $c=e$, where the degree is 3. If $c=M_i$ and e is a piece, $\mathcal{H}(c,e)$ sits in degree 0 if $t_i<p$ and in degree 8 if $t_i>p$. Degree 0 needs $U_1\geq 3+\delta+X\geq 4$. Degree 8 needs a step down on the run from t_i to p, so $D_1\geq 1$, and then $U_1=7D_1-5+\delta+X\geq 3$. Both contradict $U_1+D_1\leq 3$, and a piece c with a multiple e is the same. Let $c=M_i$ and $e=M_k$. For $i=k$ the degree is 3, so $U-7D=\delta+X\geq 1$; $D\geq 1$ would give $U\geq 8$, and $D=0$ would make the values increase from t_i to p and then from p to t_i. For $i\neq k$, $\mathcal{H}(c,e)$ sits in degree 0 if $t_k>t_i$ and in degree 8 if $t_k<t_i$. Degree 0 needs $U=3+\delta+X+7D$, so $D=0$ and $U\geq 4$; but then the first run lies below p and the second above, so the multiples are distinct and $U\leq 3$. Degree 8 needs $D\geq 1$, since with $D=0$ the values would satisfy $t_i<p<t_k$; then $7D-U=5-\delta-X\leq 4$ gives $D=1$ and $U\geq 3$, so the runs have at least four steps and share a multiple, which the last paragraph of the proof of Theorem 4.38(ii) excludes.

(ii) By the proof of Lemma 4.36(i), $\mathcal{H}(a,b)$ is the tensor product of four groups H^1 and sits in degree 4, of dimension $D(\zeta,\zeta')$; the ratio is $(-1,-1,-1,-1)$, with $D=256$, or a permutation of $(i,-i,-1,-1)$, with $D=64$.[17] A coboundary with a component in $\mathcal{H}(a,b)$ is a sum of terms of Lemma 4.24(iii) along a path $a\to\cdots\to b$ of morphisms of degree one, one of them an element u and the others components, that lowers the shift by two. By Lemma 4.36(iii) no multiple lies on it, and by Lemma 4.36(i) each step between distinct pieces lowers the shift by at least one. So the path consists of a component of Ext degree three from a to b, which lies in $\mathcal{H}^3(a,b)=0$, and a class of $\mathcal{H}^1(a,a)$ or $\mathcal{H}^1(b,b)$; or of two steps $a\to c\to b$ of Ext degree two, possibly with a class of $H^1(A,\mathcal{O}_A)$ at one of the three pieces. The output has degree one on every surface. Without the class of H^1 the two steps have degrees adding up to one on every surface, so they move disjoint pairs of coordinates and have degree zero elsewhere; a step of Ext degree two changes the parity, so its two moved entries have product -1 and are (i,i) or $(-i,-i)$, and ζ'/ζ has no entry -1. With the class of H^1, Lemma 4.28 gives a drop of one on a surface where all three leaves have positive degree, and the output has degree at least two there.

Now let a term of d_E on $\mathcal{H}^4(a,b)$ carry the components $y_1,\dots,y_m$, $m\geq 1$, between pieces by (i), and land in $\mathcal{H}(c,e)$; let K be the set of coordinates fixed by the ratio of c and e, and $X\geq m$ the drop of the chains. The output has Ext degree $5+X$. Let n_j be the number of the y_i of positive degree on B_j; the class x has degree one there, so Lemma 4.28 bounds the drop on B_j by $\max(0,n_j-1)$. For $j\notin K$ the output has degree 1 on B_j, so the degrees of the y_i there add up to at most $\max(0,n_j-1)$, which forces $n_j=0$. For $j\in K$ the y_i undo the entry $\zeta_j'/\zeta_j\neq 1$, so $n_j\geq 1$, and the output has degree at least $1+n_j-(n_j-1)=2$ there, hence exactly 2. So $5+X=(4-|K|)+2|K|$, that is $X=|K|-1$. The components change the parity X times modulo two, and together their ratio is ζ/ζ' on K and 1 elsewhere, so

$$\prod_{j\in K}\zeta_j'/\zeta_j=(-1)^{|K|-1}.$$

[17] A ratio that moves every coordinate has entries in $\{i,-1,-i\}$ and product 1. With a -1 among them it is $(-1,-1,-1,-1)$ or has exactly two entries -1 and the entries i and $-i$, twelve ratios; together thirteen, and $256+12\cdot 64=1024$. The other eight ratios have all four entries in $\{i,-i\}$ and $D=16$; (ii) makes no statement about them.

For the ratio $(-1,-1,-1,-1)$ the left side is $(-1)^{|K|}$. For a permutation of $(i,-i,-1,-1)$ the left side is real only if K contains both or neither of the coordinates moved by $\pm i$, and then it is again $(-1)^{|K|}$. So no term is nonzero.

(iii) The classes of degree two have Ext degree 5. By (i) a term of d_E on them lands in Ext degree $6+X$, with $X \geq 1$ the drop of the chains between pieces. For $X \geq 2$ this is at least 8, which needs $c = e$ and degree 3. For $X = 1$ the pieces c, e have shifts differing by four, hence the same parity, so their ratio moves at least two coordinates and $\mathcal{H}(c,e)$ has no degree above 6.

(iv) By (i) a term lands in Ext degree $4+X$, with $X \geq 1$ the drop of the chains between pieces. For $X \geq 4$ the degree is at least 8, which needs $c = e$. For $X = 3$ the degree is 7, which needs a ratio moving one coordinate, hence opposite parities, while $d_c - d_e = 4$ is even. ∎

The groups of (ii) are reached by no coboundary and left by no term in every convolution, so they survive in $\mathrm{Ext}^2(E,E)$ as soon as the shifts give them the spread two. That happens along every edge of a Cayley graph that crosses between two shift values of one parity, and a cut of that graph is never light.

Lemma 4.40 (Cuts of a Cayley graph)**.** *Let G be a finite abelian group, $S \subset G \setminus \{0\}$ a generating set and $w\colon S \to \mathbb{R}_{>0}$ a weight, extended by 0 to G. For every subset $T \subset G$ with $T \neq \emptyset, G$,*

$$\sum_{c \in T,\ e \notin T} w(e-c) \ \geq \ \min_{H} |H| \cdot w(S \setminus H),$$

the minimum over the proper subgroups H of G. For the group G_0 of the 64 ratios of pieces of one parity, with S the thirteen ratios of Lemma 4.39(ii) and $w = D$, the minimum is 1024, attained only at $H = 0$, so every such cut weighs at least 1024.

Proof. The left side is $\sum_{s \in S} w(s)\,|(T+s) \setminus T|$. Let H be the stabiliser of T, a proper subgroup since $T \neq \emptyset, G$. Then T and $T+s$ are unions of cosets of H, and for $s \notin H$ the set $(T+s) \setminus T$ is nonempty, since $|T+s| = |T|$, so it has at least $|H|$ elements. This gives the inequality. In G_0, written additively as the quadruples in $(\mathbb{Z}/4)^4$ of sum 0, the ratios of type $(i,-i,-1,-1)$ generate, since the differences of two of them give every $e_j - e_k$. The weights are 256 once and 64 otherwise, and $|S \cap H| \leq |H| - 1$. So $|H|\,w(S \setminus H)$ is 1024 for $H = 0$, at least $|H|\,(1024 - 256 - 64(|H|-2))$, which is 1536, 2560 and 3072, for $|H| = 2, 4, 8$, and at least $64|H| \geq 1024$ for $|H| \geq 16$, because some generator lies outside H.[18] ∎

Theorem 4.41 (One parity at two or three values)**.** *Keep Lemma 4.36, and suppose that the shifts of the pieces L_ζ with $\prod \zeta = \epsilon$ take exactly two values s and $s-2$, or exactly three values $s+2$, s and $s-2$, for a sign ϵ and an integer s. Then, for every order of the pieces, all shifts of the other pieces, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E) \ \geq \ 1024.$$

In particular E does not meet the polarised criterion.

Proof. Let T be the set of pieces of parity ϵ at the shift s; it is neither empty nor all of them. Two pieces of parity ϵ, one in T and the other not, have shifts differing by exactly two. When their ratio is one of the thirteen of Lemma 4.39(ii), the group $\mathcal{H}^4$ from the

[18] The orders of the subgroups of $G_0 \cong (\mathbb{Z}/4)^3$ are powers of two. A cut at one vertex weighs exactly 1024, so the bound is sharp.

piece of the larger shift to the other is reached by no coboundary and killed by no term, so its classes inject into $\mathrm{Ext}^2(E,E)$, and those of different pairs lie in different summands $\mathcal{H}(a,b)$. Their dimensions add up to the weight of the cut $(T,\complement T)$ of the Cayley graph of Lemma 4.40, since the pieces of parity ϵ form a coset of G_0; so they add up to at least 1024. ∎

Theorem 4.42 (Shifts three apart)**.** *Keep Lemma 4.36, and suppose that the pieces L_ζ with $\prod\zeta=\epsilon$ all have the shift s and the others have shifts in $\{s-3,s+3\}$. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 64\cdot 1072 - 128\cdot 8 = 67584.$$

Proof. Every piece W of parity $-\epsilon$ and every piece V of parity ϵ have shifts differing by three. By Lemma 4.39(iii) the classes of degree two of the group from the higher to the lower of the two lie in $\mathcal{H}^5$, and d_E vanishes on them. For each W these groups have dimensions adding up to 1072.[19] A coboundary with a component in such a group runs along a path of morphisms of degree one that lowers the shift by three; no multiple lies on it, by Lemma 4.36(iii), and a step between distinct pieces lowers the shift by at least one, while no piece has a shift strictly between those of W and V. So the path is the component between W and V together with a class of $H^1(A,\mathcal{O}_A)$ at one of them, and the components of all coboundaries in these groups come from $\bigoplus_a \mathcal{H}^1(a,a)$ over the 128 pieces, of dimension $128\cdot 8=1024$. The $64\cdot 1072 = 68608$ classes are cocycles, and at most 1024 dimensions of them are coboundaries. ∎

When one parity sits at a single shift, the groups $\mathcal{H}^3$ of Theorem 4.37 survive next to it unless pieces of the other parity crowd it from both sides.

Proposition 4.43 (A gap of four)**.** *Keep Lemma 4.36, let the pieces L_ζ with $\prod\zeta=\epsilon$ all have the shift y, and let T be the set of pieces of parity $-\epsilon$ at the shift $y+1$. If T is not empty and no piece has the shift $y-1$ or $y+3$, then for every order, every choice of the shifts of the M_i and every placement of the t_i, the groups $\mathcal{H}^3(W,V)$ with $W\in T$ and V differing from W in exactly three coordinates are reached by no coboundary and killed by no term, and*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 640\,|T| \;\geq\; 640.$$

The same holds with $y+1$, $y-1$, $y+3$ replaced by $y-1$, $y+1$, $y-3$ and the groups $\mathcal{H}^3(V,W)$.

Proof. By Theorem 4.37(i), which holds for every convolution, no coboundary reaches these groups. By Lemma 4.39(iv) a term of d_E on $\mathcal{H}^3(W,V)$ runs along chains of components between pieces into W and out of V that lower the shift by one or two in all; the first would pass through a piece at the shift $y+2$ or $y+3$, the second through one at $y-1$ or $y-2$. The shifts $y\pm 2$ have parity ϵ, whose pieces are all at y, and $y-1$, $y+3$ are excluded. Each $W\in T$ has 28 partners V, all at the shift y, with groups of dimensions adding up to 640. The second statement has the same proof, with the chains into V and out of W in place of those into W and out of V. ∎

[19] Let the ratio move k coordinates. For $k=1$ it is -1 in one coordinate, $D=4$, and $\mathcal{H}^5$ is H^1 tensored with the part of degree four of $H^*(\mathcal{O})$ on the other three surfaces, of dimension $\binom{6}{4}=15$: four partners with 60 each. For $k=2$ it is (i,i) or $(-i,-i)$, $D=4$, and the part of degree three on the other two surfaces has dimension $\binom{4}{3}=4$: twelve partners with 16 each. For $k=3$ the part of degree two on the last surface has dimension one, and the 28 partners give 640 as in Theorem 4.37. For $k=4$ the group sits in degree 4. So $240+192+640=1072$.

Corollary 4.44 (Five consecutive shifts)**.** *Keep Lemma 4.36. If the shifts of the* 128 *pieces* L_ζ *lie in five consecutive integers, then, for every order, every choice of the shifts of the* M_i *and every placement of the* t_i*,* $\dim \mathrm{Ext}^2(E,E) \geq 640$*; no such convolution meets the polarised criterion.*

Proof. Let the values be $s, \dots, s-4$, with one parity at values in $\{s, s-2, s-4\}$ and the other at values in $\{s-1, s-3\}$. If the second parity takes both its values, or the first takes two values two apart or all three, Theorem 4.41 applies. Otherwise the second parity sits at one value y and the first at one value or at $\{s, s-4\}$. One value each, at an odd distance at most three: Theorem 4.37 or Theorem 4.42. The first at $\{s, s-4\}$: for $y = s-1$ the pieces at $y+1 = s$ are not empty and $y-1$, $y+3$ are empty, and for $y = s-3$ the pieces at $y-1 = s-4$ are not empty and $y+1$, $y-3$ are empty, so Proposition 4.43 applies. The least of the bounds is 640. ∎

Within six consecutive values two arrangements are left, up to exchanging the parities: each parity at one shift, five apart, and the parities at $\{s, s-4\}$ and $\{s-1, s-5\}$. The first is settled by the diagonal, whose classes at a shift with no other piece nearby can be reached only through all three multiples, and the second by the groups of Theorem 4.37 once more.

Lemma 4.45 (The diagonal at a lonely shift)**.** *Keep Lemma 4.36, and let the pieces* L_ζ *with* $\prod \zeta = \epsilon$ *all have the shift* s *and the others the shift* $s-g$*, for a sign* ϵ *and an odd integer* $g \geq 5$*. Let* $y \in \mathcal{H}^2(a,a)$ *be a diagonal class at a piece* a*, and let a term of* $d_E(y)$ *in Lemma 4.24(iii) be nonzero. Then* a *is the only piece on its path, the path passes through* M_1*,* M_2 *and* M_3*, and the shifts of* a*,* M_1*,* M_2*,* M_3 *are pairwise incongruent modulo* 4*.*

Proof. Write the path as $c \to \cdots \to a \to \cdots \to e$, with the chain of components into a and the chain out of a not both empty, so that $c \neq e$, since the components of a convolution go up the order of the pieces. Let L_1 and L_2 be the first and the last piece on it and $X = d_{L_1} - d_{L_2}$; before L_1 and after L_2 the path is a run of distinct multiples, with U steps up and D steps down the value ϕ in all. As in the proof of Lemma 4.39(i), now with $\delta = 0$, the term has Ext degree $3 + X - U + 7D$ by Lemma 4.36(ii). A sequence of components from one piece to another lowers the shift by Lemma 4.36(iii), and the pieces have the shifts s and $s-g$; so $X \in \{0, g\}$, and $X = 0$ exactly when a is the only piece on the path.

If c and e are pieces, then $U = D = 0$ and $X = g$, and $\mathcal{H}^{3+g}(c,e) = 0$ since $3 + g \geq 8$ and $c \neq e$. If c is a multiple and e a piece, then $\mathcal{H}(c,e)$ sits in degree 0 or 8. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$: for $X = 0$ this gives $D = 1$ and $U = 2$; for $X = g$ it gives $D = 0$ and $U = g - 5$, which is at least one because the run before L_1 is not empty and at most three, so $g = 7$; but then the path lowers the shift by seven from L_1 to e, by a single component, of Ext degree eight between distinct pieces, or by a run through multiples, and neither is possible.[20] A piece c with a multiple e is the same with the runs exchanged. If c and e are multiples, both runs are nonempty and together pass through at most three multiples, so $U + D \leq 3$ and $U + D \geq 2$. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$; $D = 0$ gives $U = X - 5 \geq 2$, so $X = g = 7$, which is excluded as before; $D = 1$ gives $U = 2 + X \leq 2$, so $X = 0$ and $U = 2$; $D \geq 2$ gives $U \geq 9$.

[20] Between distinct pieces $\mathcal{H}^8 = 0$. A run from a piece through $m \leq 3$ distinct multiples to a piece has $m+1$ steps, at least one up and one down, and lowers the shift by $7D' - U'$ with $U' + D' = m + 1$: by 6, 5, 4 for $D' = 1$, by 13, 12 for $D' = 2$ and by 20 for $D' = 3$, never by seven.

So in every case $X = 0$ and $U+D = 3$: the path consists of a and the three multiples, with three steps. A step up raises the shift by one and a step down lowers it by seven, so every step adds one modulo 4, and the four nodes of the path have the shifts d_c, d_c+1, d_c+2, d_c+3 modulo 4 in the order of the path.[21] ■

Proposition 4.46 (Single shifts)**.** *Keep Lemma 4.36, and suppose that the pieces L_ζ with $\prod \zeta = \epsilon$ all have the shift s and the others all have the shift $s - g$, for a sign ϵ and an odd integer g. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E, E) \;\geq\; 64 \cdot 28 = 1792.$$

Proof. Exchanging the parities, we may take $g > 0$. For $g = 1$ and $g = 3$ this is Theorem 4.37 and Theorem 4.42. Let $g \geq 5$. If a term of d_E were nonzero on a diagonal class at a piece of shift s and another on a diagonal class at a piece of shift $s - g$, then by Lemma 4.45 the shifts of the three multiples would be pairwise incongruent modulo 4 and miss both the residue of s and that of $s - g$, which differ, since g is odd. So at one of the two shifts d_E vanishes on the 28 diagonal classes of each of the 64 pieces there, and by Proposition 4.29, which holds for $n = 4$, these 1792 classes inject into $\mathrm{Ext}^2(E, E)$.[22] ■

Proposition 4.47 (Interleaved shifts)**.** *Keep Lemma 4.36, and suppose that the shifts of the pieces L_ζ with $\prod \zeta = \epsilon$ take exactly the two values s and $s - 4$, and those of the others exactly the two values $s - 1$ and $s - 5$. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim \mathrm{Ext}^2(E, E) \geq 640$.*

Proof. Let A and A' be the pieces of parity ϵ at the shifts s and $s-4$, and B and B' those of parity $-\epsilon$ at $s-1$ and $s-5$; all four are nonempty. Let W and V differ in exactly three coordinates, with $d_W = d_V + 1$, that is $W \in A$ and $V \in B$, or $W \in A'$ and $V \in B'$. By Theorem 4.37(i) no coboundary reaches $\mathcal{H}^3(W, V)$, and by Lemma 4.39(iv) a term of d_E on it passes through a piece at the shift $d_W + 1$ or $d_W + 2$ or at $d_V - 1$ or $d_V - 2$. These are $s+1$, $s+2$, $s-2$, $s-3$ for $W \in A$ and $s-3$, $s-2$, $s-6$, $s-7$ for $W \in A'$, and no piece has these shifts. So the groups inject into $\mathrm{Ext}^2(E, E)$.

Let Γ be the group of the ratios of two pieces, of order 128, which acts simply transitively on the pieces, and $S_3 \subset \Gamma$ the 28 ratios that move exactly three coordinates and change the parity, with the weight D, which adds up to 640 (Theorem 4.37). With $X = A \cup B'$ the pairs above are exactly the edges of the Cayley graph of (Γ, S_3) between X and its complement $A' \cup B$; so the dimensions add up to the weight of that cut, and X is neither empty nor everything. Written additively, as the quadruples in $(\mathbb{Z}/4)^4$ of even sum, the differences of two ratios of type $(i, -i, -1, 1)$ give every $e_j - e_k$, which generate the quadruples of sum 0 modulo 4, and the elements of S_3 have sum 2 modulo 4; so S_3 generates Γ. In Lemma 4.40, $H = 0$ gives 640; for $|H| = 2, 4, 8, 16$ at most $|H| - 1$ elements of S_3 lie in H, at most four of them of weight 64, so $|H|\, w(S_3 \setminus H)$ is at least $2 \cdot 576$, $4 \cdot 448$, $8 \cdot 336$ and $16 \cdot 208$; for $|H| = 32$ some $r \in S_3$ lies outside H, and so does $-r$, which gives weight at least 32, or 64

[21] With the values $t_1 < t_2 < t_3$ and p, the path with $U = 3$ visits the four nodes in the order of their values; a path with $U = 2$ and $D = 1$ ending in degree eight visits the largest j values in increasing order and then the others, $j = 1, 2, 3$. So each placement has exactly four such paths, and for each the shifts of the multiples are determined by d_a.

[22] The proof does not look at the classes between the two parities. For $g = 5$ these are $64 \cdot 16 = 1024$ classes of degree two in groups $\mathcal{H}^7$, on which d_E vanishes by Lemma 4.39(i); but the coboundaries of the loops in $\mathcal{H}^1(a, a)$, and, when two multiples have suitable shifts, the products along runs $W \to M_i \to M_j \to V$, reach them, and counting alone gives nothing there.

if $r = -r$; and for $|H| = 64$ the weight outside H is at least 16. So every such cut weighs at least 640.[23] ∎

Corollary 4.48 (Six consecutive shifts)**.** *Keep Lemma 4.36. If the shifts of the* 128 *pieces L_ζ lie in six consecutive integers, then, for every order, every choice of the shifts of the M_i and every placement of the t_i,* $\dim \mathrm{Ext}^2(E,E) \geq 640$*; no such convolution meets the polarised criterion.*

Proof. By Corollary 4.44 we may assume that the shifts take both values s and $s-5$ of the window $s, \dots, s-5$; let the parity at s have its values in $\{s, s-2, s-4\}$ and the other in $\{s-1, s-3, s-5\}$. If one of them takes two values two apart, or three, Theorem 4.41 applies. Otherwise the first takes $\{s\}$ or $\{s, s-4\}$ and the second $\{s-5\}$ or $\{s-1, s-5\}$. For $\{s\}$ and $\{s-5\}$ this is Proposition 4.46, and for $\{s, s-4\}$ and $\{s-1, s-5\}$ it is Proposition 4.47. For $\{s\}$ and $\{s-1, s-5\}$, Proposition 4.43 applies in its second form with $y = s$: the shift $s-1$ is taken and $s+1$, $s-3$ are not. For $\{s, s-4\}$ and $\{s-5\}$ it applies in its first form with $y = s-5$: the shift $s-4$ is taken and $s-6$, $s-2$ are not. The least of the bounds is 640. ∎

For the arrangements outside these results the diagonal still carries a large part of the obstruction.

Proposition 4.49 (Cup products on the diagonal at $n = 4$)**.** *Keep Lemma 4.36, so that D^2 has dimension* $131 \cdot 28 = 3668$*. For every convolution the cup products vanish on the* 84 *diagonal classes of the M_i, and on the* 3584 *diagonal classes of the L_ζ they have a kernel of dimension at least* 400*. So their kernel on D^2 has dimension at least* 484*, and a convolution meeting the polarised criterion needs products of length at least three that remove at least* 380 *diagonal classes.*

Proof. As in the proof of Proposition 4.30, the cup products in $\mathcal{H}(a,b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, they vanish on the classes of the M_i, and for L_ζ, $L_{\zeta'}$ and a type τ of degree two on the four surfaces they vanish on the part of type τ unless ζ'/ζ fixes the coordinates in the support of τ. So a class whose part of type τ depends only on the coordinates of ζ in the support of τ has vanishing cup products. The four types $(2,0,0,0)$ have dimension one and four values of the coordinate, the six types $(1,1,0,0)$ have dimension four and sixteen values of the two coordinates: $4 \cdot 4 + 6 \cdot 16 \cdot 4 = 400$. By Proposition 4.29, which holds for $n = 4$, the kernel of d_E on D^2 injects into $\mathrm{Ext}^2(E,E)$, and it is at most 104 only if the longer products remove 380 dimensions of these 484. ∎

Figure 10 and Figure 11 draw the arrangements and the bounds.

Remark 4.50 (What the route gives). At $n = 3$ the route through convolutions of these pieces is decided. The diagonal obstruction of Theorem 4.31 leaves room only for the fourfold products along one path, and the explicit convolution of Proposition 4.33 solves the Maurer–Cartan equation. The obstruction that decides the route lies off the diagonal, in classes that no product of any length can reach: those between pieces that differ in every coordinate and have adjacent shifts (Lemma 4.34), and, at pieces whose shifts are not adjacent to the shift of the paths, the classes of type $(1,1,0)$. No order and no choice of shifts or twists avoids both, and Theorem 4.35 excludes every convolution of the 35 pieces; in the explicit convolution the first kind alone gives 2560 classes against the 57 allowed. In each placement exactly two of the eight sign patterns carry products, and the

[23] The subgroups of $\Gamma \cong (\mathbb{Z}/4)^3 \times \mathbb{Z}/2$ have orders that are powers of two. A cut at a single vertex weighs exactly 640, so the bound is sharp.

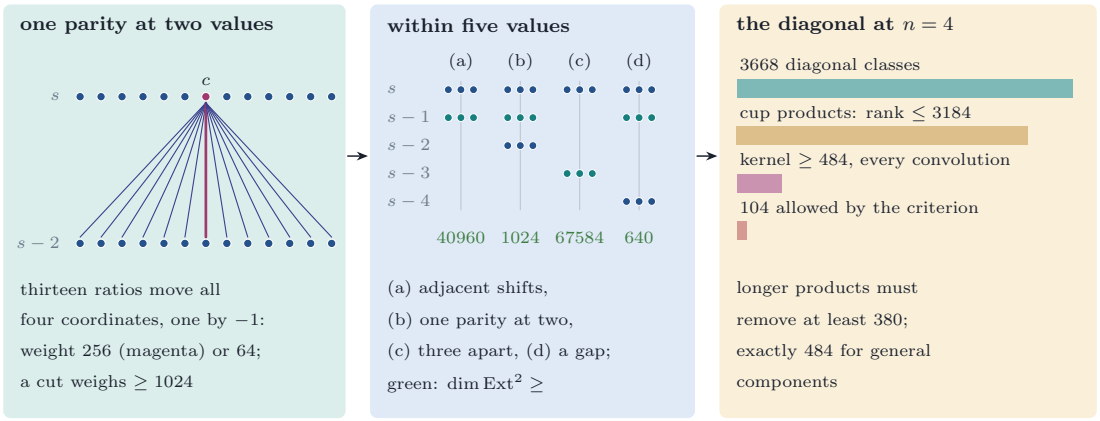


FIGURE 10. The shift arrangements of the pieces L_ζ on E_0^8. Left: one parity at two values two apart; a piece c is joined to the thirteen pieces whose ratio moves every coordinate, one by -1, and the groups $\mathcal{H}^4$ of the edges crossing between the values survive, weighing at least 1024 (Lemma 4.39(ii), Lemma 4.40). Middle: the four kinds of arrangement within five consecutive values, with the result that excludes each and its bound on $\dim \mathrm{Ext}^2$. Right: the diagonal at $n = 4$; the cup products leave at least 484 of the 3668 classes, and the criterion allows 104 (Proposition 4.49).

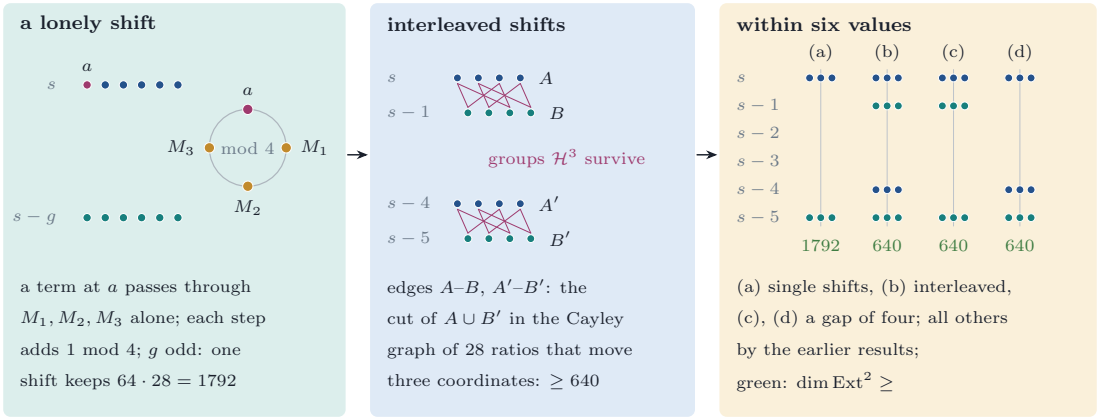


FIGURE 11. The arrangements within six consecutive values on E_0^8. Left: the parities at single shifts s and $s - g$, g odd and at least five; a term on a diagonal class at a piece a runs through M_1, M_2, M_3 and no other piece, and since each step adds one to the shift modulo 4 the four nodes have distinct residues, so the multiples serve at most one of the two shifts and the 1792 diagonal classes at the other survive (Lemma 4.45, Proposition 4.46). Middle: the interleaved shifts; the groups $\mathcal{H}^3$ between A and B and between A' and B' survive and form a cut of the Cayley graph of the 28 ratios that move three coordinates, of weight at least 640 (Proposition 4.47). Right: the four arrangements left within six values by the results for five values, each with its bound on $\dim \mathrm{Ext}^2$ (Corollary 4.48).

two are exchanged by a symmetry of the construction.[24] The member E_0^6 with this Weil structure has trivial discriminant, where the Weil classes are algebraic on every member by Markman's theorem [Mar25a, Mar25b], so a convolution meeting the criterion here would have given a second proof of a known case rather than a new one. At $n = 4$ the classes between pieces that differ in three coordinates and have adjacent shifts are again reached by nothing, in every convolution, and when the pieces L_ζ occupy two shifts, as in the explicit convolution, no term of d_E leaves them, so that $\dim \mathrm{Ext}^2(E, E) \geq 40960$ against the 104 allowed (Theorem 4.37). With three consecutive shifts the differential reaches these classes, but only through cup products into the groups between the highest and the lowest shift, and at least 22528 of them survive (Theorem 4.38). The groups of spread three, and those of spread two whose ratio moves every coordinate, one of them by -1, are untouched as well, in every convolution (Lemma 4.39): the latter cross every cut between two shift values of one parity, and weigh at least 1024 there (Theorem 4.41); with one parity at a single shift, those of spread three, or of spread one beside a gap, survive in their turn (Theorem 4.42, Proposition 4.43). Together they exclude every convolution whose pieces L_ζ have their shifts within five consecutive values (Corollary 4.44). The two arrangements left within six values fall as well: with each parity at a single shift, at any odd distance, the diagonal classes at one of the two shifts can be reached only through all three multiples, and the multiples serve at most one of them (Lemma 4.45, Proposition 4.46); the interleaved shifts leave the groups of Theorem 4.37 on a cut (Proposition 4.47). So every convolution whose shifts lie in six consecutive values fails (Corollary 4.48). For wider arrangements the cup products still leave 484 diagonal classes, so products of length at least three would have to remove 380 (Proposition 4.49), and products of length three and four among the L_ζ pass every test of Theorem 4.31. Whether some convolution of the 131 pieces whose shifts do not lie in six consecutive values, and whose parities do not each sit at one shift, meets the criterion at $n = 4$ is open. Markman's theorem does not reach the eightfolds of Weil type, so such a convolution would give a new case. Figure 6 shows the excess count and the surviving path, and Figure 7 the budget of diagonal classes.

Markman's candidate in dimension eight. Above $n = 3$ there is one concrete candidate of this kind. Markman proposes it in [Mar25b, Section 12] and notes that it remains to be checked against the weakened criterion. Its Chern character is a finite computation in the ring $\mathbb{Q}[\Theta]/(\Theta^5)$. We carry the computation out: it places the candidate at a rational point of the secant plane of [I, Definition 14.4] and reduces what has to be checked to a single condition, which Theorem 4.61 decides in the negative.

Setup 4.51 (Markman's object)**.** Let (X, Θ) be a principally polarised abelian fourfold, $d \geq 3$ odd and squarefree, $K = \mathbb{Q}(\sqrt{-d})$ and $N = (d + 9)/2$. Let D_i, $i \in \mathbb{Z}/N\mathbb{Z}$, be N translates of Θ in general position: any two, three or four of them meet transversally, in a surface, a curve and $\Theta^4 = 24$ points, and any five have empty intersection. Put $Z_i = D_i \cap D_{i+1}$, $Z = \bigcup_i Z_i$, let $C_i = Z_i \cap Z_{i+1} = D_i \cap D_{i+1} \cap D_{i+2}$ be the curves along which consecutive components meet, and let $\nu\colon \widetilde{Z} \to Z$ be the partial normalisation of Z at $m = (d + 9)(2d - 1)$ of its isolated points of self-intersection, the points of $Z_i \cap Z_j$ for i, j at cyclic distance at least three. Put

$$\mathcal{F} = [\mathcal{O}_X \longrightarrow \nu_* \mathcal{O}_{\widetilde{Z}}] \otimes \mathcal{O}_X(3\Theta),$$

[24] The automorphism of A that is -1 on the coordinate z_{n+1} and the identity on the others changes α into $-\alpha$, fixes θ and permutes the L_ζ, changing ζ_1 into $-\zeta_1$; composed with the shift [1] it changes every σ_i into $-\sigma_i$ and keeps everything else.

the complex placed in degrees 0 and 1.

Proposition 4.52 (Markman's candidate on the secant plane)**.** *In Setup 4.51, with $t = \Theta$ and u_t, v_t as in* [I, (30)] *for $n = 4$:*

(i) $\chi(\mathcal{O}_{Z_i}) = 14$, $\chi(\mathcal{O}_{C_i}) = -36$, *the isolated self-intersection points of Z number* $12N(N-5) = (d+9)(3d-3)$, *and* $\chi(\mathcal{O}_{\widetilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(27-d)$;

(ii) $\operatorname{ch}(\nu_*\mathcal{O}_{\widetilde{Z}}) = N\Theta^2 - 2N\Theta^3 + \chi(\mathcal{O}_{\widetilde{Z}})\,[\mathrm{pt}]$;

(iii) $\operatorname{ch}(\mathcal{F}) = u_t + 3v_t$, *that is*

$$\operatorname{ch}(\mathcal{F}) = 1 + 3\Theta - \tfrac{d}{2}\Theta^2 - \tfrac{d}{2}\Theta^3 + \tfrac{d^2}{24}\Theta^4,$$

the rational point $(a,b) = (1,3)$ of the secant plane P_Θ; with the twist $\mathcal{O}_X(\Theta)$ in place of $\mathcal{O}_X(3\Theta)$ the degree two component would instead be $-\frac{d+8}{2}\Theta^2$, so the twist in [Mar25b, Section 12] *must be read as 3Θ;*

(iv) $\mathrm{N}_{K/\mathbb{Q}}(3+\sqrt{-d}) = d+9 = 2N$, *with N the number of divisors, and by* [I, Theorem 14.7] *the Bogomolov discriminant of $\mathcal{F}$ is $\Delta(\mathcal{F}) = (d+9)\Theta^2$;*

(v) *the objects $\Phi(\mathcal{F}\boxtimes\mathcal{F})$ and $\Phi(\mathcal{F}\boxtimes\mathcal{F}^\vee)$ on $X\times\widehat{X}$ have ranks $\int_X \operatorname{ch}(\mathcal{F})^2 = 8d(d-9)$ and $\int_X \operatorname{ch}(\mathcal{F})\operatorname{ch}(\mathcal{F}^\vee) = 8d(d+9)$, up to sign, both nonzero for squarefree d; the same formula gives $-8q$ for the pair of secant sheaves of* [Mar25b, Section 11] *on a threefold, which is his rank $8q$ up to the shift.*

Consequently $\mathcal{F}\boxtimes\mathcal{F}$ satisfies conditions (1), (2b) and (2c) of Markman's strategy [Mar25b, Section 4]*: the transform has nonzero rank, its normalised Chern character is invariant under the special Mumford–Tate group by* [Mar25b, Corollary 9.2]*, and its Weil component is nonzero by* [Mar25b, Proposition 10.2]*, the relevant coefficient being* $\mathrm{N}(c)$ *with $c = \frac{1}{2}(1+3/\sqrt{-d}) \neq 0$. What remains is condition (2a) in its weakened form: for $E = \Phi(\mathcal{F}\boxtimes\mathcal{F})$, the map ev_E must kill every class in $H^1(T)\oplus H^0(\bigwedge^2 T)$ that preserves $\kappa(E)$ to first order.*

Proof. On a principally polarised abelian fourfold $\chi(\mathcal{O}_X(k\Theta)) = k^4$ and $\Theta^4 = 24$. The Koszul complex of r transversal translates of Θ gives ch of the structure sheaf of their intersection as $(1-e^{-\Theta})^r$, so $\operatorname{ch}(\mathcal{O}_{Z_i}) = \Theta^2 - \Theta^3 + \frac{7}{12}\Theta^4$, $\operatorname{ch}(\mathcal{O}_{C_i}) = \Theta^3 - \frac{3}{2}\Theta^4$ and ch of four divisors is Θ^4. Integrating gives $\chi(\mathcal{O}_{Z_i}) = 14$, $\chi(\mathcal{O}_{C_i}) = -36$ and 24 points; the same numbers are the Koszul sums $0 - 2\cdot 1 + 16$, $0 - 3 + 48 - 81$ and $0 - 4 + 96 - 324 + 256$ of the values $\chi(\mathcal{O}_X(-k\Theta)) = k^4$.

(i), (ii). At every point Z is locally a union of coordinate subspaces of $\mathbb{C}^4$: along C_i two surfaces $\{x_1 = x_2 = 0\}\cup\{x_2 = x_3 = 0\}$, at a point of $C_i\cap C_{i+1}$ three of them, at an isolated point $\{x_1 = x_2 = 0\}\cup\{x_3 = x_4 = 0\}$. The ideals of such an arrangement are monomial and distribute over sums, so the Mayer–Vietoris complex

$$0 \to \mathcal{O}_Z \to \bigoplus_i \mathcal{O}_{Z_i} \to \bigoplus_{i<j} \mathcal{O}_{Z_i\cap Z_j} \to \bigoplus_{i<j<k} \mathcal{O}_{Z_i\cap Z_j\cap Z_k} \to \cdots$$

with scheme-theoretic intersections is exact. Pairs at cyclic distance one meet in C_i, N of them; pairs at distance at least two meet in the 24 points of four divisors, $N(N-3)/2$ of them; the only triples with nonempty intersection are the consecutive ones $Z_i\cap Z_{i+1}\cap Z_{i+2} = C_i\cap C_{i+1}$, 24 points each, N of them; and no four components meet, since that would involve at least five divisors. Hence

$$\operatorname{ch}(\mathcal{O}_Z) = N\operatorname{ch}(\mathcal{O}_{Z_i}) - N\operatorname{ch}(\mathcal{O}_{C_i}) - \tfrac{N(N-3)}{2}\Theta^4 + N\Theta^4,$$

whose degree two part is $N\Theta^2$, degree three part $-N\Theta^3 - N\Theta^3 = -2N\Theta^3$, and degree four part $\chi(\mathcal{O}_Z)[\mathrm{pt}]$ with $\chi(\mathcal{O}_Z) = 14N + 36N - 12N(N-3) + 24N = 110N - 12N^2$. The

isolated singular points are those of pairs at distance at least three, of which there are $N(N-5)/2$, giving $12N(N-5) = 3(d+9)(d-1)$ points; the pairs at distance two lie on the double curves and are not isolated. At an isolated point the two branches meet transversally, and separating them gives $0 \to \mathcal{O}_Z \to \nu_*\mathcal{O}_{\widetilde{Z}} \to k(p) \to 0$, so each of the m normalisations raises χ by one and leaves the lower degrees of ch unchanged. With $N = (d+9)/2$, $\chi(\mathcal{O}_{\widetilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(55 - 3(d+9) + 2d - 1) = (d+9)(27-d)$.

(iii) $\mathrm{ch}(\mathcal{F}) = e^{3\Theta}(1 - \mathrm{ch}(\nu_*\mathcal{O}_{\widetilde{Z}}))$. The degree two part is $(\frac{9}{2} - N)\Theta^2 = -\frac{d}{2}\Theta^2$, which is where $N = (d+9)/2$ enters; with the twist $k\Theta$ it would be $(\frac{k^2}{2} - N)\Theta^2$, equal to $-\frac{d}{2}\Theta^2$ exactly for $k^2 = 9$, and $k = -3$ is excluded by the degree three part. The degree three part is $(\frac{27}{6} - 3N + 2N)\Theta^3 = (\frac{9}{2} - N)\Theta^3 = -\frac{d}{2}\Theta^3$, and the degree four part is $(\frac{81}{24} - \frac{9N}{2} + 6N - \frac{\chi(\mathcal{O}_{\widetilde{Z}})}{24})\Theta^4 = \frac{81+36N-\chi(\mathcal{O}_{\widetilde{Z}})}{24}\Theta^4 = \frac{d^2}{24}\Theta^4$ by (i). Comparing with [I, (30)] at $n = 4$, $u_t = 1 - \frac{d}{2}t^2 + \frac{d^2}{24}t^4$ and $v_t = t - \frac{d}{6}t^3$, gives $\mathrm{ch}(\mathcal{F}) = u_t + 3v_t$.

(iv) This follows from $\mathrm{N}(3 + \sqrt{-d}) = 9 + d$ and [I, (31)] with $(a, b) = (1, 3)$.

(v) Orlov's equivalence is $\Phi = (\mathrm{id} \times \Phi_{\mathcal{P}})^{-1} \circ \mu^*$ with $\mu(x, y) = (x + y, y)$, so the restriction of $\Phi(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$ to $\{x\} \times \widehat{X}$ is the Fourier–Mukai transform of $\tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2$, whose derived fibre at $L \in \widehat{X}$ is $R\Gamma(X, \tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L^{\pm 1})$; the rank of a perfect complex is the Euler characteristic of its derived fibre, and $\chi(\tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L) = \int_X \mathrm{ch}(\mathcal{F}_1)\,\mathrm{ch}(\mathcal{F}_2)$ by Riemann–Roch, translations and $L \in \mathrm{Pic}^0$ acting trivially on cohomology. With $\mathrm{ch}(\mathcal{F}^\vee) = u_t - 3v_t$, the involution τ changing the sign of the odd degrees, $\int u_t^2 = 8d^2$, $\int u_t v_t = 0$ and $\int v_t^2 = -8d$ give the two values. On a threefold $\int (u_t + v_t)^2 = 2\int u_t v_t = -8q$.

For the final assertion, condition (2b) is the $\mathrm{Spin}(V_{\mathbb{Q}})_B$-invariance of $\kappa(E)$, which [Mar25b, Corollary 9.2] derives from $\mathrm{ch}(\mathcal{F}) \in B$ and the nonvanishing of the rank. Condition (2c) is [Mar25b, Proposition 10.2], which for K imaginary quadratic reduces to the nonvanishing of the component of $\mathrm{ch}(\mathcal{F}) \otimes \mathrm{ch}(\mathcal{F})$ in $\ell_W \otimes \ell_{\overline{W}}$, and writing $\mathrm{ch}(\mathcal{F}) = c\,\ell_W + \overline{c}\,\ell_{\overline{W}}$ with $\ell_W = e^{\sqrt{-d}\Theta}$ gives that coefficient as $c\overline{c}$. ∎

Remark 4.53 (The remaining condition). Proposition 4.52 does not verify Markman's candidate. It verifies every statement about the candidate that concerns classes, and it shows that the one remaining statement concerns the object. For $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$ on the eightfold $X \times \widehat{X}$ of Weil type, with $\kappa(E) = \mathrm{ch}(E)e^{-c_1(E)/\operatorname{rk} E}$, that statement reads: every $x \in H^1(T) \oplus H^0(\bigwedge^2 T)$ with $x \lrcorner \kappa(E) = 0$ satisfies $\mathrm{ev}_E(x) = 0$ in $\mathrm{Ext}^2(E, E)$. The $H^2(\mathcal{O})$ summand imposes nothing, because $\operatorname{rk} E \neq 0$ makes $z \cdot \kappa(E) \neq 0$ for $z \neq 0$. By [I, Theorem 9.20] the condition fails as soon as E has a rank one direct summand whose class is not a multiple of the polarisation, so the first question about E is whether it is indecomposable. Markman's object at $n = 3$ is reflexive of rank $8q$ away from a divisor, and he proves the condition by descending to a quotient on which σ_E is injective on an invariant subspace [Mar25b, Proposition 11.5]. Whether the same can be done for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ is a question about the Ext groups of one object on one eightfold, not about any Hodge class. Theorem 4.56 below moves the question from the eightfold to the fourfold and from E to $\mathcal{F}$ itself: the condition is equivalent to the single identity (19) in $\mathrm{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2)$, the other half of it holding by construction. Theorem 4.61 shows that the identity fails, so the answer is negative.

Reduction to the fourfold. The condition of Remark 4.53 concerns an object on an eightfold. It is equivalent to a condition on the factor $\mathcal{F}$ on the fourfold, and there it splits into two conditions, one of which holds for Markman's object because the object is constructed uniformly over the moduli of principally polarised fourfolds. The reduction is not special

to that object. It works for every secant object on every abelian n-fold with $n \geq 3$, and its ingredients are three computations in the Hodge algebra of a single abelian variety, which we carry out by hand. We first fix conventions.

Setup 4.54 (The weakened criterion for one object)**.** Let Y be an abelian variety and F a nonzero perfect complex on Y. Write $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\bigwedge^2 T_Y)$ as in (10), with $x = (z, v, \pi)$ acting on $H^*(Y, \mathbb{C})$ by $x\lrcorner$, and likewise $HT^1(Y) = H^{0,1}(Y) \oplus H^0(T_Y)$, with (z, θ) acting by $z\wedge$ and by the interior product $\theta\lrcorner$. Normalise the Atiyah class $\mathrm{At}(F) \in \mathrm{Ext}^1(F, F \otimes \Omega^1_Y)$ so that $\mathrm{ch}(F) = \mathrm{tr}\exp \mathrm{At}(F)$; for a line bundle L it is then $c_1(L)$, as in (8).[25] The evaluation map of [I, Theorem 9.19] is, under HKR,

$$\mathrm{ev}_F(z, v, \pi) = z \cdot \mathrm{id}_F + v\lrcorner \mathrm{At}(F) + \tfrac{1}{2}\, \pi\lrcorner \mathrm{At}(F)^2 \ \in\ \mathrm{Ext}^2(F, F), \tag{15}$$

the contraction of the polyvector with the exponential Atiyah class [Hua21, Theorem A], [BF08]; its restriction to $H^1(T_Y)$ is the obstruction map, whose kernel is the space of first order deformations of Y to which F extends [Tod09], [Mar25a, Section 8.3]. Put

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F) = \{x \in HT^2(Y) : x\lrcorner\, \mathrm{ch}(F) = 0\},$$

the Hochschild classes preserving $\mathrm{ch}(F)$ to first order. We say that F *satisfies the weakened criterion* if ev_F vanishes on $\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$. By the commutative square $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,)\lrcorner\, \mathrm{ch}(F)$ of [BF08, Proposition 6.2.1 and Corollary 6.3.2], in the form quoted in [Mar25b, Lemma 11.3],

$$\mathrm{ev}_F(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)) = \ker \sigma_F \cap \mathrm{im}\, \mathrm{ev}_F,$$

so the weakened criterion is exactly the hypothesis of [Mar25b, Question 11.4], that the restriction of σ_F to the image of ev_F is injective. For a split object this is Proposition 4.18(ii).

Lemma 4.55 (The B-field transform)**.** *Let $B \in H^{1,1}(Y)$. For $x = (z, v, \pi) \in HT^2(Y)$ put*

$$e^B x = (z + v\lrcorner B + \tfrac{1}{2}\, \pi\lrcorner B^2,\ v + \pi\lrcorner B,\ \pi), \tag{16}$$

where $\pi\lrcorner B \in \mathrm{Hom}(H^{1,0}, H^{0,1}) = H^1(T_Y)$ is the contraction of one slot of π with B: for $\pi = \partial_a \wedge \partial_b$ it sends p_b to $\partial_a\lrcorner B$, p_a to $-\partial_b\lrcorner B$ and the other basis vectors to zero. Then e^B is a linear automorphism of $HT^2(Y)$ with inverse e^{-B}, and:

(i) *$x\lrcorner(w\, e^B) = ((e^B x)\lrcorner w)\, e^B$ for every even class $w \in H^{\mathrm{ev}}(Y, \mathbb{C})$;*

(ii) *for a line bundle L with $c_1(L) = B$, $\mathrm{ev}_{F\otimes L}(x) = \mathrm{ev}_F(e^B x)$ and $\mathrm{Ann}_{HT^2}(\mathrm{ch}(F \otimes L)) = e^{-B}\, \mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$; hence F satisfies the weakened criterion if and only if $F \otimes L$ does.*

The computation in (ii) is formal: it uses only that the twisted object has Atiyah class $\mathrm{At}(F) + B \cdot \mathrm{id}_F$ and Chern character $\mathrm{ch}(F)e^B$. It therefore applies verbatim to the twisted sheaves of [Mar25b, Section 2]*, with B rational. In particular, for F of nonzero rank r, the criterion for the twisted object $F \otimes L^{-1}$ with class $\kappa(F) = \mathrm{ch}(F)e^{-c_1(F)/r}$ is the same condition as the criterion for F with class $\mathrm{ch}(F)$.*

Proof. The computation takes place in a graded algebra on which the interior products $\iota_a = \partial_a\lrcorner$ act as odd derivations. It uses only that w is even; that B is even and central with $\iota_a\iota_b B = 0$, which holds for B of type $(1,1)$ and ∂_a, ∂_b of type $(1,0)$; and that the odd scalar classes $\iota_a B$ anticommute with every odd element. In $H^*(Y, \mathbb{C})$ these properties are clear. They also hold in $\bigoplus_{p,q} \mathrm{Ext}^p(F, F \otimes \Omega^q_Y) = \mathrm{Ext}^*(F, F) \otimes \bigwedge^* H^{1,0}$, on which the polyvectors act through the second factor by odd derivations, because the classes coming

[25] This is the convention of [Tod09] and of [Mar25a, footnote 14]. Buchweitz and Flenner use the opposite sign, which replaces $\mathrm{At}(F)$ by $-\mathrm{At}(F)$; this leaves $\mathrm{At}(F)^2$ unchanged and changes only the sign of the middle term of ev_F, which does not affect the vanishing statements below.

from $H^*(Y,\mathbb{C})$ are graded central in the Yoneda algebra. First, $\iota_a(B^k) = k(\iota_a B)B^{k-1}$, since $\iota_a B$ is odd and B is even, so $\iota_a e^B = (\iota_a B)e^B$. For w even this gives $v\lrcorner(we^B) = (v\lrcorner w)e^B + w\sum_a v(p_a)(\iota_a B)e^B = (v\lrcorner w + (v\lrcorner B)w)e^B$, and $z(we^B) = (zw)e^B$. For the bivector, with $\pi\lrcorner = \iota_a\iota_b$,

$$\iota_b(we^B) = (\iota_b w)e^B + w(\iota_b B)e^B,$$

and applying ι_a, using that $\iota_b w$ and $\iota_b B$ are odd,

$$\iota_a\iota_b(we^B) = \big[\iota_a\iota_b w + (\iota_a B)\iota_b w - (\iota_b B)\iota_a w + w\big((\iota_a\iota_b B) - (\iota_b B)(\iota_a B)\big)\big]e^B.$$

The middle two terms are $(\pi\lrcorner B)\lrcorner w$ for the element $\pi\lrcorner B$ of $\mathrm{Hom}(H^{1,0},H^{0,1})$ displayed in the statement, acting as a derivation, and the last bracket is $\frac{1}{2}\iota_a\iota_b(B^2) = \iota_a((\iota_b B)B)$. Summing the three actions gives (i). Applying (i) with B replaced by $-B$ and then by B, and using $(\pi\lrcorner B)\lrcorner B = 2(\iota_a B)(\iota_b B) = \pi\lrcorner B^2$ when $\iota_a\iota_b B = 0$, shows $e^{-B}e^B = \mathrm{id}$.

(ii) The Atiyah class of $F\otimes L$ is $\mathrm{At}(F) + B\cdot\mathrm{id}_F$, the Atiyah class being additive on tensor products [BF03, Section 3], and the two summands commute, so $\exp\mathrm{At}(F\otimes L) = \exp\mathrm{At}(F)\cdot e^B$. By (i) in the algebra $\mathrm{Ext}^*(F,F)\otimes\bigwedge^* H^{1,0}$, with $w = \exp\mathrm{At}(F)$,

$$x\lrcorner\exp\mathrm{At}(F\otimes L) = \big((e^B x)\lrcorner\exp\mathrm{At}(F)\big)e^B,$$

and the right hand side has the same component in $\mathrm{Ext}^2(F,F)\otimes\bigwedge^0$ as $(e^B x)\lrcorner\exp\mathrm{At}(F)$, because $e^B - 1 = B + \frac{1}{2}B^2 + \cdots$ raises the exterior degree. This is $\mathrm{ev}_{F\otimes L}(x) = \mathrm{ev}_F(e^B x)$. By (i) in $H^*(Y,\mathbb{C})$ with $w = \mathrm{ch}(F)$, $x\lrcorner\mathrm{ch}(F\otimes L) = 0$ if and only if $(e^B x)\lrcorner\mathrm{ch}(F) = 0$, which is the statement on the annihilators. Hence $\mathrm{ev}_{F\otimes L}(\mathrm{Ann}_{HT^2}(\mathrm{ch}(F\otimes L))) = \mathrm{ev}_F(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F))$. ∎

Theorem 4.56 (Reduction to the factor)**.** *Let (X,t) be a polarised abelian variety of dimension $n\geq 3$, let $d\geq 1$, and let F be a perfect complex on X with $\mathrm{ch}(F) = a\,u_t + b\,v_t$ for some $(a,b)\in\mathbb{Q}^2\setminus\{0\}$, in the notation of* [I, (30)]*. Then:*

(i) *No nonzero class in $HT^1(X)$ preserves $\mathrm{ch}(F)$: the map $HT^1(X)\to H^*(X,\mathbb{C})$, $y\mapsto y\lrcorner\mathrm{ch}(F)$, is injective. Its two lowest components form the matrix $\left(\begin{smallmatrix} a & b\\ b & -ad\end{smallmatrix}\right)$ of determinant $-(a^2d+b^2) = -\mathrm{N}_{K/\mathbb{Q}}(b+a\sqrt{-d})\neq 0$.*

(ii) *The classes in $HT^2(X)$ preserving $\mathrm{ch}(F)$ to first order are*

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F) = \{(0,v,0) : v\lrcorner t = 0\}\ \oplus\ \{(\tfrac{d}{2}\,\pi\lrcorner t^2,\ 0,\ \pi) : \pi\in H^0(\textstyle\bigwedge^2 T_X)\},\qquad(17)$$

the polarised first order deformations of (X,t), of dimension $n(n+1)/2$, and the Poisson directions each compensated by a class in $H^{0,2}$, of dimension $n(n-1)/2$; the total is $n^2 = \dim\mathcal{D}_{n,n}$, and the space does not depend on (a,b).

(iii) *F satisfies the weakened criterion if and only if*
   (a) *$v\lrcorner\mathrm{At}(F) = 0$ in $\mathrm{Ext}^2(F,F)$ for every $v\in H^1(T_X)$ with $v\lrcorner t = 0$, that is, F extends to first order along every polarised deformation of (X,t); and*
   (b) *$\mathrm{At}(F)^2 = -d\,t^2\cdot\mathrm{id}_F$ in $\mathrm{Ext}^2(F,F\otimes\Omega_X^2)$.*

   The trace of (b) is automatic: $\mathrm{tr}\,\mathrm{At}(F)^2 = 2\,\mathrm{ch}_2(F) = -ad\,t^2 = \mathrm{tr}(-d\,t^2\cdot\mathrm{id}_F)$, so (b) is the vanishing of the trace free part of the square of the Atiyah class.

(iv) *Let F_1,F_2 be two such complexes, $G = F_1\boxtimes F_2$ on $X\times X$ and $E = \Phi(G)$ on $X\times\widehat{X}$, with Φ Orlov's equivalence as in Proposition 4.52(v). Then $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G) = \mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_1)\otimes 1\oplus 1\otimes\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_2)$, of dimension $2n^2$, and*

$$E\text{ satisfies the weakened criterion}\iff G\text{ does}\iff F_1\text{ and }F_2\text{ do.}$$

Proof. Choose a basis $p_1,\dots,p_n$ of $H^{1,0}(X)$ and a basis $q_1,\dots,q_n$ of $H^{0,1}(X)$ with $t = \sum_a p_a\wedge q_a$, which is possible because t is a positive class of type $(1,1)$; let ∂_a be the dual

basis of T, so that $\iota_a p_b = \delta_{ab}$ and $\iota_a q_b = 0$. Write $\operatorname{ch}(F) = \sum_k c_k t^k$ with $c_k\, k! = a(-d)^{k/2}$ for k even and $c_k\, k! = b(-d)^{(k-1)/2}$ for k odd, so that

$$c_{k+1}\,(k+1)! = -d\,c_{k-1}\,(k-1)! \qquad (k \geq 1). \tag{18}$$

Two facts about the Hodge algebra are used throughout: for $v \in H^1(T)$ acting as a derivation, $v \lrcorner t^k = k\,(v \lrcorner t)\,t^{k-1}$; and for a bivector π of type $(2,0)$ in the tangent directions, $\pi \lrcorner t = 0$ because t has a single $(1,0)$ slot in each term, so $\pi \lrcorner t^k = \binom{k}{2}(\pi \lrcorner t^2)\,t^{k-2}$ by the footnote to (10). Finally, multiplication by t is injective on $H^{0,1}$ when $n \geq 2$ and on $H^{0,2}$ when $n \geq 3$: $q_a \wedge t$ contains the monomial $p_c q_c q_a$ for any $c \neq a$, and $q_a q_b \wedge t$ contains $p_c q_c q_a q_b$ for any $c \notin \{a,b\}$, and distinct monomials are independent.

(i) For $y = (z,\theta)$ the class $y \lrcorner \operatorname{ch}(F)$ has components of type $(k-1,k)$ equal to $c_{k-1} z\, t^{k-1} + c_k\, \theta \lrcorner t^k = (c_{k-1} z + k\, c_k\, (\theta \lrcorner t))t^{k-1}$. For $k = 1$ this is $az + b\,(\theta \lrcorner t)$ and for $k = 2$ it is $\big(bz - ad\,(\theta \lrcorner t)\big)t$, by (18). Both z and $\theta \lrcorner t = \sum_a \theta^a q_a$ lie in $H^{0,1}$, on which t acts injectively, so $y \lrcorner \operatorname{ch}(F) = 0$ forces $(z, \theta \lrcorner t)$ into the kernel of the displayed matrix, whose determinant is $-a^2 d - b^2 < 0$. Hence $z = 0$ and $\theta \lrcorner t = 0$, and the latter gives $\theta = 0$ because t is nondegenerate.

(ii) For $x = (z, v, \pi)$ the component of $x \lrcorner \operatorname{ch}(F)$ of type $(k-1, k+1)$ is

$$c_{k-1} z\, t^{k-1} + c_k\, v \lrcorner t^k + c_{k+1}\, \pi \lrcorner t^{k+1} = \Big(c_{k-1} z + k\, c_k\, W + \tbinom{k+1}{2} c_{k+1}\, P\Big) t^{k-1},$$

with $W = v \lrcorner t$ and $P = \pi \lrcorner t^2$, both in $H^{0,2}$. By (18), $\binom{k+1}{2} c_{k+1} = -\frac{d}{2} c_{k-1}$, so the bracket is $c_{k-1} Z + k\, c_k W$ with $Z = z - \frac{d}{2} P$. For $k = 1$ and $k = 2$ this gives $aZ + bW$ and $(bZ - adW)\,t$, and t is injective on $H^{0,2}$ because $n \geq 3$; the same determinant forces $Z = W = 0$. Conversely $Z = W = 0$ kills every component. So $\operatorname{Ann}_{HT^2}(\operatorname{ch} F)$ is the set of (z, v, π) with $v \lrcorner t = 0$ and $z = \frac{d}{2}\, \pi \lrcorner t^2$, which is (17); the two summands are independent because the first has $\pi = 0$ and the second is the graph of a linear map from the π component. A $v \in \operatorname{Hom}(H^{1,0}, H^{0,1})$ has $v \lrcorner t = \sum_a v(p_a) \wedge q_a$, which vanishes exactly when the matrix of v is symmetric; this is the tangent space of the polarised deformations, of dimension $n(n+1)/2$, as in Proposition 4.16(i). The bivectors contribute $\dim \bigwedge^2 T = n(n-1)/2$. Nothing in the description involves (a,b).

(iii) By (ii) and (15), ev_F vanishes on $\operatorname{Ann}_{HT^2}(\operatorname{ch} F)$ if and only if it vanishes on both summands. On the first it is $v \mapsto v \lrcorner \operatorname{At}(F)$, which is (a). On the second it is

$$\tfrac{d}{2}(\pi \lrcorner t^2) \cdot \mathrm{id}_F + \tfrac{1}{2}\, \pi \lrcorner \operatorname{At}(F)^2 = \tfrac{1}{2}\, \pi \lrcorner (\operatorname{At}(F)^2 + d\, t^2 \cdot \mathrm{id}_F),$$

since $\pi \lrcorner (t^2 \cdot \mathrm{id}_F) = (\pi \lrcorner t^2) \cdot \mathrm{id}_F$. Now $\Omega^2_X = \mathcal{O}_X \otimes \bigwedge^2 H^{1,0}$, so $\operatorname{Ext}^2(F, F \otimes \Omega^2_X) = \operatorname{Ext}^2(F,F) \otimes \bigwedge^2 H^{1,0}$, and the contractions with all $\pi \in \bigwedge^2 (H^{1,0})^\vee$ vanish on an element of this space if and only if the element is zero, the pairing being perfect. This is (b). For the trace, $\operatorname{ch}(F) = \operatorname{tr} \exp \operatorname{At}(F)$ gives $\operatorname{ch}_2(F) = \frac{1}{2} \operatorname{tr} \operatorname{At}(F)^2$, and $\operatorname{ch}_2(F) = c_2 t^2 = -\frac{ad}{2} t^2$, while $\operatorname{tr}(\mathrm{id}_F) = \operatorname{rk} F = c_0 = a$.

(iv) The Hochschild cohomology of a product is the tensor product, $HT^2(X \times X) = HT^2 \otimes HT^0 \oplus HT^1 \otimes HT^1 \oplus HT^0 \otimes HT^2$, the Atiyah class of G is $\operatorname{At}(F_1) \boxtimes 1 + 1 \boxtimes \operatorname{At}(F_2)$ [BF03, Proposition 3.14], [Mar25a, (8.4.1)], so $\exp \operatorname{At}(G) = \exp \operatorname{At}(F_1) \boxtimes \exp \operatorname{At}(F_2)$, and $\operatorname{Ext}^2(G,G)$ is the corresponding Künneth sum of $\operatorname{Ext}^i(F_1,F_1) \otimes \operatorname{Ext}^j(F_2,F_2)$. A polyvector on a factor contracts only the forms of that factor, so for $x_1 \in HT^i(X)$ and $x_2 \in HT^j(X)$ with $i + j = 2$,

$$(x_1 \otimes x_2) \lrcorner \operatorname{ch}(G) = \pm (x_1 \lrcorner \operatorname{ch} F_1) \otimes (x_2 \lrcorner \operatorname{ch} F_2), \qquad \operatorname{ev}_G(x_1 \otimes x_2) = \pm \operatorname{ev}_{F_1}(x_1) \otimes \operatorname{ev}_{F_2}(x_2),$$

with $\operatorname{ev}(1) = \mathrm{id}$ on HT^0 and Koszul signs that play no role; this is the decomposition [Mar25a, (8.4.2), (8.4.3)]. Let $x = x_{20} + x_{11} + x_{02}$ be a class in $\operatorname{Ann}_{HT^2}(\operatorname{ch} G)$ according to

the three summands. Since $\mathrm{ch}(F_i)$ is even, $x_{11} \lrcorner \mathrm{ch}(G)$ lies in $H^{\mathrm{odd}} \otimes H^{\mathrm{odd}}$ while the other two terms lie in $H^{\mathrm{ev}} \otimes H^{\mathrm{ev}}$, so $x_{11} \lrcorner \mathrm{ch}(G) = 0$ separately; and $x_{11} \lrcorner \mathrm{ch}(G)$ is the image of x_{11} under the tensor product of the two maps of (i), which is injective, so $x_{11} = 0$. For the even part write $x_{20} = x_1 \otimes 1$ and $x_{02} = 1 \otimes x_2$; then $(x_1 \lrcorner \mathrm{ch}\, F_1) \otimes \mathrm{ch}(F_2) + \mathrm{ch}(F_1) \otimes (x_2 \lrcorner \mathrm{ch}\, F_2) = 0$. Applying a linear form on the second factor taking the value 1 on $\mathrm{ch}(F_2)$ shows $x_1 \lrcorner \mathrm{ch}(F_1) = \lambda\, \mathrm{ch}(F_1)$ for a scalar λ, and then $x_2 \lrcorner \mathrm{ch}(F_2) = -\lambda\, \mathrm{ch}(F_2)$. But every component of $x_1 \lrcorner \mathrm{ch}(F_1)$ has Hodge type $(p, p+2)$ while $\mathrm{ch}(F_1) \neq 0$ has type (p,p); so $\lambda = 0$ and $x_i \in \mathrm{Ann}_{HT^2}(\mathrm{ch}\, F_i)$. This proves the description of $\mathrm{Ann}_{HT^2}(\mathrm{ch}\, G)$ and its dimension $2n^2$ by (ii). On it $\mathrm{ev}_G(x_1 \otimes 1 + 1 \otimes x_2) = \pm \mathrm{ev}_{F_1}(x_1) \otimes \mathrm{id}_{F_2} \pm \mathrm{id}_{F_1} \otimes \mathrm{ev}_{F_2}(x_2)$, whose two terms lie in different Künneth summands and vanish exactly when $\mathrm{ev}_{F_i}(x_i) = 0$, as $\mathrm{id}_{F_i} \neq 0$. So G satisfies the criterion if and only if F_1 and F_2 do.

For the transport under Φ we use the commutative diagram of [Mar25a, Lemma 8.3.10]: an equivalence $\Phi\colon D^b(A) \to D^b(B)$ of derived categories of abelian varieties induces isomorphisms $\Phi^{HT}\colon HT^2(A) \to HT^2(B)$ and $\Phi^{H}\colon H^*(A,\mathbb{C}) \to H^*(B,\mathbb{C})$ with $\Phi^H(x \lrcorner \mathrm{ch}\, G) = \Phi^{HT}(x) \lrcorner \mathrm{ch}\, \Phi(G)$ [CRV12], and $\Phi \circ \mathrm{ev}_G = \mathrm{ev}_{\Phi(G)} \circ \Phi^{HT}$ on $\mathrm{Hom}(G, G[2]) \to \mathrm{Hom}(\Phi G, \Phi G[2])$ [Hua21, Theorem A]; on abelian varieties the Todd class is trivial, so there is no Duflo correction [Mar25a, Remark 8.3.11]. The first square gives $\Phi^{HT}(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, G)) = \mathrm{Ann}_{HT^2}(\mathrm{ch}\, E)$ and the second $\mathrm{ev}_E(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, E)) = \Phi(\mathrm{ev}_G(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, G)))$, with Φ an isomorphism on Ext^2. So E satisfies the criterion if and only if G does. ∎

Figure 12 draws the Künneth stack and the two conditions at $n = 4$.

Corollary 4.57 (The remaining condition on the fourfold)**.** *In Setup 4.51 let $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$. Then E satisfies the weakened criterion of* [Mar25b, Question 11.4] *if and only if*

$$\mathrm{At}(\mathcal{F})^2 = -d\,\Theta^2 \cdot \mathrm{id}_{\mathcal{F}} \qquad \text{in } \mathrm{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2). \tag{19}$$

Equivalently, writing $\mathrm{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$ with $A_a \in \mathrm{Ext}^1(\mathcal{F},\mathcal{F})$ in a basis p_a of $H^{1,0}(X)$, and $\Theta = \sum_a p_a \wedge q_a$ with $q_a \in H^{0,1}(X)$, all the commutators $A_aA_b - A_bA_a \in \mathrm{Ext}^2(\mathcal{F},\mathcal{F})$ are the scalars $\pm 2d\,(q_a \wedge q_b) \cdot \mathrm{id}_{\mathcal{F}}$ prescribed by Θ^2, the sign being fixed by the Koszul rule.

Proof. By Proposition 4.52(iii), $\mathrm{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$, so Theorem 4.56 applies with $n = 4$, $(a,b) = (1,3)$ and $F_1 = F_2 = \mathcal{F}$, and E satisfies the criterion if and only if $\mathcal{F}$ satisfies (a) and (b). It remains to show that (a) holds. The data of Setup 4.51 are a principally polarised fourfold (X,Θ) and N points $x_i \in X$ with $D_i = \tau_{x_i}^*\Theta$. The conditions imposed on the D_i, that the intersections of two, three and four of them be smooth of the expected codimension and that five of them be disjoint, are open in the parameter space of such data over the local universal deformation S of (X,Θ), by the openness of smoothness for a proper flat family and the properness of the projection. The isolated points of Z, being transversal intersections of four divisors, form a finite étale cover of the open set where the conditions hold, so the m points to be normalised can be followed locally; the partial normalisation separates two smooth branches at each of them and commutes with base change. The construction therefore produces a perfect complex $\mathcal{F}_S$ on the universal family over a neighbourhood of the base point in S, whose fibre at the base point is $\mathcal{F}$ and whose base has Kodaira–Spencer image the whole space of polarised deformations, of dimension $10 = n(n+1)/2$. The obstruction $v \lrcorner \mathrm{At}(\mathcal{F})$ to extending $\mathcal{F}$ along v vanishes for every v along which $\mathcal{F}$ extends [Tod09], and this is (a).

For the reformulation, $\Omega_X^1 = \mathcal{O}_X \otimes H^{1,0}$ gives $\mathrm{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$, and $\mathrm{At}(\mathcal{F})^2 = -\sum_{a<b}(A_aA_b - A_bA_a) \otimes p_a \wedge p_b$ by the Koszul rule, while $\Theta^2 = -2\sum_{a<b}(p_a \wedge p_b)(q_a \wedge q_b)$ for $\Theta = \sum_a p_a \wedge q_a$. Comparing coefficients of $p_a \wedge p_b$ gives the commutator relations. ∎

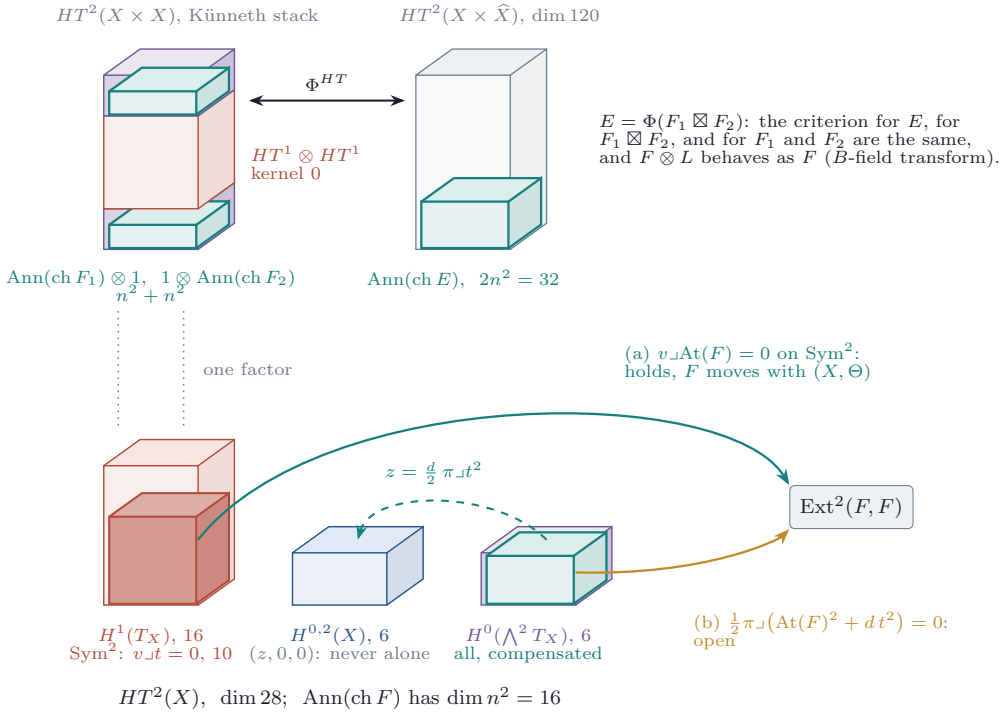


FIGURE 12. Theorem 4.56 for $n = 4$. Top: the Künneth stack of $HT^2(X \times X)$, whose middle summand $HT^1 \otimes HT^1$ meets the kernel in zero by (i), so that the classes preserving $\mathrm{ch}(F_1 \boxtimes F_2)$ are two copies of the kernel on one factor; across Orlov's isomorphism, HT^2 of the eightfold with the $2n^2$ classes preserving $\mathrm{ch}(E)$. Bottom: $HT^2(X)$ as three slabs with heights proportional to their dimensions, and inside them the n^2 classes of (17): the polarised deformations in front of $H^1(T_X)$, and the whole of $H^0(\bigwedge^2 T_X)$, each bivector tilted into $H^{0,2}$ by its compensation $\frac{d}{2}\, \pi \lrcorner t^2$; nothing in $H^{0,2}$ alone preserves $\mathrm{ch}(F)$. On the right, the two conditions that the evaluation map imposes on the two parts, (a) closed by the construction of $\mathcal{F}$ and (b) the identity (19).

Remark 4.58 (Scope of the reduction). The reduction accomplishes three things. First, it removes the eightfold: the condition of Remark 4.53, on an object of rank $8d(d-9)$ on $X \times \widehat{X}$ of Weil type, is equivalent to one identity (19) for a rank one complex on the fourfold, in the Yoneda algebra $\mathrm{Ext}^*(\mathcal{F},\mathcal{F})$. The dimension of the space of classes that has to be killed drops from $2n^2$ to n^2 and, by (a), to $n(n-1)/2 = 6$. At $n = 3$ the count $n^2 = 9$ is the nine-dimensional kernel of [Mar25a, Corollaries 8.3.12 and 8.5.2], which (17) writes out for every secant object in every dimension. Second, the half of that space made of geometric deformations imposes nothing, because Markman's construction is uniform over the moduli of principally polarised fourfolds. Third, by Lemma 4.55 the two forms of the criterion, for E with $\mathrm{ch}(E)$ and for the twisted object with $\kappa(E)$, coincide, so nothing is lost by working with the untwisted object.

What remains is the identity itself, which is a Heisenberg relation: the components of the Atiyah class in the n tangent directions must commute up to the scalars prescribed by Θ^2, and the Chern character forces only the trace of the relation. The identity says exactly that $\mathcal{F}$ deforms, as an object of the deformed category, along the Poisson directions of X compensated by the gerbe classes $\frac{d}{2}\,\pi \lrcorner \Theta^2$; these are the noncommutative first order deformations of X with a gerby correction, in the language of [Mar25a, Section 8.5]. It holds in the one case where the method is known to work. For Markman's sheaves $\mathcal{F}_1 = I_{\bigcup C_i}(\Theta)$ on the Jacobian threefold of [Mar25b, Section 11.1], with $\mathrm{ch} = u_\Theta + v_\Theta$ and $d = q$, the kernel of the obstruction map equals the annihilator of the Chern character [Mar25a, Corollary 8.3.12], which is stronger than the weakened criterion; so by Theorem 4.56(iii) these sheaves satisfy $\mathrm{At}(\mathcal{F}_1)^2 = -q\,\Theta^2 \cdot \mathrm{id}$ on the threefold. For the rank one complex of Setup 4.51 on the fourfold, the identity is a computation of Yoneda products in $\mathrm{Ext}^*(\mathcal{F},\mathcal{F})$, and the computation can be made local. Below we carry it out at one point of the support and find that the identity fails.

Failure at an isolated double point. The identity (19) can be tested locally. The evaluation map is an algebra homomorphism [Mar25a, (8.3.3)], so on a bivector $\theta \wedge \theta'$ it is the Yoneda product of the classes

$$A_\theta = \mathrm{ev}_{\mathcal{F}}(\theta) = \theta \lrcorner \mathrm{At}(\mathcal{F}) \in \mathrm{Ext}^1(\mathcal{F},\mathcal{F}), \qquad \theta \in H^0(T_X),$$

which are the first order deformations of $\mathcal{F}$ along the translations of X. The product of two global classes restricts to the product of their germs. At a point where the support of $\mathcal{F}$ has two smooth branches of codimension two meeting transversally, the germs multiply nontrivially; since the compensating classes $z \cdot \mathrm{id}_{\mathcal{F}}$ have zero germs, the identity fails. Markman's construction has such points and cannot avoid them.

Lemma 4.59 (Localisation)**.** *Let F be a perfect complex on a smooth projective variety Y, let $\mathcal{E}xt^m(F,F)$ be the sheaf associated with $U \mapsto \mathrm{Hom}_{D(U)}(F|_U, F|_U[m])$, and let $e\colon \mathrm{Ext}^m(F,F) \to H^0(Y, \mathcal{E}xt^m(F,F))$ send a morphism $F \to F[m]$ to its germs. Then e is compatible with Yoneda products; $e(z \cdot \mathrm{id}_F) = 0$ for $z \in H^m(\mathcal{O}_Y)$, $m \geq 1$; and for $\theta \in H^0(T_Y)$ the germ of A_θ at p is the class of the jet extension $0 \to F_p \to J_\theta(F_p) \to F_p \to 0$ of the germ F_p along the derivation θ of $\mathcal{O}_{Y,p}$, which is $\mathcal{O}_{Y,p}$-linear in θ.*

Proof. Composition of morphisms commutes with restriction to open sets; this is the first assertion. For the second, $z \cdot \mathrm{id}_F$ restricted to an affine open U is $z|_U \cdot \mathrm{id}_{F|_U}$, and $z|_U \in H^m(U, \mathcal{O}_U) = 0$. For the third, the Atiyah class of F is the class of the first jet sequence $0 \to F \otimes \Omega^1_Y \to J^1(F) \to F \to 0$ [BF03, Section 3], and its restriction to U is the jet sequence of $F|_U$. Contracting with θ is pushing out along $\mathrm{id} \otimes \theta\colon F \otimes \Omega^1 \to F$,

which produces the extension $J_\theta(F)$ whose underlying sheaf is $F \oplus F$ with $f \cdot (s, s') = (fs + \theta(f)s', fs')$. Linearity in θ is linearity of the contraction. ∎

Lemma 4.60 (Two branches through a point)**.** *Let $\mathcal{O}$ be the local ring of $\mathbb{C}^4$ at the origin or its completion $\mathbb{C}[[x_1, \dots, x_4]]$; let $P_1 = \{x_1 = x_2 = 0\}$ and $P_2 = \{x_3 = x_4 = 0\}$, two smooth surfaces meeting transversally at the origin; put $N_1 = \langle \partial_1, \partial_2 \rangle$ and $N_2 = \langle \partial_3, \partial_4 \rangle$, the normal spaces of P_1 and of P_2 at the origin, so that $\bigwedge^2 T_0 = \bigwedge^2 N_1 \oplus N_1 \otimes N_2 \oplus \bigwedge^2 N_2$. For a germ of vector field θ and a perfect complex F of $\mathcal{O}$-modules write $a_\theta \in \operatorname{Ext}^1_{\mathcal{O}}(F, F)$ for the class of the jet extension along θ.*

(i) *For $F = I = I_{P_1} \cap I_{P_2}$, the ideal of the union, $\operatorname{Ext}^2_{\mathcal{O}}(I, I) \cong N_1 \otimes N_2$, a vector space of dimension four on which the maximal ideal acts trivially, and $a_\theta a_{\theta'}$ is, up to a fixed nonzero scalar, the component of $\theta(0) \wedge \theta'(0)$ in $N_1 \otimes N_2$. In particular $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in \bigwedge^2 N_1 \oplus \bigwedge^2 N_2$.*

(ii) *For $F = K = [\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, the complex in degrees 0 and 1 whose cohomology is I in degree 0 and the residue field k in degree 1, $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in N_1 \otimes N_2$.*

In both cases the bilinear map $(\theta, \theta') \mapsto a_\theta a_{\theta'}$ is nonzero, and it depends only on $\theta(0) \wedge \theta'(0)$.

Proof. Write $A = \mathbb{C}[x_1, x_2]$, $B = \mathbb{C}[x_3, x_4]$, $R = A \otimes_{\mathbb{C}} B$, with maximal ideals $\mathfrak{m}_A$, $\mathfrak{m}_B$ at the origin, and $k = \mathbb{C}$. The ideal of the union is $I = \mathfrak{m}_A R \cap \mathfrak{m}_B R = \mathfrak{m}_A \otimes_{\mathbb{C}} \mathfrak{m}_B$, because the two ideals are generated by disjoint sets of variables. Everything below is homogeneous, the Ext groups in degree two are of finite length, and finite length modules and jet classes are unchanged by the flat base change from R to $\mathcal{O}$; so we compute over R. For finitely generated modules M, M' over A and N, N' over B, the tensor product of free resolutions resolves $M \otimes_{\mathbb{C}} N$ and the Künneth formula gives

$$\operatorname{Ext}^n_R(M \otimes N, M' \otimes N') = \bigoplus_{i+j=n} \operatorname{Ext}^i_A(M, M') \otimes_{\mathbb{C}} \operatorname{Ext}^j_B(N, N'), \tag{20}$$

compatibly with Yoneda products up to the Koszul sign, since a product of chain maps on the tensor product of resolutions is the tensor product of the products.

The basic case is a single plane. The Koszul complex $0 \to A \xrightarrow{(-x_2, x_1)^t} A^2 \xrightarrow{(x_1, x_2)} \mathfrak{m}_A \to 0$ is a free resolution, so $\mathfrak{m}_A$ has projective dimension one and $\operatorname{Ext}^i_A(\mathfrak{m}_A, -) = 0$ for $i \geq 2$; and $\operatorname{Ext}^1_A(\mathfrak{m}_A, \mathfrak{m}_A) = \operatorname{coker}(\operatorname{Hom}(A^2, \mathfrak{m}_A) \to \operatorname{Hom}(A, \mathfrak{m}_A)) = \mathfrak{m}_A/\mathfrak{m}_A^2$, a two-dimensional space killed by $\mathfrak{m}_A$. On a complex of free modules with differential d the jet extension along a derivation D is the complex with terms $P_i \oplus P_i$ and differential $\left(\begin{smallmatrix} d & -D(d) \\ 0 & d \end{smallmatrix}\right)$, $D(d)$ the entrywise derivative, because the twisted module $J_D(P_i)$ is free on the elements $(e, 0)$ and $(0, e)$ and $d(0, e) = \sum_l d_{le}(0, e_l) - \sum_l D(d_{le})(e_l, 0)$; so the jet class is represented by the degree -1 chain map $-D(d)$. For $D = \partial_i$ on the Koszul complex this is $e \mapsto \delta_{i2} f_1 - \delta_{i1} f_2$ on the generator e of the last term, whose image in $\mathfrak{m}_A/\mathfrak{m}_A^2$ is $\delta_{i2} x_1 - \delta_{i1} x_2$:

$$\alpha_1 := a_{\partial_1} = -[x_2], \qquad \alpha_2 := a_{\partial_2} = [x_1],$$

a basis of $\operatorname{Ext}^1_A(\mathfrak{m}_A, \mathfrak{m}_A)$, which we identify with N_1 by $\partial_i \mapsto \alpha_i$. Similarly β_3, β_4 form a basis of $\operatorname{Ext}^1_B(\mathfrak{m}_B, \mathfrak{m}_B)$, identified with N_2.

(i) By (20) with $M = M' = \mathfrak{m}_A$, $N = N' = \mathfrak{m}_B$,

$$\operatorname{Ext}^2_R(I, I) = \operatorname{Ext}^1_A \otimes \operatorname{Ext}^1_B = N_1 \otimes N_2,$$

the other two summands vanishing by the projective dimension, and it is killed by $\mathfrak{m}_A$ and $\mathfrak{m}_B$. The derivation ∂_1 acts on the factor A, so the jet extension of $I = \mathfrak{m}_A \otimes \mathfrak{m}_B$ along it is $J_{\partial_1}(\mathfrak{m}_A) \otimes \mathfrak{m}_B$ and $a_{\partial_1} = \alpha_1 \otimes 1$; likewise $a_{\partial_2} = \alpha_2 \otimes 1$, $a_{\partial_3} = 1 \otimes \beta_3$, $a_{\partial_4} = 1 \otimes \beta_4$. Hence $a_{\partial_i} a_{\partial_j} = \pm \alpha_i \otimes \beta_j$ for $i \leq 2 < j$, the four of them a basis of $N_1 \otimes N_2$, while

$a_{\partial_1}a_{\partial_2}=\alpha_1\alpha_2\otimes 1=0$ and $a_{\partial_3}a_{\partial_4}=0$ because $\operatorname{Ext}^2_A(\mathfrak{m}_A,\mathfrak{m}_A)=0$. For germs $\theta=\sum_i c^i\partial_i$ and $\theta'=\sum_j c'^j\partial_j$ the linearity of Lemma 4.59 and the triviality of the module structure give $a_\theta a_{\theta'}=\sum_{i,j}c^i(0)c'^j(0)\,a_{\partial_i}a_{\partial_j}$, the image of $\theta(0)\wedge\theta'(0)$ under the projection of $\bigwedge^2 T_0$ onto $N_1\otimes N_2$ followed by the isomorphism $\partial_i\otimes\partial_j\mapsto\pm\alpha_i\otimes\beta_j$.

(ii) Let $Q=\mathcal{O}_{P_1}\oplus\mathcal{O}_{P_2}$. The map $\mathcal{O}\to Q$ has kernel I and cokernel $\mathcal{O}/(I_{P_1}+I_{P_2})=k$, so K sits in an exact triangle $I\xrightarrow{\iota}K\xrightarrow{\pi}k[-1]\xrightarrow{\varepsilon}I[1]$. The jet extension along θ is an exact functor, natural in its argument, so the jet classes commute with ι and π; and for the residue field a^k_θ depends only on $\theta(0)$, because $\operatorname{Ext}^1(k,k)$ is killed by $\mathfrak{m}$ and the class is linear in θ. Therefore

$$\pi[2]\circ a^K_\theta a^K_{\theta'}=a^k_\theta a^k_{\theta'}\circ\pi=\pi^*\big(a^k_{\theta(0)}a^k_{\theta'(0)}\big),\qquad \pi^*\colon\operatorname{Ext}^2(k,k)\to\operatorname{Ext}^1(K,k),$$

and it suffices to show that the right hand side is nonzero. The long exact sequence of the triangle for $\operatorname{Hom}(-,k)$ reads

$$\operatorname{Hom}(I,k)\xrightarrow{\varepsilon^*}\operatorname{Ext}^2(k,k)\xrightarrow{\pi^*}\operatorname{Ext}^1(K,k),$$

so $\ker\pi^*=\operatorname{im}\varepsilon^*$. Now $k=k_A\otimes k_B$ and, by (20), $\operatorname{Ext}^*(k,k)$ is the tensor product of $\operatorname{Ext}^*_A(k,k)$ and $\operatorname{Ext}^*_B(k,k)$. On the Koszul resolution $A\to A^2\to A$ of k_A the jet chain maps of ∂_1,∂_2 are $-D(d)$ as above, and their composite $A\to A$ is the scalar -1, whose class in $\operatorname{Ext}^2_A(k,k)=\operatorname{Hom}(A,k)=k$ is nonzero; the degree one classes γ_1,γ_2 of ∂_1,∂_2 are the basis $-e_1^*,-e_2^*$ of $\operatorname{Ext}^1_A(k,k)=\operatorname{Hom}(A^2,k)$, and their product spans $\operatorname{Ext}^2_A(k,k)$. With the analogous classes γ_3,γ_4 over B, the products $\gamma_1\gamma_2\otimes 1$, $\gamma_i\otimes\gamma_j$ $(i\le 2<j)$ and $1\otimes\gamma_3\gamma_4$ form a basis of $\operatorname{Ext}^2(k,k)=k\oplus(k^2\otimes k^2)\oplus k$, so $\partial_i\wedge\partial_j\mapsto a^k_{\partial_i}a^k_{\partial_j}$ identifies $\bigwedge^2(N_1\oplus N_2)$ with $\operatorname{Ext}^2(k,k)$, and $a^k_{\theta(0)}a^k_{\theta'(0)}$ is the image of $\theta(0)\wedge\theta'(0)$. The class ε is the Künneth product $\varepsilon_A\otimes\varepsilon_B$ of the classes of $0\to\mathfrak{m}_A\to A\to k\to 0$ and its B-analogue, because K is the two-step truncation of the tensor product of the complexes $[A\to k_A]$ and $[B\to k_B]$ and ε is the inclusion of the omitted term $k_A\otimes k_B$. Hence ε^* maps $\operatorname{Hom}(I,k)=\operatorname{Hom}(\mathfrak{m}_A,k)\otimes\operatorname{Hom}(\mathfrak{m}_B,k)$ into $\operatorname{Ext}^1_A(k,k)\otimes\operatorname{Ext}^1_B(k,k)=N_1\otimes N_2$, and onto it, since $\operatorname{Hom}(\mathfrak{m}_A,k)\to\operatorname{Ext}^1_A(k,k)$ is an isomorphism: it is the connecting homomorphism of the sequence, injective because $\operatorname{Hom}(A,k)\to\operatorname{Hom}(\mathfrak{m}_A,k)$ is zero and surjective because $\operatorname{Ext}^1_A(A,k)=0$. So $\ker\pi^*=N_1\otimes N_2$ and $\pi^*(\theta(0)\wedge\theta'(0))\neq 0$ exactly when $\theta(0)\wedge\theta'(0)\notin N_1\otimes N_2$. ∎

Theorem 4.61 (Markman's candidate does not satisfy the weakened criterion)**.** *In Setup 4.51, for every odd squarefree $d\ge 3$, every choice of the translates D_i in general position and every choice of the m points to be normalised:*

(i) *the identity (19) fails: there is a constant bivector $\pi\in H^0(\bigwedge^2 T_X)$ with $\frac12\pi\lrcorner(\operatorname{At}(\mathcal{F})^2+d\,\Theta^2\cdot\mathrm{id}_{\mathcal{F}})\neq 0$ in $\operatorname{Ext}^2(\mathcal{F},\mathcal{F})$;*

(ii) *$\mathcal{F}$ does not satisfy the weakened criterion: the class $x_{\mathcal{F}}=(\frac{d}{2}\pi\lrcorner\Theta^2,0,\pi)$ has $x_{\mathcal{F}}\lrcorner\operatorname{ch}(\mathcal{F})=0$ and $\operatorname{ev}_{\mathcal{F}}(x_{\mathcal{F}})\neq 0$;*

(iii) *for every perfect complex F_2 on X with Chern character on the secant plane, in particular for $F_2=\mathcal{F}$ and $F_2=\mathcal{F}^\vee$, the object $E=\Phi(\mathcal{F}\boxtimes F_2)$ on the eightfold $X\times\widehat{X}$ of Weil type does not satisfy the weakened criterion of* [Mar25b, Question 11.4]*: the class $x=\Phi^{HT}(x_{\mathcal{F}}\otimes 1)$ satisfies $x\lrcorner\operatorname{ch}(E)=0$ and $\operatorname{ev}_E(x)\neq 0$, so σ_E is not injective on the image of ev_E.*

Proof. (i) The surfaces Z_i and Z_j at cyclic distance at least three meet in the 24 points of $D_i\cap D_{i+1}\cap D_j\cap D_{j+1}$, and there are $12N(N-5)=(d+9)(3d-3)$ such points, of which $m=(d+9)(2d-1)$ are normalised; the remaining $(d+9)(d-2)\ge 12$ are not. Let p be one of the latter. The four divisors through p meet transversally, so their

local equations $x_1,\dots,x_4$ form a regular system of parameters of $\mathcal{O}_{X,p}$ and identify its completion with $\mathbb{C}[[x_1,\dots,x_4]]$, with $D_i=\{x_1=0\}$, $D_{i+1}=\{x_2=0\}$, $D_j=\{x_3=0\}$ and $D_{j+1}=\{x_4=0\}$; hence $Z_i=P_1$ and $Z_j=P_2$ in the notation of Lemma 4.60. The completion is flat, so a class whose completed germ is nonzero is nonzero. No other D_k passes through p, since five divisors have empty intersection, so no other component of Z does. Near p the map ν is an isomorphism, $\nu_*\mathcal{O}_{\widetilde{Z}}=\mathcal{O}_Z$, and $\mathcal{F}$ is the ideal sheaf $I_Z=I_{P_1}\cap I_{P_2}$ twisted by the locally trivial $\mathcal{O}_X(3\Theta)$: the germ $\mathcal{F}_p$ is the module I of Lemma 4.60(i).

Let $\theta,\theta'\in H^0(T_X)$ be constant vector fields whose values at p are ∂_1 and ∂_3; they exist because the constant fields span every tangent space. Put $\pi=\theta\wedge\theta'$. Since $\mathrm{ev}_{\mathcal{F}}$ is an algebra homomorphism, $\mathrm{ev}_{\mathcal{F}}(\pi)=\frac{1}{2}\pi\lrcorner\,\mathrm{At}(\mathcal{F})^2=A_\theta A_{\theta'}$. By Lemma 4.59 its germ at p is $a_\theta a_{\theta'}$, which is nonzero by Lemma 4.60(i), because $\partial_1\wedge\partial_3$ has nonzero component in $N_1\otimes N_2$. The germ of $\frac{d}{2}(\pi\lrcorner\,\Theta^2)\cdot\mathrm{id}_{\mathcal{F}}$ at p is zero by Lemma 4.59. So the germ at p of $\frac{1}{2}\pi\lrcorner(\mathrm{At}(\mathcal{F})^2+d\Theta^2\cdot\mathrm{id}_{\mathcal{F}})$ is $a_\theta a_{\theta'}\neq 0$, and the class itself is nonzero.

(ii) By Theorem 4.56(ii) the class $x_{\mathcal{F}}$ preserves $\mathrm{ch}(\mathcal{F})$, and by (15) $\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})$ is the class of (i).

(iii) Both $\mathcal{F}$ and $\mathcal{F}^\vee$ are admissible as F_2, since $\mathrm{ch}(\mathcal{F}^\vee)=u_\Theta-3v_\Theta$ lies on the secant plane. By Theorem 4.56(iv), E satisfies the criterion only if both factors do, and $\mathcal{F}$ does not. The class x of the statement is the transport of $x_{\mathcal{F}}\otimes 1\in\mathrm{Ann}_{HT^2}(\mathrm{ch}(\mathcal{F}\boxtimes F_2))$, on which ev is $\pm\,\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})\otimes\mathrm{id}_{F_2}\neq 0$; since Φ is an isomorphism on Ext^2, $\mathrm{ev}_E(x)\neq 0$. The last assertion follows from the commutative square of Setup 4.54: $\sigma_E(\mathrm{ev}_E(x))=x\lrcorner\,\mathrm{ch}(E)=0$ with $\mathrm{ev}_E(x)\neq 0$. ∎

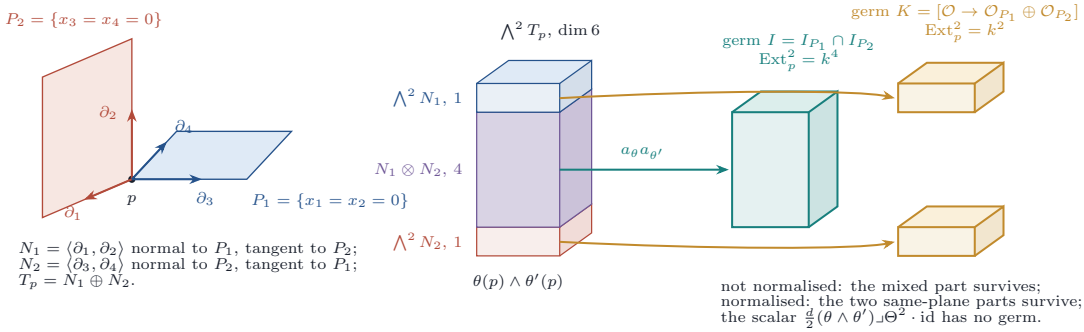
Figure 13 draws the local computation at the double point.

Remark 4.62 (Normalised points). At a normalised point the germ of $\mathcal{F}$ is the complex K of Lemma 4.60(ii), and the same argument, with $\theta(p)=\partial_1$ and $\theta'(p)=\partial_2$, produces a nonzero germ. The failure therefore does not depend on which points are separated. Separating all of them would not help even if the construction allowed it: normalising every isolated point changes $\chi(\mathcal{O}_{\widetilde{Z}})$ to $110N-12N^2+12N(N-5)=50N$, while the secant plane requires $(d+9)(27-d)=72N-4N^2$, and the two agree only for $N=11/2$. The two local computations are complementary: at a point of type (i) the obstruction lies in $N_1\otimes N_2$, at a point of type (ii) in $\bigwedge^2 N_1\oplus\bigwedge^2 N_2$, and together these exhaust $\bigwedge^2 T_p$.

Corollary 4.63 (A local obstruction in every dimension)**.** *Let (X,t) be a polarised abelian variety of dimension $n\geq 4$ and F a perfect complex on X with $\mathrm{ch}(F)=au_t+bv_t$, $(a,b)\neq(0,0)$. Suppose that at some point p the germ F_p is isomorphic, in the derived category of $\mathcal{O}_{X,p}$-modules, to the ideal of the union of two smooth germs of codimension two meeting transversally along a germ of codimension four, or, when $n=4$, to the complex $[\mathcal{O}\to\mathcal{O}_{P_1}\oplus\mathcal{O}_{P_2}]$ of two such germs. Then F does not satisfy the weakened criterion, and neither does $\Phi(F\boxtimes F_2)$ on $X\times\widehat{X}$ for any secant F_2. In particular, in every dimension $n\geq 4$, the union Z of the intersections $D_i\cap D_{i+1}$ of consecutive translates of a polarisation in general position contains, for every pair (i,j) of nonconsecutive indices, a subvariety $Z_i\cap Z_j$ of codimension four along which Z has two transversal branches; any secant object of the form $[\mathcal{O}_X\to\nu_*\mathcal{O}_{\widetilde{Z}}]\otimes L$ in which ν is an isomorphism at a general point of one such $Z_i\cap Z_j$ fails the criterion.*

Proof. In formal coordinates $x_1,\dots,x_n$ at p with the two branches $\{x_1=x_2=0\}$ and $\{x_3=x_4=0\}$, which exist because the four local equations of the branches form part of a regular system of parameters, the ideal is $I\otimes_{\mathbb{C}}\mathbb{C}[x_5,\dots,x_n]$ with I as in Lemma 4.60, and

At an isolated double point of the support the translation classes multiply to a nonzero germ.



At every isolated double point of Z, of either kind, some constant bivector π has $\mathrm{ev}_{\mathcal{F}}(\pi) + \frac{d}{2}(\pi\lrcorner\Theta^2)\,\mathrm{id} \neq 0$; the class $\pi + \frac{d}{2}\,\pi\lrcorner\Theta^2$ preserves $\mathrm{ch}(\mathcal{F})$, so the weakened criterion fails for $\mathcal{F}$ and for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$.

FIGURE 13. Lemma 4.60 and Theorem 4.61. Left: two smooth surfaces of a fourfold meeting transversally at p, with the normal spaces N_1 and N_2, each tangent to the other branch. Centre: the bivectors at p, split into $\bigwedge^2 N_1$, $N_1 \otimes N_2$ and $\bigwedge^2 N_2$, with heights proportional to the dimensions. Right: the local Ext^2 of the germ of $\mathcal{F}$ at p in the two cases, and the part of the bivectors that the products of the translation classes carry onto it: the mixed part when the point is not normalised, the two same-plane parts when it is. The compensating scalar has no germ, so in either case the identity (19) fails at p.

(20) with the third factor gives $\mathrm{Ext}^2(I_p, I_p) = N_1 \otimes N_2 \otimes \mathcal{O}_{W,p}$, where W is the intersection of the branches. The jet classes along $\partial_5, \dots, \partial_n$ vanish, since these derivations act on a free factor, and the products of the classes along $\partial_1, \dots, \partial_4$ are as in the lemma. Reducing modulo the maximal ideal of $\mathcal{O}_{W,p}$ shows that $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ has a nonzero component in $N_1 \otimes N_2$. When $n = 4$ and F_p is the complex $[\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, Lemma 4.60(ii) applies directly. The rest of the argument is that of Theorem 4.61, using Theorem 4.56 for $n \geq 3$. For the last assertion, $Z_i \cap Z_j = D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$ is a transversal intersection of four divisors, of codimension four, and at a general point of it no fifth divisor passes, so Z has exactly the two branches Z_i and Z_j there. ∎

Remark 4.64 (Scope of the local obstruction). Theorem 4.61 answers the question posed in [Mar25b, Section 12]: the secant objects proposed there do not satisfy the weakened criterion of [Mar25b, Question 11.4], for any discriminant. The reason is local and geometric. That question asks whether the semiregularity map may be required to be injective only on the image of the evaluation map. For a secant object this image contains, by Theorem 4.56, the products $A_\theta A_{\theta'}$ of the translation classes shifted by the scalars $\frac{d}{2}(\theta \wedge \theta')\lrcorner\Theta^2$, and the semiregularity map kills them all. For the criterion to hold these shifted products must vanish, and they cannot: at an isolated double point of the support, the translation classes of the two branches multiply to a nonzero local class, while a scalar has no local class at all. The fourth ingredient of [I, Remark 14.16] can accordingly be stated more sharply. A secant object of positive rank on a fourfold that satisfies the weakened criterion must have, at every point, germs whose translation classes multiply to zero in the local Ext^2.

By Lemma 4.60 this excludes every support with two transversal branches of codimension two through a point, whether or not the branches are separated, and by Corollary 4.63 it excludes the chain of consecutive intersections of translates in every dimension. An object supported on a smooth surface, or on surfaces meeting only along curves, is not excluded by this argument, but no such object with Chern character on the secant plane is known.

4.4. **Smooth supports and four discriminants.** Theorem 4.61 and Corollary 4.63 exclude every secant object whose support has two transversal branches of codimension two through a point. This leaves supports that are smooth or that meet themselves only along curves, and Remark 4.64 ends with the observation that no such object is known. In this subsection we explain why the local argument cannot reach the smooth case, and then determine what a smooth support can be. The answer is narrow: on the fourfold the invariants of the support are forced, and the discriminant is confined to a finite list.

We begin with the fact that at a smooth point there is nothing local left to compute.

Proposition 4.65 (No local obstruction at a smooth point)**.** *Let Y be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$ and let $S \subset Y$ be smooth of codimension two. Then*

$$\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y, \qquad \mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S) = N_{S/Y}, \qquad \mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) = 0 \quad (q \geq 2),$$

and the same holds for $\mathcal{I}_S \otimes L$ for any line bundle L. Consequently, for $F = \mathcal{I}_S \otimes L$ both sides of (19) have vanishing germs in $\mathcal{E}xt^2$, and Lemma 4.59 decides nothing.

Proof. Twisting by L changes neither side, so take $F = \mathcal{I}_S$. The sheaf $\mathcal{I}_S$ is torsion free of rank one, and its dual is $\mathcal{O}_Y$ because S has codimension two; hence every local endomorphism of $\mathcal{I}_S$ is multiplication by a function, and $\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y$. Apply $\mathcal{H}om(-, \mathcal{I}_S)$ to

$$0 \to \mathcal{I}_S \to \mathcal{O}_Y \to \mathcal{O}_S \to 0. \tag{21}$$

Since $\mathcal{E}xt^q(\mathcal{O}_Y, \mathcal{I}_S) = 0$ for $q \geq 1$, the long exact sequence gives

$$\mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) \cong \mathcal{E}xt^{q+1}(\mathcal{O}_S, \mathcal{I}_S) \qquad (q \geq 1). \tag{22}$$

Now apply $\mathcal{H}om(\mathcal{O}_S, -)$ to (21). Because S is smooth of codimension two, hence a local complete intersection, the Koszul resolution gives $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_Y) = 0$ for $k \neq 2$ and $\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) = \omega_S \otimes \omega_Y^{-1} = \omega_S$, while $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_S) = \bigwedge^k N_{S/Y}$ for $0 \leq k \leq 2$ and zero otherwise. Computing both from the same Koszul complex shows that the map

$$\rho \colon \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) = \omega_S \longrightarrow \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S) = \textstyle\bigwedge^2 N_{S/Y} = \omega_S$$

induced by $\mathcal{O}_Y \to \mathcal{O}_S$ is the identity: in local coordinates in which $S = \{x_1 = x_2 = 0\}$, both groups are computed as $M/(x_1, x_2)M$ for $M = \mathcal{O}_Y$ and $M = \mathcal{O}_S$, and the map is reduction modulo (x_1, x_2), which is the identity on $\mathcal{O}_S$.

Consider the long exact sequence of $\mathcal{H}om(\mathcal{O}_S, -)$ in degrees two and three. Since ρ is surjective and $\mathcal{E}xt^3(\mathcal{O}_S, \mathcal{O}_Y) = 0$, the segment

$$\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) \xrightarrow{\rho} \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S) \to \mathcal{E}xt^3(\mathcal{O}_S, \mathcal{I}_S) \to \mathcal{E}xt^3(\mathcal{O}_S, \mathcal{O}_Y) = 0$$

gives $\mathcal{E}xt^3(\mathcal{O}_S, \mathcal{I}_S) = 0$, so $\mathcal{E}xt^2(\mathcal{I}_S, \mathcal{I}_S) = 0$ by (22). For $k \geq 3$ both $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_Y)$ and $\mathcal{E}xt^k(\mathcal{O}_S, \mathcal{O}_S)$ vanish, so $\mathcal{E}xt^{k+1}(\mathcal{O}_S, \mathcal{I}_S) = 0$ and $\mathcal{E}xt^k(\mathcal{I}_S, \mathcal{I}_S) = 0$. In degrees one and two, since ρ is injective, the segment

$$0 = \mathcal{E}xt^1(\mathcal{O}_S, \mathcal{O}_Y) \to N_{S/Y} \to \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{I}_S) \to \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_Y) \xrightarrow{\rho} \mathcal{E}xt^2(\mathcal{O}_S, \mathcal{O}_S),$$

in which $N_{S/Y} = \mathcal{E}xt^1(\mathcal{O}_S, \mathcal{O}_S)$, gives $\mathcal{E}xt^2(\mathcal{O}_S, \mathcal{I}_S) \cong N_{S/Y}$, and this is $\mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S)$ by (22).

For the last assertion, the right side of (19) has vanishing germs by Lemma 4.59, and the left side lies in $\mathrm{Ext}^2(F, F\otimes\Omega_Y^2)$, whose sheaf of germs in the $\mathcal{E}xt^2$ direction is $\mathcal{E}xt^2(F,F)\otimes\Omega_Y^2 = 0$. ∎

Remark 4.66 (The global form of the criterion). Proposition 4.65 is the precise reason why the argument of Theorem 4.61 stops at a smooth support, and it also shows what takes its place. Since $\mathcal{E}xt^2 = 0$, the local-to-global spectral sequence for $\mathrm{Ext}^2(\mathcal{I}_S, \mathcal{I}_S)$ has only two surviving rows, $H^1(S, N_{S/Y})$ and $H^2(Y, \mathcal{O}_Y)$, and they sit in different steps of the filtration: the first is the quotient that records the geometry of S, the second the subgroup of scalars. A product $A_\theta A_{\theta'}$ of translation classes therefore has a well defined image in $H^1(S, N_{S/Y})$, while the scalar on the right of (19) has image zero there. The criterion thus splits into two conditions of different kinds: a vanishing in $H^1(S, N_{S/Y})$, which is a statement about the normal bundle and not about any point, and an identity in $H^2(Y, \mathcal{O}_Y)$. At a double point the first condition already fails locally. At a smooth support it is a global question, and what is decided here is which surfaces S are available at all.

The invariants of a smooth support are far from free. Since the Todd class of an abelian variety is trivial, Grothendieck–Riemann–Roch converts the Chern character of the object into the complete set of numerical invariants of the surface, and three classical inequalities then constrain them.

Theorem 4.67 (The invariants of a smooth support)**.** *Let (X,Θ) be a principally polarised abelian fourfold, let $d>0$, let $b\in\mathbb{Z}_{>0}$, and let $\mathcal{F} = \mathcal{I}_S(b\Theta)$ be a coherent sheaf with*

$$\mathrm{ch}(\mathcal{F}) = u_\Theta + b\, v_\Theta$$

and $S\subset X$ smooth. Write $N = \frac{1}{2}\mathrm{N}_{K/\mathbb{Q}}(b+\sqrt{-d}) = \frac{1}{2}(b^2+d)$, $\iota\colon S\hookrightarrow X$ and $\lambda = \iota^*\Theta$. Then S is a surface and

$$\begin{aligned} &[S] = N\Theta^2, && \iota_*K_S = \tfrac{4bN}{3}\,\Theta^3, && \chi(\mathcal{O}_S) = 4N(2b^2-N),\\ &K_S^2 = 12N(4b^2-N), && e(S) = 12N(4b^2-3N), && K_S\cdot\lambda = 32bN, \end{aligned} \tag{23}$$

with $\lambda^2 = 24N$ and $[S]^2 = 24N^2$. Moreover K_S is globally generated, hence nef; S is minimal of general type; and

$$\tfrac{4}{9}b^2 \;\le\; N \;\le\; b^2, \qquad \textit{equivalently} \qquad -\tfrac{1}{9}b^2 \;\le\; d \;\le\; b^2. \tag{24}$$

The lower bound is the Hodge index theorem and is implied by $d>0$; the upper bound is the Bogomolov–Miyaoka–Yau inequality. In terms of b and d the five invariants read

$$\chi(\mathcal{O}_S) = (b^2+d)(3b^2-d), \quad K_S^2 = 3(b^2+d)(7b^2-d), \quad e(S) = 3(b^2+d)(5b^2-3d), \tag{25}$$

and the defect in the Bogomolov–Miyaoka–Yau inequality is

$$3e(S) - K_S^2 = 24\,(b^4-d^2), \tag{26}$$

so that the inequality amounts exactly to $d\le b^2$. Finally, K_S is never numerically proportional to λ.

Proof. Since $\mathrm{ch}_0(\mathcal{F}) = 1$ and $c_1(\mathcal{F}) = b\Theta$, the sheaf $\mathcal{F}(-b\Theta) = \mathcal{I}_S$ has $\mathrm{ch}_0 = 1$, $\mathrm{ch}_1 = 0$, so S has codimension at least two, and

$$\mathrm{ch}(\mathcal{O}_S) = 1 - \mathrm{ch}(\mathcal{I}_S) = 1 - \mathrm{ch}(\mathcal{F})\, e^{-b\Theta}.$$

Put $z = b+\sqrt{-d}$, so that $z\bar z = 2N$. Since $u_\Theta = \Re e^{\sqrt{-d}\Theta}$ and $\sqrt{d}\,v_\Theta = \Im e^{\sqrt{-d}\Theta}$, we have $\mathrm{ch}(\mathcal{F}) = \Im(z\,e^{\sqrt{-d}\Theta})/\sqrt{d}$, hence $\mathrm{ch}(\mathcal{F})e^{-b\Theta} = \Im(z\,e^{-\bar z\Theta})/\sqrt{d}$, whose coefficient of Θ^k is

$(-1)^k\Im(z\overline{z}^k)/(\sqrt{d}\,k!) = (-1)^{k+1}\,2N\Im(z^{k-1})/(\sqrt{d}\,k!)$. With $\Im z = \sqrt{d}$, $\Im z^2 = 2b\sqrt{d}$ and $\Im z^3 = (3b^2-d)\sqrt{d}$ this gives

$$\mathrm{ch}_2(\mathcal{O}_S) = \tfrac{b^2+d}{2}\Theta^2 = N\Theta^2, \quad \mathrm{ch}_3(\mathcal{O}_S) = -\tfrac{2bN}{3}\Theta^3, \quad \mathrm{ch}_4(\mathcal{O}_S) = \tfrac{3b^4+2db^2-d^2}{24}\Theta^4.$$

The first identity says S has codimension exactly two, so it is a surface, with class $N\Theta^2$. Because T_X is trivial, $\mathrm{td}(T_X) = 1$, and Grothendieck–Riemann–Roch for ι reads $\mathrm{ch}(\iota_*\mathcal{O}_S) = \iota_*\,\mathrm{td}(T_S)$. Comparing degrees three and four,

$$\mathrm{ch}_3(\mathcal{O}_S) = \iota_*\tfrac{c_1(T_S)}{2} = -\tfrac{1}{2}\iota_*K_S, \qquad \mathrm{ch}_4(\mathcal{O}_S) = \iota_*\tfrac{c_1^2+c_2}{12} = \chi(\mathcal{O}_S)\,[\mathrm{pt}],$$

the second by Noether's formula. With $\Theta^4 = 24\,[\mathrm{pt}]$ these give $\iota_*K_S = \frac{4bN}{3}\Theta^3$ and, after substituting $d = 2N - b^2$, $\chi(\mathcal{O}_S) = 3b^4 + 2db^2 - d^2 = 4N(2b^2 - N)$. Then $K_S\cdot\lambda = \int_X \iota_*K_S\cdot\Theta = \frac{4bN}{3}\cdot 24 = 32bN$ and $\lambda^2 = \int_X [S]\cdot\Theta^2 = 24N$.

For the last two invariants, the normal bundle sequence and the triviality of T_X give $c(T_S)c(N_{S/X}) = 1$, so $c_1(N_{S/X}) = K_S$ and $c_2(N_{S/X}) = K_S^2 - e(S)$; the self-intersection formula $[S]^2 = \int_S c_2(N_{S/X})$ then reads

$$24N^2 = K_S^2 - e(S),$$

while Noether's formula reads $12\chi(\mathcal{O}_S) = K_S^2 + e(S)$. Solving the two gives the stated K_S^2 and $e(S)$.

The conormal sequence presents Ω^1_S as a quotient of $\Omega^1_X|_S = \mathcal{O}_S^{\oplus 4}$, so Ω^1_S is globally generated, and therefore so is $K_S = \det\Omega^1_S$, a quotient of $\bigwedge^2\mathcal{O}_S^{\oplus 4}$. A globally generated line bundle is nef. A globally generated rank two bundle on a surface has $c_2 \geq 0$, the class c_2 being represented by the vanishing scheme of a general section, which is empty or finite; hence

$$e(S) = c_2(\Omega^1_S) = 12N(4b^2 - 3N) \;\geq\; 0,$$

so $N \leq \frac{4}{3}b^2 < 4b^2$ and therefore $K_S^2 = 12N(4b^2 - N) > 0$. A nef divisor of positive self-intersection is big, so $\kappa(S) = 2$ and, K_S being nef, S is minimal of general type. The Bogomolov–Miyaoka–Yau inequality $c_1^2 \leq 3c_2$ therefore applies and reads $K_S^2 \leq 3e(S)$. Substituting $N = \frac{1}{2}(b^2+d)$ in (23) gives (25), hence $3e(S) - K_S^2 = 24(b^4 - d^2)$, and the inequality is $d \leq b^2$, that is $N \leq b^2$.

For the lower bound, λ is ample on S, so the Hodge index theorem gives $(K_S\cdot\lambda)^2 \geq K_S^2\,\lambda^2$, that is $(32bN)^2 \geq 12N(4b^2 - N)\cdot 24N$, which simplifies to $9N \geq 4b^2$. Since $N = \frac{1}{2}(b^2+d)$ with $d > 0$, this holds automatically. Equality in the Hodge index theorem occurs exactly when K_S is numerically proportional to λ, hence exactly when $9N = 4b^2$, and this never happens, because $9N = \frac{9}{2}(b^2+d) > 4b^2$ for $d > 0$. ∎

Figure 14 draws the Bogomolov–Miyaoka–Yau defect over the quarter plane of (b, d).

Corollary 4.68 (Four discriminants on the fourfold)**.** *Let (X,Θ) be a principally polarised abelian fourfold and $d > 0$ squarefree. A sheaf $\mathcal{I}_S(3\Theta)$ with Chern character $u_\Theta + 3v_\Theta$, the value of Proposition 4.52(iii), and with smooth support S exists only if $d \leq 9$, that is $d \in \{1,2,3,5,6,7\}$; if in addition $N = (9+d)/2$ is an integer, as in Setup 4.51, only*

$$d \in \{1,\,3,\,5,\,7\}, \qquad N \in \{5,\,6,\,7,\,8\}.$$

For these values the invariants are forced to be those of Table 1. In particular, for $d > 9$ no such sheaf has smooth support.

Proof. This is immediate from (24) at $b = 3$ and the arithmetic of (23), and the table evaluates (23) at $b = 3$. The value $d = 9$ is the equality case of the Bogomolov–Miyaoka–Yau inequality, where $K_S^2 = 3e(S) = 2916$ and S would have to be a quotient of the ball; it is excluded because 9 is not squarefree. ∎

d	N	$\chi(\mathcal{O}_S)$	K_S^2	$e(S)$	$K_S \cdot \lambda$	λ^2
1	5	260	1860	1260	480	120
3	6	288	2160	1296	576	144
5	7	308	2436	1260	672	168
7	8	320	2688	1152	768	192

TABLE 1. The only invariants a smooth support can have on an abelian fourfold, by Theorem 4.67 and Corollary 4.68, at $b = 3$ and with N an integer. Every entry is determined by $\mathrm{ch}(\mathcal{F})$ alone. The slope $K_S^2/\chi(\mathcal{O}_S)$ runs from 93/13 at $d = 1$ to 42/5 at $d = 7$, and reaches the Bogomolov–Miyaoka–Yau ceiling of 9 only at the excluded value $d = 9$.

Remark 4.69 (Status of the smooth case)*.* The local argument does not apply here, for a structural reason given by Proposition 4.65: the sheaf $\mathcal{E}xt^2$ of a smooth ideal vanishes, so both sides of (19) are locally zero, and the comparison of germs that decides Theorem 4.61 has nothing to compare.

The surface is rigidly constrained. At $b = 3$ its class, its canonical class, its Euler characteristic, its Euler number and its two intersection numbers with the polarisation are all determined by d, and they are compatible with existence only for $d \leq 9$. This is the content of Corollary 4.68, and it restricts the construction: for all but finitely many imaginary quadratic fields the secant construction of Setup 4.51 with smooth support is not available, whatever the geometry of the surface.

What remains is a single existence question. For $d \in \{1, 3, 5, 7\}$, does a principally polarised abelian fourfold carry a smooth surface S with the invariants of Table 1, and if so, does the image of $A_\theta A_{\theta'}$ in $H^1(S, N_{S/X})$ vanish for all $\theta, \theta' \in H^0(T_X)$? The first half is a question about surfaces in abelian fourfolds with prescribed Chern numbers, of the kind that is answered by a construction or by a classification argument; the second is a question about one cohomology group of one normal bundle. Neither is settled here. The simplest surfaces of the right class are ruled out by one entry of the table: a complete intersection of divisors in $|a\Theta|$ and $|b'\Theta|$ has class $ab'\Theta^2$, so $ab' = N$, and canonical class $(a + b')\lambda$ by adjunction, so $K_S \cdot \lambda = 24(a + b')N$; since $ab' = N \geq 5$ forces $a + b' \geq 5$, this exceeds the required $96N$. The surface, if it exists, is not cut out by two theta divisors, nor by any two divisors algebraically equivalent to multiples of Θ; nor, by Corollary 4.84 below, is its ideal sheaf presented by any two bundles whose classes lie in the subring generated by line bundles. It must also move with X to first order in every polarised direction, in the exact sense of Proposition 4.73, while the injectivity that Lemma 4.88 asks of it holds automatically when X is simple (Lemma 4.70).

A positive answer to both, for a surface that moves in this way, together with an affirmative answer to [Mar25b, Question 11.4], would produce Weil classes on abelian eightfolds for at most four discriminants; the other discriminants, every $n \geq 5$ and every field of degree larger than two would still have to be reached through [III, Proposition 2.13] and the propagation statement.

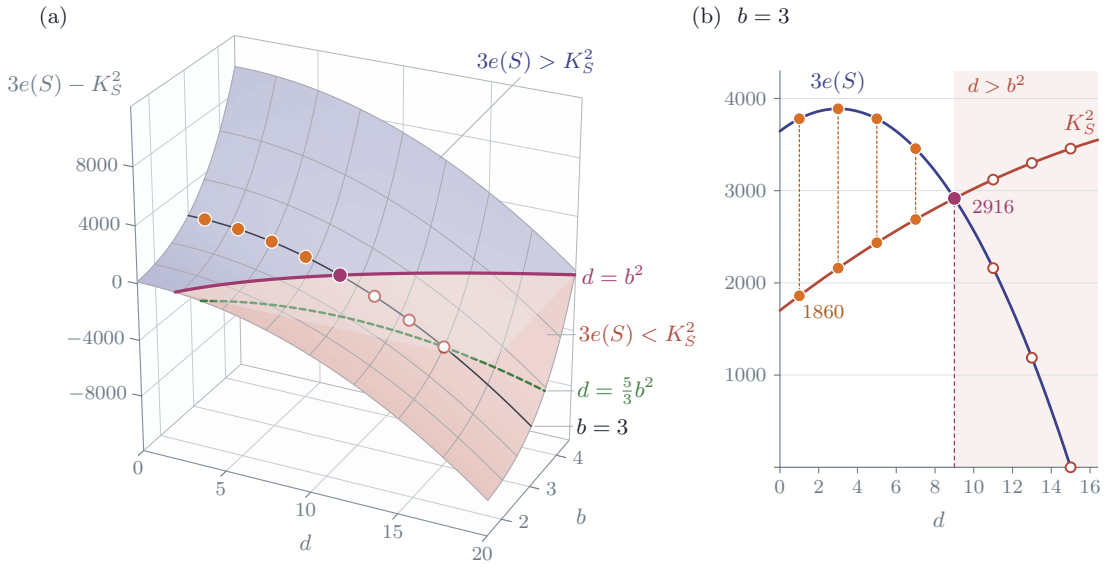


FIGURE 14. Theorem 4.67 over the quarter plane of (b, d). The height of the surface is the defect $3e(S) - K_S^2 = 24(b^4 - d^2)$ in the Bogomolov–Miyaoka–Yau inequality for the surface the theorem attaches to the pair, compressed by a signed cube root so that four orders of magnitude fit one frame; the compression is odd and monotone, so the zero locus and the sign are the true ones. The surface is drawn indigo where the defect is positive and clay where it is negative, and it crosses zero along $d = b^2$, which is therefore the whole content of the inequality. The weaker wall $d = \frac{5}{3}b^2$, where $e(S)$ itself changes sign, is drawn dashed and lies above it. On the line $b = 3$, which is the value Proposition 4.52(iii) forces on the fourfold, the four squarefree discriminants $d = 1, 3, 5, 7$ with N an integer sit above the wall and are marked with their drop lines; $d = 9$ sits exactly on it, the equality case, and $d = 11, 13, 15$ sit below and are excluded.

Two of the conditions on a smooth support are decided by the table, and the third, which Theorem 4.56(iii)(a) imposes on every candidate, has an exact form. The first is the injectivity of $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_S)$ that Lemma 4.88 below requires of every support.

Lemma 4.70 (The injectivity condition for a smooth support)**.** *Let X be a simple complex abelian fourfold, let $V = H^0(T_X)$, and let $S \subset X$ be a smooth surface. The following are equivalent:*

(i) *the restriction $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_S)$ is injective;*
(ii) *no nonzero translation-invariant holomorphic 2-form on X restricts to zero on S;*
(iii) *S is not Lagrangian for any translation-invariant holomorphic symplectic form on X;*
(iv) *the Gauss image of S in $\mathrm{Gr}(2, V) \subset \mathbb{P}(\bigwedge^2 V)$ spans $\mathbb{P}(\bigwedge^2 V)$.*

If $e(S) \neq [S]^2$, they hold. In particular they hold for every smooth support of Table 1, where $e(S) = 12N(36 - 3N)$ and $[S]^2 = 24N^2$ agree only for $5N = 36$.

Proof. The restriction $\iota^* \colon H^2(X, \mathbb{C}) \to H^2(S, \mathbb{C})$ is a morphism of Hodge structures defined over $\mathbb{Q}$, so it commutes with complex conjugation, which interchanges the $(0,2)$ and $(2,0)$ parts; hence ι^* is injective on $H^{0,2}(X) = H^2(\mathcal{O}_X)$ if and only if it is injective on

$H^{2,0}(X) = \bigwedge^2 V^\vee$, the translation-invariant 2-forms. A holomorphic 2-form on a compact surface is zero as a class in $H^{2,0}(S) = H^0(K_S)$ if and only if it is zero as a form; this is (i)⇔(ii). The restriction of $p \wedge q \in \bigwedge^2 V^\vee$ to S is the pull-back of the Plücker coordinate $p \wedge q$ along the Gauss map $\gamma\colon S \to \mathrm{Gr}(2, V)$, $s \mapsto T_s S$, as a section of $K_S = \gamma^*\mathcal{O}(1)$, so a 2-form vanishes on S if and only if the corresponding hyperplane contains $\gamma(S)$; this is (ii)⇔(iv).

A nonzero 2-form on $V \cong \mathbb{C}^4$ has rank two or four. If ω has rank four, $\omega|_S = 0$ says that every tangent plane of S is ω-isotropic, that is, S is Lagrangian for ω. If $\omega = \alpha \wedge \beta$ has rank two, with $\alpha, \beta \in V^\vee$ independent, then $\alpha|_S$ and $\beta|_S$ are independent holomorphic 1-forms on S, because S generates the simple X and a nonzero invariant 1-form restricts nontrivially to a generating subvariety, and $\alpha|_S \wedge \beta|_S = 0$. By the theorem of Castelnuovo and de Franchis [BHPV04] there is a fibration $f\colon S \to C$ onto a curve of genus at least two with $\alpha|_S$ and $\beta|_S$ pulled back from C. A fibre F of f is a curve in X on which α and β vanish, so all its tangent lines lie in the fixed plane $B = \ker\alpha \cap \ker\beta \subset V$. The abelian subvariety generated by $F - F$ is the image of a sum map from a power of F, whose differential at a general point is spanned by tangent vectors of F; so its tangent space lies in B, its dimension is at most two, and it is nonzero because F is not a point. This contradicts the simplicity of X. Hence only the rank-four case occurs, which is (ii)⇔(iii).

Finally, if S is Lagrangian for a symplectic ω, then ω identifies T_S with the annihilator of T_S in $\Omega^1_X|_S$, which is the conormal bundle, so $N_{S/X} \cong \Omega^1_S$ and $c_2(N_{S/X}) = e(S)$; since T_X is trivial, $c_2(N_{S/X}) = [S]^2$ by the self-intersection formula. So $e(S) \neq [S]^2$ excludes (iii). For the table, Theorem 4.67 gives $[S]^2 = 24N^2$ and $e(S) = 12N(36 - 3N)$ at $b = 3$, and $24N^2 = 12N(36 - 3N)$ reads $5N = 36$. For general b the two agree only when $N = \frac{4}{5}b^2$, that is $d = \frac{3}{5}b^2$, whose first integral instance is $b = 5$, $d = 15$. ∎

Remark 4.71 (The class $K_S - 4\lambda$). Theorem 4.67 records $K_S \cdot \lambda = 96N$ and $\lambda^2 = 24N$ at $b = 3$, so $K_S \cdot \lambda = 4\lambda^2$, and that K_S is never numerically proportional to λ. The sharper statement is that

$$\delta = K_S - 4\lambda \in \ker\bigl(\iota_*\colon H^2(S, \mathbb{Q}) \to H^6(X, \mathbb{Q})\bigr), \qquad \delta^2 = -12N(N-4) < 0,$$

because $\iota_* K_S = 4N\Theta^3$ and $\iota_*\lambda = \Theta \cup [S] = N\Theta^3$. So δ is a nonzero algebraic class of type $(1,1)$, orthogonal by the projection formula to the whole image of $H^2(X, \mathbb{Q})$ in $H^2(S, \mathbb{Q})$, and of negative square: the Picard number of S is at least two, and the kernel of the Gysin map on $H^2(S, \mathbb{Q})$, a sub-Hodge structure, contains an algebraic class. For general b the class is $K_S - \frac{4b}{3}\lambda$, with square $\frac{4N}{3}(4b^2 - 9N)$, and its vanishing is the equality case $9N = 4b^2$ of the Hodge index bound in Theorem 4.67. This excludes none of the four cases; it describes them.

The third condition is the one that Theorem 4.56(iii)(a) imposes on every candidate and that the construction of Setup 4.51 meets by its uniformity over the moduli of principally polarised fourfolds (Corollary 4.57): the object must extend to first order along every polarised deformation of (X, Θ). For $\mathcal{F} = \mathcal{I}_S(3\Theta)$ with S smooth this is a condition on S. If S extends along a deformation X_v of X that keeps Θ of type $(1,1)$, then so does $\mathcal{F}$. Conversely, the obstruction $v \lrcorner \mathrm{At}(\mathcal{F}) \in \mathrm{Ext}^2(\mathcal{F}, \mathcal{F})$ has, under the edge map to $H^1(S, \mathcal{E}xt^1(\mathcal{F}, \mathcal{F})) = H^1(S, N_{S/X})$ of Remark 4.66, the image of v under $H^1(T_X) \to H^1(\iota^* T_X) \to H^1(N_{S/X})$, by Lemma 4.59 applied on the overlaps of a cover; and that image is the obstruction $\mathrm{ob}_S(v)$ to extending the subscheme S along X_v. So condition (a) for $\mathcal{F}$ is the statement that S extends to first order along every polarised deformation of X, and Bloch's semiregularity map [Blo72] puts it in a form that names the

object it concerns. For $S \subset X$ smooth of codimension two with normal bundle $N = N_{S/X}$, let

$$\pi\colon H^1(S,N) \xrightarrow{\cong} H^1(S, N^\vee \otimes K_S) \xrightarrow{j} H^1(S, \iota^*\Omega^1_X \otimes K_S) \xrightarrow{\iota_*} H^3(X, \Omega^1_X)$$

be the composite of $N \cong N^\vee \otimes \bigwedge^2 N$, of the conormal inclusion $N^\vee \to \iota^*\Omega^1_X$ tensored with K_S, and of the Gysin map, with $K_X \cong \mathcal{O}_X$. This is the semiregularity map of Bloch for codimension two, and it satisfies $\pi(\mathrm{ob}_S(v)) = v \lrcorner [S]$ for every $v \in H^1(X, T_X)$.

Lemma 4.72 (Duality)**.** *Under Serre duality $H^1(X,T_X) \times H^3(X,\Omega^1_X) \to H^4(\mathcal{O}_X) = \mathbb{C}$,*

$$\{v \in H^1(X,T_X) : S \text{ extends to first order along } v\} = \pi\big(H^1(S,N)\big)^\perp.$$

Proof. S extends along v if and only if $\mathrm{ob}_S(v) = 0$, and ob_S is the composite of the restriction $r\colon H^1(X,T_X) \to H^1(S, \iota^*T_X)$ with the map induced by $\iota^*T_X \to N$. The Serre dual of r is the Gysin map $\iota_*\colon H^1(S, \iota^*\Omega^1_X \otimes K_S) \to H^3(X, \Omega^1_X \otimes K_X)$, and the Serre dual of $H^1(\iota^*T_X) \to H^1(N)$ is the map $H^1(N^\vee \otimes K_S) \to H^1(\iota^*\Omega^1_X \otimes K_S)$ induced by the conormal inclusion. So the dual of ob_S is $\iota_* \circ j$, which is π after the identification $N \cong N^\vee \otimes K_S$, and the kernel of a linear map is the annihilator of the image of its dual. ∎

Proposition 4.73 (The first-order condition is a rank condition)**.** *Let $S \subset X$ be a smooth surface with $[S] = N\Theta^2$, $N \neq 0$. Then $\pi(H^1(S,N)) \supseteq \Theta \cup H^{0,2}(X)$, a subspace of dimension six, and the following are equivalent:*

(i) *S extends to first order along every polarised deformation of (X,Θ), that is, $\mathcal{I}_S(3\Theta)$ satisfies condition (a) of Theorem 4.56(iii);*
(ii) *$\pi(H^1(S,N)) = \Theta \cup H^{0,2}(X)$;*
(iii) *$\operatorname{rank}\pi = 6$, the least value possible.*

In particular a smooth secant support is as far from semiregular in Bloch's sense as a surface with this class can be: $\dim\ker\pi = h^1(N) - 6$, with $h^1(N) = 2h^0(N) - 2\chi(\mathcal{O}_S) + 24N^2 \geq 8 - 2\chi(\mathcal{O}_S) + 24N^2$, which is at least 88, 296, 568 and 904 for $N = 5,6,7,8$.

Proof. The polarised deformations form $P = \{v : v \lrcorner \Theta = 0\}$, of dimension ten in $H^1(T_X) = V \otimes H^{0,1}(X)$. For $z \in H^{0,2}(X)$ the pairing is $\langle v, \Theta \cup z\rangle = (v \lrcorner \Theta) \cup z$, so $\Theta \cup H^{0,2}(X) \subseteq P^\perp$; the map $z \mapsto \Theta \cup z$ is injective by hard Lefschetz, so this subspace has dimension six, and $\dim P^\perp = 16 - 10 = 6$, so $P^\perp = \Theta \cup H^{0,2}(X)$. By Lemma 4.72, (i) says $P \subseteq \pi(H^1(N))^\perp$, that is $\pi(H^1(N)) \subseteq P^\perp$. On the other hand $\pi(\mathrm{ob}_S(v)) = v \lrcorner [S] = 2N\,\Theta \cup (v \lrcorner \Theta)$, and $v \mapsto v \lrcorner \Theta$ is surjective onto $H^{0,2}(X)$, because Θ identifies V with $\overline{V}^\vee$ and the contraction becomes the exterior product $V \otimes \overline{V}^\vee \to \bigwedge^2 \overline{V}^\vee$. So $\pi(H^1(N)) \supseteq \Theta \cup H^{0,2}(X)$ always, and (i), (ii), (iii) are equivalent. For $h^1(N)$, Riemann–Roch with $c_1(N) = K_S$ and $c_2(N) = [S]^2 = 24N^2$ gives $\chi(N) = 2\chi(\mathcal{O}_S) - 24N^2$, Serre duality with $N^\vee \otimes K_S \cong N$ gives $h^2(N) = h^0(N)$, and $h^0(N) \geq 4$ because the four translation vector fields map injectively to $H^0(N)$: a translation vector field tangent to S everywhere would restrict to a nowhere vanishing vector field on S, so that $e(S) = c_2(T_S) = 0$, while a secant support has $e(S) = 12N(36 - 3N) \neq 0$ for $N = 5,6,7,8$ by Theorem 4.67. ∎

Remark 4.74 (What the rank condition asks). Proposition 4.73 does not decide the condition; it converts it into the statement that the kernel of Bloch's map has codimension exactly six, as it has for the complete intersections $\Theta \cap \Theta_x$, which do move with X. The kernel of π is $j^{-1}(V^\vee \otimes \ker(H^1(K_S) \to H^3(\mathcal{O}_X)))$, and for $q(S) = 4$ it is the image in $H^1(N^\vee \otimes K_S)$ of $H^0(\Omega^1_S \otimes K_S)$ modulo $V^\vee \otimes H^0(K_S)$. A count with $\chi(\Omega^1_S \otimes K_S) = 2\chi(\mathcal{O}_S) + 24N^2$ and $h^1(T_S) \leq h^0(N) + 12$ shows the rank condition to be consistent with every inequality available, so no exclusion follows at this level; what would decide it is $\ker\pi$ for an actual surface, and no candidate surface is known.

4.5. **Depth and the local obstruction.** By Proposition 4.65, at a smooth point of the support there is nothing local to compare. This is a special case of a much more general statement, and the general statement shows exactly which supports Theorem 4.61 and Corollary 4.63 exclude and which lie beyond their reach. The dividing line is neither smoothness nor the local complete intersection property, but depth.

Theorem 4.75 (Cohen–Macaulay supports carry no local obstruction)**.** *Let Y be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$, let $Z \subset Y$ be a closed subscheme of codimension two, and let $p \in Z$. If the local ring $\mathcal{O}_{Z,p}$ is Cohen–Macaulay, then*

$$\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = 0 \qquad \text{for every } q \geq 2,$$

and the same holds for $\mathcal{I}_Z \otimes L$ for every line bundle L. Consequently both sides of (19) have vanishing germ at p, and Lemma 4.59 decides nothing there.

Proof. Twisting by L changes no $\mathcal{E}xt$ sheaf, so take $F = \mathcal{I}_Z$. Write $A = \mathcal{O}_{Y,p}$, a regular local ring of dimension m, and $I = \mathcal{I}_{Z,p}$. By hypothesis A/I is Cohen–Macaulay of dimension $m-2$, so $\operatorname{depth} A/I = m-2$, and the Auslander–Buchsbaum formula gives

$$\operatorname{pd}_A(A/I) = m - \operatorname{depth}_A(A/I) = 2.$$

From the exact sequence $0 \to I \to A \to A/I \to 0$ with A free it follows that $\operatorname{pd}_A I = 1$. A module of projective dimension one has a free resolution of length one, so $\operatorname{Ext}^q_A(I, M) = 0$ for every A-module M and every $q \geq 2$; taking $M = I$ gives the vanishing, since $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = \operatorname{Ext}^q_A(I, I)$. The last assertion follows from Lemma 4.59: the right side of (19) is $z \cdot \mathrm{id}_F$ with $z \in H^2(\mathcal{O}_Y)$ and has vanishing germs, and the left side lies in a sheaf whose $\mathcal{E}xt^2$ factor is zero. ∎

Remark 4.76 (Sharpness of the hypothesis). Two observations show that Theorem 4.75 is the right statement.

First, Cohen–Macaulayness is strictly weaker than the more familiar conditions that could be imposed instead. Among codimension two germs in $\mathbb{C}^4$, the smooth germ (x_1, x_2), two planes meeting along a line, (x_1, x_2x_3), three planes meeting along a line, $(x_1, x_2x_3x_4)$, the double plane (x_1^2, x_2) and the quadric cone $(x_1, x_2^2 - x_3x_4)$ are complete intersections. The fat plane $(x_1, x_2)^2$ and the cone over the twisted cubic are the ideals of 2×2 minors of $\left(\begin{smallmatrix} x_1 & x_2 & 0 \\ 0 & x_1 & x_2 \end{smallmatrix}\right)$ and of $\left(\begin{smallmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \end{smallmatrix}\right)$, of the expected codimension two, so they are Cohen–Macaulay by the Hilbert–Burch theorem, although neither is a local complete intersection, both needing three generators. By Theorem 4.75, $\mathcal{E}xt^2(\mathcal{I}_Z, \mathcal{I}_Z)$ vanishes at all seven. Reducedness, irreducibility and the complete intersection property are therefore all irrelevant; what matters is the projective dimension, which by Auslander–Buchsbaum is a matter of depth.

Second, the vanishing fails at the germs that the construction of Setup 4.51 produces. Two planes meeting at a point and three planes meeting pairwise at a point are reduced surface germs that become disconnected when the point is removed, so they have depth 1: at least 1 because they are reduced, and at most 1 by the connectedness theorem recalled below. A plane with an embedded point has depth 0. By Auslander–Buchsbaum the projective dimensions of $\mathcal{O}_{Z,p}$ are 3, 3 and 4, and the proof of Corollary 4.77(i) below gives $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p \neq 0$ for $q = 2$, 2 and 3. By Hartshorne's connectedness theorem, a scheme whose local ring at p is Cohen–Macaulay is connected in codimension one at p, so a support with two components through p meeting only at p is never Cohen–Macaulay there. This is precisely the configuration analysed in Lemma 4.60.

Corollary 4.77 (The local dichotomy)**.** *Let (X,t) be a polarised abelian variety of dimension $n \geq 4$ and F a perfect complex with $\operatorname{ch}(F) = au_t + bv_t$, $(a,b) \neq (0,0)$, whose germ at each point is the germ of $\mathcal{I}_Z \otimes L$ for a subscheme Z of codimension two, and let $p \in Z$.*

(i) *If Z is not Cohen–Macaulay at p, then $\mathcal{E}xt^q(\mathcal{I}_Z,\mathcal{I}_Z)_p \neq 0$ for $q = \operatorname{pd}\mathcal{I}_{Z,p} \geq 2$, so the comparison of germs of Lemma 4.59 has room at p. If moreover Z has at p two smooth branches of codimension two meeting transversally, as in Lemma 4.60 and Corollary 4.63, then F does not satisfy the weakened criterion, nor does $\Phi(F \boxtimes F_2)$ for any secant F_2.*

(ii) *If Z is Cohen–Macaulay, then there is no local obstruction at any point. Condition (b) of Theorem 4.56(iii) is then a single global condition: it holds if and only if the image of $A_\theta A_{\theta'}$ in $H^1(Z,\mathcal{E}xt^1(\mathcal{I}_Z,\mathcal{I}_Z))$ vanishes for all $\theta,\theta' \in H^0(T_X)$, the scalar on the right of (19) being forced by the trace. Condition (a), the extension of F to first order along the polarised deformations, is not affected by the depth and remains to be checked.*

In case (ii), $\mathcal{I}_Z$ has projective dimension one at every point, so it is locally the ideal of maximal minors of a Hilbert–Burch matrix; if Z is moreover smooth, then $\mathcal{E}xt^1(\mathcal{I}_Z,\mathcal{I}_Z) = N_{Z/X}$ and the invariants of Z are those of Theorem 4.67.

Proof. (i) Put $A = \mathcal{O}_{X,p}$, of dimension n, and $I = \mathcal{I}_{Z,p}$. Since A/I is not Cohen–Macaulay, $\operatorname{depth} A/I < \dim A/I \leq n-2$, so $\operatorname{pd}_A(A/I) \geq 3$ by the Auslander–Buchsbaum formula and $q = \operatorname{pd}_A I \geq 2$. In a minimal free resolution $F_\bullet \to I$ the entries of the differentials lie in the maximal ideal $\mathfrak{m}$, so $\operatorname{Ext}^q_A(I,I)$, the cokernel of $\operatorname{Hom}(F_{q-1},I) \to \operatorname{Hom}(F_q,I)$, surjects onto $\operatorname{Hom}(F_q,I) \otimes A/\mathfrak{m} \neq 0$ by Nakayama's lemma. At two transversal branches the conclusion is Theorem 4.61 and Corollary 4.63. (ii) Theorem 4.75 gives $\mathcal{E}xt^q = 0$ for $q \geq 2$ at every point, so the local-to-global spectral sequence for $\operatorname{Ext}^2(\mathcal{I}_Z,\mathcal{I}_Z)$ has only the two rows $H^1(\mathcal{E}xt^1)$ and $H^2(\mathcal{O}_X)$, and these lie in different steps of the filtration, the second being the subgroup of scalars. The right side of (19) lies in that subgroup, so its image in the quotient $H^1(\mathcal{E}xt^1)$ is zero, and (b) forces the image of the left side there to vanish. Conversely, if that image vanishes, then $\operatorname{At}(F)^2 = z \cdot \mathrm{id}_F$ for some $z \in H^2(\mathcal{O}_X) \otimes \bigwedge^2 H^{1,0}$, the map $z \mapsto z \cdot \mathrm{id}_F$ being injective because its composite with the trace is multiplication by $\operatorname{rk} F = 1$; taking traces, $z = \operatorname{tr}\operatorname{At}(F)^2 = -d\,t^2$ by the last assertion of Theorem 4.56(iii), which is (b). The statement on projective dimension was established in the proof of Theorem 4.75, and the Hilbert–Burch theorem is the structure theorem for ideals of projective dimension one over a regular local ring; the last clause is Proposition 4.65 together with Theorem 4.67. ∎

Remark 4.78 (Normalisation and depth). Corollary 4.77 explains the tension recorded in Remark 4.62. The partial normalisation performed in Setup 4.51 separates the two branches at an isolated double point, and separating them is exactly what would make the local ring Cohen–Macaulay there; the normalisation is thus an attempt to reach case (ii). Remark 4.62 shows that the attempt cannot succeed: normalising every isolated point moves $\chi(\mathcal{O}_{\widetilde{Z}})$ to $50N$, while the secant plane demands $(d+9)(27-d) = 72N - 4N^2$, and the two agree only at $N = 11/2$. Repairing the depth destroys the Chern character, and keeping the Chern character keeps a point where the depth fails. This is the precise form of the obstruction, and it explains why a construction assembled from intersections of translates of a polarisation cannot be made to work: by Corollary 4.63 such a chain always has a point of the bad type, and by Remark 4.62 it cannot be separated without leaving the plane.

4.6. **Hilbert–Burch resolutions.** Corollary 4.77 leaves one class of supports and, beside the extension along the polarised deformations, one condition on them. The class consists of the Cohen–Macaulay subschemes of codimension two, and for these the proof of Theorem 4.75 gives more than the vanishing it was used for: the ideal sheaf has projective dimension one at every point. This is the Hilbert–Burch shape, whose invariants are polynomials in the twists.

Setup 4.79 (A resolution of the support)**.** Let (X,Θ) be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $Z \subset X$ be Cohen–Macaulay of codimension two, and suppose $\mathcal{I}_Z$ admits a global resolution

$$0 \longrightarrow E_1 \xrightarrow{\varphi} E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0, \qquad \operatorname{rk} E_0 = r+1, \quad \operatorname{rk} E_1 = r, \tag{27}$$

with $E_0 = \bigoplus_{i=1}^{r+1} \mathcal{O}(e_i\Theta)$ and $E_1 = \bigoplus_{j=1}^{r} \mathcal{O}(f_j\Theta)$ sums of line bundles. Put $p_i = e_i + b$ and $q_j = f_j + b$.

Lemma 4.80 (The resolution in moments)**.** *In Setup 4.79,* $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\,v_\Theta$ *holds if and only if*

$$\sum_{i=1}^{r+1} p_i^k - \sum_{j=1}^{r} q_j^k = c_k, \qquad k = 0,1,2,3,4, \qquad (c_k) = (1,\ b,\ -d,\ -db,\ d^2). \tag{28}$$

The equation at $k=0$ is the rank count and is automatic. In addition, φ can be injective only if $\max_i e_i \geq \max_j f_j$, and $\mathcal{O}_X$ embeds in $E_0^\vee$ only if $\min_i e_i \leq 0$.

Proof. From (27), $\operatorname{ch}(\mathcal{I}_Z) = \operatorname{ch}(E_0) - \operatorname{ch}(E_1)$, so $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = \sum_i e^{p_i\Theta} - \sum_j e^{q_j\Theta}$. Comparing coefficients of $\Theta^k/k!$ with those of $u_\Theta + bv_\Theta$, which are the c_k displayed, gives (28). For the last two statements, recall that on an abelian variety $\operatorname{Hom}(\mathcal{O}(f\Theta),\mathcal{O}(e\Theta)) = H^0(\mathcal{O}((e-f)\Theta))$ vanishes unless $e \geq f$. If φ is injective, each summand of E_1 must map nonzero to E_0, which requires $\max_i e_i \geq f_j$ for every j. Dualising (27) and using $\mathcal{I}_Z^\vee = \mathcal{O}_X$ gives an injection $\mathcal{O}_X \hookrightarrow E_0^\vee$, which requires $-e_i \geq 0$ for some i. ∎

The case $r=1$ consists of the complete intersections, and it is settled for every b and every d by an argument that needs no search.

Theorem 4.81 (No complete intersection supports a secant object)**.** *Let (X,Θ) be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $d>0$ and $b \geq 1$. No complete intersection $Z = D_1 \cap D_2$ of two divisors satisfies $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\,v_\Theta$.*

Proof. Such a Z is Setup 4.79 with $r=1$, the resolution being the Koszul complex of the two equations. Put $w(x) = x^2 + d$, which is strictly positive on $\mathbb{R}$ because $d>0$, and read (28) in the combinations $c_{k+2} + d\,c_k$ for $k = 0,1,2$. Those three numbers are

$$c_2 + dc_0 = -d + d = 0, \qquad c_3 + dc_1 = -db + db = 0, \qquad c_4 + dc_2 = d^2 - d^2 = 0,$$

so that

$$\sum_i w(p_i)\,p_i^k = \sum_j w(q_j)\,q_j^k, \qquad k = 0,1,2.$$

The two sides are the moments of order 0, 1 and 2 of the positive measures $\nu_P = \sum_i w(p_i)\delta_{p_i}$ and $\nu_Q = \sum_j w(q_j)\delta_{q_j}$, so those measures have the same mass, the same mean and the same variance. At $r=1$ the measure ν_Q has one atom, hence variance zero, so ν_P has variance zero as well, and a positive measure with two atoms has variance zero only if the atoms coincide. Hence $p_1 = p_2$, and equality of the means forces $q_1 = p_1$. The resolution (27) then cancels, and $\mathcal{I}_Z \cong \mathcal{O}(e_1\Theta)$ would be a line bundle, which the ideal sheaf of a subscheme of codimension two is not.

In terms of the divisors, the same computation shows that their degrees a_1 and a_2 would have to satisfy $a_1+a_2=\frac{4}{3}b$ and $(b-a_1)(b-a_2)=\frac{1}{2}(\frac{1}{3}b^2+d)$; the quadratic with roots $b-a_1$ and $b-a_2$ then has discriminant $-\frac{2}{9}b^2-2d<0$, so the degrees are not even real. ∎

For $r\geq 2$ the moment conditions alone no longer decide the question. The Chern characters of the bundles of Setup 4.79, and of every bundle built from line bundles, lie in one lattice, and the secant plane meets that lattice only along a congruence. Write a class $\xi\in\mathbb{Q}[\Theta]/(\Theta^5)$ as $\xi=\sum_{k=0}^{4}c_k(\xi)\,\Theta^k/k!$, as in Lemma 4.80, so that $c_k(e^{j\Theta})=j^k$, and let $\mathcal{L}_\Theta$ be the subgroup generated by the classes $e^{j\Theta}=\operatorname{ch}\mathcal{O}_X(j\Theta)$, $j\in\mathbb{Z}$. It is a subring, since $e^{a\Theta}e^{b\Theta}=e^{(a+b)\Theta}$, and it contains ch of every bundle obtained from line bundles algebraically equivalent to multiples of Θ by direct sums, tensor products, duals, exterior and symmetric powers, kernels of surjections and cokernels of injections between such bundles, and extensions; in particular the kernel bundles $M_{m\Theta}=\ker\bigl(H^0(\mathcal{O}(m\Theta))\otimes\mathcal{O}_X\to\mathcal{O}_X(m\Theta)\bigr)$, of class $m^4-e^{m\Theta}$, and their duals and twists.

Lemma 4.82 (Binomial moments)**.** *Define $m_k(\xi)\in\mathbb{Q}$, $0\leq k\leq 4$, by*

$$c_i(\xi)=\sum_{k=0}^{i}S(i,k)\,k!\,m_k(\xi),\tag{29}$$

where $S(i,k)$ are the Stirling numbers of the second kind; explicitly

$$m_0=c_0,\qquad m_1=c_1,\qquad m_2=\tfrac{1}{2}(c_2-c_1),$$
$$m_3=\tfrac{1}{6}(c_3-3c_2+2c_1),\qquad m_4=\tfrac{1}{24}(c_4-6c_3+11c_2-6c_1).$$

Then $\xi\in\mathcal{L}_\Theta$ if and only if $m_k(\xi)\in\mathbb{Z}$ for $k=0,\dots,4$.

Proof. For $\xi=\sum_j a_j e^{j\Theta}$ one has $c_i(\xi)=\sum_j a_j j^i$, and $j^i=\sum_k S(i,k)\,k!\binom{j}{k}$, so (29) holds with $m_k(\xi)=\sum_j a_j\binom{j}{k}\in\mathbb{Z}$. Conversely, the five vectors $\bigl(\binom{j}{0},\dots,\binom{j}{4}\bigr)$ for $j=0,\dots,4$ form a unitriangular matrix, so every integer vector $(m_0,\dots,m_4)$ is an integer combination of them, and the corresponding $\sum_{j=0}^{4}a_je^{j\Theta}$ lies in $\mathcal{L}_\Theta$ with the prescribed moments. Since (29) is invertible over $\mathbb{Q}$, the moments determine ξ. ∎

Equivalently, sending $e^{j\Theta}$ to x^j identifies $\mathbb{Q}[\Theta]/(\Theta^5)$ with $\mathbb{Q}[y]/(y^5)$, $y=x-1$, the class $e^{j\Theta}$ going to $(1+y)^j$; then $m_k(\xi)$ is the coefficient of y^k and $\mathcal{L}_\Theta$ is $\mathbb{Z}[y]/(y^5)$.

Theorem 4.83 (The secant plane and the lattice of line bundles)**.** *For $(a,b)\in\mathbb{Z}^2$ and $d\geq 1$,*

$$m(a\,u_\Theta+b\,v_\Theta)=\Bigl(a,\ b,\ -\frac{ad+b}{2},\ \frac{3ad+2b-bd}{6},\ \frac{ad^2-11ad+6bd-6b}{24}\Bigr).$$

Consequently $a\,u_\Theta+b\,v_\Theta\in\mathcal{L}_\Theta$ if and only if

$$ad\equiv b\pmod 2,\qquad 3ad+2b-bd\equiv 0\pmod 6,$$
$$ad^2-11ad+6bd-6b\equiv 0\pmod{24}.$$

For $a=1$ and $b=3$ the moments are $\bigl(1,3,-\frac{d+3}{2},1,\frac{(d+9)(d-2)}{24}\bigr)$, and $u_\Theta+3v_\Theta\in\mathcal{L}_\Theta$ if and only if d is odd and $24\mid(d+9)(d-2)$, that is, if and only if

$$d\equiv 15\ \text{or}\ 23\pmod{24}.$$

Proof. In the variable $y=x-1$ put $L=\log(1+y)=y-\frac{1}{2}y^2+\frac{1}{3}y^3-\frac{1}{4}y^4$ modulo y^5. Under $e^{j\Theta}\mapsto x^j$ the class Θ goes to L, hence $u_\Theta\mapsto 1-\frac{d}{2}L^2+\frac{d^2}{24}L^4$ and $v_\Theta\mapsto L-\frac{d}{6}L^3$. With $L^2=y^2-y^3+\frac{11}{12}y^4$, $L^3=y^3-\frac{3}{2}y^4$ and $L^4=y^4$ modulo y^5, the coefficients of

$1, y, \dots, y^4$ in $a\,u_\Theta + b\,v_\Theta$ are the five displayed expressions, and the congruences are the integrality of the last three. For $b = 3$ the second reads $3d+6-3d = 6$, so $m_3 = 1$, and the third reads $d^2 + 7d - 18 = (d+9)(d-2)$. Modulo 8 the condition is $8 \mid (d+1)(d-2)$ with d odd, hence $d \equiv 7 \pmod 8$; modulo 3 it is $3 \mid d(d-2)$, hence $d \equiv 0, 2 \pmod 3$; together $d \equiv 15, 23 \pmod{24}$. ∎

Corollary 4.84 (Split resolutions, for every rank and every twist)**.** *Let $Z \subset X$ be Cohen–Macaulay of codimension two with a resolution $0 \to E_1 \to E_0 \to \mathcal{I}_Z \to 0$ in which $\operatorname{ch} E_0$ and $\operatorname{ch} E_1$ both lie in $\mathcal{L}_\Theta$: for instance Setup 4.79 with any rank and any twists, or E_0 and E_1 built from such line bundles and from the bundles $M_{m\Theta}$ by the operations listed above. If $\operatorname{ch}(\mathcal{I}_Z(3\Theta)) = u_\Theta + 3v_\Theta$, then $d \equiv 15$ or $23 \pmod{24}$. In particular this never happens for the four discriminants of Corollary 4.68.*

Proof. $\operatorname{ch}(\mathcal{I}_Z(3\Theta)) = e^{3\Theta}(\operatorname{ch} E_0 - \operatorname{ch} E_1) \in \mathcal{L}_\Theta$, and Theorem 4.83 applies. ∎

Remark 4.85 (What the congruence measures). The fractional part of m_4 is the invariant that separates the secant plane from the line bundles. For $d \in \{1, 3, 5, 7\}$ it is $-\frac{5}{12}, \frac{1}{2}, \frac{3}{4}, \frac{1}{3}$ modulo $\mathbb{Z}$, and any object with Chern character $u_\Theta + 3v_\Theta$ must carry a class outside $\mathcal{L}_\Theta$ with exactly that defect. Two checks. A point has $m = (0, 0, 0, 0, 1/24)$, and the partial normalisation of Setup 4.51 at $(d+9)(2d-1)$ isolated points shifts m_4 by $(d+9)(2d-1)/24$; since $(d+9)(d-2) + (d+9)(2d-1) = 3(d+9)(d-1) \equiv 0 \pmod{24}$ for odd d, the unnormalised chain, which lies in $\mathcal{L}_\Theta$, is moved onto the plane by exactly the right defect, as it must be. For a smooth support the class of the structure sheaf has $m(\mathcal{O}_S) = \big(0, 0, N, -3N, N(83-2N)/12\big)$, and $N(83-2N)/12$ is not an integer for $N \in \{5, 6, 7, 8\}$; so a smooth secant support is a surface whose structure sheaf is not in the line-bundle subring, unlike every complete intersection of translates of Θ and every degeneracy locus of a map between bundles in $\mathcal{L}_\Theta$. The Fourier–Mukai transforms of line bundles and the push-forwards of line bundles along isogenies do have classes outside $\mathcal{L}_\Theta$, which is why the semi-homogeneous bundles need the separate argument of Theorem 4.92.

The lattice constrains the classes of the two terms of a resolution; their ranks constrain the Chern classes, because a vector bundle of rank s has $c_k = 0$ for $k > s$, and on a fourfold this bites for $s \le 3$.

Proposition 4.86 (Resolutions of rank one and two)**.** *Let (X, Θ) be a principally polarised abelian fourfold, let $b \ge 1$ and $d > 0$, and let $Z \subset X$ be of codimension two with a resolution*

$$0 \longrightarrow E_1 \longrightarrow E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0$$

by vector bundles of ranks r and $r+1$, and with $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\,v_\Theta$.

(i) *$r \ge 2$, whatever the bundles. Equivalently, Z is not the zero locus of a section of a vector bundle of rank two.*

(ii) *If $r = 2$ and the Chern classes of E_1 are polynomials in Θ, as they are when $\operatorname{Hdg}^2(X) = \mathbb{Q}\Theta$ and $\operatorname{Hdg}^4(X) = \mathbb{Q}\Theta^2$, then, writing $c_1(E_1(b\Theta)) = a\Theta$ with $a \in \mathbb{Z}$,*

$$3d - b^2 - 4ab \equiv 0 \pmod{12}.$$

At $b = 3$ this says $d \equiv 3 \pmod 4$.

In particular, at the discriminants $d = 1$ and $d = 5$ of Corollary 4.68, a resolution whose Chern classes are polynomials in Θ has $r \ge 3$.

Proof. Put $E = E_1(b\Theta)$, $G = E_0(b\Theta)$ and $F = \mathcal{I}_Z(b\Theta)$, so that $0 \to E \to G \to F \to 0$ is exact and $c(G) = c(E)\,c(F)$. Newton's identities applied to $\operatorname{ch}(F) = 1 + b\Theta - \frac{d}{2}\Theta^2 -$

$\frac{db}{6}\Theta^3 + \frac{d^2}{24}\Theta^4$ give, with $N = \frac{1}{2}(b^2+d)$,

$$c(F) = 1 + b\Theta + N\Theta^2 + \tfrac{bN}{3}\Theta^3 + \tfrac{N(b^2-3d)}{12}\Theta^4.$$

Since E and G are vector bundles, $c_k(E) = 0$ for $k > r$ and $c_k(G) = 0$ for $k > r+1$.

(i) Let $r = 1$ and $\ell = c_1(E)$, so that $c(G) = (1+\ell)\,c(F)$. The rank two bundle G has $c_3(G) = \frac{bN}{3}\Theta^3 + N\ell\,\Theta^2 = 0$, and cup product with Θ^2 is an isomorphism $H^2(X,\mathbb{Q}) \to H^6(X,\mathbb{Q})$ by hard Lefschetz, so $\ell = -\frac{b}{3}\Theta$. Then

$$c_4(G) = c_4(F) + \ell\,c_3(F) = N\Big(\frac{b^2-3d}{12} - \frac{b^2}{9}\Big)\Theta^4 = -\frac{N(b^2+9d)}{36}\,\Theta^4,$$

which is not zero, since $N > 0$ and $d > 0$. A section of a rank two bundle V vanishing in codimension two exactly on Z gives the Koszul resolution $0 \to \det V^\vee \to V^\vee \to \mathcal{I}_Z \to 0$, which has $r = 1$; conversely, at $r = 1$ the map $E_1 \to E_0$ is a section of $E_0 \otimes E_1^\vee$ with zero locus Z.

(ii) Let $r = 2$, and write $c_1(E) = a\Theta$ and $c_2(E) = \frac{m}{2}\Theta^2$ with $a, m \in \mathbb{Z}$, since Θ and $\frac{1}{2}\Theta^2$ are primitive integral classes. Now $c_3(E) = c_4(E) = 0$, and the rank three bundle G has

$$c_4(G) = c_4(F) + c_1(E)\,c_3(F) + c_2(E)\,c_2(F) = 0.$$

Integrating, with $\int_X \Theta^4 = 24$, and dividing by $2N$ gives $6m = 3d - b^2 - 4ab$. By Riemann–Roch with trivial Todd class, $\chi(E) = \int_X \mathrm{ch}_4(E) = a^4 - 2a^2m + \frac{1}{2}m^2$ is an integer, so m is even, which is the congruence. At $b = 3$ it reads $3(d-3) \equiv 12a \equiv 0$ modulo 12, that is, $d \equiv 3 \pmod 4$. ∎

Remark 4.87 (What the ranks exclude). Part (i) extends Theorem 4.81 from two divisors to the zero locus of a section of any rank two bundle, stable or not, so the construction of Serre produces no secant support; it holds for every X, with no condition on $\mathrm{NS}(X)$. Part (ii) holds for arbitrary bundles whose Chern classes are polynomials in Θ, not only for sums of line bundles, and at $b = 3$ it excludes $r = 2$ whenever $d \not\equiv 3 \pmod 4$. The integrality of the Chern classes of E_0 is automatic, since $c(E_0(b\Theta)) = c(E_1(b\Theta))\,c(F)$ and $c(F)$ is integral, so it is the vanishing above the rank, and not integrality, that constrains. Neither condition restricts $r \geq 3$: a bundle of rank four has no vanishing Chern class on a fourfold, and at $r = 3$ the only condition, $c_4(E) = 0$, involves E alone.

Corollary 4.84 says where a split resolution can have the right Chern character. A stronger statement holds: no object with a split resolution satisfies the criterion, at any rank and for any discriminant (Theorem 4.89). It rests on a necessary condition that we isolate, because it applies to every object and can be tested on any candidate before any global computation is attempted.

Lemma 4.88 (Morphisms to a line bundle are killed by $H^2(\mathcal{O})$)**.** *Let (X,t) be a polarised abelian variety of dimension $n \geq 3$, let F be a perfect complex with $\mathrm{ch}(F) = au_t + bv_t$, $(a,b) \neq (0,0)$, which satisfies the weakened criterion of Setup 4.54, and let L be a line bundle with $c_1(L) \in \mathbb{Q}\,t$. Then for every $m \in \mathbb{Z}$, every morphism $p\colon F \to L[m]$, every morphism $q\colon L[m] \to F$, and every $z \in H^2(\mathcal{O}_X)$,*

$$z \cdot p = 0 \quad \textit{in } \mathrm{Hom}(F, L[m+2]), \qquad z \cdot q = 0 \quad \textit{in } \mathrm{Hom}(L[m], F[2]),$$

where $z \cdot (\,\cdot\,)$ is the $H^*(\mathcal{O}_X)$-module structure on morphisms in $D^b(X)$. In particular, if $F = \mathcal{I}_Z \otimes L'$ for a closed subscheme Z and a line bundle L' with $c_1(L') \in \mathbb{Q}t$, then the restriction $H^{n-2}(\mathcal{O}_X) \to H^{n-2}(\mathcal{O}_Z)$ is injective.

Proof. For a constant bivector π the class $x_\pi = (\frac{d}{2}\,\pi\lrcorner t^2, 0, \pi)$ preserves $\mathrm{ch}(F)$ by Theorem 4.56(ii), so $\mathrm{ev}_F(x_\pi) = 0$ by hypothesis. The Hochschild action is by natural transformations $\mathrm{id} \to \mathrm{id}[2]$ of $D^b(X)$, so for $p\colon F \to L[m]$ we have $\mathrm{ev}_{L[m]}(x_\pi)\circ p = p \circ \mathrm{ev}_F(x_\pi) = 0$, where $\mathrm{ev}_{L[m]} = \mathrm{ev}_L[m]$. By (11), $\mathrm{ev}_L(x_\pi) = c_\pi \cdot \mathrm{id}_L$ with $c_\pi = \frac{1}{2}\,\pi\lrcorner(\lambda^2 + d\,t^2) \in H^{0,2}(X)$ and $\lambda = c_1(L) = e\,t$, that is, $c_\pi = \frac{e^2+d}{2}\,\pi\lrcorner t^2$. Since $e^2 + d > 0$ and $\pi \mapsto \pi\lrcorner t^2$ is an isomorphism from $H^0(\bigwedge^2 T_X)$ onto $H^{0,2}(X)$, as in the proof of Theorem 4.56(iii), c_π runs over all of $H^2(\mathcal{O}_X)$ as π varies; as $(c_\pi \cdot \mathrm{id}_L)\circ p = c_\pi \cdot p$, this proves the assertion for p. The assertion for q follows in the same way from the square $\mathrm{ev}_F(x_\pi)\circ q = q \circ \mathrm{ev}_{L[m]}(x_\pi)$. For the last assertion, apply the first to the inclusion $p\colon \mathcal{I}_Z \otimes L' \to L'$. After twisting by L'^{-1}, the sequence $0 \to \mathcal{I}_Z \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$ gives an exact sequence

$$\mathrm{Ext}^2(\mathcal{O}_Z, \mathcal{O}_X) \longrightarrow H^2(\mathcal{O}_X) \longrightarrow \mathrm{Ext}^2(\mathcal{I}_Z, \mathcal{O}_X),$$

in which the second map is $z \mapsto z \cdot p$. So $z \cdot p = 0$ for all z means that the first map is surjective, and by Serre duality, ω_X being trivial, the first map is dual to the restriction $H^{n-2}(\mathcal{O}_X) \to H^{n-2}(\mathcal{O}_Z)$. ∎

Theorem 4.89 (No two-term complex of powers of the polarisation satisfies the criterion)**.** *Let (X, t) be a polarised abelian variety of dimension $n \geq 3$ and M an ample line bundle with $c_1(M) \in \mathbb{Q}t$. Let A and B be direct sums of tensor powers $M^{\otimes e}$, $e \in \mathbb{Z}$, let $\varphi\colon A \to B$ be any morphism, and let F be the cone of φ, or any shift of it. If $F \neq 0$ and $\mathrm{ch}(F) = au_t + bv_t$ with $(a, b) \neq (0, 0)$, then F does not satisfy the weakened criterion. In particular, no object with a resolution $0 \to E_1 \to E_0 \to F \to 0$ by sums of powers of M satisfies it, whatever the rank.*

Proof. A pair of summands $M^{\otimes e}$ of A and $M^{\otimes e}$ of B joined by a nonzero entry of φ, which is then a nonzero scalar, can be cancelled without changing the cone up to isomorphism; do this until no such pair remains. If $A = 0$ or $B = 0$ after cancelling, F is a shifted sum of line bundles, $\mathrm{ev}_F(x_\pi)$ is diagonal with the nonzero entries $c_\pi \cdot \mathrm{id}$ of the proof of Lemma 4.88, and the criterion fails as in Corollary 4.21; so suppose both are nonzero and consider the triangle $A \xrightarrow{\varphi} B \xrightarrow{q} F \to A[1]$. Let $M^{\otimes b_j}$ be a summand of B, ι_j its inclusion into B, and $q_j = q \circ \iota_j\colon M^{\otimes b_j} \to F$. Applying $\mathrm{Hom}(M^{\otimes b_j}, -)$ to the triangle gives the exact sequence

$$\mathrm{Hom}(M^{\otimes b_j}, A[2]) \xrightarrow{\varphi_*} \mathrm{Hom}(M^{\otimes b_j}, B[2]) \xrightarrow{q_*} \mathrm{Hom}(M^{\otimes b_j}, F[2]),$$

and $z \cdot q_j = q_*(z \cdot \iota_j)$ for $z \in H^2(\mathcal{O}_X)$. Now $\mathrm{Hom}(M^{\otimes b_j}, B[2]) = \bigoplus_k H^2(M^{\otimes(b_k - b_j)})$, and $H^2(M^{\otimes m}) = 0$ for $m \neq 0$: for $m > 0$ by Kodaira vanishing, ω_X being trivial, and for $m < 0$ because a negative line bundle on an abelian variety has cohomology only in degree $n \geq 3$. So $z \cdot \iota_j$ is the element with component z in the summand $k = j$ and zero elsewhere. It lies in the image of φ_* only if there is $\psi \in \mathrm{Hom}(M^{\otimes b_j}, A[2]) = \bigoplus_i H^2(M^{\otimes(a_i - b_j)})$ with $(\varphi \circ \psi)_j = \sum_i \varphi_{ji}\psi_i = z$; but $\psi_i \neq 0$ forces $a_i = b_j$, and for such i the entry φ_{ji} is a scalar, zero after the cancellation. So $z \cdot q_j \neq 0$ for every $z \neq 0$, which contradicts Lemma 4.88. The last assertion is the case of a resolution, where $F \cong \mathrm{coker}\,\varphi$ is the cone of $\varphi\colon E_1 \to E_0$. ∎

The powers of the polarisation are not the only bundles on which the compensated Poisson classes act by scalars. The same holds for every bundle whose Atiyah class is a multiple of the identity, and on an abelian variety these are the semi-homogeneous bundles of Mukai [Muk78]. The argument of Theorem 4.89 goes through for them, with the cohomology of the fibre product of two isogenies in place of the cohomology of a power of Θ. We state the result in a self-contained form.

Definition 4.90 (Isogeny blocks)**.** Let (X,Θ) be a polarised abelian variety. An *isogeny block* is a vector bundle $E=\pi_*L$, where $\pi\colon Y\to X$ is an isogeny and L is a line bundle on Y with $c_1(L)=\nu\,\pi^*\Theta$ in $H^2(Y,\mathbb{Q})$, $\nu\in\mathbb{Q}$; its *slope* is ν, so that $c_1(E)=\nu\deg(\pi)\,\Theta$. The block is *simple* if $\tau_g^*L\not\cong L$ for every nonzero $g\in\ker\pi$. The powers $\mathcal{O}(e\Theta)$ are the blocks with $\pi=\mathrm{id}$.

Lemma 4.91 (Three properties of a block)**.** *Let $E=\pi_*L$ be an isogeny block of slope ν on an abelian n-fold.*

(i) $\mathrm{At}(E)=\nu\Theta\cdot\mathrm{id}_E$*; hence for a constant bivector π the compensated class x_π of Theorem 4.56(ii) acts by* $\mathrm{ev}_E(x_\pi)=\frac{\nu^2+d}{2}\,(\pi\lrcorner\Theta^2)\cdot\mathrm{id}_E$*, and Lemma 4.88 holds with E in place of the line bundle L.*

(ii) *For a second block $E'=\pi'_*L'$ of slope ν', $\mathrm{Ext}^q(E',E)=H^q(Z,M)$ with $Z=Y'\times_X Y$, a disjoint union of abelian varieties isogenous to X, and M a line bundle whose class on each component is $(\nu-\nu')$ times the pullback of Θ. If $\nu\neq\nu'$ and $n\geq 3$ then $\mathrm{Ext}^2(E',E)=0$. If E is simple then $\mathrm{Ext}^2(E,E)=H^2(\mathcal{O}_X)\cdot\mathrm{id}_E$.*

(iii) *A simple block is μ-stable with respect to Θ; hence a nonzero morphism between simple blocks of the same slope is an isomorphism.*

Proof. (i) Since π is étale, $\pi^*\Omega^1_X=\Omega^1_Y$ and π_* is exact, so the first jet sequence of π_*L is the direct image of that of L, and $\mathrm{At}(\pi_*L)$ is the image of $\mathrm{At}(L)$ under $\pi_*\colon \mathrm{Ext}^1_Y(L,L\otimes\pi^*\Omega^1_X)\to\mathrm{Ext}^1_X(\pi_*L,\pi_*L\otimes\Omega^1_X)$. As $\mathrm{At}(L)=c_1(L)\cdot\mathrm{id}_L=\nu\,\pi^*\Theta\cdot\mathrm{id}_L$, the projection formula gives $\nu\Theta\cdot\mathrm{id}_{\pi_*L}$. Then (15) with a scalar Atiyah class gives the value of $\mathrm{ev}_E(x_\pi)$, and the proof of Lemma 4.88 used nothing else about L.

(ii) Adjunction for the finite étale π', flat base change along the cartesian square with vertices Z, Y', Y, X, and the projection formula give $\mathrm{Ext}^q_X(\pi'_*L',\pi_*L)=H^q(Y',L'^{-1}\otimes\pi'^*\pi_*L)=H^q(Z,p'^*L'^{-1}\otimes p^*L)$, with p,p' the two projections; the class of $M=p'^*L'^{-1}\otimes p^*L$ is $(\nu-\nu')$ times the pullback of Θ on every component, which is an abelian variety finite étale over Y'. A line bundle on an abelian n-fold whose class is a positive multiple of an ample class has cohomology only in degree 0, and one whose class is a negative multiple only in degree n, by Kodaira vanishing and Serre duality with trivial canonical bundle; so $H^2(Z,M)=0$ when $\nu\neq\nu'$ and $n\geq 3$. When $E'=E$, the components of $Y\times_X Y$ are the graphs of the translations by $g\in\ker\pi$, on which M restricts to $L^{-1}\otimes\tau_g^*L\in\mathrm{Pic}^0(Y)$; a nontrivial element of Pic^0 has no cohomology, so $\mathrm{Ext}^q(E,E)=\bigoplus_g H^q(\mathcal{O}_Y)$ over the g with $\tau_g^*L\cong L$, which for a simple block is $g=0$ alone, and $H^q(\mathcal{O}_Y)=H^q(\mathcal{O}_X)$ through π.

(iii) The pullback $\pi^*E=\bigoplus_{g\in\ker\pi}\tau_g^*L$ is a direct sum of line bundles of one slope, hence μ-semistable, and semistability descends along the finite étale π; so E is semistable. If $F\subset E$ is a saturated subsheaf of the same slope, then $\pi^*F\subset\pi^*E$ is a saturated subsheaf of the same slope of a direct sum of pairwise non-isomorphic line bundles, the τ_g^*L being distinct because the block is simple, hence a partial sum $\bigoplus_{g\in S}\tau_g^*L$; it is invariant under the translations by $\ker\pi$, which permute the summands transitively, so S is empty or everything and $F=0$ or $F=E$. Finally, a nonzero morphism f between μ-stable bundles of the same slope is injective and generically surjective, so the two bundles have the same rank; for blocks of the same slope ν they then have the same c_1, since $c_1(E)=\nu\,\mathrm{rk}(E)\,\Theta$, and $\det f$ is a nonzero section of a numerically trivial line bundle, hence nowhere zero. So f is an isomorphism. ∎

Theorem 4.92 (No two-term complex of simple blocks satisfies the criterion)**.** *Let (X,Θ) be a polarised abelian variety of dimension $n\geq 3$, let A and B be direct sums of simple isogeny blocks, let $\varphi\colon A\to B$ be any morphism and let F be the cone of φ or a shift of it. If $F\neq 0$ and $\mathrm{ch}(F)=a u_\Theta+b v_\Theta$ with $(a,b)\neq(0,0)$, then F does not satisfy the weakened*

criterion. In particular, no object with a resolution $0 \to E_1 \to E_0 \to F \to 0$ *by sums of simple isogeny blocks satisfies it, whatever the rank.*

Proof. Cancel, by Gaussian elimination in the two-term complex, every pair of isomorphic blocks of A and B joined by an entry of φ that is an isomorphism; this changes neither the cone nor the block structure of the remaining terms. By Lemma 4.91(iii), every remaining entry between blocks of the same slope is zero. If A or B is empty after cancelling, F is a shifted sum of blocks, on each of which $\mathrm{ev}(x_\pi)$ is a nonzero scalar by (i). Otherwise the rest is the proof of Theorem 4.89 with Lemma 4.91 in place of the vanishing of $H^2(M^{\otimes m})$. For a block E' of B with inclusion ι and $q' = q \circ \iota \colon E' \to F$, the class $z \cdot \iota \in \mathrm{Hom}(E', B[2])$ has component $z \cdot \mathrm{id}_{E'} \neq 0$ in $\mathrm{Ext}^2(E', E') = H^2(\mathcal{O}_X) \cdot \mathrm{id}_{E'}$ by (ii). It lies in the image of φ_* only if $\sum_i \varphi_{ji}\psi_i = z \cdot \mathrm{id}_{E'}$ for some $\psi_i \in \mathrm{Ext}^2(E', A_i)$, and $\psi_i \neq 0$ forces A_i to have the slope of E' by (ii), for which $\varphi_{ji} = 0$. So $z \cdot q' \neq 0$, contradicting Lemma 4.88 in the form given by (i). ∎

Remark 4.93 (Semi-homogeneous bundles). By Mukai [Muk78], the simple isogeny blocks are exactly the simple semi-homogeneous bundles, a bundle E being semi-homogeneous if for every x there is $P \in \mathrm{Pic}^0(X)$ with $\tau_x^* E \cong E \otimes P$; and every semi-homogeneous bundle on a variety with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$ is an iterated extension of such blocks and their twists by Pic^0. We do not use this. In these terms, Theorem 4.92 says that the bundles in a Hilbert–Burch resolution of a candidate cannot all have scalar Atiyah class. The obstruction is always the same: the compensated Poisson classes act on a block by the scalar $\frac{\nu^2+d}{2}\,\pi_\lrcorner\,\Theta^2$, which would vanish only for $\nu^2 = -d$, and cones of maps between blocks inherit the scalar through the summand inclusions, because Ext^2 between blocks of different slopes vanishes. A candidate must therefore be built from a bundle whose Atiyah class is not a scalar, that is, one which is not projectively flat.

Corollary 4.94 (Split resolutions and the remaining shapes)**.** *Let* (X, Θ) *be a principally polarised abelian fourfold.*

(i) *No secant object of rank one whose support has a resolution* (27) *by sums of powers of* Θ *satisfies the weakened criterion, for any* r*, any twists and any* d*, whether or not its matrix degenerates in codimension two.*

(ii) *If a secant object* $\mathcal{I}_Z(b\Theta)$ *satisfies the criterion, then* Z *has no point at which two smooth branches of codimension two meet transversally, no two-term resolution of* $\mathcal{I}_Z$ *splits into powers of* Θ *or into simple semi-homogeneous bundles, and the restriction* $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_Z)$ *is injective. If* Z *is Cohen–Macaulay,* $\mathcal{I}_Z$ *has projective dimension one. If* $b = 3$ *and* Z *is smooth, then* $d \leq 9$*, so* $d \in \{1, 3, 5, 7\}$ *when* $N = (9 + d)/2$ *is an integer as in Setup 4.51, the two terms of a resolution do not both have classes in* $\mathcal{L}_\Theta$*, and when* X *is simple the injectivity is automatic.*

(iii) *At* $b = 3$*, smooth supports and split resolutions are disjoint, for every rank and every twist of the resolution: a smooth support forces* $d \leq 9$*, and a resolution by bundles with classes in* $\mathcal{L}_\Theta$ *forces* $d \equiv 15, 23 \pmod{24}$*, hence* $d \geq 15$*, so that, even without (i), no support could have both properties.*

Proof. Part (i) is Theorem 4.89 with $M = \mathcal{O}_X(\Theta)$, the resolution (27) presenting $\mathcal{I}_Z(b\Theta)$ as the cone of $\varphi \otimes \mathcal{O}(b\Theta)$. In (ii), the branches are excluded by Corollary 4.77(i), projective dimension one comes from Corollary 4.77(ii), the non-splitting from Theorem 4.89 and Theorem 4.92, the injectivity from the last assertion of Lemma 4.88, the bound on d from Corollary 4.68, the statement on the two terms from Corollary 4.84, and the automatic injectivity from Lemma 4.70. Part (iii) compares Corollary 4.68 with Corollary 4.84: no $d \leq 9$ is congruent to 15 or 23 modulo 24. ∎

5. Weil classes of CM fields and the polarised criterion

5.1. **Weil tori of a CM field and the pure form.** The argument of Theorem 3.5 uses the field only through the eigenspaces of its action, and it extends to every CM field. For $m \geq 2$ the Hodge ring of a very general Weil torus is larger than for an imaginary quadratic field, since products of Weil classes over a set of embeddings closed under conjugation are Hodge classes there, and a Weil torus may carry subvarieties of codimension $2n$; but it carries none of codimension n, and that is all the argument needs.

Setup 5.1 (Weil tori of a CM field)**.** Keep [I, Setup 17.1], let $n \geq 2$ and let A be of (F, n, δ)-Weil type, with $V_{\mathbb{C}} = \bigoplus_{\sigma} V_{\sigma}$ as there. Complex conjugation carries $V_{\overline{\sigma}}$ onto V_{σ}. A complex structure on $V_{\mathbb{R}}$ commuting with F, with $H^{1,0} \cap V_{\sigma}$ of dimension n for every σ, is the same as a choice, for each real place τ, of complementary n-dimensional subspaces $X_{\tau}, Y_{\tau} \subset V_{\sigma_{\tau}}$: then $H^{1,0} \cap V_{\sigma_{\tau}} = X_{\tau}$, $H^{0,1} \cap V_{\sigma_{\tau}} = Y_{\tau}$, and on $V_{\overline{\sigma}_{\tau}}$ the two parts are the conjugates of Y_{τ} and X_{τ}. Call the complex torus obtained in this way a Weil torus of F, and let S_F be the connected complex manifold of these choices, of dimension $2mn^2$. Its tangent space at A is $\mathrm{Hom}_F(H^{1,0}, H^{0,1})$, the polarised family $\mathcal{D}_F$ is the submanifold on which E is of type $(1,1)$, and S_F contains non-algebraic tori. For $\sigma = \sigma_{\tau}$ the line $\bigwedge^{2n} V_{\sigma} = \bigwedge^{n} X_{\tau} \otimes \bigwedge^{n} Y_{\tau}$ is of type (n,n), and likewise for $\overline{\sigma}_{\tau}$, so W_F consists of Hodge classes on every member of S_F. For a set Σ of embeddings write $\alpha_{\Sigma} = \prod_{\sigma \in \Sigma} \alpha_{\sigma}$, with α_{σ} spanning $\bigwedge^{2n} V_{\sigma}$.

Lemma 5.2 (Hodge classes and subvarieties of a very general Weil torus)**.** *Let T be a very general member of S_F and $d = 2mn = \dim T$.*

(i) *Every Hodge class of T lies in the span of the α_{Σ}. In particular $\mathrm{Hdg}^k(T) = 0$ unless n divides k, and $\mathrm{Hdg}^n(T) = W_F(T)$.*

(ii) *Let κ_0 be the Kähler class of a hermitian form on the tangent space that is block diagonal for the splitting of $H^{1,0}$ into the $H^{1,0} \cap V_{\sigma}$. Then $\int_T \alpha_{\Sigma}\, \kappa_0^{d-n|\Sigma|} = 0$ unless Σ is closed under conjugation.*

(iii) *Every irreducible analytic subset of T other than a point and T itself has codimension ns with s even, hence at least $2n$.*

Proof. (i) As in the proof of Lemma 3.2: a rational class of Hodge type at a very general member is of Hodge type on all of S_F, and the Mumford–Tate group M of a very general member lies in $G = \mathrm{Res}_{F/\mathbb{Q}} \mathrm{GL}_F(V)$, since F acts by endomorphisms. Here $G(\mathbb{R}) = \prod_{\tau} \mathrm{GL}(V_{\sigma_{\tau}})$ acts transitively on S_F, and the $\mathrm{Ad}\, G(\mathbb{R})$-span of the complex structure of one member is an ideal of $\bigoplus_{\tau} \mathfrak{gl}_{2n}(\mathbb{C})$ whose projection to every factor contains a non-central element, i on X_{τ} and $-i$ on Y_{τ}. An ideal of a sum of simple real Lie algebras is a sum of some of them, and each $\mathfrak{sl}_{2n}(\mathbb{C})$ is simple as a real Lie algebra, so the span contains $\bigoplus_{\tau} \mathfrak{sl}_{2n}(\mathbb{C})$ and M contains $\mathrm{Res}_{F/\mathbb{Q}} \mathrm{SL}_F(V)$. Over $\mathbb{C}$ this group is $\prod_{\sigma} \mathrm{SL}(V_{\sigma})$ acting on $\bigwedge^{*} V_{\mathbb{C}} = \bigotimes_{\sigma} \bigwedge^{*} V_{\sigma}$, and the invariants of $\mathrm{SL}(V_{\sigma})$ in $\bigwedge^{*} V_{\sigma}$ are $\bigwedge^{0}$ and $\bigwedge^{2n}$. So the invariants are spanned by the α_{Σ}, of degree $2n|\Sigma|$, and a rational class in the span of the α_{σ} lies in W_F.

(ii) Choose the hermitian form so that $\kappa_0 = \sqrt{-1} \sum_{\sigma} \sum_{k} \xi_{\sigma,k} \wedge \overline{\xi}_{\sigma,k}$ with $\xi_{\sigma,k}$ a basis of $H^{1,0} \cap V_{\sigma}$; each term lies in $V_{\sigma} \otimes V_{\overline{\sigma}}$. A product of classes has nonzero integral only if its degree in every V_{ρ} is $2n$. The factor α_{Σ} has degree $2n$ in V_{ρ} for $\rho \in \Sigma$ and 0 otherwise, and every term of a power of κ_0 has the same degree in V_{ρ} as in $V_{\overline{\rho}}$. For $\rho \in \Sigma$ that degree must be 0, so $V_{\overline{\rho}}$ must be filled by α_{Σ}, that is $\overline{\rho} \in \Sigma$.

(iii) Let Z be irreducible of codimension c with $0 < c < d$. For a Kähler class κ, $\int_T [Z]\, \kappa^{d-c} = \int_Z \kappa^{d-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$, and by (i) $c = ns$

and $[Z]$ is a combination of the α_Σ with $|\Sigma| = s$. Complex conjugation permutes the embeddings without fixed points, so a set closed under it has even size; if s were odd, (ii) would give $\int_T [Z]\,\kappa_0^{d-c} = 0$. ∎

Proposition 5.3 (Coherent sheaves on a very general Weil torus of F)**.** *Let T be a very general member of S_F. Every coherent analytic sheaf $\mathcal{F}$ on T has $\mathrm{ch}_k(\mathcal{F}) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \le k \le 2n-1$.*

Proof. The Chern classes of [Gri10] are Hodge classes, additive, and compatible with restriction to open subsets, so by Lemma 5.2(i) only $k = n$ needs an argument. First let $\mathcal{F}$ be supported on a proper analytic subset W. Its restriction to $T \setminus W$ is zero, so $\mathrm{ch}_n(\mathcal{F})$ comes from $H^{2n}_W(T, \mathbb{Q})$, which is isomorphic to the Borel–Moore homology $H_{2d-2n}(W, \mathbb{Q})$; by Lemma 5.2(iii) every component of W has codimension at least $2n$, so W has real dimension at most $2d - 4n < 2d - 2n$ and that group vanishes. In general, the torsion of $\mathcal{F}$ is of this kind, and so is the cokernel of the inclusion of a torsion-free sheaf in its reflexive hull. Every coherent sheaf has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$, so as in the proof of Proposition 3.4 a torsion-free sheaf has a Jordan–Hölder filtration whose factors Q_j are stable for every Kähler class, and $\mathrm{ch}_n(Q_j) = \mathrm{ch}_n(Q_j^{**})$. By Bando and Siu [BS94], $\int_T c_2(Q_j^{**})\,\kappa_0^{d-2} \ge 0$, with equality exactly when Q_j^{**} is a projectively flat bundle. Here $c_2(Q_j^{**})$ lies in $\mathrm{Hdg}^2(T)$, which is zero for $n \ge 3$ and is $W_F(T)$ for $n = 2$, where the integral vanishes by Lemma 5.2(ii) with $|\Sigma| = 1$. So Q_j^{**} is projectively flat with $c_1 = 0$, hence flat, and $\mathrm{ch}_n(Q_j) = 0$. Summing over the factors and adding the torsion gives $\mathrm{ch}_n(\mathcal{F}) = 0$. ∎

Theorem 5.4 (The pure form is never met, for any CM field)**.** *Let $n \ge 2$, let A be of (F, n, δ)-Weil type, and let E be a perfect complex on A whose Chern character stays of Hodge type on every member of S_F and has $\mathrm{ch}_n(E) \ne 0$; for instance $\mathrm{ch}(E) = c + N\omega$ with $c \in \mathbb{Q}$, $N \ne 0$ and ω a nonzero rational Weil class. Then σ_E is not injective, and $\dim \mathrm{Ext}^2(E, E) > r(\mathrm{ch}\,E)$. In particular no complex of the pure form $\mathrm{ch}(E) = N\omega_{s_0}$ in* [I, Theorem 17.7(v)]*, with or without a rank, has injective semiregularity map, whatever the CM field.*

Proof. The proof of Theorem 3.5 applies with S_F in place of S: an injective σ_E would let E deform, by the theorem of Buchweitz and Flenner in Pridham's form and Kuranishi's method, to a bounded complex of holomorphic vector bundles on every member of a neighbourhood of A in S_F, with Chern character the parallel transport of $\mathrm{ch}(E)$; at a very general member ch_n of that complex vanishes by Proposition 5.3, while it is the transport of $\mathrm{ch}_n(E) \ne 0$. A class $c + N\omega$ is of Hodge type on all of S_F by Setup 5.1. The inequality follows from the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$, since $\dim \mathrm{Ext}^2(E, E) \ge \mathrm{rk}\,\mathrm{ev}_E \ge r(\mathrm{ch}\,E)$ and equality in both forces ev_E onto with kernel the annihilator, and then σ_E injective, as in Theorem 3.9(iv). ∎

For $m = 1$ this is Theorem 3.5, and for $m = 1$, $n = 2$ the vanishing of Proposition 5.3 is Voisin's theorem [Voi02b, Theorem 10]. So the criterion of [I, Theorem 17.7(v)] needs a polynomial P in the classes η_t that is not constant, for every CM field and every $n \ge 2$. At a quartic field and $n = 2$ this pushes the least possible value of $\dim \mathrm{Ext}^2(E, E)$ for a complex meeting it from 80 to 88 (Corollary 5.15).

5.2. **Quartic CM fields at $n = 2$.** By [III, Theorem 2.70(ii)] the only form of (P2) on a minimal route to the conjecture is the one for the split families of the CM fields of degree at least four, and by [III, Proposition 2.13(i)] the Weil classes of any family of such a field in a higher dimension give those of its families at $n = 2$. For a quartic field these are

families of abelian eightfolds, the smallest families of a field of degree at least four. This subsection determines how far the available mechanisms reach there.

Divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, which has codimension two when δ is trivial (Proposition 5.6, Corollary 5.7). For biquadratic F, Markman's theorem reaches them on a further locus of dimension four (Proposition 5.8). For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial δ, Schoen's cycles on Prym varieties of cyclic covers reach them on a locus of dimension at most six, at whose very general points the Hodge group is all of G_F (Remark 5.9). The Casimir class of the quadratic form on the weight two part R_F below, the natural class of degree four attached to it, is algebraic exactly when the Weil classes are (Proposition 5.10), and correspondences with abelian varieties whose simple factors have dimension at most seven cannot help at a member with full Hodge group (Proposition 5.12). For a general shape the polarised criterion asks $\dim \mathrm{Ext}^2(E,E) = 112$ of one complex (Theorem 5.13). No Orlov product of sheaves on a principally polarised fourfold with real multiplication meets the criterion through a dimension count, whatever the quartic field (Theorem 5.22), and no Orlov product of complexes does so unless both factors have nonzero self-Ext groups in some degree at most -3 (Remark 5.24). Those products are in any case candidates only for a twisted form of the criterion, the form of Markman's reduction (Remark 5.23). Objects with pure spinor characters give no Weil class at all (Proposition 5.25). Together these results fix the loci that divisors, scalar extension and correspondences reach inside $\mathcal{D}_F$, and the numbers that one complex has to attain on the rest.

Throughout, F is a quartic CM field and F_0 its real quadratic subfield, with real places τ_1, τ_2, and (V, H) is of $(F,2)$-Weil type with discriminant δ; so $m = n = 2$, the members are abelian eightfolds, and $\mathcal{D}_F$ has dimension $mn^2 = 8$. By [I, Proposition 17.3(i)], $H^1(A,\mathbb{C}) = \bigoplus_\sigma V_\sigma$ over the four embeddings σ of F, with $\dim V_\sigma^{1,0} = \dim V_\sigma^{0,1} = 2$. Put

$$R_F = \textstyle\bigwedge_F^2 V \;\subseteq\; \bigwedge_{\mathbb{Q}}^2 V = H^2(A,\mathbb{Q}),$$

the $\mathbb{Q}$-subspace with complexification $\bigoplus_\sigma \bigwedge^2 V_\sigma$; for $m = 1$ it is the space R of [I, Proposition 15.18]. It is an F-vector space of dimension 6, of $\mathbb{Q}$-dimension 24, with Hodge numbers $(4, 16, 4)$. Fix a generator Ω of the F-line $W_F = \bigwedge_F^4 V$, so that $W_F = \{a\Omega : a \in F\}$, and define a symmetric F-bilinear form q on R_F by $x \wedge_F y = q(x,y)\,\Omega$. Let h be the hermitian form on R_F induced by H, $h(x_1 \wedge x_2, y_1 \wedge y_2) = \det\big(H(x_i, y_j)\big)$, and h_4 the one induced on $\bigwedge_F^4 V$; in a basis $e_1, \dots, e_4$ with $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$, $h_4(\Omega,\Omega)$ is $\det H$, so its class in $F_0^\times / \mathrm{N}_{F/F_0}(F^\times)$ is δ. With dual bases (ω_k), (ω_k^*) of F for the trace form, as in [I, Theorem 17.5], put $x \star y = \sum_k (\omega_k x) \cup (\omega_k^* y)$ for $x, y \in R_F$. Then

$$x \star y = q(x,y)\,\Omega, \tag{30}$$

because the complexification of $(\omega_k x) \cup (\omega_k^* y)$ is $\sum_{\sigma,\sigma'} \sigma(\omega_k)\sigma'(\omega_k^*)\, x_\sigma \wedge y_{\sigma'}$, the sum over k of the coefficient is 1 for $\sigma = \sigma'$ and 0 otherwise, as in the proof of [I, Theorem 17.5(ii)], and $\sum_\sigma x_\sigma \wedge y_\sigma$ is the complexification of $q(x,y)\Omega$. Finally let $\theta_j \in H^{1,1}(A)$ be the component of the polarisation that pairs V_σ with $V_{\overline{\sigma}}$ for the two embeddings $\sigma, \overline{\sigma}$ over τ_j. Then $\eta_t = \sum_j \tau_j(t)\,\theta_j$ for the classes η_t of [I, Proposition 17.3(iii)], and $\mathbb{C}[\theta_1, \theta_2]$ is the complexification of the $\mathbb{Q}$-algebra generated by the η_t.

Proposition 5.5 (Hodge classes, and a quaternion algebra acting on R_F)**.** *Let A be a member of an $(F, 2, \delta)$-family.*

(i) *If $\mathrm{Hg}(A) = G_F$, the space of Hodge classes of A has dimension 2 in degree two, spanned by the η_t, and 7 in degree four, spanned by the three products of two η_t and by $W_F(A)$, which meets the span of those products in 0.*

(ii) *The map* $\phi\colon R_F \to R_F$ *defined by* $q(x,y) = h(x,\phi y)$ *satisfies* $\phi(ax) = \overline{a}\,\phi(x)$ *for* $a \in F$ *and* $\phi^2 = h_4(\Omega,\Omega)^{-1}$*, and it commutes with* $\mathrm{SU}(V,H)$*. So* ϕ *and the action of* F *are Hodge endomorphisms of* R_F *at every member, and the Hodge classes in* R_F *form a module over the algebra* $\mathcal{Q}_\delta = F \oplus F\phi$*.*

(iii) $\mathcal{Q}_\delta$ *is the cyclic quaternion algebra* $(F/F_0,\delta)$ *over* F_0*, which is split if and only if* δ *is trivial. If* $\mathrm{Hg}(A) = G_F$*, then* $\mathcal{Q}_\delta$ *is the whole algebra of Hodge endomorphisms of* R_F*.*

(iv) *If* δ *is trivial, then* $R_F = W \otimes_{F_0} F$ *for a rational sub-Hodge structure* W*, stable under* F_0*, of dimension* 12 *and with Hodge numbers* $(2,8,2)$*.*

Proof. (i) As in the proof of [I, Proposition 17.3(iii)], over $\mathbb{C}$ the group G_F is $\mathrm{SL}(U_1)\times\mathrm{SL}(U_2)$ with $\dim U_j = 4$, and $H^1(A,\mathbb{C}) = \bigoplus_j (U_j \oplus U_j^\vee)$. The invariants of $\mathrm{SL}(U)$ in $\bigwedge^a U \otimes \bigwedge^b U^\vee$ form a line if $a = b$ or $\{a,b\} = \{0,4\}$, and are zero otherwise. In degree two this leaves the bidegree $(1,1)$ in one factor: two lines, spanned by θ_1 and θ_2. In degree four it leaves $(2,2)$ in one factor or $(1,1)$ in both, three lines spanned by θ_1^2, $\theta_1\theta_2$ and θ_2^2, which are nonzero because $\dim U_j = 4$; and $(4,0)$ or $(0,4)$ in one factor, four lines spanning $W_F \otimes \mathbb{C}$. These lie in different summands, so $W_F(A)$ meets the span of the products in 0.

(ii) Choose an H-orthogonal basis $e_1,\dots,e_4$ of V and put $a_i = H(e_i,e_i) \in F_0$. Replacing Ω by $\lambda\Omega$ replaces ϕ by $\overline{\lambda}^{-1}\phi$ and $h_4(\Omega,\Omega)$ by $\mathrm{N}(\lambda)h_4(\Omega,\Omega)$, so we may take $\Omega = e_1\wedge e_2\wedge e_3\wedge e_4$, with $h_4(\Omega,\Omega) = a_1a_2a_3a_4$. For $J = \{j < j'\}$ put $e_J = e_j \wedge e_{j'}$ and $a_J = a_j a_{j'}$, and let J^c be the complement, with $e_{J^c}\wedge e_J = \epsilon_J\Omega$. The e_J form an h-orthogonal F-basis with $h(e_J,e_J) = a_J$, and $q(e_K,e_J)$ is ϵ_J for $K = J^c$ and 0 otherwise, so $\phi(e_J) = \epsilon_J a_{J^c}^{-1} e_{J^c}$. Two-forms commute, so $\epsilon_{J^c} = \epsilon_J$ and $\phi^2(e_J) = (a_J a_{J^c})^{-1} e_J = h_4(\Omega,\Omega)^{-1}e_J$. Semilinearity holds because q is linear and h antilinear in the second variable. An element of $\mathrm{SU}(V,H)$ preserves h and fixes Ω, hence preserves q and commutes with ϕ, and it commutes with F. The Hodge group of every member lies in G_F, because the circle defining the Hodge structure preserves E, commutes with F and acts on each V_σ with determinant $z^2z^{-2} = 1$; so ϕ and F commute with it, and they are Hodge endomorphisms.

(iii) The relations $\phi a = \overline{a}\phi$ and $\phi^2 = h_4(\Omega,\Omega)^{-1}$ present $F \oplus F\phi$ as the cyclic algebra $(F/F_0, h_4(\Omega,\Omega)^{-1})$, which is $(F/F_0,\delta)$ because a square in F_0 is a norm; and a cyclic algebra $(F/F_0,c)$ is split exactly when c is a norm from F. It has $\mathbb{Q}$-dimension 8, since ϕ is not F-linear. If $\mathrm{Hg}(A) = G_F$, the Hodge endomorphisms of R_F are the commutant of G_F. Over $\mathbb{C}$, $R_F \otimes \mathbb{C} = \bigoplus_j (\bigwedge^2 U_j \oplus \bigwedge^2 U_j^\vee)$, where $\bigwedge^2 U_j \cong \bigwedge^2 U_j^\vee$ is irreducible for $\mathrm{SL}(U_j)$ and trivial for the other factor, so the commutant is $M_2(\mathbb{C}) \times M_2(\mathbb{C})$, of dimension 8.

(iv) If δ is trivial, $h_4(\Omega,\Omega) = \mathrm{N}(b)$ for some $b \in F$, and $\psi = b\phi$ satisfies $\psi^2 = b\overline{b}\,\phi^2 = 1$. Its fixed space W is an F_0-subspace and a sub-Hodge structure, ψ being a Hodge endomorphism, and an element $\beta \in F$ with $\overline{\beta} = -\beta \neq 0$ maps it isomorphically onto the (-1)-eigenspace, since $\psi\beta = -\beta\psi$. So $R_F = W \oplus \beta W = W \otimes_{F_0} F$, and W has half the dimension and half the Hodge numbers of R_F. ∎

Proposition 5.6 (What divisor classes reach)**.** *Let* A *be a member of an* $(F,2,\delta)$*-family.*

(i) *The following are equivalent:* (a) $W_F(A)$ *lies in the subalgebra of* $H^*(A,\mathbb{Q})$ *generated by* $\mathrm{NS}(A)_\mathbb{Q}$*;* (b) $W_F(A)$ *meets that subalgebra in a nonzero class;* (c) R_F *contains a nonzero rational Hodge class. When they hold,* R_F *contains a rational Hodge class* t *with* $q(t,t) \neq 0$*, and* $W_F(A) = \{(ft)\star t : f \in F\}$*, a space of polynomials with rational coefficients in divisor classes.*

(ii) *For a nonzero* $\mathcal{Q}_\delta$*-submodule* $M \subseteq R_F$ *let* $\mathcal{D}(M) \subseteq \mathcal{D}_F$ *be the set of points at which* M *consists of Hodge classes, and let* $\mathrm{NL}(R_F) \subseteq \mathcal{D}_F$ *be the set of points at which*

(c) *holds. Then* $\mathrm{NL}(R_F) = \bigcup_M \mathcal{D}(M)$*, a countable union. Each nonempty* $\mathcal{D}(M)$ *is a closed complex submanifold, isomorphic to a product of two bounded symmetric domains of type IV and dimension* $4-d$ *each, where* $d = \dim_F M$*, and* $d = 1$ *occurs only if* δ *is trivial. So if* δ *is trivial,* $\mathrm{NL}(R_F)$ *is the union of the nonempty* $\mathcal{D}(M)$ *with* $d = 1$*, each of dimension exactly* 6*; if* δ *is not trivial, every* $\mathcal{D}(M)$ *has dimension at most* 4*.*

(iii) $\mathrm{NL}(R_F)$ *contains the base point* s_0 *of* [I, Theorem 17.5]*, and for every* δ *it contains the locus of products* $B_1 \times B_2$ *of two abelian fourfolds of* $(F,1)$*-Weil type, which has dimension* 4*. The split locus, of members isogenous to* $X \otimes_{\mathcal{O}_{F_0}} \mathcal{O}_F$ *with* X *an abelian fourfold with* $F_0 \subseteq \mathrm{End}^0(X)$*, has dimension* 6*, lies in the family of trivial discriminant, and is contained in* $\mathrm{NL}(R_F)$*.*

Proof. (i) Since $W_F(A) \neq 0$, (a) implies (b). Let $F^\times$ act on $H^*(A,\mathbb{Q})$ by $\alpha \mapsto \iota(\alpha)^*$, which on $\bigotimes_\sigma \bigwedge^{a_\sigma} V_\sigma$ is the character $\prod_\sigma \sigma^{a_\sigma}$. The isotypic parts for the characters σ^2 in degree two and σ^4 in degree four are R_F and W_F. The projections p_R and p_W onto them are rational, the two sets of characters being Galois orbits, and they are morphisms of Hodge structures, being $\mathbb{Q}$-linear combinations of the $\iota(\alpha)^*$. The characters occurring in H^2 are the products $\rho\rho'$, and the only way to write σ^4 as a product of two of them is $\sigma^2 \cdot \sigma^2$, so $p_W(D \cup D') = q(p_R D, p_R D')\,\Omega$ for $D, D' \in H^2(A,\mathbb{Q})$. If a nonzero $w \in W_F$ equals $\sum_i D_i \cup D_i'$ with $D_i, D_i' \in \mathrm{NS}(A)_\mathbb{Q}$, then $w = p_W(w) = \sum_i q(p_R D_i, p_R D_i')\,\Omega$, so some $p_R D_i$ is a nonzero rational Hodge class in R_F, and (b) implies (c). Conversely, let $t \in R_F$ be a nonzero rational Hodge class. The part of type $(1,1)$ of $\bigwedge^2 V_\sigma$ is $V_\sigma^{1,0} \wedge V_\sigma^{0,1}$, and H is definite of opposite signs on the two factors by [I, Proposition 17.3(i)], so h is negative definite on it at each real place, and $h(t,t) \in F_0$ is totally negative. By Proposition 5.5(ii), $q(t,\phi t) = h(t,\phi^2 t) = h_4(\Omega,\Omega)^{-1} h(t,t) \neq 0$, so one of the rational Hodge classes t, ϕt, $t+\phi t$, call it t', has $q(t',t') \neq 0$. Each $\omega_k f t'$ and each $\omega_k^* t'$ is a rational Hodge class in H^2, hence a divisor class, and by (30) $(ft') \star t' = f\, q(t',t')\,\Omega$, which runs over W_F as f runs over F. So (c) implies (a), with the stated formula.

(ii) The Hodge classes in R_F form a $\mathcal{Q}_\delta$-module, so $\mathrm{NL}(R_F)$ is the union of the $\mathcal{D}(M)$, and there are countably many M. Fix a real place τ, let σ be an embedding over it, and put $R_\tau = R_F \otimes_{F_0,\tau} \mathbb{R}$, a complex vector space of dimension 6 through $F \otimes_{F_0,\tau} \mathbb{R} = \mathbb{C}$, identified with $\bigwedge^2 V_\sigma$. Since $\det H$ has sign $(-1)^n = 1$ at τ, the map ϕ induces an antilinear ϕ_τ on R_τ with $\phi_\tau^2 = \lambda^2$, $\lambda > 0$. Its eigenspace $W_\tau = \{x : \phi_\tau x = \lambda x\}$ is a real form of R_τ and a real sub-Hodge structure with $W_\tau \otimes \mathbb{C} \cong \bigwedge^2 V_\sigma$, of Hodge numbers $(1,4,1)$, on which $q = \lambda h$ is real of signature $(2,4)$, positive on the plane Π_τ of real classes of type $(2,0)+(0,2)$ and negative on those of type $(1,1)$. The group $\mathrm{SU}(V_\tau, H_\tau)$ acts on W_τ preserving q, through the isogeny $\mathrm{SU}(2,2) \to \mathrm{SO}(2,4)^0$, and the map from $\mathcal{D}_{2,2}$ to the space of oriented positive planes in W_τ that sends a point to Π_τ is the equivariant isomorphism of the symmetric domains of this group onto one component, $\mathcal{D}^{\mathrm{IV}}(W_\tau)$, of dimension 4. For a quadratic space P of signature $(2,k)$ write likewise $\mathcal{D}^{\mathrm{IV}}(P)$, of dimension k. Let M have F-dimension d. Then $M_\tau = M \otimes_{F_0,\tau} \mathbb{R}$ is stable under ϕ and F, so $M_\tau = P_\tau \otimes \mathbb{C}$ with $P_\tau = M_\tau \cap W_\tau$ of real dimension d, and M consists of Hodge classes exactly when every P_τ is of type $(1,1)$, that is, orthogonal to Π_τ. So $\mathcal{D}(M) = \prod_\tau \mathcal{D}^{\mathrm{IV}}(P_\tau^\perp)$, which is empty unless every P_τ is negative definite and then has dimension $2(4-d)$. If $d = 1$, then $M = Fx$ with $\phi x = bx$ for some $b \in F$, and $\phi^2 x = \bar{b}b\,x$ gives $h_4(\Omega,\Omega)^{-1} = \mathrm{N}(b)$, so δ is trivial. If δ is trivial, $\mathcal{Q}_\delta \cong M_2(F_0)$, every nonzero module contains one with $d = 1$, and $\mathcal{D}(M) \subseteq \mathcal{D}(M')$ for $M' \subseteq M$.

(iii) The classes $\delta_i(f)$ of [I, Theorem 17.5(ii)] have complexification in $\bigoplus_\sigma \bigwedge^2 V_\sigma$, so they lie in R_F, and they are Hodge classes at s_0. For a product, $V = V_1 \oplus V_2$ with H-orthogonal F-planes of signature $(1,1)$ at each place, and $\bigwedge^2_F V_1 \subseteq R_F$ is $W_F(B_1)$, of type $(1,1)$ by [I, Proposition 17.3(i)]. Every δ is the product of the discriminants of two such planes, for instance of $\mathrm{diag}(1,-\delta)$ and $\mathrm{diag}(1,-1)$, and each factor varies in a family of dimension $m = 2$. On the split locus $H^1(A,\mathbb{Q}) = U \otimes_{F_0} F$, with $U = H^1(X,\mathbb{Q})$ and the complex structure of X on the first factor. The polarisation class e of X lies in $\bigwedge^2_{F_0} U$, the hermitian form of A is c times the sesquilinear extension of the F_0-bilinear form e, with $c \in F$ totally imaginary, and so $\det H = c^4 \operatorname{Pf}(e)^2$ is a square in F_0 and δ is trivial. The class $e \otimes 1 \in (\bigwedge^2_{F_0} U) \otimes_{F_0} F = R_F$ is fixed by $\mathrm{Hg}(A) = \mathrm{Hg}(X) \subseteq \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{Sp}(U,e)$, so it is a nonzero Hodge class. The polarised abelian fourfolds with real multiplication by F_0 form families of dimension $[F_0 : \mathbb{Q}] \cdot 3 = 6$. ∎

Corollary 5.7 (A base locus of codimension two)**.** *In every $(F,2,\delta)$-family, $\Sigma_F \supseteq \mathrm{NL}(R_F)$, and at every point of $\mathrm{NL}(R_F)$ the Weil classes are the polynomials in divisor classes of Proposition 5.6(i). If δ is trivial, $\mathrm{NL}(R_F)$ is a countable union of closed complex submanifolds of $\mathcal{D}_F$ of codimension exactly two, and it contains the split locus; if δ is not trivial, they have codimension at least four, and the product loci have codimension four. In both cases $\mathrm{NL}(R_F)$ contains the base point s_0 of* [I, Theorem 17.5] *and is stable under $G_F(\mathbb{Q})$. At a point outside $\mathrm{NL}(R_F)$ no nonzero Weil class is a polynomial in divisor classes.*

Proof. Everything except the stability is Proposition 5.6. An element $g \in G_F(\mathbb{Q})$ commutes with F and with ϕ, so it carries $\mathcal{Q}_\delta$-submodules to $\mathcal{Q}_\delta$-submodules, and $g\,\mathcal{D}(M) = \mathcal{D}(gM)$. ∎

[I, Theorem 17.5] gives one CM point, whose $G_F(\mathbb{Q})$-orbit is dense but countable. When δ is trivial, Corollary 5.7 replaces it by a locus of codimension two on which the Weil classes are balanced products of divisor classes, as they are at s_0; nothing is claimed about the rest of the Hodge ring at its points. It does not decide $\mathbf{W}(F,2,\delta)$: $\mathrm{NL}(R_F)$ is a countable union of proper closed submanifolds, hence meagre, and [I, Theorem 17.7(iii)] allows Σ_F to be meagre. The last sentence of the corollary is the analogue of [I, Proposition 15.18]: no argument through divisor classes reaches a member outside $\mathrm{NL}(R_F)$. The codimension is two rather than $h^{2,0}(R_F) = 4$ (compare [I, Remark 15.21]) because Hodge classes in R_F come in $\mathcal{Q}_\delta$-modules, and a module of F-dimension one costs one condition at each real place. Figure 15 shows where the Weil classes of such a family are known to be algebraic.

Proposition 5.8 (Scalar extension from an imaginary quadratic subfield)**.** *Let $F = KF_0$ with K imaginary quadratic; these are the biquadratic quartic CM fields, and the only ones containing an imaginary quadratic field. Let C be an abelian fourfold of $(K,2,\delta)$-Weil type and $B_F = C \otimes_{\mathcal{O}_K} \mathcal{O}_F$, which is of $(F,2,\iota(\delta))$-Weil type by* [III, Proposition 2.13(ii)]*; write (V_C, H_C) for the hermitian space of C. For $f \in F$ let $g_f \in \mathrm{Hom}(B_F, C) \otimes \mathbb{Q}$ be the element inducing $v \mapsto v \otimes f$ from $H^1(C,\mathbb{Q})$ to $H^1(B_F,\mathbb{Q}) = H^1(C,\mathbb{Q}) \otimes_K F$.*

(i) *$W_F(B_F)$ lies in the $\mathbb{Q}$-span of the classes $g_f^* w$, $f \in F$, $w \in W_K(C)$, which has dimension 10. Hence $W_F(B_F)$ is algebraic, by Markman's theorem $\mathbf{W}(K,2,\delta)$* [Mar25a] *(*[III, Proposition 2.12(ii)]*).*

(ii) *The members B_F form a locus of dimension 4 in the $(F,2,\iota(\delta))$-family. At those with $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ the space R_F contains no nonzero Hodge class, so this locus is not contained in $\mathrm{NL}(R_F)$; it meets $\mathrm{NL}(R_F)$, for instance where $C = C_1 \times C_2$*

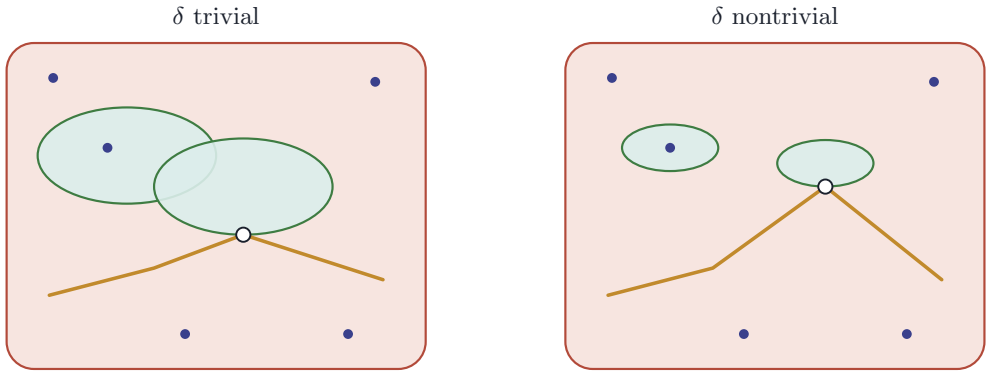


FIGURE 15. Where the Weil classes of an $(F, 2, \delta)$-family are known to be algebraic, F a quartic CM field; each panel is the period domain $\mathcal{D}_F$, of dimension 8. The grass regions are components of the Noether–Lefschetz locus $\mathrm{NL}(R_F)$, on which W_F consists of polynomials in divisor classes (Proposition 5.6, Corollary 5.7): of dimension exactly 6 when δ is trivial, where the quaternion algebra $\mathcal{Q}_\delta$ of Proposition 5.5 is split, and at most 4 otherwise. The ochre curve is the locus of scalar extensions of Weil fourfolds, of dimension 4 when F is biquadratic, on which Markman's theorem applies (Proposition 5.8); it is not contained in $\mathrm{NL}(R_F)$ but meets it (ring) where the fourfold is a product of Weil surfaces. The indigo points are CM points; the base point s_0 of [I, Theorem 17.5] lies on $\mathrm{NL}(R_F)$. For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial δ the Prym locus of Remark 5.9, of dimension at most 6, is not drawn. On the rest of the domain, the very general member included, the algebraicity of the Weil classes is the open propagation statement.

with C_1, C_2 abelian surfaces of Weil type, and then B_F is a product of two abelian fourfolds of $(F,1)$-Weil type.

Proof. (i) The map $v \mapsto v \otimes f$ commutes with the complex structures, that of B_F being the one of C tensored with F, so g_f exists. Over $\mathbb{C}$, for each embedding κ of K let V_κ be the κ-eigenspace of K on $H^1(C,\mathbb{C})$ and $U_\kappa = F \otimes_{K,\kappa} \mathbb{C} \cong \mathbb{C}^2$, the sum of the lines L_σ, $\sigma|\kappa$, on which F acts through σ. The κ-part of $H^1(B_F,\mathbb{C})$ is $V_\kappa \otimes U_\kappa$ and its σ-part is $V_\kappa \otimes L_\sigma$, so the σ-component $\bigwedge^4 V_\kappa \otimes L_\sigma^{\otimes 4}$ of $W_F(B_F) \otimes \mathbb{C}$ lies in $\bigwedge^4 V_\kappa \otimes \mathrm{Sym}^4 U_\kappa \subseteq \bigwedge^4 (V_\kappa \otimes U_\kappa)$. The map g_f^* sends a generator α_κ of $\bigwedge^4 V_\kappa$ to $\alpha_\kappa \otimes f_\kappa^{\otimes 4}$, with f_κ the image of f in U_κ. The fourth powers of the elements of the Zariski dense subset F of U_κ span $\mathrm{Sym}^4 U_\kappa$, so the complexified span of the $g_f^* w$ is $\bigoplus_\kappa \bigwedge^4 V_\kappa \otimes \mathrm{Sym}^4 U_\kappa$, of dimension $2 \cdot 5$, and it contains $W_F(B_F) \otimes \mathbb{C}$. A rational subspace whose complexification lies in that of a rational subspace lies in it, and pull-backs of algebraic classes are algebraic.

(ii) The map $C \mapsto B_F$ embeds the four-dimensional domain of the K-family into $\mathcal{D}_F$. As rational Hodge structures $H^1(B_F,\mathbb{Q}) \cong H^1(C,\mathbb{Q})^2$, so $\mathrm{Hg}(B_F) = \mathrm{Hg}(C)$, and $R_F \otimes \mathbb{C} = \bigoplus_\sigma \bigwedge^2 V_\kappa \otimes L_\sigma^{\otimes 2}$, with $\bigwedge^2 V_\kappa$ irreducible and nontrivial for $\mathrm{SL}(V_\kappa)$; so when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$, R_F has no invariants. For $C = C_1 \times C_2$ one has $B_F = (C_1 \otimes_{\mathcal{O}_K} \mathcal{O}_F) \times (C_2 \otimes_{\mathcal{O}_K} \mathcal{O}_F)$, and Proposition 5.6(iii) applies. ∎

Proposition 5.8 gives no new case of the Hodge conjecture: B_F is isogenous to $C \times C$, the classes are Hodge classes there, and when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ [III, Proposition 2.24] already gives the conjecture for $C \times C$ from [Mar25a].

Remark 5.9 (Prym varieties of cyclic covers). For three cyclotomic fields published work gives a further locus, at whose very general points the Hodge group is the whole of G_F. Let $N \geq 3$, let $\widetilde{C} \to C$ be an étale cyclic cover of degree N of a smooth projective curve of genus $g \geq 2$, and let $P \subseteq J(\widetilde{C})$ be the abelian subvariety on whose H^1 the covering group acts through characters of exact order N. Then $F = \mathbb{Q}(\zeta_N)$ acts on P, and every embedding of F occurs in $H^{1,0}(P)$ with multiplicity $g-1$ [Sch88, Lemma 1.6a], [PZ25, Lemma 5.1], so P is of $(F, g-1)$-Weil type ([I, Remark 17.2]). The hermitian form is split, because the part of $H_1(\widetilde{C},\mathbb{Z})$ on which the covering group acts through characters of exact order N has a hyperbolic $\mathbb{Z}[\zeta_N]$-basis [Loo97, Proposition 2.2]; so $\delta = [(-1)^{g-1}]$, which is trivial for $g = 3$. Schoen proved that the Weil classes of P are algebraic [Sch88, Theorem 2.0 with $r = 0$, Corollary 3.1], and Patel and Zhang give a proof by unramified geometric class field theory [PZ25, Theorems 1.1, 1.2 and 5.3, Remark 3.7]. By [Loo97, Theorem 2.4], for every $g \geq 2$, the monodromy group of the family of the P over a finite cover of the moduli of such covers contains a subgroup of finite index of an arithmetic subgroup of G_F. Every factor $\mathrm{SU}(g-1, g-1)$ of $G_F(\mathbb{R})$ is noncompact, so by Borel's density theorem [Bor60] its Zariski closure contains G_F; the Hodge group of every member lies in G_F, so by André's theorem [And92, Theorem 1] a very general P has $\mathrm{Hg}(P) = G_F$, and by [I, Proposition 17.3(iii)] the Hodge conjecture holds for it in every degree. All of this is a consequence of [Sch88, Loo97, And92], and we make no claim of priority.

For $\varphi(N) = 4$, that is for $F = \mathbb{Q}(\zeta_5) = \mathbb{Q}(\zeta_{10})$, $\mathbb{Q}(\zeta_8)$ and $\mathbb{Q}(\zeta_{12})$, and $g = 3$, the P are abelian eightfolds of $(F, 2, 1)$-Weil type. With their $G_F(\mathbb{Q})$-translates, which lie in Σ_F by [I, Theorem 17.7(ii)], they fill a countable union of analytic subvarieties of $\mathcal{D}_F$, the *Prym locus* of F, of positive dimension, since the monodromy is infinite, and of dimension at most $3g - 3 = 6$ [Sch88, pp. 25–26]. At a very general point of it $\mathrm{Hg} = G_F$, so R_F has no nonzero Hodge class there, each $\bigwedge^2 V_\sigma$ being a nontrivial irreducible representation of $\mathrm{SL}(V_\sigma)$, and the Hodge group is larger than that of a scalar extension of a Weil

fourfold: such points lie outside $\mathrm{NL}(R_F)$, outside the locus of Proposition 5.8, and are not CM points. They are the only members with $\mathrm{Hg} = G_F$ at which we know the Weil classes of a quartic CM field to be algebraic. They do not decide $\mathbf{W}(F,2,1)$: the count $3g-3 = 6 < 8 = \frac{1}{2}\varphi(N)(g-1)^2 = \dim \mathcal{D}_F$ of [Sch88, pp. 25–26] makes the union meagre, as $\mathrm{NL}(R_F)$ is, and [I, Theorem 17.7(iii)] allows Σ_F to be meagre. Nor does Proposition 5.12 exclude them: the cycles come from $J(\widetilde{C})$, which has P as a factor. For imaginary quadratic fields the count is $3n \geq n^2$ for $n \leq 3$, and there the Prym varieties do reach whole families: Schoen proved the algebraicity for the Weil fourfolds and for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-3})$ [Sch88, Theorem 3.2], [Sch98], and Koike for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-1})$ [Koi04, Theorem 2.1, Corollary 2.1].

Proposition 5.10 (The Casimir class)**.** *Let A be a member of an $(F,2,\delta)$-family and $c \in F^\times$, and let*

$$C_q \in R_F \otimes_{\mathbb{Q}} R_F \subseteq H^2(A,\mathbb{Q}) \otimes H^2(A,\mathbb{Q}) \subseteq H^4(A \times A, \mathbb{Q})$$

be the Casimir element $\sum_j u_j \otimes u_j^*$ of the nondegenerate symmetric form $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$ on R_F, for a basis (u_j) and its dual basis (u_j^*). Then C_q is a Hodge class, $\Delta^* C_q = 6c\,\Omega$ for the diagonal $\Delta\colon A \to A \times A$, and C_q lies in the $\mathbb{Q}$-span of the classes f^*w, $w \in W_F(A)$, $f(x,y) = \iota(a)x + \iota(b)y$ with $a,b \in \mathcal{O}_F$. Hence $W_F(A)$ is algebraic if and only if C_q is.

Proof. For an F-basis (x_I) of R_F and its q-dual basis (x_I^q), the elements $\omega_k x_I$ and $c\,\omega_k^* x_I^q$ form dual bases for $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$. So by (30) $\Delta^* C_q = \sum_I x_I \star (c\,x_I^q) = \sum_I c\,\Omega = 6c\,\Omega$, and if C_q is algebraic so is $W_F(A)$, by [I, Proposition 17.3(ii)]. Conversely, $\mathrm{SU}(V,H)$ preserves q, so C_q is invariant under the diagonal action of G_F, which contains $\mathrm{Hg}(A \times A)$, and C_q is a Hodge class. Since q pairs $\bigwedge^2 V_\sigma$ only with itself, the complexification of C_q lies in $\bigoplus_\sigma \bigwedge^2 V_\sigma \otimes \bigwedge^2 V_\sigma$, inside the part $\bigoplus_\sigma \bigwedge^4(V_\sigma \otimes \mathbb{C}^2)$ of $H^4(A \times A, \mathbb{C})$ on which $F^\times$, acting diagonally, has the characters σ^4. The group G_F acts on V_σ through $\mathrm{SL}(V_\sigma)$, whose invariants in $\bigwedge^4(V_\sigma \otimes \mathbb{C}^2)$ are $\bigwedge^4 V_\sigma \otimes \mathrm{Sym}^4\mathbb{C}^2$ by Cauchy's formula, so the G_F-invariants in that part form the rational space $W_F(A) \otimes_F \mathrm{Sym}^4_F(F^2)$, of dimension 20. The map f^* sends w to $w \otimes (a,b)^{\otimes 4}$, and the fourth powers of the elements of $\mathcal{O}_F^2$ span $\mathrm{Sym}^4_F(F^2)$; so the classes f^*w span that space, and C_q is algebraic if $W_F(A)$ is. ∎

Remark 5.11 (What the exceptional isogeny gives). By the proof of Proposition 5.6(ii), at each real place R_F becomes $W_\tau \otimes \mathbb{C}$ with W_τ of K3 type, $\mathrm{SU}(V_\tau, H_\tau)$ acting through $\mathrm{SU}(2,2) \to \mathrm{SO}(2,4)$, and when δ is trivial the W_τ are the real components of the rational structure W of Proposition 5.5(iv). A route of Kuga–Satake type would start from this weight two structure and would have to return to degree four, on A or on $A \times A$; by Proposition 5.10 the natural class it meets there, the Casimir class of q, is algebraic exactly when the Weil classes are. So this mechanism restates $\mathbf{W}(F,2,\delta)$ at each member and does not reduce it.

Proposition 5.12 (No correspondence from a small abelian variety)**.** *Let A be a member with $\mathrm{Hg}(A) = G_F$, and let X be an abelian variety such that $\mathrm{Hg}(X)$ has no normal subgroup isogenous to G_F; by the proof, this holds when every simple factor of X has dimension at most 7, in particular when $\dim X \leq 7$, and it holds when X is of CM type. Then $\mathrm{Hg}(A \times X) = \mathrm{Hg}(A) \times \mathrm{Hg}(X)$, and $\mathrm{Hdg}^*(A^a \times X^b) = \mathrm{Hdg}^*(A^a) \otimes \mathrm{Hdg}^*(X^b)$ for all $a,b \geq 0$. So the class of an algebraic cycle Z on $A^a \times X^b$ is $\sum_i \beta_i \otimes \xi_i$ with $\beta_i \in \mathrm{Hdg}^*(A^a)$ and linearly independent $\xi_i \in \mathrm{Hdg}^*(X^b)$; the correspondence Z carries $H^*(X^b,\mathbb{Q})$ into the span of the β_i; and if the Hodge conjecture holds for X^b, every β_i is algebraic.*

Proof. The group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}}\,\mathrm{SU}(V,H)$ is almost simple over $\mathbb{Q}$. By Goursat's lemma the kernel N of the projection $\mathrm{Hg}(A\times X)\to\mathrm{Hg}(X)$ maps onto a normal subgroup of $\mathrm{Hg}(A)=G_F$. If that subgroup is G_F, then $\mathrm{Hg}(A\times X)$ contains $G_F\times 1$ and is the product. Otherwise N is finite, $\mathrm{Hg}(X)$ has a normal subgroup H isogenous to G_F, contrary to the hypothesis, and the cocharacter μ defining the Hodge structure of X projects through $H\to G_F$ to that of A. This situation also cannot occur when every simple factor of X has dimension at most 7, or when X is of CM type, as follows. Over $\mathbb{C}$, H is $\mathrm{SL}_4\times\mathrm{SL}_4$, and μ projects in each factor to $\mathrm{diag}(\frac12,\frac12,-\frac12,-\frac12)$ modulo the centre. Since μ acts on $H^1(X,\mathbb{C})$ with two weights differing by one, the weights of its projection spread by at most one on every irreducible constituent for H; for the representation of SL_4 of highest weight $a\varpi_1+b\varpi_2+c\varpi_3$ the spread is $a+2b+c$, and spreads add over the two factors. So every nontrivial constituent is the standard representation of one factor or its dual. These four are permuted transitively by $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, as the embeddings of F are; so a simple factor Y of X on whose H^1 the group H acts nontrivially has $\dim_{\mathbb{Q}} H^1(Y,\mathbb{Q})\geq 16$, that is $\dim Y\geq 8$. A torus has no such H. Next, $\mathrm{Hg}(A^a\times X^b)$ is $\mathrm{Hg}(A)\times\mathrm{Hg}(X)$ acting factorwise on $H^*(A^a)\otimes H^*(X^b)$, and the invariants of a product of groups in a tensor product of representations of the factors are the tensor product of the invariants. Finally, the intersection pairing is perfect on $\mathrm{Hdg}^*(X^b)$, the invariants being orthogonal to the other isotypic parts, so there are $\xi_j^*\in\mathrm{Hdg}^*(X^b)$ with $\int\xi_i\xi_j^*=\delta_{ij}$; they are algebraic when the conjecture holds for X^b, and then $\beta_j=pr_{A^a*}(Z\cdot pr_{X^b}^*\xi_j^*)$ is algebraic. ∎

So a correspondence with such an X can produce a Weil class of A only from an algebraic cycle on some $A^a\times X^b$ that already yields one on A^a. At a member with full Hodge group this excludes every abelian variety of dimension at most seven, among them the fourfolds and sixfolds of Weil type of the imaginary quadratic fields on which Markman's theorems are proved, and every CM abelian variety. The same argument, with real points in place of dimensions, excludes rational Hodge structures Y of K3 type: by Zarhin's theorem [Zar83] the Hodge group of such a Y is the restriction of scalars of an orthogonal or unitary group over the endomorphism field of its transcendental part, noncompact at exactly one real place, so its real Lie algebra has at most one simple factor isomorphic to $\mathfrak{su}(2,2)$, while that of G_F has two; hence $\mathrm{Hg}(A\times Y)=\mathrm{Hg}(A)\times\mathrm{Hg}(Y)$. This last statement rests on Zarhin's theorem, which we quote without proof.

Theorem 5.13 (The polarised criterion for a quartic field at $n=2$)**.** *Let A be a member of an $(F,2,\delta)$-family, let $\omega=\sum_\sigma u_\sigma\alpha_\sigma$, with α_σ spanning $\bigwedge^4 V_\sigma$ and every $u_\sigma\neq 0$, as holds for every nonzero rational Weil class, let $p\in\mathbb{C}[\theta_1,\theta_2]$, put $\gamma=\omega+p$, and let $r(\gamma)$ be the rank of the contraction $HT^2(A)\to H^*(A,\mathbb{C})$, $\xi\mapsto\xi\lrcorner\gamma$. Write $T_\sigma=(V_\sigma^{1,0})^\vee\subseteq T_0A$.*

(i) $\dim HT^2(A)=28+64+28=120$, *and*

$$\mathrm{Ann}_{HT^2}(\omega)=\bigoplus\nolimits_\sigma \mathrm{Hom}(V_\sigma^{1,0},V_\sigma^{0,1})\ \oplus\ \bigoplus\nolimits_{\{\sigma,\sigma'\},\,\sigma\neq\sigma'} T_\sigma\otimes T_{\sigma'},$$

of dimension $0+16+24$: the F-linear first order deformations of A and the bivectors mixing two embeddings. So $r(\omega)=80$. Unlike the case $m=1$ of Theorem 2.5(i), no nonzero element of $H^{0,2}(A)$ kills ω.

(ii) *$r(\gamma)\leq 112$ for every p. For a general p, $r(\gamma)=112$, and whenever $r(\gamma)=112$ the annihilator of γ is the tangent space $T_{[A]}\mathcal{D}_F$ of the family, of dimension 8, inside $H^1(T_A)$. The value 112 is also taken at a general element, with nonzero component in $W_F\otimes\mathbb{C}$, of the complexified Hodge ring of a member with $\mathrm{Hg}(A)=G_F$.*

(iii) *$r(\gamma)\geq 80$ for every p, with equality exactly when p is a constant.*

(iv) *The rank* $r(\gamma)$ *takes exactly the fifteen values* 80, 84, 87, 88, 91, 92, 94, 95, 96, 104, 107, 108, 110, 111, 112*; Theorem 5.14 computes it for every* p*. On the monomials* $\theta_1^i\theta_2^j$ *with a general coefficient it takes the six values* 80, 84, 88, 104, 108, 112.

(v) *Let* E *be a perfect complex on* A *with* $\operatorname{ch}(E) = N\omega_1 + P$*, where* $N \neq 0$*,* ω_1 *is a nonzero rational Weil class and* P *is a polynomial with rational coefficients in the classes* η_t*. Then* $\dim \operatorname{Ext}^2(E,E) \geq r(\operatorname{ch} E) \geq 80$*. If* $\dim \operatorname{Ext}^2(E,E) = r(\operatorname{ch} E)$*, then* ev_E *is onto with kernel* $\operatorname{Ann}_{HT^2}(\operatorname{ch} E)$ *and* σ_E *is injective, so that* $\mathbf{W}(F,2,\delta)$ *follows by* [I, Theorem 17.7(v)].

Proof. (i) $HT^2(A) = \bigwedge^2 H^{0,1} \oplus H^{0,1} \otimes T_0A \oplus \bigwedge^2 T_0A$, with $H^{0,1}$ and T_0A of dimension 8. As in the proof of Theorem 2.5(i) the three summands map into different Hodge components, so the annihilator is the sum of the three annihilators; and the torus $(\mathbb{C}^\times)^4$ acting on each V_σ by scalars separates the images further. A block $V_\rho^{0,1} \wedge V_{\rho'}^{0,1}$ of $\bigwedge^2 H^{0,1}$ kills α_σ exactly when $\sigma \in \{\rho, \rho'\}$, and otherwise maps injectively into a weight space that no other pair of a block and an embedding reaches; there are four embeddings, so some σ lies outside $\{\rho, \rho'\}$, and no nonzero element of $\bigwedge^2 H^{0,1}$ kills ω. A block $\operatorname{Hom}(V_\rho^{1,0}, V_{\rho'}^{0,1})$ of $H^1(T_A)$ kills α_σ for $\sigma \neq \rho$; it kills α_ρ when $\rho' = \rho$, because $V_\rho^{0,1}$ is already exhausted in α_ρ, and maps it injectively into its own weight space when $\rho' \neq \rho$. A bivector in $T_\rho \otimes T_{\rho'}$ with $\rho \neq \rho'$ kills every α_σ, whose two factors of type $(1,0)$ both lie in $V_\sigma^{1,0}$, and $\bigwedge^2 T_\sigma$ maps α_σ to a nonzero multiple of a generator of $\bigwedge^2 V_\sigma^{0,1}$. The dimensions are 0, $4 \cdot 4$ and $6 \cdot 4$.

(ii) The classes θ_j and the rational Weil classes stay of Hodge type along $\mathcal{D}_F$, so each is killed by $T_{[A]}\mathcal{D}_F \subseteq H^1(T_A)$, as in Proposition 3.11(i); hence $r(\gamma) \leq 120 - 8$, and when equality holds the annihilator, of dimension 8, is $T_{[A]}\mathcal{D}_F$. Take a basis of $H^1(A,\mathbb{C})$ adapted to the V_σ and to the Hodge decomposition, in which the polarisation pairs the basis of $V_\sigma^{1,0}$ with that of $V_{\overline{\sigma}}^{0,1}$. The torus acting diagonally on it contains a subtorus that multiplies each θ_j by a scalar and each α_σ by an arbitrary scalar, and contraction is equivariant; so $r(\omega + p)$ depends on ω only through a rescaling of the variables of p, and each statement of the theorem holds for every ω once it holds for one. The rank is lower semicontinuous in p, and it equals 112 at $p = \theta_1^2\theta_2^2$ by Theorem 5.14(iii) below, so it equals 112 for a general p. Every element of the complexified Hodge ring of a member with $\operatorname{Hg}(A) = G_F$ is invariant under G_F, so its components stay of Hodge type along $\mathcal{D}_F$ and it is killed by $T_{[A]}\mathcal{D}_F$; its rank is therefore at most 112. The ring contains $\omega + \theta_1^2\theta_2^2$, since it contains $W_F \otimes \mathbb{C}$ and the classes θ_1, θ_2, whose span is defined over $\mathbb{Q}$, and the rank is lower semicontinuous; so it is 112 at a general element of the ring, whose component in $W_F \otimes \mathbb{C}$ is not zero.

(iii) and (iv) are Theorem 5.14(iii) and (iv) below.

(v) By the commutative square $\sigma_E \circ \operatorname{ev}_E = (\,\cdot\,) \lrcorner \operatorname{ch}(E)$ of Setup 4.54, which holds on every abelian variety, $\dim \operatorname{Ext}^2(E,E) \geq \operatorname{rk} \operatorname{ev}_E \geq r(\operatorname{ch} E)$, and equality forces ev_E to be onto with kernel the annihilator and σ_E to be injective, as in Theorem 3.9(iv); and $r(\operatorname{ch} E) \geq 80$ by (iii), since the complexification of $\operatorname{ch}(E)$ is a γ as above with $\omega = N\omega_1$. The last implication is [I, Theorem 17.7(v)], whose character is exactly of this form. ∎

At $m = 1$ and $n = 2$ the corresponding numbers are 28, 12 and 24 (Theorem 3.9); in both cases the general value is the dimension of HT^2 minus that of the family. For the pure shape the argument of Theorem 2.5(ii) and (iv) transfers with the annihilator of Theorem 5.13(i): an injective σ_E would force $A_\theta A_{\theta'} = 0$ for $\theta \in T_\rho$, $\theta' \in T_{\rho'}$, $\rho \neq \rho'$, and a smooth component of the support of codimension at least two along which E is locally free of positive rank would be tangent to three of the four T_ρ. The pure shape is excluded, for every CM field, by Theorem 5.4.

The rank $r(\gamma)$ can be computed for every p. In the basis of the proof of Theorem 5.13(ii) write $x_{\sigma,1}, x_{\sigma,2}$ for the basis of $V_\sigma^{1,0}$, $y_{\sigma,1}, y_{\sigma,2}$ for that of $V_\sigma^{0,1}$, and $\partial_{\sigma,1}, \partial_{\sigma,2}$ for the dual basis of T_σ, so that $\theta_j = \sum_{\sigma|\tau_j} \sum_k x_{\sigma,k}\, y_{\overline{\sigma},k}$ and $\alpha_\sigma = x_{\sigma,1}x_{\sigma,2}y_{\sigma,1}y_{\sigma,2}$, and use the torus of that proof to make every u_σ equal to 1. Expand $p = \sum_{i,j=0}^4 e_{ij}\, \theta_1^i\theta_2^j/(i!\, j!)$, so that e_{ij} is a derivative of p at 0, and put:

- $\mu(p) = 0$ if $\partial_1\partial_2 p = 0$ and $\mu(p) = 1$ otherwise, the derivatives taken in $\mathbb{C}[\theta_1, \theta_2]$;
- $\rho_1(p)$ the dimension of the span of $\partial_1 p$ and $\partial_1^2 p$ modulo θ_1^3, which is the rank of the 2×15 matrix whose columns are the pairs $(e_{i,j}, e_{i+1,j})$, $1 \le i \le 3$, $0 \le j \le 4$; and $\rho_2(p)$ likewise, with the two indices exchanged;
- for $t = 1, 2$, with $\sigma, \overline{\sigma}$ the embeddings over τ_t, $W_t \subseteq HT^2(A)$ the span of the eight elements $y_{\sigma,1} \wedge y_{\sigma,2}$, $y_{\overline{\sigma},1} \wedge y_{\overline{\sigma},2}$, $\partial_{\sigma,1} \wedge \partial_{\sigma,2}$, $\partial_{\overline{\sigma},1} \wedge \partial_{\overline{\sigma},2}$ and $y_{\sigma,k} \otimes \partial_{\overline{\sigma},l}$, $y_{\overline{\sigma},k} \otimes \partial_{\sigma,l}$ with $k \ne l$, and $R_t(p)$ the rank of $\xi \mapsto \xi \lrcorner \gamma$ on W_t;
- $h_1 = e_1e_3 - e_2^2$, $h_2 = e_2e_4 - e_3^2$, $h_3 = e_1e_4 - e_2e_3$ on $\mathbb{C}^4$, and $V_\pm \subset \mathbb{C}^4$ the set where $h_3 = \pm e_3$, $2h_2 = \pm e_4$ and $2h_1 + 1 = \pm e_2$.

Theorem 5.14 (The rank of the criterion at a quartic field)**.** *In this notation:*

(i) $r(\gamma) = 64 + 16\mu(p) + 4\rho_1(p) + 4\rho_2(p) + R_1(p) + R_2(p)$.

(ii) $R_t(p) \in \{7, 8\}$. *If* $R_1(p) = 7$, *then* $(e_{10}, e_{20}, e_{30}, e_{40})$ *lies on* $V_+ \cup V_-$ *and* $\rho_1(p) = 2$; *conversely, if* $\mu(p) = 0$ *and* $(e_{10}, \dots, e_{40})$ *lies on* $V_+ \cup V_-$, *then* $R_1(p) = 7$, *as for* $p = \theta_1^2/2$. *The same holds for* R_2 *with* $(e_{01}, \dots, e_{04})$. *Each* $V_\pm$ *contains the surface of points* $(w\lambda \pm \frac{3}{2\lambda},\ w\lambda^2 \pm \frac{1}{2},\ w\lambda^3,\ w\lambda^4)$, $w \in \mathbb{C}$, $\lambda \in \mathbb{C}^\times$.

(iii) $r(\gamma) \ge 80$, *with equality exactly when* p *is a constant, and* $r(\gamma)$ *takes exactly the fifteen values*

$$80,\ 84,\ 87,\ 88,\ 91,\ 92,\ 94,\ 95,\ 96,\ 104,\ 107,\ 108,\ 110,\ 111,\ 112,$$

the values 87, 91, 94, 95, 107, 110 *and* 111 *only when* R_1 *or* R_2 *is* 7. *The value* 100 *is not taken.*

(iv) *For* $\omega = \sum_\sigma u_\sigma \alpha_\sigma$ *with arbitrary nonzero* u_σ, *the same holds with* e_{ij} *replaced by* $\kappa_1^i\kappa_2^j e_{ij}$, *where* $\kappa_t^4 = (u_\sigma u_{\overline{\sigma}})^{-1}$ *for the embeddings over* τ_t. *The four choices of* κ_t *fix* $V_+ \cup V_-$, *and* μ, ρ_1, ρ_2 *do not depend on them.*

Proof. (i) Let S be the torus of the basis that fixes θ_1, θ_2 and every α_σ. If $x_{\sigma,k}$ has weight $a_{\sigma,k}$, then $y_{\sigma,k}$ has weight $-a_{\overline{\sigma},k}$ and $\partial_{\sigma,k}$ has weight $-a_{\sigma,k}$, and the only relations are $a_{\sigma,1} + a_{\sigma,2} = a_{\overline{\sigma},1} + a_{\overline{\sigma},2}$, one for each real place; so S has dimension six. Attach to $y_{\sigma,k}$ the label $(\overline{\sigma}, k)$ and to $\partial_{\sigma,k}$ the label (σ, k). A basis element of HT^2 is a product of two of these and has weight $-(a_P + a_Q)$ for its two labels P, Q, and two such weights agree exactly when the pairs of labels coincide or are $\{(\sigma,1),(\sigma,2)\}$ and $\{(\overline{\sigma},1),(\overline{\sigma},2)\}$. So HT^2 splits into 34 weight spaces: the eight lines spanned by $y_{\sigma,k} \otimes \partial_{\overline{\sigma},k}$; sixteen spaces of dimension four, one for each pair of labels over different real places; eight of dimension four, one for each pair $\{(\sigma,k),(\overline{\sigma},l)\}$ over one place; and W_1, W_2. The class γ is invariant under S, so contraction carries each weight space into the same weight of $H^*(A, \mathbb{C})$, and $r(\gamma)$ is the sum of the ranks on the 34 spaces.

Two rules compute these ranks. For a label $P = (\rho, k)$ with ρ over τ_t put $y_P = y_{\overline{\rho},k}$, the element of $H^{0,1}$ that carries it. Contraction by $\partial_{\rho,k}$ is a derivation that maps θ_t to $\pm y_P$ and the other $\theta_{t'}$ to 0, so it maps $f(\theta_1, \theta_2)$ to $\pm y_P\, (\partial_t f)(\theta_1, \theta_2)$. Hence a basis element with labels P, Q carries p to $\pm y_P y_Q\, (Dp)(\theta_1, \theta_2)$, where D is 1, ∂_t or $\partial_t\partial_{t'}$ according as none, one or both labels are carried by elements of T_0A, and t, t' are their places. Multiplication by $y_P y_Q$ removes from θ_1 and θ_2 the terms that contain y_P or y_Q, and the products $\theta_1^a\theta_2^b$

of what remains are linearly independent as long as a and b do not exceed the numbers of remaining terms. Every monomial of $y_P y_Q f(\theta_1,\theta_2)$ contains, with each $x_{\sigma,i}$, its partner $y_{\overline\sigma,i}$. The second rule is that the images of ω below contain some $x_{\sigma,i}$ without its partner, so they are independent of every image of p.

A line $y_{\sigma,k}\otimes\partial_{\overline\sigma,k}$ carries p to $\pm y_{\sigma,k}^2(\cdots)=0$ and $\alpha_{\overline\sigma}$ to $\pm y_{\sigma,k}x_{\overline\sigma,k'}y_{\overline\sigma,1}y_{\overline\sigma,2}\neq 0$, $k'\neq k$: rank one. On a space of the second kind, with labels $P=(\rho,k)$ over τ_1 and $Q=(\rho',l)$ over τ_2, the four basis elements are $y_P\wedge y_Q$, $y_P\otimes\partial_{\rho',l}$, $y_Q\otimes\partial_{\rho,k}$ and $\partial_{\rho,k}\wedge\partial_{\rho',l}$. They carry ω to $y_Py_Q(\alpha_\rho+\alpha_{\rho'})$, $y_P\,(\partial_{\rho',l}\lrcorner\,\alpha_{\rho'})$, $y_Q\,(\partial_{\rho,k}\lrcorner\,\alpha_\rho)$ and 0, the factors y_P and y_Q killing $\alpha_{\overline\rho}$ and $\alpha_{\overline{\rho}'}$; these contain $x_{\rho,k'}$ or $x_{\rho',l'}$ with $k'\neq k$, $l'\neq l$, without its partner, and the first three are independent, having two, one and one such factors x at different places. The bivector carries γ to $\pm y_Py_Q(\partial_1\partial_2 p)(\theta_1,\theta_2)$, with three terms left in each θ_t, and $\partial_1\partial_2 p$ has degree at most three in each variable; so this image is zero exactly when $\partial_1\partial_2 p=0$: rank $3+\mu(p)$. On a space of the third kind, with labels $P=(\sigma,k)$ and $Q=(\overline\sigma,l)$ over τ_1, the element $y_P\wedge y_Q$ carries ω to $y_Py_Q(\alpha_{\sigma'}+\alpha_{\overline{\sigma}'})$, σ' and $\overline{\sigma}'$ the embeddings over τ_2, and the other three elements kill ω: $y_P\otimes\partial_{\overline\sigma,l}$ and $y_Q\otimes\partial_{\sigma,k}$ carry $\alpha_{\overline\sigma}$ and α_σ into multiples of $y_{\overline\sigma,k}y_{\overline\sigma,1}y_{\overline\sigma,2}=0$ and $y_{\sigma,l}y_{\sigma,1}y_{\sigma,2}=0$, and the bivector meets no α. They carry p to $\pm y_Py_Q(\partial_1 p)$, twice, and to $\pm y_Py_Q(\partial_1^2 p)$, with two terms left in θ_1 and four in θ_2. So the rank is 1 plus the dimension of the span of $\partial_1 p$ and $\partial_1^2 p$ modulo θ_1^3; in the basis $\theta_1^a\theta_2^j/(a!\,j!)$, $a\le 2$, their coefficient vectors are $(e_{a+1,j})$ and $(e_{a+2,j})$, and the rank is $1+\rho_1(p)$; over τ_2 it is $1+\rho_2(p)$. The total is $8+16(3+\mu)+4(1+\rho_1)+4(1+\rho_2)+R_1+R_2$.

(ii) Over τ_1 write x_k,X_k,y_k,Y_k for $x_{\sigma,k},x_{\overline\sigma,k},y_{\sigma,k},y_{\overline\sigma,k}$, so that $\theta_1=\sum_k(x_kY_k+X_ky_k)$, $\alpha_\sigma=x_1x_2y_1y_2$ and $\alpha_{\overline\sigma}=X_1X_2Y_1Y_2$, and put $c=y_1y_2$, $C=Y_1Y_2$, $a_k=x_ky_1y_2Y_k$, $A_k=X_ky_kY_1Y_2$, $b=x_1x_2y_1y_2Y_1Y_2$ and $B=X_1X_2y_1y_2Y_1Y_2$. Let α' be the part of ω over τ_2, so that $\alpha'\theta_2=0$, and put $\langle e_i\rangle=\sum_j e_{ij}\theta_2^j/j!$. Since c kills the terms X_ky_k of θ_1, $c\,\theta_1^i/i!$ is c, $a_1{+}a_2$, $-b$ and 0 for $i=0$, 1, 2 and $i\ge 3$, so that $c\gamma=B{+}c(\alpha'{+}\langle e_0\rangle){+}(a_1{+}a_2)\langle e_1\rangle{-}b\langle e_2\rangle$; in the same way, with ∂ acting as a left derivation, the other seven basis elements of W_1 carry γ to

$$\begin{aligned}
C\gamma &= b+C(\alpha'+\langle e_0\rangle)+(A_1+A_2)\langle e_1\rangle-B\langle e_2\rangle,\\
\partial_{\sigma,1}\lrcorner\,\partial_{\sigma,2}\lrcorner\,\gamma &= -c+C\langle e_2\rangle+(A_1+A_2)\langle e_3\rangle-B\langle e_4\rangle,\\
\partial_{\overline\sigma,1}\lrcorner\,\partial_{\overline\sigma,2}\lrcorner\,\gamma &= -C+c\langle e_2\rangle+(a_1+a_2)\langle e_3\rangle-b\langle e_4\rangle,\\
\pm y_k\,\partial_{\overline\sigma,k'}\lrcorner\,\gamma &= A_k+c\langle e_1\rangle+(a_1+a_2)\langle e_2\rangle-b\langle e_3\rangle,\\
\pm Y_k\,\partial_{\sigma,k'}\lrcorner\,\gamma &= a_k+C\langle e_1\rangle+(A_1+A_2)\langle e_2\rangle-B\langle e_3\rangle,
\end{aligned}$$

where $\{k,k'\}=\{1,2\}$ and the sign is + for $k=1$. The differences of the last two pairs are A_1-A_2 and a_1-a_2. In a basis containing A_1-A_2, a_1-a_2, $c\alpha'$ and $C\alpha'$, these four coordinates form the identity matrix on the two differences and on $c\gamma$, $C\gamma$, and vanish on the other four images. So R_1 is 4 plus the rank of the two images of the bivectors and of the halves of the sums of the last two pairs, $\frac12(A_1+A_2)+c\langle e_1\rangle+(a_1+a_2)\langle e_2\rangle-b\langle e_3\rangle$ and $\frac12(a_1+a_2)+C\langle e_1\rangle+(A_1+A_2)\langle e_2\rangle-B\langle e_3\rangle$. In the coordinates along c, a_1+a_2, b, C, $-(A_1+A_2)$, B times $\theta_2^j/j!$, and up to the signs of some columns, the matrix of these four vectors is $K=(K^0\,|\,\mathrm{diag}(H,H))$: the columns with $j=0$ form, with $e_k=e_{k0}$,

$$K^0=\begin{pmatrix} e_1 & e_2 & -e_3 & 0 & -\frac12 & 0\\ e_2 & e_3 & -e_4 & -1 & 0 & 0\\ 0 & \frac12 & 0 & e_1 & -e_2 & -e_3\\ -1 & 0 & 0 & e_2 & -e_3 & -e_4\end{pmatrix},$$

and those with $j \geq 1$ form $\operatorname{diag}(H, H)$, with H the 2×12 matrix of the pairs $(e_{i,j}, e_{i+1,j})$, $1 \leq i \leq 3$, $1 \leq j \leq 4$. So $R_1 = 4 + \operatorname{rk} K$. Let (a, b, c, d) annihilate the columns of K^0. The first two columns give $d = ae_1 + be_2$ and $c = -2(ae_2 + be_3)$, and the other four then say that (a, b) is orthogonal to the rows of

$$Q = \begin{pmatrix} e_3 & e_4 \\ -e_1e_2 & e_2^2 - 2e_1e_3 - 1 \\ 2e_2^2 - e_1e_3 - \frac{1}{2} & e_2e_3 \\ e_1e_4 - 2e_2e_3 & e_2e_4 - 2e_3^2 \end{pmatrix},$$

so $\operatorname{rk} K^0 = 2 + \operatorname{rk} Q$. If Q were zero, then $e_3 = e_4 = 0$ by its first row, $e_2^2 = \frac{1}{4}$ by its third, and the second row would have the entry $-\frac{3}{4}$. So $\operatorname{rk} K \geq 3$ and $R_1 \geq 7$. If $R_1 = 7$, then $\operatorname{rk} Q = 1$, and the 2×2 minors of Q vanish exactly on $V_+ \cup V_-$. Indeed, with $P = 2h_1 + 1$ and m_{ij} the minor of rows i and j, expanding gives

$$m_{12} = e_2h_3 - e_3P, \quad 2m_{13} = e_4P - 2e_2h_2, \quad m_{14} = 2e_3h_2 - e_4h_3,$$
$$2m_{23} = e_2^2 - P^2, \quad m_{24} = Ph_3 - e_2e_3, \quad 2m_{34} = e_3^2 - h_2 - 4h_1h_2.$$

On $V_\pm$, $(P, h_3, 2h_2) = \pm(e_2, e_3, e_4)$, which makes the first five vanish, and the last is $e_3^2 - h_2 - 2(\pm e_2 - 1)h_2 = e_3^2 + h_2 - e_2e_4 = 0$. Conversely, let all six vanish. If $e_2 = e_3 = e_4 = 0$ then $P = 1$ and $m_{23} \neq 0$; so $(P, h_3, 2h_2) = \lambda(e_2, e_3, e_4)$ for some λ, by the first three. If $e_2 \neq 0$, then $m_{23} = 0$ gives $\lambda = \pm 1$. If $e_2 = 0$, then $P = 0$, so $e_1e_3 = -\frac{1}{2}$, and $h_3 = e_1e_4 = \lambda e_3$ and $2h_2 = -2e_3^2 = \lambda e_4$ give $e_4 = -2\lambda e_3^2$ and then $\lambda^2 = 1$. In both cases the point lies on V_λ. If moreover $\rho_1(p) \leq 1$, then $h_1 = h_2 = h_3 = 0$, and the minors of the second and third rows and of the first and third rows become $(e_2^2 - 1)/2$ and, once $e_2^2 = 1$, $e_4/2$; so $e_4 = 0$, $e_3^2 = e_2e_4 = 0$ and $e_1e_3 = 0 \neq e_2^2$, against $h_1 = 0$. Hence $\rho_1(p) = 2$. If $\mu(p) = 0$, then $e_{ij} = 0$ for $i, j \geq 1$, so $H = 0$ and $R_1 = 6 + \operatorname{rk} Q$, which is 7 on $V_\pm$; the point $(0, 1, 0, 0)$ of $p = \theta_1^2/2$ lies on V_-. The parametrised points satisfy $h_1 = \pm\frac{1}{2}w\lambda^2 - \frac{1}{4}$, $h_2 = \pm\frac{1}{2}w\lambda^4$ and $h_3 = \pm w\lambda^3$, which are the equations of $V_\pm$. The space W_2 is the same with the indices exchanged.

(iii) A nonzero e_{ij} with $i, j \geq 1$ enters both Hankel matrices, so $\mu = 1$ forces $\rho_1, \rho_2 \geq 1$, and by (ii) $R_t = 7$ forces $\rho_t = 2$. The tuples $(\mu, \rho_1, \rho_2, R_1, R_2)$ allowed by these two conditions give exactly the fifteen values, and 80 only for $(0, 0, 0, 8, 8)$; $\rho_1 = \rho_2 = 0$ means that $e_{ij} = 0$ unless $i = j = 0$, that is, p is constant. Each value is taken. For $p = 0$, θ_1, θ_1^2, $\theta_1^2 + \theta_2$, $\theta_1^2 + \theta_2^2$, $\theta_1\theta_2$, $\theta_1\theta_2^2$ and $\theta_1^2\theta_2^2$ the points $(e_{10}, \dots, e_{40})$ and $(e_{01}, \dots, e_{04})$ are 0, $(1, 0, 0, 0)$ or $(0, 2, 0, 0)$, which lie on neither $V_\pm$, as $2h_1 + 1$ is 1, 1 and -7 there; so $R_1 = R_2 = 8$, and (μ, ρ_1, ρ_2) is $(0, 0, 0)$, $(0, 1, 0)$, $(0, 2, 0)$, $(0, 2, 1)$, $(0, 2, 2)$, $(1, 1, 1)$, $(1, 1, 2)$ and $(1, 2, 2)$, giving 80, 84, 88, 92, 96, 104, 108 and 112. For $p = \theta_1^2/2 + q$ with q one of 0, θ_2, θ_2^2, $\theta_2^2/2$ the point $(0, 1, 0, 0)$ lies on V_- and $\mu = 0$, so $R_1 = 7$, and $R_2 = 7$ exactly for $q = \theta_2^2/2$; this gives 87, 91, 95 and 94. For $p = \theta_1^2/2 + \theta_1\theta_2 + q$ with q one of 0, θ_2^2, $\theta_2^2/2$, the only nonzero e_{ij} with $i, j \geq 1$ is $e_{11} = 1$, so the columns of $\operatorname{diag}(H, H)$ are multiples of $(1, 0, 0, 0)$ and $(0, 0, 1, 0)$; at $e = (0, 1, 0, 0)$ the columns of K^0 span the space of $(1, 0, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 1, 0, -1)$, so $\operatorname{rk} K = 3$ and $R_1 = 7$, and likewise $R_2 = 7$ for $q = \theta_2^2/2$. With $\mu = 1$ and $(\rho_1, \rho_2) = (2, 1)$, $(2, 2)$, $(2, 2)$ this gives 107, 111 and 110.

(iv) The element of the torus of the proof of Theorem 5.13(ii) that multiplies $x_{\sigma,k}$ by s_σ and $y_{\sigma,k}$ by t_σ, with $s_\sigma t_{\overline{\sigma}} = \kappa_t$ for σ over τ_t, multiplies θ_t by κ_t and α_σ by $(s_\sigma t_\sigma)^2$, and these can be chosen to make every $u_\sigma(s_\sigma t_\sigma)^2$ equal to 1 exactly when $\kappa_t^4 = (u_\sigma u_{\overline{\sigma}})^{-1}$. It carries $\omega + p$ to the normalised class with coefficients $\kappa_1^i\kappa_2^j e_{ij}$, and contraction is equivariant. Replacing κ_t by $-\kappa_t$ changes the signs of e_1 and e_3, which fixes V_+ and V_-, and replacing it

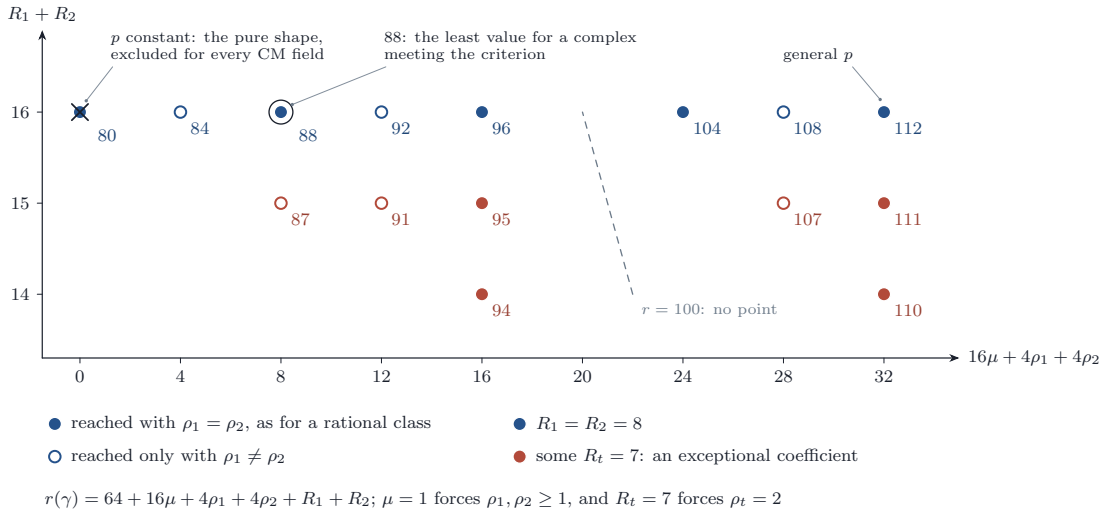


FIGURE 16. The rank $r(\gamma)$ of Theorem 5.14, over the plane of the Hankel part $16\mu + 4\rho_1 + 4\rho_2$ and of $R_1 + R_2$. The fifteen points are the tuples allowed by the two conditions of the theorem, and each carries its value. Filled points are reached with $\rho_1 = \rho_2$, as the character of a complex with rational coefficients is (Corollary 5.15(ii)); red points need an exceptional coefficient at one place at least. The value 80 is the pure shape, excluded for every CM field by Theorem 5.4; so 88 is the least rank a complex meeting the criterion can have. The level $r = 100$ meets no point.

by $\sqrt{-1}\,\kappa_t$ exchanges them; scaling the columns and the second row of the Hankel matrices does not change their ranks. ■

The count of (i), drawn in Figure 16, is the analogue of the dimension counts of Theorem 3.9 for a field of degree four: sixty-four of the rank come from the Weil class alone, the Hankel terms measure how far p is from an exponential in each variable, and the two eight-dimensional spaces W_t carry the exceptional coefficients, the analogue of the exceptional ratio of Remark 3.12. Indeed $\rho_1(p) \leq 1$ exactly when $(e_{10}, \dots, e_{40})$ lies on the cone over the twisted cubic, that is, when the part of p in θ_1 alone is, up to a constant, a multiple of $e^{\lambda\theta_1} - 1$, of θ_1 or of θ_1^4.

Corollary 5.15 (The least object the criterion asks for)**.** *Let E be as in Theorem 5.13*(v).

(i) $\dim \mathrm{Ext}^2(E, E) \geq 81$.

(ii) *The coefficients of P satisfy $e_{ji} = \iota(e_{ij})$ for the nontrivial automorphism ι of F_0, so $\rho_1 = \rho_2$, and $r(\mathrm{ch}\, E)$ is one of* 80, 88, 94, 95, 96, 104, 110, 111, 112.

(iii) *If $\dim \mathrm{Ext}^2(E, E) = r(\mathrm{ch}\, E)$, then P is not constant and $\dim \mathrm{Ext}^2(E, E)$ is one of* 88, 94, 95, 96, 104, 110, 111, 112. *The value* 88 *requires $\mu = 0$ and $\rho_1 = \rho_2 = 1$, and it is the rank of $N\omega_1 + \eta_s$ for every $s \in F_0^\times$.*

Proof. (i) By Theorem 5.13(v), $\dim \mathrm{Ext}^2(E, E) \geq r(\mathrm{ch}\, E) \geq 80$. If $r(\mathrm{ch}\, E) = 80$, then P is a constant by Theorem 5.14(iii), so $\mathrm{ch}(E) = c + N\omega_1$ and Theorem 5.4 makes the inequality strict. (ii) Write $\eta_t = \tau_1(t)\theta_1 + \tau_2(t)\theta_2$ as before Proposition 5.5. A polynomial with rational coefficients in the η_t becomes, after exchanging θ_1 and θ_2, the polynomial whose coefficients are the images under ι of the old ones, so $e_{ji} = \iota(e_{ij})$; the Hankel matrix

of ρ_2 is then the image under ι of that of ρ_1, which has the same rank. The values are those of Theorem 5.14(iii) with $\rho_1 = \rho_2$, the scaling of (iv) there changing neither μ nor the ρ_t. (iii) If P were constant, Theorem 5.4 would give $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$. For $P = \eta_s$ the pairs are $(\tau_t(s), 0)$, so $\rho_1 = \rho_2 = 1$, $\mu = 0$, and $R_t = 8$ by Theorem 5.14(ii). ■

The Weil tori of Setup 5.1 exclude more than the pure shape. A polynomial in θ_1, θ_2 can stay of Hodge type on all of S_F, and then the argument of Theorem 5.4 applies to it.

Proposition 5.16 (The characters that the Weil tori exclude)**.** *Let E be as in Theorem 5.13*(v)*, and suppose that P lies in the span of the $\theta_1^i\theta_2^j$ with $i,j \in \{0,4\}$, as for $\mathrm{ch}(E) = c + N\omega_1 + b\,\theta_1^4 + \iota(b)\,\theta_2^4$ with $b \in F_0$. Then σ_E is not injective and $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$. Consequently, if $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$, then $P = c + f(\theta_1) + f^\iota(\theta_2)$ with $c \in \mathbb{Q}$, where $f(\theta)$ is a nonzero multiple either of θ or of $e^{\lambda\theta} - 1$ for some $\lambda \neq 0$, and f^ι is obtained by applying ι to the coefficients of f.*

Proof. In the basis before Theorem 5.14, θ_1 is a sum of four products of two classes, one from V_σ and one from $V_{\overline{\sigma}}$ for the two embeddings over τ_1, in disjoint sets of basis vectors. So θ_1^4 is 4! times their product, a nonzero element of $\bigwedge^4 V_\sigma \otimes \bigwedge^4 V_{\overline{\sigma}}$, that is a nonzero multiple of $\alpha_\sigma\alpha_{\overline{\sigma}}$, and likewise for θ_2^4. Hence $\mathrm{ch}(E) = N\omega_1 + P$ lies in the span of the classes α_Σ, which are of type (p,p) on every member of S_F (Setup 5.1); being rational, it is a Hodge class on every member. Its component of degree four is $N\omega_1 \neq 0$, since P has none, and Theorem 5.4 applies. If $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$, then $\mu = 0$ and $\rho_1 = \rho_2 = 1$ by Corollary 5.15(iii), so P has no mixed monomials, and by the paragraph after Theorem 5.14 its part in θ_1 alone is a multiple of $e^{\lambda\theta_1} - 1$, of θ_1 or of θ_1^4; its part in θ_2 alone is obtained by ι, by Corollary 5.15(ii). In the third case P lies in the span of 1, θ_1^4 and θ_2^4, which the first part excludes. ■

The shape $c + b\,\theta_1^4 + \iota(b)\,\theta_2^4$ has rank 88, and with a multiple of $\theta_1^4\theta_2^4$ added it has rank 104, by Theorem 5.14: $\rho_1 = \rho_2 = 1$, μ is 0 or 1, and $R_1 = R_2 = 8$, since $(e_{10}, \dots, e_{40}) = (0,0,0,24b)$ with $b \neq 0$ does not meet $V_+ \cup V_-$. Proposition 5.16 removes both. What remains at 88 is a Chern character whose polynomial part is exponential or linear in each θ_t separately, such as $c + \eta_s + N\omega_1$. For these the classes θ_t are not of Hodge type on the Weil tori, and the argument needs a family of complex tori along which the character stays of Hodge type and whose members are not algebraic. The next proposition, the analogue for a quartic field of Proposition 3.11(iii) and (iv), shows that there is none: the Hodge locus of such a character is the polarised family itself.

Proposition 5.17 (The Hodge locus of a character at a quartic field)**.** *Keep the notation before Theorem 5.14, with $\gamma = \omega + p$ and every $u_\sigma \neq 0$, and for the embeddings $\sigma, \overline{\sigma}$ over τ_t put $\omega_t = u_\sigma\alpha_\sigma + u_{\overline{\sigma}}\alpha_{\overline{\sigma}}$. Call τ_t a* mid place *of p if p has a monomial $\theta_1^i\theta_2^j$ with nonzero coefficient whose exponent of θ_t is 1, 2 or 3, and call a mid place* exceptional *if the only such monomial is θ_t^2 itself and its coefficient c_t in p satisfies $c_t^2\theta_t^4 = \frac{3}{4}\,\omega_t^2$, the ratio of Proposition 3.11(iv).*

(i) *The annihilator of γ in $H^1(T_A)$ is the space of F-linear $\xi \in \bigoplus_\sigma \mathrm{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1})$ with $\xi \lrcorner \theta_t = 0$ at every mid place, plus, at each exceptional place, a line spanned by an element of $\mathrm{Hom}(V_\sigma^{1,0}, V_{\overline{\sigma}}^{0,1}) \oplus \mathrm{Hom}(V_{\overline{\sigma}}^{1,0}, V_\sigma^{0,1})$ with both components nonzero. Its dimension is $8 + 4a + b$, where a is the number of places that are not mid places and b the number of exceptional places.*

(ii) *The connected component through A of the Hodge locus of γ, in the space of complex structures on the real torus underlying A, is S_F when neither place is a mid place,*

that is when p lies in the span of the $\theta_1^i\theta_2^j$ with $i,j\in\{0,4\}$; and it is $\mathcal{D}_F$ when both places are mid places and neither is exceptional. In the second case all its members are abelian varieties polarised by $\theta_1+\theta_2$.

(iii) *If both places are exceptional, then near A the Hodge locus is the orbit of A under $G_1\times G_2$, where $G_t\cong\mathrm{Spin}(4,3)$ is the connected stabiliser of $X_t=\omega_t+c_t\theta_t^2$ in the real linear group of $V_\sigma\oplus V_{\overline\sigma}$, and it has dimension 10. A very general member T of it has $\mathrm{NS}(T)=0$ and $\mathrm{Hdg}^2(T)\otimes\mathbb{C}$ contained in the span of X_1 and X_2, and on every member, for every Kähler class κ, $\int X_t\,\kappa^6$ is nonzero with the sign of c_t, in the normalisation in which θ_t is a real class with θ_t^2 positive.*

(iv) *If γ is the complexified character of Theorem 5.13(v), then both places are mid places or neither is, and both are exceptional or neither is; and an exceptional γ has $r(\gamma)=94$, or 110 when p contains $\theta_1^4\theta_2^4$.*

Proof. (i) Grade $\bigwedge^* H^1$ by the degrees in the four V_σ, and write $\xi_{\sigma\sigma'}$ for the component of ξ in $\mathrm{Hom}(V_{\sigma'}^{1,0},V_\sigma^{0,1})$, which raises the degree in V_σ by one and lowers that in $V_{\sigma'}$ by one. The monomial $\theta_1^i\theta_2^j$ has degree i in the two embeddings over τ_1 and j in those over τ_2, and α_σ has degree 4 in V_σ. By the proof of Theorem 5.13(i), $\xi_{\sigma\sigma}\lrcorner\omega=0$, and for $\sigma\neq\sigma'$ the only term of $\xi_{\sigma\sigma'}\lrcorner\omega$ is $u_{\sigma'}\xi_{\sigma\sigma'}\lrcorner\alpha_{\sigma'}$, of degree 3 in $V_{\sigma'}$ and 1 in V_σ, which determines $\xi_{\sigma\sigma'}$. A term $\xi_{\rho\rho'}\lrcorner\theta_1^i\theta_2^j$ has that degree only if σ,σ' lie over the same place τ_t, (i,j) is $(2,0)$ or $(0,2)$ with the 2 at t, and $(\rho,\rho')=(\sigma',\sigma)$. So every component $\xi_{\sigma\sigma'}$ with σ,σ' over different places vanishes, and the equations split into the F-linear part and, at each place, the pair $\xi'=\xi_{\overline\sigma\sigma}$, $\xi''=\xi_{\sigma\overline\sigma}$.

For F-linear ξ put $D_t=\xi\lrcorner\theta_t\in V_\sigma^{0,1}\otimes V_{\overline\sigma}^{0,1}$; the map $\xi\mapsto(D_1,D_2)$ is onto, with kernel $T_{[A]}\mathcal{D}_F$. Then $\xi\lrcorner\theta_1^i\theta_2^j=i\,\theta_1^{i-1}\theta_2^jD_1+j\,\theta_1^i\theta_2^{j-1}D_2$, the monomials lie in different degrees, and the two terms have different Hodge types at the two places, so each vanishes. Now $\theta_t^3D_t=0$ for every D_t, as a degree count shows, while multiplication by θ_t^k is injective on $V_\sigma^{0,1}\otimes V_{\overline\sigma}^{0,1}$ for $k\le2$: $\theta_t^2y_{\sigma,k}y_{\overline\sigma,l}$ is 2 times the product of $y_{\sigma,k}y_{\overline\sigma,l}$ with the two terms of θ_t that avoid $y_{\sigma,k}$ and $y_{\overline\sigma,l}$, and different (k,l) give different monomials. Multiplication by $\theta_{t'}^j$, $j\le4$, acts on the other factor of $\bigwedge^*H^1=\bigwedge^*(V_\sigma\oplus V_{\overline\sigma})\otimes\bigwedge^*(V_{\sigma'}\oplus V_{\overline{\sigma'}})$ and is nonzero there. So the conditions on the F-linear part are exactly $D_t=0$ at the mid places, a space of dimension $16-4(2-a)=8+4a$.

At a place τ_t let c_t be the coefficient of θ_t^2 in p. The terms involving ξ'' are $u_{\overline\sigma}\xi''\lrcorner\alpha_{\overline\sigma}+c_t\,\xi'\lrcorner\theta_t^2$, of degree 3 in $V_{\overline\sigma}$ and 1 in V_σ, the same with σ and $\overline\sigma$ and with ξ' and ξ'' exchanged, and $i\,\theta_t^{i-1}\theta_{t'}^j(\xi'\lrcorner\theta_t)$ and its analogue for ξ'' for every other monomial with exponent $i\in\{1,2,3\}$ of θ_t. Here $\xi'\lrcorner\theta_t$ lies in the line $\bigwedge^2V_{\overline\sigma}^{0,1}$, on which multiplication by $\theta_t^{i-1}\theta_{t'}^j$ is injective for $i\le3$ by the same count. If such another monomial occurs, $\xi'\lrcorner\theta_t=\xi''\lrcorner\theta_t=0$, and then $\xi'=\xi''=0$, the maps $\xi'\mapsto\xi'\lrcorner\alpha_\sigma$ and $\xi''\mapsto\xi''\lrcorner\alpha_{\overline\sigma}$ being injective. Otherwise $\xi''\lrcorner\theta_t^2=2\theta_t\,(\xi''\lrcorner\theta_t)$ is a multiple of $\theta_t\,y_{\sigma,1}y_{\sigma,2}$, which is $\xi_0'\lrcorner\alpha_\sigma$ for one element ξ_0', and likewise with σ and $\overline\sigma$ exchanged; so ξ' and ξ'' are multiples of ξ_0' and ξ_0''. Explicitly, $\xi_0'(x_{\sigma,1})=-y_{\overline\sigma,2}$ and $\xi_0'(x_{\sigma,2})=y_{\overline\sigma,1}$ give $\xi_0'\lrcorner\alpha_\sigma=\theta_t\,y_{\sigma,1}y_{\sigma,2}$ and $\xi_0'\lrcorner\theta_t=2y_{\overline\sigma,1}y_{\overline\sigma,2}$, and symmetrically for ξ_0''. Writing $\xi'=a\xi_0'$ and $\xi''=b\xi_0''$, the two remaining equations are $u_\sigma a+4c_tb=0$ and $4c_ta+u_{\overline\sigma}b=0$, and their determinant vanishes exactly when $16c_t^2=u_\sigma u_{\overline\sigma}$. Since $\theta_t^4=24\,\alpha_\sigma\alpha_{\overline\sigma}$ and $\omega_t^2=2u_\sigma u_{\overline\sigma}\alpha_\sigma\alpha_{\overline\sigma}$, that is the condition $c_t^2\theta_t^4=\frac34\omega_t^2$, and the solutions then form a line on which both components are nonzero.

(ii) By Proposition 3.11(i), applied to each homogeneous component, the Hodge locus of γ in the chart at A is the linear space $\mathrm{Ann}_{H^1(T_A)}(\gamma)$. If neither place is a mid place, that space is the space of all F-linear ξ, which is also the part of S_F in the chart; γ is

of Hodge type on all of S_F, as in the proof of Proposition 5.16, and at every member of S_F its component of degree four is ω, whose annihilator in $H^1(T)$ is again the space of F-linear deformations (Theorem 5.13(i), whose proof uses only the decomposition into the V_σ). So S_F is open in the Hodge locus; it is closed in the space of complex structures, since commuting with F is a closed condition and the dimensions of the $H^{1,0} \cap V_\sigma$ are locally constant on it; and it is connected. If both places are mid places and neither is exceptional, the annihilator is $T_{[A]}\mathcal{D}_F$ by (i), at every member of $\mathcal{D}_F$, since the conditions of (i) are stated in terms of the flat classes θ_t and ω_t; and the argument of Proposition 3.11(iii) shows that $\mathcal{D}_F$ is open, closed and connected in the Hodge locus.

(iii) At each place X_t is, in the 8-dimensional space $V_\sigma \oplus V_{\overline{\sigma}}$ with the form θ_t, the class of Remark 3.12, and the argument there applies word for word, with the factor $\mathfrak{su}(2,2)$ at τ_t of the Lie algebra of G_F in place of the Hodge group: the stabiliser is $\mathfrak{so}(4,3)$, with an orbit of dimension 5 in the Grassmannian. The other components of γ are constants, multiples of θ_t^4, which is 24 times the volume form of $V_\sigma \oplus V_{\overline{\sigma}}$, and of $\theta_1^4\theta_2^4$, all fixed by $G_1 \times G_2$. So the orbit of A lies in the Hodge locus, and it has dimension 10, which by (i) is that of the locus near A. A rational class of Hodge type at a very general member is of Hodge type on the whole orbit, hence killed by the complex structures of all its members, which lie in $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ and generate an ideal projecting nontrivially to both simple factors, that is all of it. The invariants of $\mathfrak{so}(7,\mathbb{C})$ in the exterior algebra of its spin representation are 1, X_t and the volume form (Remark 3.12), so in degree two there are none and in degree four only X_1 and X_2. For the sign, a Kähler class is κ_h for a positive hermitian form h on the tangent space; only the terms of κ_h^6 that contain the whole volume of the other place pair with X_t, and they give $\det(h_{t'})$ times $\kappa_{h'}^2$ for the Schur complement h' of $h_{t'}$ in h, a positive hermitian form on the tangent space at τ_t. So $\int X_t\kappa_h^6$ is a positive multiple of the integral of Remark 3.12 for h', which has the sign of c_t; and $G_1 \times G_2$ fixes X_t and carries Kähler cones to Kähler cones.

(iv) An automorphism of $\overline{\mathbb{Q}}$ inducing ι on F_0 fixes the rational class γ and exchanges θ_1 with θ_2 and ω_1 with ω_2, so it carries the conditions at τ_1 to those at τ_2. An exceptional p lies in the span of 1, θ_t^2, θ_t^4 and $\theta_1^4\theta_2^4$ with $c_t \neq 0$, so $\rho_1 = \rho_2 = 2$, $\mu = 1$ exactly when $\theta_1^4\theta_2^4$ occurs, and $(e_{10}, \dots, e_{40})$ lies on $V_+ \cup V_-$, so $R_1 = R_2 = 7$ by Theorem 5.14(ii); the formula of Theorem 5.14(i) gives 94 or 110. ∎

Corollary 5.18 (The Weil tori at rank 88)**.** *Let E be as in Theorem 5.13*(v) *with* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\, E) = 88$*, so that* $\mathrm{ch}(E) = N\omega_1 + c + f(\theta_1) + f^\iota(\theta_2)$ *as in Proposition 5.16. Then the connected component through A of the Hodge locus of* $\mathrm{ch}(E)$ *is $\mathcal{D}_F$, and every complex torus reached from A by a deformation along which* $\mathrm{ch}(E)$ *stays of Hodge type is an abelian variety of the family. So no argument that deforms E itself to a torus that is not algebraic can exclude it; the characters that such an argument excludes at* 88 *are exactly those of Proposition 5.16.*

Proof. The coefficient e_{10} of f is nonzero, both for a multiple of θ and for one of $e^{\lambda\theta} - 1$, and $e_{01} = \iota(e_{10})$, so both places are mid places, with the monomials θ_1 and θ_2, and neither is exceptional. Proposition 5.17(ii) applies. ∎

As after Proposition 3.11, this concerns E itself: an autoequivalence of $D^b(A)$ changes the character, and the proposition then applies to the new one. For the exceptional characters of Proposition 5.17(iii), of rank 94 or 110, the proof of Theorem 5.4 can be started, since the locus contains tori without divisors, but it breaks down as in Remark 3.12: the classes X_t have integrals of constant sign against every Kähler class, so Bando and Siu give no vanishing: they bound the signs of the coefficients of ch_2 of a torsion-free

sheaf along X_1 and X_2, which a complex, an alternating sum, need not respect. So at a quartic field the Weil tori and their exceptional analogues exclude exactly the characters of Proposition 5.16, and what the criterion still asks for, at 88 and above, is the construction of an object.

The natural candidates are Markman's secant objects [Mar25c], in the twisted form in which he states his reduction. For them a B-field twist of $\mathrm{ch}(E)$, the class κ of Theorem 5.22, is flat, and under a non-degeneracy condition its component in H^4 is a polynomial in the η_t plus a nonzero Weil class [Mar25c, Proposition 10.2.1, Theorem 1.1.2]. But $\mathrm{ch}(E)$ itself is not flat, and κ is not of the form $N\omega + P(\eta)$ in higher degrees (Remark 5.23), so they are candidates only for a twisted form of Theorem 5.13(v). The dimension count that works for imaginary quadratic fields (Theorem 3.19) can be tested on them exactly. Let X be a principally polarised abelian fourfold with real multiplication by the ring of integers $\mathcal{O}_{F_0}$, write $\theta_1, \theta_2 \in H^2(X, \mathbb{R})$ for the components of the polarisation at the two real places of F_0, so that $\theta = \theta_1 + \theta_2$ and $\theta_j^3 = 0$, and let $\mathcal{R}$ be the $\mathbb{Q}$-subalgebra of $H^{\mathrm{ev}}(X, \mathbb{Q})$ spanned by the rational combinations of the $\theta_1^a \theta_2^b$, $0 \le a, b \le 2$. A class $v = \sum_{a,b} V_{ab}\, \theta_1^a \theta_2^b/(a!\, b!)$ of $\mathcal{R}$ has a 3×3 matrix V with entries in $F_0 \otimes \mathbb{R}$.

Proposition 5.19 (The secant space of a quartic field)**.** *Let $F = F_0(\sqrt{-q})$ with $q \in F_0$ totally positive, let $t \in F_0$, $s_j = \sqrt{-1}\sqrt{\tau_j(q)}$, and for $\varepsilon \in \{\pm 1\}^2$ put $e_\varepsilon = \exp(\sum_j (\tau_j(t) + \varepsilon_j s_j)\theta_j)$.*

(i) *The span $S(t,q)$ of the four e_ε is a rational subspace of $\mathcal{R}$ of dimension $4 = 2^{[F_0:\mathbb{Q}]}$, secant to the spinor variety. For $t = 0$ it is the secant space of* [Mar25c, §11.1, (11.1.2)] *for the polarisation θ and q; replacing θ by $f\theta$, $f \in F_0^\times$, replaces q by qf^2, and a rational F_0-B-field t transports Markman's F-structure by the rational automorphism of $X \times \widehat{X}$ acting on $H^*(X)$ as e^{θ_t}, $\theta_t = \sum_j \tau_j(t)\theta_j$, and replaces $S(0,q)$ by $S(t,q) = e^{\theta_t} S(0,q)$.*

(ii) *For $v = \sum_\varepsilon w_\varepsilon e_\varepsilon \in S(t,q)$ let $\rho = \operatorname{rk} V$ and $N_w = \#\{\varepsilon : w_\varepsilon \neq 0\}$. The ranks of contraction from $HT^k(X)$ into v are 1, 8 and $2\rho + 4N_w$ for $k = 0, 1, 2$, with $(\rho, N_w) \in \{(1,4), (2,4), (2,2)\}$, and $N_w = 2$ only for biquadratic F and v in the secant plane of one of its two imaginary quadratic subfields* [Mar25c, §11.2.1].

(iii) *For $v = e^{\theta_t}(a\,u + v_f + c\,w)$ in the basis $u = (1 - \frac{1}{2}q_1\theta_1^2)(1 - \frac{1}{2}q_2\theta_2^2)$, $v_f = \theta_f - \frac{1}{2}\theta_{\bar{f}q}\theta_1\theta_2$, $w = \theta_1\theta_2$ of $e^{-\theta_t} S(t,q)$, one has $\chi(v,v) = \int_X v^\vee v = 4(\mathrm{N}(q)a^2 + c^2 + \mathrm{Tr}(\bar{q}f^2))$, which is positive definite.*

Proof. (i) The Galois group permutes the four exponentials as it permutes the CM types of F, so their span is rational; since $\theta_j^3 = 0$, $e^{\pm s_j\theta_j} = (1 - \frac{1}{2}\tau_j(q)\theta_j^2) \pm s_j\theta_j$, and the four classes are independent. The conjugates in $\mathfrak{so} \cap \mathrm{End}^0(X \times \widehat{X})$ of the standard structure have eigenvalues $t \pm s$, and the pure spinors of their eigenspaces are the e_ε.

(ii) Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with U_j the eigenspaces of F_0, and $HT^*(X)$ and $H^*(X)$ are tensor products over j, compatibly with contraction. By Lemma 4.55(i), $\xi \lrcorner e^{\lambda\theta_j} = y^\lambda(\xi) e^{\lambda\theta_j}$, and the blocks $HT_1^a \otimes HT_2^b$ have independent images because they shift the two Hodge bidegrees by (a,b) while v is balanced. The block $(1,1)$ contributes $4N_w$, by the invertibility of the Hadamard matrix of the signs; the blocks $(2,0)$ and $(0,2)$ contribute ρ each; the blocks $(1,0)$ and $(0,1)$ contribute 4 each in degree one. Reality gives $w_{-\varepsilon} = \overline{w_\varepsilon}$, and when F is not biquadratic the Galois group acts transitively on the CM types, so $N_w = 4$.

(iii) One has $\int_X e^{c_1\theta_1 + c_2\theta_2} = c_1^2 c_2^2$, and χ is invariant under e^{θ_t}; the form is positive definite because q is totally positive. ∎

The integral classes of $\mathcal{R}$ are described by one congruence, and the description does not depend on the field.

Lemma 5.20 (The integral classes of $\mathcal{R}$)**.** *Let F_0 be a real quadratic field with ring of integers $\mathcal{O}$ and different $\mathfrak{d}$, and let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}$. A class $v \in \mathcal{R}$ with matrix*

$$V = \begin{pmatrix} r & \bar{h} & \bar{g} \\ h & c & \bar{k} \\ g & k & m \end{pmatrix}, \qquad r, c, m \in \mathbb{Q}, \quad h, g, k \in F_0,$$

lies in $H^{\mathrm{ev}}(X, \mathbb{Z})$ if and only if $r, c, m \in \mathbb{Z}$, $h, g, k \in \mathcal{O}$ and $c \equiv g$ modulo $\mathfrak{d}$.

Proof. Integrality can be tested one prime p at a time, and after a faithfully flat extension of $\mathbb{Z}_p$. Let $M = H_1(X, \mathbb{Z})$ and let $\langle\,,\rangle$ be the polarisation form on M, that is, θ. The Rosati involution fixes $\mathcal{O}$, so $\langle\,,\rangle = \mathrm{Tr}_{F_0/\mathbb{Q}}\,\psi$ for a unique $\mathcal{O}$-bilinear alternating form ψ with values in $\mathfrak{d}^{-1}$, and ψ is perfect because θ is principal. Write $\mathcal{O} = \mathbb{Z}[\nu]$ and $\delta = \nu - \bar{\nu}$, a generator of $\mathfrak{d}$ with $\bar{\delta} = -\delta$, and put $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$, a product of at most two discrete valuation rings. The form $\delta\psi$ on $M_p = M \otimes \mathbb{Z}_p$ is perfect with values in $\mathcal{O}_p$, so M_p has a symplectic $\mathcal{O}_p$-basis e, f, e^*, f^*. Hence $M_p = \mathcal{O}_p \otimes_{\mathbb{Z}_p} W$, with W the $\mathbb{Z}_p$-span of this basis, and $\langle\,,\rangle = B \otimes \omega_0$, where ω_0 is the standard symplectic form of W and $B(\alpha, \beta) = \mathrm{Tr}(\delta^{-1}\alpha\beta)$.

Let Λ be the ring of integers of a finite extension of $\mathbb{Q}_p$ over which F_0 splits, and fix an embedding of F_0 in its field of fractions. The map $\alpha \otimes \lambda \mapsto (\alpha\lambda, \bar{\alpha}\lambda)$ identifies $\mathcal{O}_p \otimes_{\mathbb{Z}_p} \Lambda$ with the lattice of pairs $(x, y) \in \Lambda^2$ with $x \equiv y$ modulo δ, whose basis is $(1, 1), (\delta, 0)$. Let $\varepsilon_1, \varepsilon_2$ be the coordinates and $u_1, \dots, u_4$ the basis of $W^\vee$ dual to e, f, e^*, f^*. The dual lattice has basis ε_2, $(\varepsilon_1 - \varepsilon_2)/\delta$, so $H^1(X, \Lambda)$ has the basis

$$Y_i = \varepsilon_2 \otimes u_i, \qquad Z_i = \delta^{-1}(X_i - Y_i), \qquad \text{where } X_i = \varepsilon_1 \otimes u_i,$$

and $H^*(X, \Lambda)$ is the exterior algebra on the Y_i and Z_i. In these coordinates $B = \delta^{-1}(\varepsilon_1^2 - \varepsilon_2^2)$, and the components of $\theta = B \otimes \omega_0$ in the two eigenspaces of $\mathcal{O}$, the first being the one on which $\mathcal{O}$ acts through the fixed embedding, are

$$\theta_1 = \delta^{-1}(X_1X_3 + X_2X_4), \qquad \theta_2 = -\delta^{-1}(Y_1Y_3 + Y_2Y_4).$$

Substitute $X_i = Y_i + \delta Z_i$ in $v = \sum_{a,b} V_{ab}\,\theta_1^a\theta_2^b/(a!\,b!)$ and expand in the monomials in the Y_i and Z_i. Up to sign, every coefficient is one of

$$r,\ m,\ h,\ k,\ c,\ g,\ g - c,\ \delta h,\ \delta k,\ \delta g,\ \delta^2 g,\ \frac{g - c}{\delta},\ \frac{h - \bar{h}}{\delta},\ \frac{k - \bar{k}}{\delta},\ \frac{2c - g - \bar{g}}{\delta^2},$$

and each of r, m, h, k, c, g and $(g - c)/\delta$ occurs, so an integral class satisfies the conditions at p. Conversely, suppose that they hold. Every $x \in \mathcal{O}_p$ has $x - \bar{x} \in \delta\mathcal{O}_p$, which disposes of $h - \bar{h}$ and $k - \bar{k}$; and $g = c + \delta g'$ with $g' \in \mathcal{O}_p$ gives $2c - g - \bar{g} = -\delta(g' - \bar{g}') \in \delta^2\Lambda$. So every coefficient lies in Λ. When p is unramified in F_0, δ is a unit at p and the congruence is empty. ∎

Proposition 5.21 (Integral secant classes of small Euler characteristic)**.** *Let $F_0 = \mathbb{Q}(\sqrt{D})$ with $D > 1$ squarefree, let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and let v be an integral class in a space $S(t, q)$ of Proposition 5.19, with ρ and N_w as there.*

(i) *If $\rho = 1$, then $\chi(v, v) \neq 4$.*

(ii) *If $\rho = 2$ and $N_w = 4$, then $\chi(v, v) \neq 6$.*

Proof. By Lemma 5.20 the entries of V lie in $\mathcal{O} = \mathcal{O}_{F_0}$, the diagonal ones in $\mathbb{Z}$, and $c \equiv g$ modulo $\mathfrak{d}$; conjugation is trivial on $\mathcal{O}/\mathfrak{d}$.

(i) Write $V = \lambda\alpha\bar{\alpha}^T$ with $\alpha \in \mathcal{O}^3$, and let $\mathfrak{a}$ be the ideal generated by the entries of α; replacing α by a multiple, we may take $\mathfrak{a}$ integral and prime to $\mathfrak{d}$. The entries $\lambda\alpha_i\bar{\alpha}_j$ lie in $\mathcal{O}$ and generate $\lambda\mathfrak{a}\bar{\mathfrak{a}} = \lambda\mathrm{N}(\mathfrak{a})\mathcal{O}$, so $\mu = \lambda\mathrm{N}(\mathfrak{a}) \in \mathbb{Z}$. With $D' = \alpha_1^2 - \alpha_0\alpha_2 \in \mathfrak{a}^2$ one has $\chi = 4\lambda^2\mathrm{N}(D')$, and $\mathrm{N}(D') \in \mathrm{N}(\mathfrak{a})^2\mathbb{Z}$, so $\chi = 4$ forces $\mu = \pm 1$ and $(D') = \mathfrak{a}^2$. Since conjugation is trivial modulo $\mathfrak{d}$, $\alpha_1\bar{\alpha}_1 - \alpha_2\bar{\alpha}_0 \equiv D'$ modulo $\mathfrak{a}\mathfrak{d}$, and the congruence $c \equiv g$ gives $\lambda D' \in \mathfrak{d}\,\bar{\mathfrak{a}}^{-1}$. The ideal of $\lambda D'$ is $\mathfrak{a}\bar{\mathfrak{a}}^{-1}$, so $\mathfrak{a} \subseteq \mathfrak{d}$, which is impossible for $\mathfrak{a}$ prime to $\mathfrak{d}$ and $\mathfrak{d} \neq (1)$.

(ii) The matrix $\operatorname{adj} V$ is Hermitian of rank one, so $\operatorname{adj} V = \mu\,\bar{l}\,l^T$ with $l \in \mathcal{O}^3$, $l^T V = 0$ and $\mu \in \mathbb{Q}^\times$. Put $\Delta = l_1^2 - 4l_0l_2$ and $m = \mu^2\mathrm{N}(\Delta)$. Replacing l by λl, $\lambda \in F_0^\times$, replaces μ by $\mu/\mathrm{N}(\lambda)$ and Δ by $\lambda^2\Delta$, so m depends only on V. As the entries of $\operatorname{adj} V$ are integral, μ is integral at every prime at which l is primitive, and l can be made primitive at the primes above any given p, because $\mathcal{O} \otimes \mathbb{Z}_{(p)}$ is a principal ideal domain; so $m \in \mathbb{Z}$. Let J be the matrix with rows $(0,0,1)$, $(0,-2,0)$, $(1,0,0)$. Since $\int_X \theta_1^2\theta_2^2 = 4$, one has $\chi(v,v) = \operatorname{tr}(JVJ\bar{V})$; and $\operatorname{adj} J$ has rows $(0,0,2)$, $(0,-1,0)$, $(2,0,0)$, so that $l^T(\operatorname{adj} J)\,l = -\Delta$ and

$$\sigma_2(JVJ\bar{V}) = \operatorname{tr}\operatorname{adj}(JVJ\bar{V}) = \mu^2\,\Delta\bar{\Delta} = m.$$

At a real place, $V = PWQ^T$ with $W = (w_\varepsilon)$ and P, Q the 3×2 matrices with columns $(1, \pm s_1, s_1^2)$ and $(1, \pm s_2, s_2^2)$, when $t = 0$; multiplication by e^{θ_t} replaces V by $UV\bar{U}^T$ with U unipotent and $U^TJU = J$, which conjugates $JVJ\bar{V}$, so we may take $t = 0$. As $(1,z,z^2)\,J\,(1,z',z'^2)^T = (z-z')^2$ and $w_{-\varepsilon} = \overline{w_\varepsilon}$, the nonzero eigenvalues of $JVJ\bar{V}$ are $16\mathrm{N}(q)\big(|w_{++}| \pm |w_{+-}|\big)^2$, so

$$\chi^2 - 4m = \big(64\mathrm{N}(q)\,|w_{++}|\,|w_{+-}|\big)^2 \geq 0,$$

with equality exactly when $N_w = 2$. Modulo $\mathfrak{d}$ one has $\bar{V} \equiv V$, $\operatorname{tr}(JV) = g - 2c + \bar{g} \equiv 0$ and $\sigma_2(JV) = \operatorname{tr}\big(\operatorname{adj}(V)\operatorname{adj}(J)\big) = \mu\,l^T(\operatorname{adj} J)\,\bar{l} \equiv -\mu\Delta$, with l primitive at the primes dividing $\mathfrak{d}$. Hence

$$\chi \equiv \operatorname{tr}\big((JV)^2\big) = \operatorname{tr}(JV)^2 - 2\sigma_2(JV) \equiv 2\mu\Delta \pmod{\mathfrak{d}}.$$

Now let $\chi = 6$ and $N_w = 4$. Then $1 \leq m \leq 8$, because Δ is totally negative, and m satisfies three conditions.

(a) $4m \equiv 36$ modulo $\mathfrak{d} \cap \mathbb{Z}$, which is $D\mathbb{Z}$ if $D \equiv 1$ modulo 4 and $2D\mathbb{Z}$ otherwise. As D is squarefree, this says that $m \equiv 9$ modulo the odd part D_o of D.

(b) $m \equiv 0$ or 1 modulo 4. Take l primitive at 2, so that $\mu \in \mathbb{Z}_{(2)}$. Since $\mathrm{N}(x+4y) \equiv \mathrm{N}(x)$ modulo 4 for $x, y \in \mathcal{O} \otimes \mathbb{Z}_{(2)}$, $\mathrm{N}(\Delta) \equiv \mathrm{N}(l_1^2) = \mathrm{N}(l_1)^2$, and $m \equiv \big(\mu\mathrm{N}(l_1)\big)^2$ modulo 4.

(c) If $D \equiv 2$ modulo 4, then m is odd. The prime $\mathfrak{p}$ above 2 is ramified, $v_\mathfrak{p}(2) = 2$ and $v_\mathfrak{p}(\sqrt{D}) = 1$, so $v_\mathfrak{p}(\mathfrak{d}) = 3$ while $v_\mathfrak{p}(6) = 2$; the congruence $6 \equiv 2\mu\Delta$ modulo $\mathfrak{p}^3$ forces $v_\mathfrak{p}(\mu\Delta) = 0$, so μ and $\mathrm{N}(\Delta)$ are odd.

If $D \equiv 1$ modulo 4 and $D \neq 5$, then $D \geq 13$ and (a) has no solution in $[1,8]$. If $D \equiv 3$ modulo 4, (a) gives $m \in \{3,6\}$ for $D = 3$, $m = 2$ for $D = 7$ and nothing for $D \geq 11$, and (b) excludes these values. If $D = 2D'$, then (b) and (c) give $m \equiv 1$ modulo 4, which with (a) says $m \equiv 9$ modulo $4D'$, and this has no solution in $[1,8]$ unless $D' = 1$. There remain $D = 5$, with $m = 4$, and $D = 2$, with $m \in \{1,5\}$.

For $\mathbb{Q}(\sqrt{5})$, $m = 4$ means that Δ is -2 times a unit or minus the square of a unit, since 2 is inert; and $\Delta \equiv l_1^2$ modulo 4, while neither -1 nor twice a unit is a square modulo

4 in $\mathbb{Z}[\varphi]$. For $\mathbb{Q}(\sqrt{2})$, (c) makes μ odd, so $\mu = \pm 1$ and $\mathrm{N}(\Delta) \in \{1, 5\}$; the norms of the elements of $\mathbb{Z}[\sqrt{2}]$ prime to 2 are ± 1 modulo 8, so $\mathrm{N}(\Delta) = 1$ and $\Delta = -(1+\sqrt{2})^{2k}$, which is 3 or $1 + 2\sqrt{2}$ modulo 4, and neither is a square modulo 4 in $\mathbb{Z}[\sqrt{2}]$. ■

Theorem 5.22 (No dimension-count certificate for quartic CM fields)**.** *Let $F = F_0(\sqrt{-q})$ be a quartic CM field with real quadratic subfield F_0, let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and give $A = X \times \widehat{X}$ the action of F of* [Mar25c, §11.1]*, transported by the B-field e^{θ_t} for a rational $t \in F_0$. Let F_1, F_2 be perfect complexes on X with $\mathrm{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves, whose Chern characters lie in spaces $S(t', q')$ of Proposition 5.19, and let $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \mathrm{ch}(E)e^{\ell/2}$. Put $v_1 = \mathrm{ch}(F_1)$ and $v_2 = \mathrm{ch}(F_2^\vee)$.*

(i) *$r(\kappa) = r(\mathrm{ch}\, E) = r^2(v_1) + 64 + r^2(v_2)$, where $r^2(v_i) \in \{12, 18, 20\}$ is given by Proposition 5.19(ii); so $r(\kappa) \le 104$, and $r(\kappa) \in \{100, 102, 104\}$ unless F is biquadratic and some v_i has $N_w = 2$.*

(ii) *$\dim \mathrm{Ext}^2(E, E) > r(\kappa)$. So the injectivity of σ_E cannot be obtained from a dimension count for any such E.*

(iii) *If $t = 0$ and both Chern characters lie in $S(0, q)$, then κ is invariant under G_F, and so is a Hodge class on every member of the $(F, 2, 1)$-family. This is the case $[F : \mathbb{Q}] = 4$ of* [Mar25c, Proposition 10.2.1(1)] *and of the inclusion stated before* [Mar25c, Lemma 10.1.3]*, with other conventions for Φ and for the sign of the twist.*

Proof. (i) By Lemma 4.55(i), multiplication by $e^{\ell/2}$ conjugates contraction by an automorphism of HT^2, so $r(\kappa) = r(\mathrm{ch}\, E)$; and, as in the proof of Theorem 4.56(iv), Φ identifies $r(\mathrm{ch}\, E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$. Write r^k for the rank of contraction from $HT^k(X)$. The block $HT^1 \otimes HT^1$ maps into $H^{\mathrm{odd}} \otimes H^{\mathrm{odd}}$ by the tensor product of the two contractions from HT^1, of rank $r^1(v_1)r^1(v_2) = 64$, and the other two blocks map into $H^{\mathrm{ev}} \otimes H^{\mathrm{ev}}$. If $(x_1 \lrcorner v_1) \otimes v_2 + v_1 \otimes (x_2 \lrcorner v_2) = 0$, then $x_1 \lrcorner v_1 = \lambda v_1$ for a scalar λ, and $\lambda = 0$ because contraction from HT^2 raises $q - p$ by two on classes of type (p, q) while $v_1 \neq 0$ is of type (p, p); so those two blocks contribute $r^2(v_1) + r^2(v_2)$, and $r^2(v_i) = 2\rho + 4N_w \in \{12, 18, 20\}$ by Proposition 5.19(ii), with 12 only for biquadratic F.

(ii) By the Künneth formula, as in Proposition 3.20, $\dim \mathrm{Ext}^2(E, E) = e_0 e_2' + e_1 e_1' + e_2 e_0'$ with $e_k = \dim \mathrm{Ext}^k(F_1, F_1) \ge r^k(v_1)$, by the commutative square $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,) \lrcorner \mathrm{ch}(F)$, and likewise for F_2. So $\dim \mathrm{Ext}^2(E, E) \ge r(\kappa)$, with equality only if each F_i is minimal, $\dim \mathrm{Ext}^k(F_i, F_i) = 1, 8, r^2(\mathrm{ch}\, F_i)$ for $k = 0, 1, 2$. For such F_i, Serre duality gives $\chi(F_i, F_i) = r^2 - 14$, that is 4, 6 or -2 for $(\rho, N_w) = (1, 4)$, $(2, 4)$, $(2, 2)$. The last is impossible because χ is positive on $S(t', q')$. The Chern character of a perfect complex on a complex torus is integral, since by the Künneth theorem the topological K-theory of a real torus is generated by products of classes pulled back from its circle factors, whose Chern characters are integral. So the first two values are excluded by Proposition 5.21.

(iii) Over $\mathbb{R}$, $H^1(X, \mathbb{R}) = U_1 \oplus U_2$, with U_j the eigenspace of F_0 for τ_j, and the U_j are orthogonal for θ. As in the proof of Proposition 3.21, the correspondence $Z \mapsto \mathrm{ch}(\Phi(Z))e^{\ell/2}$, the Poincaré class $\ell = \ell_1 + \ell_2$, the classes η and γ and the exponentials e_ε all decompose as tensor products over j of classes of even degree, so that the image of $e_\varepsilon \boxtimes e_{\varepsilon'}^\vee$ is the tensor product over j of the images of $e^{\varepsilon_j s_j \theta_j} \boxtimes e^{-\varepsilon_j' s_j \theta_j}$ in the j-th factor. On that factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with d replaced by $\tau_j(q)$, and the identities of Proposition 3.21 at $n = 2$, which are identities in the exterior algebra of a symplectic space with d a symbol, show that each image is a multiple of $e^{\pm(s_j/2\tau_j(q))\eta_j}$ or of a Weil

class $(\gamma_j \mp s_j\ell_j)^2$ of the j-th factor. These are invariant under the special unitary group SU_j of the j-th factor, and $G_F(\mathbb{R}) = SU_1 \times SU_2$ acts factor by factor; so the rational class κ is invariant under $G_F(\mathbb{R})$, which is Zariski dense in G_F, and the Hodge group of every member of the family lies in G_F. ∎

Remark 5.23 (The shape of the secant characters). Let $t = 0$ in Theorem 5.22(iii), and write $\eta = \eta_1 + \eta_2$, $\gamma = \gamma_1 + \gamma_2$ and $\ell = \ell_1 + \ell_2$ for the decompositions over the real places used in its proof: $\eta_j = \tau_j(q)\theta_j + \widehat{\theta}_j$ and $\gamma_j = \tau_j(q)\theta_j - \widehat{\theta}_j$, with θ_j and $\widehat{\theta}_j$ here the pull-backs to A of the components of the polarisations of X and $\widehat{X}$. The classes η_t of A are the combinations of η_1 and η_2, and the Weil classes at the two embeddings over τ_j are the multiples of $(\gamma_j \mp s_j\ell_j)^2$. Let $\kappa_{(a,b)}$ be the component of κ of degree a at τ_1 and b at τ_2. Although κ is flat, it does not lie in $\mathbb{C}[\eta_1, \eta_2] + W_F \otimes \mathbb{C}$ once κ_4 has a nonzero Weil part, unlike the classes of Proposition 3.21. Indeed, by the proof of Theorem 5.22(iii), κ is a sum of products X_1X_2 in which each X_j is a multiple of $e^{\pm c_j\eta_j}$, $c_j = s_j/(2\tau_j(q))$, or of $(\gamma_j \mp s_j\ell_j)^2$. The components $\kappa_{(4,0)}$ and $\kappa_{(4,8)}$ receive exactly the terms whose second factor is an exponential, through its parts of degree 0 and 8, and $c_2^4 = 1/(16\tau_2(q)^2)$ for both signs; so

$$\kappa_{(4,8)} = \frac{\kappa_{(4,0)}\,\eta_2^4}{384\,\tau_2(q)^2},$$

and symmetrically. The polynomials in η_1, η_2 of this bidegree are the multiples of $\eta_1^2\eta_2^4$ and W_F has no component there, while $\kappa_{(4,0)}$ has a nonzero Weil part as soon as κ_4 has one, since a nonzero rational Weil class has nonzero components at all four embeddings. The class $\mathrm{ch}(E) = \kappa e^{-\ell/2}$ itself is not flat when $\mathrm{rk}(E) \neq 0$: for Y in the Lie algebra of G_F, $D_Y \mathrm{ch}(E) = -\frac{1}{2}(D_Y\ell)\,\mathrm{ch}(E)$, whose part of degree two is $-\frac{1}{2}\,\mathrm{rk}(E)\,D_Y\ell$, and ℓ is not flat. So these objects, when their rank is not zero, are candidates only for a twisted form of Theorem 5.13(v), with hypothesis a flat twisted character κ with $\kappa_4 = N\omega + P_4$, which needs the extension of Theorem 2.2 to twisted objects [Per26] and the extraction of untwisted cycles of [Mar25b, Section 2], as after Theorem 3.19. This is the form in which Markman states his reduction: he uses the normalised class $\mathrm{ch}(E)\exp(-c_1(E)/\,\mathrm{rk}\,E)$, which is $\kappa e^{-\kappa_1/\kappa_0}$ when $\mathrm{rk}\,E \neq 0$, asserts the shape of a polynomial plus a nonzero Weil class only for its component in H^4, and takes as propagation hypothesis that this component stays algebraic [Mar25c, Proposition 10.2.1, Corollary 10.2.3].

Remark 5.24 (Negative Ext groups). The hypothesis $\mathrm{Ext}^{<0}(F_i, F_i) = 0$ in Theorem 5.22(ii) can be replaced by a condition on the amplitude. Let F_1, F_2 be any perfect complexes with characters in spaces $S(t', q')$, let $e_k = \dim \mathrm{Ext}^k(F_1, F_1)$ and $e'_k = \dim \mathrm{Ext}^k(F_2, F_2)$, and put $\nu_i = \sum_{j\geq 3}(-1)^j \dim \mathrm{Ext}^{-j}(F_i, F_i)$. Then still $\dim \mathrm{Ext}^2(E, E) \geq r(\kappa)$, and equality forces, for each i, $\dim \mathrm{Ext}^k(F_i, F_i) = 1, 8, r^2(v_i)$ for $k = 0, 1, 2$, $\mathrm{Ext}^{-1}(F_i, F_i) = \mathrm{Ext}^{-2}(F_i, F_i) = 0$ and $\nu_i = (\chi(v_i, v_i) - r^2(v_i) + 14)/2$. This gives $\nu_i \geq 2$ for $(\rho, N_w) = (1, 4)$ and $(2, 2)$, and $\nu_i \geq 1$ for $(2, 4)$ unless $F_0 = \mathbb{Q}(\sqrt{3})$ and $\chi(v_i, v_i) = 4$, where $\nu_i = -1$, a case that needs integral classes of that kind. So equality needs, for both i, some $\mathrm{Ext}^{-j}(F_i, F_i) \neq 0$ with $j \geq 4$ even, or $j \geq 3$ odd in the exceptional case, and it fails as soon as one F_i has its cohomology sheaves in an interval of length at most 3, at most 2 in the exceptional case. These conditions are necessary, not sufficient. The strict inequality $\dim \mathrm{Ext}^2(E', E') > r(\mathrm{ch}\,E')$ passes to the images E' of such an E under derived equivalences, push-forward and pull-back along isogenies, and twists by line bundles or B-fields.

Indeed, $\dim \mathrm{Ext}^2(E, E) = \sum_{a\in\mathbb{Z}} e_a e'_{2-a}$ by the Künneth formula, and $e_a = e_{4-a}$ by Serre duality. The terms $a = 0, 1, 2$ give at least $r(\kappa)$, with equality only if e_k and e'_k are the

ranks r^k for $k \leq 2$, as in the proof of Theorem 5.22(ii); the terms $a = 3, 4, -1, -2$ are $e_1 e'_{-1}$, $e_0 e'_{-2}$, $e_{-1} e'_1$ and $e_{-2} e'_0$, which must vanish. Then $\chi(F_i, F_i) = 2e_0 - 2e_1 + e_2 + 2\sum_{j\geq 1}(-1)^j e_{-j}$ gives the value of ν_i. The number $\chi(v, v)$ is a positive even integer: positive by Proposition 5.19(iii), and even because $\int_X x^2$ is even for $x \in H^4(X, \mathbb{Z}) = \bigwedge^4 H^1(X, \mathbb{Z})$. For $(1, 4)$, $\chi \in 4\mathbb{Z}$ by the proof of Proposition 5.21(i), and $\chi \neq 4$; for $(2, 2)$, $r^2 = 12$. For $(2, 4)$, $r^2 = 20$, and in the proof of Proposition 5.21(ii) $\chi^2 - 4m > 0$ with $m \geq 1$, so $\chi \geq 4$, and $\chi \neq 6$. If $\chi = 4$, then $m \in \{1, 2, 3\}$, and conditions (a) to (c) of that proof, with 36 replaced by 16, say that $m \equiv 4$ modulo the odd part of D, so that $D \in \{2, 3, 6\}$; that $m = 1$; and, for $D \equiv 2$ modulo 4, that m is even, because $v_{\mathfrak{p}}(4) = 4 \geq 3$ here gives $v_{\mathfrak{p}}(\mu\Delta) \geq 1$. Only $D = 3$ remains. A complex with cohomology sheaves in an interval of length l has $\mathrm{Ext}^{-j}(F, F) = 0$ for $j > l$. Equivalences preserve Ext algebras and, through the induced isomorphisms of HT^* and H^*, ranks of contraction ([CRV12], as in the proof of Theorem 4.56(iv)); for an isogeny g, $g^* g_* E = \bigoplus_{x \in \ker g} t_x^* E$, so $\mathrm{Ext}^2(g_* E, g_* E)$ contains $\mathrm{Ext}^2(E, E)$ while $r(\mathrm{ch}\, g_* E) = r(\mathrm{ch}\, E)$, and likewise for g^*; twists are Lemma 4.55(i).

What Theorem 5.22 and Remark 5.24 leave for $\mathbf{W}(F, 2, \delta)$ is one of three things: an Orlov product both of whose factors have nonzero Ext groups in some even degree at most -4 (or odd degree at most -3 when $F_0 = \mathbb{Q}(\sqrt{3})$); a kernel on $X \times X$ that is not an external product; or a proof of the injectivity of σ_E for some Orlov product that does not go through a dimension count, which is what Markman's question [Mar25c, Question 11.2.2] asks for. The next proposition shows that a kernel whose transform has a pure spinor as its character, for instance a line bundle on the graph of a homomorphism, gives no Weil class.

Proposition 5.25 (Pure spinors give no Weil class)**.** *Keep the notation of Remark 5.23 on $A = X \times \widehat{X}$ with $t = 0$, let $V_j \subseteq H^1(A, \mathbb{R})$ be the part at τ_j, and call a class flat if it is invariant under $G_F(\mathbb{R})$. Let $v = c\, e^{\beta} \Omega_N$ be a real pure spinor, with $c \in \mathbb{R}^\times$, β a real 2-form, $N \subseteq H^1(A, \mathbb{R})$ and Ω_N a generator of $\det N$. If v is flat, then N is one of 0, V_1, V_2, $H^1(A, \mathbb{R})$, and $v \in \mathbb{R}[\eta_1, \eta_2]$. This applies to the characters of line bundles on translates of abelian subvarieties, of points and of simple semi-homogeneous bundles, and of their images under derived equivalences and isogenies, twisted by any real B-field; for instance to Φ of a line bundle on the graph of a homomorphism $X \to X \times X$. The component in $W_F \otimes \mathbb{R}$ of such a class vanishes in every degree: no such object has a character, twisted as stated, of the form required by Theorem 5.13(v) or by its twisted form, whatever the quartic field.*

Proof. Over $\mathbb{R}$, $G_F(\mathbb{R}) = SU_1 \times SU_2$ with SU_j acting on V_j and trivially on the other part, and $V_j \otimes \mathbb{C}$ is the sum of the standard representation of $SU_j \otimes \mathbb{C} \cong \mathrm{SL}_4$ and its dual, which are not isomorphic (proof of Proposition 5.5(i)). So V_1 and V_2 are irreducible and not isomorphic, and 0, V_1, V_2, $H^1(A, \mathbb{R})$ are the only stable subspaces. The flat classes are $\mathrm{Inv}_1 \otimes \mathrm{Inv}_2$, where by the same proof $\mathrm{Inv}_j \subseteq \bigwedge^* V_j$ is spanned by the powers of η_j and the two Weil classes $(\gamma_j \mp s_j \ell_j)^2$; in degree two it is $\mathbb{R}\eta_j$, and η_j^4 spans $\bigwedge^8 V_j$.

If v is real and pure over $\mathbb{C}$, then $N = \{\xi : \xi \wedge v = 0\}$ is defined over $\mathbb{R}$, and so are c and the class of β modulo $N \wedge H^1$. For $g \in G_F(\mathbb{R})$, $gv = v$ gives $gN = N$, and $g\Omega_N = \det(g|_N)\Omega_N = \Omega_N$ because the connected semisimple group $G_F(\mathbb{R})$ has no nontrivial character. Hence $(e^{g\beta - \beta} - 1)\Omega_N = 0$, whose part of lowest degree gives $g\beta - \beta \in N \wedge H^1$. So N is one of the four stable subspaces, and the class of β in $\bigwedge^2(H^1/N)$ is invariant; by complete reducibility it is the image of an invariant 2-form $a_1\eta_1 + a_2\eta_2$, and since $e^{n \wedge x}\Omega_N = \Omega_N$ for $n \in N$, $v = c\, e^{a_1\eta_1 + a_2\eta_2}\Omega_N$, with Ω_N a multiple of 1, η_1^4, η_2^4 or $\eta_1^4\eta_2^4$. The

characters listed are pure spinors, and derived equivalences, isogenies and twists preserve purity; this is the classical description recalled in Setup 3.26 and after Corollary 3.17. ■

Markman's candidate for a quartic field. Markman gives one explicit pair of sheaves for a biquadratic field [Mar25c, Example 11.2.7, Lemma 11.2.8] and asks whether the second can be chosen so that its semiregularity map is injective on the image of the evaluation map [Mar25c, Question 11.2.2]. For one object on the fourfold this is the weakened criterion of Setup 4.54, and the local method of Theorem 4.61 decides it for his pair: both sheaves fail it, at points of the curves that the construction glues in. Two local computations are needed, one for a sheaf that is locally free on a smooth support and one for the ideal of a curve in a smooth divisor. As in Lemma 4.60, for a germ of vector field θ and a perfect complex F of modules over a local ring we write $a_\theta \in \operatorname{Ext}^1(F,F)$ for the class of the jet extension along θ.

Lemma 5.26 (Free germs on smooth supports)**.** *Let $\mathcal{O}$ be the local ring of a smooth complex variety Y at a point p, or its completion, let $C \ni p$ be a smooth subvariety of codimension $c \geq 1$, put $N = T_pY/T_pC$, and let F be a perfect complex of $\mathcal{O}$-modules quasi-isomorphic to a shift of $\mathcal{O}_{C,p}^{\oplus r}$, $r \geq 1$. Then $\operatorname{Ext}^2_{\mathcal{O}}(F,F)$ is a free $\mathcal{O}_{C,p}$-module isomorphic to $M_r(\mathcal{O}_{C,p}) \otimes \bigwedge^2 N$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a sign that does not depend on θ, θ', the element $\mathrm{id} \otimes (\bar\theta \wedge \bar\theta')$, where $\bar\theta, \bar\theta'$ are the images of $\theta(p), \theta'(p)$ in N. In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p)$, $\theta'(p)$ and T_pC span a space of dimension $\dim C + 2$.*

Proof. Choose formal coordinates $x_1, \dots, x_m$ at p with $C = \{x_1 = \cdots = x_c = 0\}$. Ext groups of finitely generated modules and jet extensions commute with the flat base change from $R = \mathbb{C}[x_1, \dots, x_m]$ to $\mathcal{O}$, and a shift changes neither the Ext algebra nor the jet classes, so we may take $F = (R/(x_1, \dots, x_c))^{\oplus r}$ over R. Write $R = A_1 \otimes \cdots \otimes A_c \otimes B$ with $A_i = \mathbb{C}[x_i]$ and $B = \mathbb{C}[x_{c+1}, \dots, x_m]$; then $F = k_1 \otimes \cdots \otimes k_c \otimes B^{\oplus r}$, with k_i the residue field of A_i at the origin. On the resolution $A_i \xrightarrow{x_i} A_i$ of k_i the jet chain map of ∂_i is $-\partial_i(x_i) = -1$, as in the proof of Lemma 4.60, so its class γ_i spans $\operatorname{Ext}^1_{A_i}(k_i, k_i) = k_i$, and $\operatorname{Ext}^j_{A_i}(k_i, k_i) = 0$ for $j \geq 2$. The module $B^{\oplus r}$ is free, so its Ext algebra is $M_r(B)$ in degree zero and its jet classes vanish. By (20), applied factor by factor, $\operatorname{Ext}^*_R(F,F)$ is the graded tensor product of the exterior algebra on $\gamma_1, \dots, \gamma_c$ with $M_r(B)$, and $a_{\partial_i} = \gamma_i \otimes \mathrm{id}$ for $i \leq c$, while $a_{\partial_i} = 0$ for $i > c$. Hence $\operatorname{Ext}^2_R(F,F) = \bigwedge^2\langle\gamma_1, \dots, \gamma_c\rangle \otimes M_r(B)$, and $B = \mathcal{O}_C$ in these coordinates. For $\theta = \sum_i f^i\partial_i$ and $\theta' = \sum_i f'^i\partial_i$ the linearity of Lemma 4.59 gives

$$a_\theta a_{\theta'} = \sum_{i,j\leq c} f^i f'^j\, \gamma_i\gamma_j \otimes \mathrm{id} = \sum_{i<j\leq c} \left(f^i f'^j - f^j f'^i\right) \gamma_i\gamma_j \otimes \mathrm{id},$$

because the γ_i anticommute and square to zero. The functions $x_1, \dots, x_c$ act by zero, so modulo the maximal ideal of $\mathcal{O}_C$ the coefficients are those of $\theta(p) \wedge \theta'(p)$ on $\partial_i \wedge \partial_j$, $i < j \leq c$, which are the coordinates of $\bar\theta \wedge \bar\theta'$ in $\bigwedge^2 N$. A section of a free module with nonzero reduction is nonzero. ■

Lemma 5.27 (A curve in a divisor)**.** *Let $\mathcal{O}$, Y and p be as in Lemma 5.26, let $D \ni p$ be a smooth divisor, let $C \subset D$ be a smooth subvariety of codimension two in D through p, and let F be quasi-isomorphic to a shift of the ideal $I_{C\subset D}$ of C in $\mathcal{O}_D$, regarded as an $\mathcal{O}$-module. Put $\nu = T_pY/T_pD$ and $N' = T_pD/T_pC$. Then $\operatorname{Ext}^2_{\mathcal{O}}(F,F) \cong \mathcal{O}_{C,p} \otimes \nu \otimes N'$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a fixed sign, the image of $\theta(p) \wedge \theta'(p)$ under*

$$\textstyle\bigwedge^2 T_pY \longrightarrow \bigwedge^2(T_pY/T_pC) \longrightarrow \bigwedge^2(T_pY/T_pC)/\bigwedge^2 N' \cong \nu \otimes N'.$$

In particular $a_\theta a_{\theta'} \neq 0$ *whenever* $\theta(p) \wedge \theta'(p)$ *does not lie in* $\bigwedge^2 T_pD + T_pC \wedge T_pY$.

Proof. With formal coordinates in which $D = \{x_1 = 0\}$ and $C = \{x_1 = x_2 = x_3 = 0\}$, and the same reduction to the polynomial ring, write $R = \mathbb{C}[x_1] \otimes \mathbb{C}[x_2, x_3] \otimes B$ with $B = \mathbb{C}[x_4, \dots, x_m]$. Then $I_{C\subset D} = k \otimes \mathfrak{m} \otimes B$, where k is the residue field of $\mathbb{C}[x_1]$ and $\mathfrak{m} = (x_2, x_3)$. As in the proof of Lemma 4.60, $\mathfrak{m}$ has projective dimension one, $\operatorname{Ext}^1(\mathfrak{m}, \mathfrak{m}) = \mathfrak{m}/\mathfrak{m}^2$ has the basis $\alpha_2 = a_{\partial_2}$, $\alpha_3 = a_{\partial_3}$ and is killed by $\mathfrak{m}$, and $\operatorname{Ext}^2(\mathfrak{m}, \mathfrak{m}) = 0$; the factor k contributes γ_1 in degree one, as in Lemma 5.26, and B contributes nothing in positive degree. By (20)

$$\operatorname{Ext}^2_R(I_{C\subset D}, I_{C\subset D}) = \operatorname{Ext}^1(k,k) \otimes \operatorname{Ext}^1(\mathfrak{m}, \mathfrak{m}) \otimes B = \langle \gamma_1 \rangle \otimes \langle \alpha_2, \alpha_3 \rangle \otimes B,$$

the other summands vanishing by the projective dimensions, and $a_{\partial_1} a_{\partial_j} = -a_{\partial_j} a_{\partial_1} = \pm \gamma_1 \otimes \alpha_j$ for $j = 2, 3$, while $a_{\partial_2} a_{\partial_3} = 1 \otimes \alpha_2\alpha_3 = 0$, $a^2_{\partial_1} = 0$ and $a_{\partial_i} = 0$ for $i \geq 4$. As before, $a_\theta a_{\theta'}$ reduces to the component of $\theta(p) \wedge \theta'(p)$ on $\partial_1 \wedge \langle \partial_2, \partial_3 \rangle$. In these coordinates ∂_1 spans ν, the images of ∂_2, ∂_3 span N', and the component is the image in $\bigwedge^2(T_pY/T_pC)/\bigwedge^2 N'$. Its kernel on $\bigwedge^2 T_pY$ is spanned by $\partial_2 \wedge \partial_3$ and the $\partial_i \wedge \partial_j$ with $j \geq 4$, which is $\bigwedge^2 T_pD + T_pC \wedge T_pY$. ∎

Now keep the notation of Proposition 5.19, with X an abelian fourfold with real multiplication by F_0 and $\theta = \theta_1 + \theta_2$ a polarisation in $\bigwedge^2_{F_0} H^1(X, \mathbb{Q})$. Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with U_j the eigenspace of F_0 at τ_j, and the tangent space splits accordingly as $T = T_1 \oplus T_2$, with T_j of dimension two, dual to $U_j \cap H^{1,0}$; translation identifies every T_pX with T. Let π_j span $\bigwedge^2 T_j$ and put

$$x_j = \left(\tfrac{1}{2}\tau_j(q)\, \pi_j \lrcorner\, \theta_j^2,\ 0,\ \pi_j\right) \in HT^2(X), \qquad j = 1, 2. \tag{31}$$

The six classes $(0, v, 0)$ with $v \in \operatorname{Hom}(H^{1,0}, H^{0,1})$ preserving the decomposition by places and $v \lrcorner \theta_1 = v \lrcorner \theta_2 = 0$ are the first order deformations of X with its real multiplication and polarisation; we call them the polarised F_0-linear deformations.

Theorem 5.28 (The local obstruction for a quartic secant character)**.** *In this notation, let* $q \in F_0$ *be totally positive and* $t \in F_0$.

(i) $x_j \lrcorner w = 0$ *for every* $w \in S(0, q)$, *and* $(e^{-\theta_t} x_j) \lrcorner w = 0$ *for every* $w \in S(t, q)$, $j = 1, 2$. *If* $w \in S(t, q)$ *has* $(\rho, N_w) = (2, 4)$, *then* $\operatorname{Ann}_{HT^2}(w)$ *is spanned by* $e^{-\theta_t} x_1$, $e^{-\theta_t} x_2$ *and the six polarised* F_0*-linear deformations, and its bivector components span exactly* $\langle \pi_1, \pi_2 \rangle$.

(ii) *Let* F *be a perfect complex on* X *with* $\operatorname{ch}(F) \in S(t, q)$, *and let* θ, θ' *be translation invariant vector fields with* $\theta \wedge \theta' = \pi_j$. *If* $a_\theta a_{\theta'} \neq 0$ *in* $\operatorname{Ext}^2(F_p, F_p)$ *at some point* p, *then* $\operatorname{ev}_F(e^{-\theta_t} x_j) \neq 0$ *and* F *does not satisfy the weakened criterion. Neither do* $\Phi(F \boxtimes G)$ *and* $\Phi(G \boxtimes F)$ *on* $X \times \widehat{X}$, *for any nonzero perfect complex* G *on* X, *with* Φ *as in Theorem 4.56(iv).*

(iii) *The hypothesis of* (ii) *holds at* p, *for* $j = 1$ *or for* $j = 2$, *whenever* F_p *is a shift of a free module of positive rank over the local ring of a smooth curve through* p, *since* T_pC *lies in at most one of* T_1, T_2; *it holds for both* j *when* F_p *is a shift of a free module over the residue field. It holds for* j *when* F_p *is a shift of the ideal of a smooth curve* C *in a smooth divisor* D *with* $T_pC \not\subseteq T_j$ *and* $T_j \not\subseteq T_pD$.

Proof. (i) Let $t = 0$. By the proof of Lemma 4.55(i), for a bivector π of type $(2, 0)$ in the tangent directions and a class B of type $(1, 1)$, $\pi \lrcorner e^B = \frac{1}{2}(\pi \lrcorner B^2)\, e^B$. The interior products by vectors of T_j are odd derivations that vanish on $\bigwedge^* U_{j'}$ for $j' \neq j$, so $\pi_j \lrcorner (ab) = (\pi_j \lrcorner a)\, b$ for a in the subalgebra generated by U_j and b an even class generated by $U_{j'}$. Applied

to $e_\varepsilon = e^{\varepsilon_1 s_1 \theta_1} e^{\varepsilon_2 s_2 \theta_2}$, with $s_j^2 = -\tau_j(q)$, this gives $\pi_j \lrcorner e_\varepsilon = -\frac{1}{2}\tau_j(q)(\pi_j \lrcorner \theta_j^2)\, e_\varepsilon$ for all four signs, and so $x_j \lrcorner e_\varepsilon = 0$. A class $(0, v, 0)$ with v preserving the places and killing θ_1 and θ_2 acts as a derivation that kills every polynomial in θ_1, θ_2, hence all of $S(0, q)$. For general t, Lemma 4.55(i) gives $y \lrcorner (we^{\theta_t}) = ((e^{\theta_t} y) \lrcorner w) e^{\theta_t}$, so $e^{-\theta_t}$ carries $\mathrm{Ann}_{HT^2}(w)$ onto $\mathrm{Ann}_{HT^2}(we^{\theta_t})$; it fixes the six classes $(0, v, 0)$, because $v \lrcorner \theta_t = 0$, and by (16) it does not change bivector components. The eight classes are independent: the six have no bivector component and are independent, and the other two have the independent components π_1, π_2. When $(\rho, N_w) = (2, 4)$ the rank of contraction from $HT^2(X)$, of dimension 28, is $2\rho + 4N_w = 20$ by Proposition 5.19(ii), so the annihilator has dimension 8 and the eight classes span it.

(ii) By (i), $y = e^{-\theta_t} x_j = (z, v, \pi_j)$ lies in $\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$, and by (15) $\mathrm{ev}_F(y) = z \cdot \mathrm{id}_F + v \lrcorner \mathrm{At}(F) + \frac{1}{2}\pi_j \lrcorner \mathrm{At}(F)^2$. The last term is $A_\theta A_{\theta'}$, as in the proof of Theorem 4.61, and its germ at p is $a_\theta a_{\theta'} \neq 0$ by Lemma 4.59. The germs of the first two terms vanish: $z \cdot \mathrm{id}_F$ by Lemma 4.59, and $v \lrcorner \mathrm{At}(F)$ because it is a composite $F \to F \otimes \Omega^1_X[1] \to F \otimes \Omega^1_X \otimes T_X[2] \to F[2]$ in which the middle arrow is $\mathrm{id} \otimes v$ with $v \in \mathrm{Ext}^1(\mathcal{O}_X, T_X) = H^1(T_X)$, and v restricts to zero on every affine open set. So $\mathrm{ev}_F(y) \neq 0$ with $y \lrcorner \mathrm{ch}(F) = 0$, and F fails the weakened criterion. For $G \boxtimes F$ and $F \boxtimes G$ the class $y \otimes 1$, resp. $1 \otimes y$, preserves the Chern character, and its evaluation $\pm \mathrm{ev}_F(y) \otimes \mathrm{id}_G$ is nonzero in its Künneth summand, as in the proof of Theorem 4.56(iv); the transport under Φ is the one given there.

(iii) Since T_j is a plane, the image of π_j in $\bigwedge^2(T/T_pC)$ is nonzero exactly when $T_j \cap T_pC = 0$. For a curve this says $T_pC \not\subseteq T_j$, which fails for at most one j because $T_1 \cap T_2 = 0$; for a point it always holds. Lemma 5.26 then gives $a_\theta a_{\theta'} \neq 0$. In the last case the image of π_j in $\bigwedge^2(T/T_pC)$ is the nonzero decomposable vector of the plane $(T_j + T_pC)/T_pC$, and it lies in $\bigwedge^2 N'$ only if that plane is N', that is, only if $T_j + T_pC = T_pD$, which would put T_j inside T_pD. So Lemma 5.27 applies. ∎

At a point where F is locally free both sides of the local identity vanish, as at a smooth point of a surface in Proposition 4.65; the test sees only where the sheaf drops dimension or fails to be locally free on its support. Markman's construction produces exactly such points.

Proposition 5.29 (The classes of Markman's candidate)**.** *Let X be the Jacobian of a curve of genus four with its principal polarisation Θ, with real multiplication by a real quadratic field F_0 and an automorphism g with $g^* = \hat{\eta}(f)$ on $H^1(X, \mathbb{Q})$, where $f \in F_0$ has norm one and $f^2 \neq 1$; let $q > 0$ be rational, so that $F = F_0(\sqrt{-q})$ is biquadratic, and write $f_j = \tau_j(f)$. Let $\alpha_0 = \Theta - \frac{q}{6}\Theta^3$, the character of the sheaf $E_0 = F_q$ of* [Mar25a, Example 8.2.4]*, and $\beta' = g^*\Theta - \frac{q}{6}(g^{-1})^*\Theta^3$, a multiple of which is the character of the sheaf E' of* [Mar25c, Example 11.2.7]*. With $A_j = 1 - \frac{q}{2}\theta_j^2$ and $\theta = \Theta$:*

(i) *$\alpha_0 = \theta_1 A_2 + \theta_2 A_1$ and $\beta' = f_1^2 \theta_1 A_2 + f_2^2 \theta_2 A_1$, both in $S(0, q)$. With $s = \sqrt{-q}$, the coefficients of β' on the four e_ε are $w_{++} = -w_{--} = (f_1^2 + f_2^2)/4s$ and $w_{+-} = -w_{-+} = (f_1^2 - f_2^2)/4s$, so β' has $(\rho, N_w) = (2, 4)$, while α_0 lies on the secant plane of $\mathbb{Q}(\sqrt{-q})$ and has $(\rho, N_w) = (2, 2)$.*

(ii) *The ranks of contraction from $HT^2(X)$ are 12 for α_0 and 20 for β'. The annihilator of β' is that of Theorem 5.28(i); that of α_0 contains, besides it, the classes $(\frac{q}{2}\pi \lrcorner \Theta^2, 0, \pi)$ for every bivector π, by Theorem 4.56(ii) with $t = \Theta$ and $d = q$.*

(iii) *$\int_X \alpha_0 \beta' = -4q\, \mathrm{Tr}_{F_0/\mathbb{Q}}(f^2) \neq 0$, which is Markman's formula for the rank of $\Phi(E_0 \boxtimes E')$* [Mar25c, Lemma 11.2.8]*; $\chi(\beta', \beta') = 4q\, \mathrm{Tr}_{F_0/\mathbb{Q}}(f^4)$ and $\chi(\alpha_0, \alpha_0) = 8q$; and the rank*

of contraction from $HT^2(X\times X)$ into $\alpha_0\boxtimes\beta'$ is $12+64+20=96$, as Theorem 5.22(i) predicts.

Proof. (i) The automorphism g acts on U_j by f_j, hence on θ_j by f_j^2, and $f_1f_2=1$. Since $\theta_j^3=0$, $\Theta^3=3\theta_1^2\theta_2+3\theta_1\theta_2^2$ and $(g^{-1})^*\Theta^3=3f_1^{-4}f_2^{-2}\theta_1^2\theta_2+3f_1^{-2}f_2^{-4}\theta_1\theta_2^2=3f_2^2\theta_1^2\theta_2+3f_1^2\theta_1\theta_2^2$, which gives the two expressions. With $t=0$ and $e_\varepsilon=A_1A_2-q\varepsilon_1\varepsilon_2\theta_1\theta_2+s(\varepsilon_1\theta_1A_2+\varepsilon_2\theta_2A_1)$, the class $\sum_\varepsilon w_\varepsilon e_\varepsilon$ equals β' exactly when $\sum w_\varepsilon=\sum\varepsilon_1\varepsilon_2w_\varepsilon=0$, $s\sum\varepsilon_1w_\varepsilon=f_1^2$ and $s\sum\varepsilon_2w_\varepsilon=f_2^2$, whose solution is the one stated. All four coefficients are nonzero because $f_1^2=f_2^2$ with $f_1f_2=1$ would force $f^2=1$. For α_0 the same computation with $f_1=f_2=1$ gives $w_{+-}=w_{-+}=0$. The matrix V of Proposition 5.19 has rank two in both cases.

(ii) The ranks are $2\rho+4N_w$, by Proposition 5.19(ii); $\alpha_0=v_\Theta$ in the notation of [I, (30)], with $d=q$.

(iii) By $\int_X\theta_1^2\theta_2^2=4$, $\int\alpha_0\beta'=-2q(f_1^2+f_2^2+f_1^{-2}+f_2^{-2})=-4q\,\mathrm{Tr}(f^2)$, which is negative; the Euler pairings are Proposition 5.19(iii) in the basis given there. The last number is Theorem 5.22(i) with $r^2(\alpha_0)=12$ and $r^2(\beta')=20$. ∎

Corollary 5.30 (Markman's candidate does not satisfy the weakened criterion)**.** *In Proposition 5.29, let E' be the sheaf of* [Mar25c, Example 11.2.7]*, obtained by gluing N general translates of g^*F' to a line bundle on a curve C' at the points where C' meets them, and assume that C' is reduced.*

(i) *E' does not satisfy the weakened criterion, so it does not answer* [Mar25c, Question 11.2.2].

(ii) *E_0 does not satisfy the weakened criterion.*

(iii) *For every nonzero perfect complex G on X, neither $\Phi(E_0\boxtimes G)$ nor $\Phi(G\boxtimes E')$ nor $\Phi(G\boxtimes E'^\vee)$ satisfies it; in particular Markman's object $\Phi(E_0\boxtimes E')$ does not.*

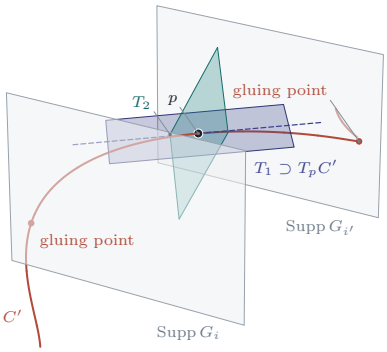
Proof. (i) Let p be a smooth point of C' that lies on none of the translates. Near p the sheaf E' is the line bundle on C', so $E'_p\cong\mathcal{O}_{C',p}$, and $\mathrm{ch}(E')$ is a multiple of $\beta'\in S(0,q)$. Theorem 5.28(ii) and (iii) apply with the j for which $T_pC'\not\subseteq T_j$.

(ii) In [Mar25a, Example 8.2.4] the sheaf $E_0=F_q$ is the push-forward, from the theta divisor $\Theta\subset X$, of the ideal twisted by $\mathcal{O}(\Theta)$ of the union of a translate W of the surface W_2 and q translates $C_1,\dots,C_q$ of the curve W_1, each meeting W in one point, and its character is α_0. Let p be a point of C_1 outside W, the other C_i and the finite singular locus of Θ. Near p, E_0 is the ideal of the smooth curve C_1 in the smooth divisor Θ. Choose translation invariant fields with $\theta(p)\notin T_p\Theta$ and $\theta'(p)\in T_p\Theta\setminus T_pC_1$. Then $\theta(p)\wedge\theta'(p)$ has nonzero image in $\nu\otimes N'$, and Lemma 5.27 gives $a_\theta a_{\theta'}\neq0$. By Proposition 5.29(ii) the class $x=(\frac{q}{2}\,\pi\lrcorner\Theta^2,0,\pi)$ with $\pi=\theta\wedge\theta'$ preserves α_0, and the argument of Theorem 5.28(ii) gives $\mathrm{ev}_{E_0}(x)\neq0$.

(iii) The first two cases are the last assertion of Theorem 5.28(ii), with the class of (ii) for E_0. For $E'^\vee$, the germ at p is the dual of $\mathcal{O}_{C'}$, a shift of a line bundle on C' because C' is a local complete intersection of codimension three near p, and $\mathrm{ch}(E'^\vee)=-\mathrm{ch}(E')$ lies in $S(0,q)$. ∎

Figure 17 draws the support of Markman's sheaf near a point of the glued curve, with the two eigenplanes of the real multiplication.

Remark 5.31 (What is left of Markman's question). The question [Mar25c, Question 11.2.2] asks for a sheaf F_2 on X whose character lies in $S(0,q)$ and forms with α_0 a pair as in [Mar25c, Lemma 11.2.1], and whose semiregularity map is injective on the image of $HT^2(X)$ in $\mathrm{Ext}^2(F_2,F_2)$. Corollary 5.30 does not settle it. It decides it for the sheaf E'



$T_0X = T_1 \oplus T_2$: the eigenplanes of F_0.
T_pC' lies in at most one of them; here in T_1.
$T_2 \cap T_pC' = 0$, so π_2 has nonzero image in $\bigwedge^2 N$, $N = T_pX/T_pC'$, and the germ of $\mathrm{ev}_{E'}(x_2)$ at p is that image:
$x_2 \lrcorner \mathrm{ch}(E') = 0$ and $\mathrm{ev}_{E'}(x_2) \neq 0$.

		pairs $\{k,l\}$ with $a_{\partial_k} a_{\partial_l} \neq 0$						
		12	13	14	23	24	34	pd
(i)	$\mathcal{O}_C$, a line bundle on a smooth curve	■	■		■			3
(ii)	$k(p)$, a skyscraper	■	■	■	■	■	■	4
(iii)	a curve ideal in a smooth divisor	■	■					2
(iv)	a point ideal in a smooth divisor	■	■	■	■	■	■	3
(v)	$\mathcal{O}_D$, a smooth divisor							1
(vi)	two planes meeting at a point		■	■	■	■		2
(vii)	maximal Cohen–Macaulay on a node							1
(viii)	the glued germ of Example 11.2.7				■	■	■	3
(ix)	reflexive, non-Cohen–Macaulay, node	■		■	■		■	2
(x)	$\mathcal{O}_C^2$, rank two on a smooth curve	■	■		■			3

coordinates adapted to the germ: a filled cell is a nonzero Yoneda square in the local Ext^2; at a point of C' the pairs of normal directions are filled, and one of π_1, π_2 projects onto them.

FIGURE 17. Theorem 5.28 and Corollary 5.30. Left: the support of Markman's sheaf E' near a point p of the glued curve C', between two of the translated supports that the curve crosses at its gluing points. The tangent space at p is the sum of the eigenplanes T_1 and T_2 of the real multiplication; the tangent line of C' lies in at most one of them, here in T_1, so π_2 has nonzero image in $\bigwedge^2 N$, $N = T_pX/T_pC'$, which is the germ of $\mathrm{ev}_{E'}(x_2)$ at p. Right: for ten germs at the origin of $\mathbb{C}^4$, the pairs $\{k,l\}$ of coordinate directions whose jet classes have nonzero product in the local Ext^2, and the projective dimension; rows (i), (ii) and (x) are Lemma 5.26, row (iii) is Lemma 5.27, row (vi) is Lemma 4.60.

of the example, and for every choice of N, of the translates and of the gluing, because the obstruction sits at any smooth point of the glued curve. By Theorem 5.28, a sheaf that answers the question, and likewise a first factor with character on the secant plane that meets the criterion, has no point at which it is locally free on a smooth curve or on a point, no point at which it is the ideal of a curve in a smooth divisor in the position of Theorem 5.28(iii), and at a point where it is locally free on a smooth surface S the tangent plane T_pS must meet both T_1 and T_2, by Lemma 5.26 with $c = 2$. Corrections of the character by sheaves on curves, the device of [Mar25c, Example 11.2.7], are therefore excluded. Sheaves that are locally free on smooth divisors, and the global part of the identity $\mathrm{ev}_F(e^{-\theta_t} x_j) = 0$, are not reached by this test. Even a positive answer would give an object with $\dim \mathrm{Ext}^2 \geq 97$, by Theorem 5.22(ii) and Proposition 5.29(iii), so its semiregularity could not come from a dimension count.

5.3. **Sextic CM fields.** The dimension count of Theorem 5.22 does not extend to CM fields of degree six, for a reason of parity (Remark 5.33). Let $F = F_0(\sqrt{-q})$ be a sextic CM field with totally real cubic subfield F_0, real places τ_1, τ_2, τ_3 and $q \in F_0$ totally positive, and let X be a principally polarised abelian sixfold with real multiplication by $\mathcal{O}_{F_0}$, so that $H^1(X, \mathbb{C}) = U_1 \oplus U_2 \oplus U_3$ with U_j the eigenspace of τ_j, of dimension four and Hodge type $(2,2)$. Let $\theta_j \in H^{1,1}(X)$ be the component of the polarisation

on U_j, so that $\theta = \sum_j \theta_j$, $\theta_j^3 = 0$ and $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$. For $t \in F_0$ and $\varepsilon \in \{\pm 1\}^3$ put $s_j = \sqrt{-1}\,\sqrt{\tau_j(q)}$ and $e_\varepsilon = \exp\bigl(\sum_j(\tau_j(t) + \varepsilon_j s_j)\theta_j\bigr)$, and let $S(t,q)$ be the span of the eight e_ε. As in Proposition 5.19(i) it is a rational subspace of dimension $8 = 2^{[F_0:\mathbb{Q}]}$ of the algebra generated by the θ_j, the Galois group permuting the e_ε as it permutes the CM types of F; for $t = 0$ it is the secant space of Markman's construction for F on $X \times \widehat{X}$ [Mar25c, §11.1], and $S(t,q) = e^{\sum_j \tau_j(t)\theta_j} S(0,q)$. A real class $v = \sum_\varepsilon w_\varepsilon e_\varepsilon$ has $w_{-\varepsilon} = \overline{w}_\varepsilon$. Write N_w for the number of nonzero w_ε; $A_{jj'}$ for the number of sign pairs (σ,σ') such that $w_\varepsilon \neq 0$ for some ε with $\varepsilon_j = \sigma$ and $\varepsilon_{j'} = \sigma'$; ρ_j for the rank of the 2×4 matrix (w_ε) whose rows are indexed by ε_j and whose columns by the two other signs; and, for $\{j,j',j''\} = \{1,2,3\}$ and $\sigma' = \pm 1$, $M_{\sigma'}^{(j;j')}$ for the 2×2 matrix $(w_\varepsilon)_{\varepsilon_j,\varepsilon_{j''}}$ with $\varepsilon_{j'} = \sigma'$ fixed. Let $r^k(v)$ be the rank of the contraction $HT^k(X) \to H^*(X,\mathbb{C})$, $\xi \mapsto \xi \lrcorner v$.

Proposition 5.32 (The count in degree six)**.** *Let $v \in S(t,q)$ be a nonzero real class.*

(i) $r^1(v) = 12$,

$$r^2(v) = 4\,(A_{12}+A_{13}+A_{23}) + \rho_1 + \rho_2 + \rho_3, \qquad r^3(v) = 8N_w + 2\sum_{j\neq j'} \bigl(\operatorname{rk} M_+^{(j;j')} + \operatorname{rk} M_-^{(j;j')}\bigr).$$

So $r^2(v) \le 54$ and $r^3(v) \le 112$, with equality when every w_ε is nonzero and every $M_{\sigma'}^{(j;j')}$ is invertible, which holds outside a proper Zariski closed subset of $S(t,q)$. When F contains an imaginary quadratic field K and v lies in the secant plane of K, so that $N_w = 2$, one has $(r^2, r^3) = (30, 40)$.

(ii) $\chi(v,v) = \int_X v^\vee v = -64\,\mathrm{N}_{F_0/\mathbb{Q}}(q)\sum_\varepsilon |w_\varepsilon|^2 < 0$. *For a CM field of degree $2m$, with X of dimension $2m$ and 2^m exponentials, the same computation gives $(-4)^m \mathrm{N}(q)\sum_\varepsilon |w_\varepsilon|^2$: the Euler form on the secant space is definite, of sign $(-1)^m$.*

(iii) *Let F_1, F_2 be perfect complexes on X with $\operatorname{Ext}^{<0}(F_i,F_i) = 0$ and characters v_i in spaces $S(t_i,q_i)$, let $A = X \times \widehat{X}$ carry the action of F of* [Mar25c, §11.1]*, transported by the exponential of a rational F_0-B-field, and put $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \operatorname{ch}(E)e^{\ell/2}$. Then $r(\kappa) = r(\operatorname{ch} E) = r^2(v_1) + 144 + r^2(v_2) \le 252$, while every class that stays of Hodge type over $\mathcal{D}_F$ has rank at most $\dim HT^2(A) - \dim \mathcal{D}_F = 276 - 12 = 264$. Moreover $\dim \operatorname{Ext}^2(E,E) \ge r(\kappa)$, with equality if and only if each F_i has*

$$\dim \operatorname{Ext}^k(F_i,F_i) = 1,\ 12,\ r^2(v_i),\ e_3,\ r^2(v_i),\ 12,\ 1 \quad (k = 0,\dots,6),$$

where $e_3 = 2r^2(v_i) - 22 - \chi(v_i,v_i)$, and this profile is compatible with the bound $e_3 \ge r^3(v_i)$ exactly when $\chi(F_i,F_i) \le -(r^3(v_i) - 2r^2(v_i) + 22)$, that is $\chi(F_i,F_i) \le -26$ for v_i general in its secant space, and $\chi(F_i,F_i) \le -2$ when $N_w = 2$.

(iv) *If $t_1 = t_2 = 0$ and $q_1 = q_2 = q$, then κ is invariant under G_F, and so is a Hodge class on every member of the $(F,2,1)$-family.*

Proof. (i) Choose a basis x_{j1}, x_{j2} of $U_j^{1,0}$ and y_{j1}, y_{j2} of $U_j^{0,1}$ with $\theta_j = x_{j1}y_{j1} + x_{j2}y_{j2}$, and let $\omega_j = y_{j1}y_{j2}$ span $\bigwedge^2 U_j^{0,1}$. Then $H^*(X,\mathbb{C})$ and $HT^*(X) = \bigwedge^*(H^{0,1} \oplus T)$ are the tensor products over j of the corresponding algebras of the three factors, $HT^k(X) = \bigoplus_{a_1+a_2+a_3=k} HT_1^{a_1} \otimes HT_2^{a_2} \otimes HT_3^{a_3}$, and contraction is compatible with the factorisation; $v = \sum_\varepsilon w_\varepsilon \bigotimes_j e^{\lambda_j(\varepsilon_j)\theta_j}$ with $\lambda_j(\pm 1) = \tau_j(t) \pm s_j$, two distinct numbers. The blocks of $HT^k(X)$ shift the three Hodge bidegrees of $H^*(X)$ by different amounts, so their images are independent and $r^k(v)$ is the sum of their ranks. Inside a block the bookkeeping rests on two facts about one factor: the two classes $e^{\lambda_j(\pm 1)\theta_j}$ are linearly independent,

and so are the subspaces $U_j^{0,1}e^{\lambda_j(\pm1)\theta_j}$ of $H^*(U_j)$, since $u\,e^{\lambda\theta_j}=u+\lambda\,u\theta_j$ for $u\in U_j^{0,1}$; whereas $\omega_j e^{\lambda\theta_j}=\omega_j$ for every λ, so that the summands of a block through HT_j^2 share the factor ω_j and are combined, not juxtaposed. On one factor, contraction into $e^{\lambda\theta_j}$ is as follows. For $\xi\in HT_j^1$, $\xi\lrcorner e^{\lambda\theta_j}=y^\lambda(\xi)\,e^{\lambda\theta_j}$ with $y^\lambda(\xi)=\xi^{0,1}+\lambda\,\xi_T\lrcorner\theta_j\in U_j^{0,1}$, as in Lemma 4.55(i); y^λ is onto, its kernel K_j^λ has dimension two, and $HT_j^1=K_j^{\lambda_j(+1)}\oplus K_j^{\lambda_j(-1)}$ because $\lambda_j(+1)\neq\lambda_j(-1)$ and $\xi_T\lrcorner\theta_j$ determines ξ_T. For $\xi\in HT_j^2$, since $\omega_j\theta_j=0$, one finds $\omega_j\lrcorner e^{\lambda\theta_j}=\omega_j$, $(y_{ja}\otimes\partial_{x_{jb}})\lrcorner e^{\lambda\theta_j}=\pm\lambda\,\omega_j$ for $a\neq b$ and 0 for $a=b$, and $(\partial_{x_{j1}}\wedge\partial_{x_{j2}})\lrcorner e^{\lambda\theta_j}=\pm\lambda^2\omega_j$: the image of HT_j^2 is the line $\mathbb{C}\omega_j$, $\xi\lrcorner e^{\lambda\theta_j}=c(\xi,\lambda)\,\omega_j$, and the pairs $(c(\xi,\lambda_j(+1)),c(\xi,\lambda_j(-1)))$ fill $\mathbb{C}^2$. For $k\geq3$, HT_j^k kills $e^{\lambda\theta_j}$, because every element of it contains ω_j or $\partial_{x_{j1}}\wedge\partial_{x_{j2}}$ together with a further factor, and the further factor kills ω_j.

Now count. The block HT_j^1 acts on v by $\xi\mapsto\sum_\varepsilon w_\varepsilon\,y^{\lambda_j(\varepsilon_j)}(\xi)\bigotimes_{j'}e^{\lambda_{j'}(\varepsilon_{j'})\theta_{j'}}$; on the summand $K_j^{\lambda_j(-\sigma)}$ only the ε with $\varepsilon_j=\sigma$ contribute, and they map it isomorphically onto $U_j^{0,1}$ tensored with a nonzero class as soon as one such w_ε is nonzero. Both signs occur in the support of a nonzero real w, since $w_{-\varepsilon}=\overline{w}_\varepsilon$, so the block has rank 4 and $r^1(v)=12$. The block $HT_j^1\otimes HT_{j'}^1$ is the sum of the four summands $K_j^{\lambda_j(-\sigma)}\otimes K_{j'}^{\lambda_{j'}(-\sigma')}$, each of dimension four; the summand indexed by (σ,σ') maps isomorphically onto $U_j^{0,1}\otimes U_{j'}^{0,1}$ tensored with the class $\sum_{\sigma''}w_\varepsilon e^{\lambda_{j''}(\sigma'')\theta_{j''}}$, which is nonzero exactly when the pair (σ,σ') occurs in the support, and to zero otherwise; so the block has rank $4A_{jj'}$. The block HT_j^2 maps ξ to $\sum_\sigma c(\xi,\lambda_j(\sigma))\,z_\sigma$ with $z_\sigma=\omega_j\otimes\sum_\mu w_{\sigma\mu}e_\mu$, where μ runs over the four sign patterns of the other two places and the e_μ are independent; as $(c(\xi,\lambda_j(+1)),c(\xi,\lambda_j(-1)))$ fills $\mathbb{C}^2$, the rank is the dimension of the span of z_+,z_-, which is ρ_j. This gives $r^2(v)$. In degree three the block $HT_1^1\otimes HT_2^1\otimes HT_3^1$ is the sum of eight summands $\bigotimes_j K_j^{\lambda_j(-\varepsilon_j)}$, of dimension eight, on each of which exactly one term of v acts, isomorphically when $w_\varepsilon\neq0$ and by zero otherwise: rank $8N_w$. The block $HT_j^2\otimes HT_{j'}^1$ factors through $\mathbb{C}^2\otimes HT_{j'}^1$ by $c\otimes\mathrm{id}$, which is onto, and on $\mathbb{C}^2\otimes K_{j'}^{\lambda_{j'}(-\sigma')}$ the map is $M_{\sigma'}^{(j;j')}\otimes y^{\lambda_{j'}(\sigma')}$ followed by tensoring with independent classes, of rank $2\,\mathrm{rk}\,M_{\sigma'}^{(j;j')}$; the two values of σ' give independent images. The blocks HT_j^3 contribute nothing. This gives $r^3(v)$. The maximal values are $48+6=54$ and $64+6\cdot8=112$; the conditions for them are the nonvanishing of the w_ε and of the determinants of the $M_{\sigma'}^{(j;j')}$, which are polynomials in the coordinates of v, nonzero on $S(t,q)$. If $F\supseteq K$ and v lies in the secant plane of K, the support of w is a pair $\{\varepsilon_0,-\varepsilon_0\}$, the two CM types of F induced from K; then $A_{jj'}=2$, $\rho_j=2$, $N_w=2$ and every $M_{\sigma'}^{(j;j')}$ has one nonzero entry, so $(r^2,r^3)=(24+6,16+24)$.

(ii) Since $\theta_j^3=0$, $\theta^{2m}=\frac{(2m)!}{2^m}\theta_1^2\cdots\theta_m^2$, and $\int_X\theta^{2m}/(2m)!=1$ gives $\int_X\theta_1^2\cdots\theta_m^2=2^m$ and $\int_X\exp(\sum_j c_j\theta_j)=\prod_j c_j^2$. The dual of $e^{\lambda\theta_j}$ is $e^{-\lambda\theta_j}$, so $\int_X e_\varepsilon^\vee e_{\varepsilon'}=\prod_j(\lambda_j(\varepsilon'_j)-\lambda_j(\varepsilon_j))^2$, which is 0 unless $\varepsilon'=-\varepsilon$, where it is $\prod_j(2s_j)^2=(-4)^m\mathrm{N}(q)$. Hence $\chi(v,v)=(-4)^m\mathrm{N}(q)\sum_\varepsilon w_\varepsilon w_{-\varepsilon}=(-4)^m\mathrm{N}(q)\sum_\varepsilon|w_\varepsilon|^2$.

(iii) As in the proof of Theorem 5.22(i), $r(\kappa)=r(\mathrm{ch}\,E)$ by Lemma 4.55(i), and Φ identifies $r(\mathrm{ch}\,E)$ with the rank of contraction from $HT^2(X\times X)=\bigoplus_{a+b=2}HT^a(X)\otimes HT^b(X)$ into $v_1\boxtimes v_2$; the block $HT^1\otimes HT^1$ contributes $r^1(v_1)r^1(v_2)=144$ and the other two contribute $r^2(v_1)+r^2(v_2)$, by the type argument given there. The tangent space $T_{[A]}\mathcal{D}_F\subseteq H^1(T_A)$, of dimension $mn^2=12$, kills every class that stays of Hodge

type along $\mathcal{D}_F$, as in Proposition 3.11(i), and $\dim HT^2(A) = 66 + 144 + 66$. By the Künneth formula, with $\mathrm{Ext}^{<0} = 0$, $\dim \mathrm{Ext}^2(E,E) = e_0e_2' + e_1e_1' + e_2e_0'$ where $e_k = \dim \mathrm{Ext}^k(F_1,F_1)$ and $e_k' = \dim \mathrm{Ext}^k(F_2,F_2)$; the commutative square $\sigma \circ \mathrm{ev} = (\,\cdot\,)\lrcorner \mathrm{ch}$ of Setup 4.54, valid in every degree, gives $e_k \geq r^k(v_1)$ and $e_k' \geq r^k(v_2)$, and $e_0, e_0' \geq 1$. So $\dim \mathrm{Ext}^2(E,E) \geq r^2(v_2) + 144 + r^2(v_1)$, with equality if and only if $e_0 = e_0' = 1$, $e_1 = e_1' = 12$, $e_2 = r^2(v_1)$ and $e_2' = r^2(v_2)$. Serre duality on the sixfold gives $e_k = e_{6-k}$, so $\chi(F_i,F_i) = 2(e_0 - e_1 + e_2) - e_3 = 2r^2(v_i) - 22 - e_3$, and by Riemann–Roch on an abelian variety $\chi(F_i,F_i) = \chi(v_i,v_i)$, which is (ii); this determines e_3, and $e_3 \geq r^3(v_i)$ is the stated inequality. With the maximal ranks it reads $\chi \leq -(112 - 108 + 22) = -26$, and with $N_w = 2$ it reads $\chi \leq -(40 - 60 + 22) = -2$.

(iv) The proof of Theorem 5.22(iii) applies with three factors in place of two: for $t = 0$ the classes $e_\varepsilon \boxtimes e_\varepsilon^\vee$, the correspondence $Z \mapsto \mathrm{ch}(\Phi(Z))e^{\ell/2}$ and the Poincaré class decompose as tensor products over j of classes of even degree, on each factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with d replaced by $\tau_j(q)$, and the identities of Proposition 3.21 at $n = 2$ show that each image is a multiple of an exponential of η_j or of a Weil class of the j-th factor, invariant under the special unitary group of that factor; $G_F(\mathbb{R})$ is the product of the three, and is Zariski dense in G_F. Statement (iv) is the case $[F : \mathbb{Q}] = 6$ of [Mar25c, Proposition 10.2.1(1)], with the inclusion $\widetilde{\phi}(B \otimes B) \subset A$ of [Mar25c, p. 33], which Markman proves for every CM field in his conventions for Φ and for the twist; we make no claim of priority. ∎

Remark 5.33 (The special role of degree four). In Theorem 5.22 the equality $\dim \mathrm{Ext}^2(E,E) = r(\kappa)$ determines the whole Ext profile of each factor, because on a fourfold Ext^2 is the middle degree and Serre duality supplies the rest; the Euler characteristic of a minimal factor is then the small positive integer $r^2 - 14$, the Euler form on the secant space being positive definite, and the arithmetic of Proposition 5.21 excludes it. On a sixfold the same equality fixes Ext^0, Ext^1 and Ext^2 and leaves the middle group Ext^3 free, and the Euler form has the opposite sign: $\chi(F_i,F_i) = -64\mathrm{N}(q)\sum |w_\varepsilon|^2$ is a negative integer that e_3 absorbs. So no dimension count excludes a minimal Orlov product over a sextic field, and none does over any CM field of degree $2m \geq 6$, where $\mathrm{Ext}^3, \ldots, \mathrm{Ext}^m$ are free and the Euler form has sign $(-1)^m$. What Proposition 5.32(iii) leaves for such an object is the profile displayed there, with $\chi(F_i,F_i) \leq -26$ for a general secant character, and the flat twisted character of (iv), which stays of Hodge type over the whole twelve-dimensional period domain. Whether a sheaf with that profile exists on a sixfold with real multiplication by a cubic field is not known; nothing in Theorem 5.22 bears on it. A certificate of the kind of Theorem 3.19 for a sextic field would be an Orlov product of such sheaves, with $\dim \mathrm{Ext}^2(E,E) = r(\kappa) \leq 252$ on the twelvefold, whose twisted character has a nonzero component in the Weil classes $W_F(A)$; flatness does not give that component, and Lemma 5.35 computes it. In degree six as in degree four, the Orlov products never reach the rank of a class of general shape: $r(\kappa) \leq 252$ against 264, and $r(\kappa) \leq 104$ against 112.

The count of Proposition 5.32(iii) is a condition on the Euler characteristic of the character alone, so it can be tested on the lattice of integral characters, as Proposition 5.21 tests the quartic one. In degree four the test excludes every minimal character. In degree six it can exclude an integral character only if $-26 < \chi(v,v) < 0$ (Proposition 5.34(i),(ii)), and it excludes neither of the integral classes $\mathrm{Re}\, e^{\sqrt{-1}\,\theta}$ and $\mathrm{Im}\, e^{\sqrt{-1}\,\theta}$ (Proposition 5.34(iv)). For $t = 0$, the case in which Proposition 5.32(iv) makes κ flat, put $B_j = 1 - \frac{1}{2}\tau_j(q)\,\theta_j^2$, so

that $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, and for $c_0, c_3 \in \mathbb{Q}$ and $f, g \in F_0$ let

$$v(c_0, f, g, c_3) = c_0 B_1B_2B_3 + \sum_j \tau_j(f)\,\theta_j B_k B_l + \sum_j \tau_j(g)\,B_j\theta_k\theta_l + c_3\,\theta_1\theta_2\theta_3,$$

where $\{j,k,l\} = \{1,2,3\}$ in each sum.

Proposition 5.34 (Integral flat characters in degree six)**.** *Keep the notation of Proposition 5.32.*

(i) *For every perfect complex F on a complex torus of dimension six, $\chi(F,F)$ is even. So the condition of Proposition 5.32(iii) holds for every F whose character has $N_w = 2$.*

(ii) *For every nonzero real $v \in S(t,q)$, $r^3(v) - 2r^2(v) \le 4$, with equality only if $(r^2(v), r^3(v)) = (54, 112)$. So for every shape the condition of Proposition 5.32(iii) is implied by $\chi(F_i, F_i) \le -26$.*

(iii) *The rational points of $S(0,q)$ are the classes $v(c_0,f,g,c_3)$, and*

$$\chi(v,v) = -8\big(\mathrm{N}(q)\,c_0^2 + \mathrm{Tr}(\mathrm{N}(q)\,q^{-1}f^2) + \mathrm{Tr}(q\,g^2) + c_3^2\big).$$

(iv) *For $q = 1$, the classes $\mathrm{Re}\,e^{\sqrt{-1}\,\theta} = v(1,0,-1,0)$ and $\mathrm{Im}\,e^{\sqrt{-1}\,\theta} = v(0,1,0,-1)$ are integral points of $S(0,1)$ with $\chi = -32$ and $N_w = 2$, so they satisfy the condition of Proposition 5.32(iii).*

Proof. (i) The Chern character $v = \mathrm{ch}(F)$ is integral, as in the proof of Theorem 5.22(ii), and $\chi(F,F) = \int_X v^\vee v$ by Riemann–Roch. Write $v = \sum_k v_{2k}$, so that $\int_X v^\vee v = \sum_k (-1)^k \int_X v_{2k} v_{12-2k}$. The terms k and $6-k$ are equal, since $(-1)^k = (-1)^{6-k}$ and classes of even degree commute, so they contribute an even number when $k \ne 3$. For $k = 3$ write $v_6 = \sum_I a_I e_I$ in the basis of $\bigwedge^6 H^1(X,\mathbb{Z})$ given by the products e_I of six elements of a basis of $H^1(X,\mathbb{Z})$; since $e_Ie_I = 0$ and $e_Ie_J = e_Je_I$, $v_6^2 = 2\sum_{I<J} a_Ia_Je_Ie_J$ is even. For $N_w = 2$ the condition of Proposition 5.32(iii) is $\chi(F,F) \le -2$, and $\chi(F,F)$ is a negative even integer by Proposition 5.32(ii).

(ii) For j and $\sigma = \pm 1$ let $W_{j,\sigma}$ be the 2×2 slice $(w_\varepsilon)_{\varepsilon_j = \sigma}$, with rows and columns indexed by the signs at the two other places. The matrices $M_\sigma^{(j';j)}$ are $W_{j,\sigma}$ and its transpose, so $\sum_{j\ne j'}(\mathrm{rk}\,M_+^{(j;j')} + \mathrm{rk}\,M_-^{(j;j')}) = 2\sum_{j,\sigma}\mathrm{rk}\,W_{j,\sigma}$. Since $w_{-\varepsilon} = \overline{w}_\varepsilon$, $W_{j,-}$ is the complex conjugate of $W_{j,+}$ with its rows and its columns reversed, so the two slices have the same rank k_j. Let S be the support of w and $N = N_w = |S|$. By Proposition 5.32(i),

$$r^3(v) - 2r^2(v) = 8N - 8\,(A_{12} + A_{13} + A_{23}) + \sum_j (8k_j - 2\rho_j). \tag{32}$$

The set S is stable under $\varepsilon \mapsto -\varepsilon$, which exchanges the two slices along each place; so N is even, each slice has $N/2$ nonzero entries, $k_j \ge 1$, and $A_{jj'} \in \{2,4\}$, the projection of S to the signs at j and j' being stable under negation. Also $1 \le \rho_j \le 2$ and $k_j \le 2$, and a slice whose nonzero entries lie in one row or one column has rank one. If $\rho_j = 1$, then $w_\varepsilon = a_{\varepsilon_j} W(\varepsilon')$, with ε' the signs at the two other places, so every slice along a place $j' \ne j$ is $a \otimes$ (a row of W) and $k_{j'} \le 1$. Now take the four values of N in turn.

If $N = 8$, every $A_{jj'}$ is 4 and (32) is $-32 + \sum_j (8k_j - 2\rho_j)$. When every ρ_j is 2 this is at most $-32 + 3\cdot 12 = 4$, with equality exactly when every k_j is 2, and then $r^2 = 48 + 6 = 54$ and $r^3 = 64 + 48 = 112$. When some ρ_j is 1, the two other k are at most 1 and (32) is at most $-32 + 14 + 6 + 6 = -6$.

If $N = 6$, S misses one pair $\{\pm\varepsilon_0\}$. As ε_0 and $-\varepsilon_0$ differ at every place, they never have the same signs at two places, so every sign pair at two places occurs in S, every $A_{jj'}$ is 4, and (32) is at most $48 - 96 + 3\cdot 14 = -6$.

If $N=4$, $S=\{\pm\varepsilon,\pm\varepsilon'\}$, and after replacing ε' by $-\varepsilon'$ if necessary, ε and ε' differ at exactly one place j. For the two other places k and l one has $A_{kl}=2$ and $A_{jk}=A_{jl}=4$. The two nonzero entries of a slice along k are $w_{\pm\varepsilon}$ and $w_{\pm\varepsilon'}$ with the same sign, whose indices agree at l; they lie in one row or one column, so $k_k=1$, and likewise $k_l=1$. Hence (32) is at most $32-80+8\,(2+1+1)-6=-22$.

If $N=2$, $S=\{\pm\varepsilon_0\}$, every $A_{jj'}$ is 2, every k_j is 1, and every ρ_j is 2, the two slices along j having their nonzero entries at different positions; so (32) is $16-48+24-12=-20$, the value $40-60$ of Proposition 5.32(i).

So $r^3-2r^2\le 4$, with equality only at $(54,112)$, and the threshold r^3-2r^2+22 of Proposition 5.32(iii) is at most 26.

(iii) Since $\theta_j^3=0$ and $s_j^2=-\tau_j(q)$, $e^{\pm s_j\theta_j}=B_j\pm s_j\theta_j$, so the eight exponentials and the eight products $\prod_j\beta_j$ with $\beta_j\in\{B_j,\theta_j\}$ span the same complex space, the change of basis being the tensor cube of an invertible 2×2 matrix. The Galois group permutes the τ_j, the θ_j and the B_j compatibly, so $\sum_a C_a\prod_j\beta_j$, indexed by the subsets a of $\{1,2,3\}$ with $\beta_j=\theta_j$ for $j\in a$, is rational exactly when $C_{\pi(a)}=\sigma(C_a)$ for every σ inducing π; the stabiliser of $\{j\}$, and of its complement, is the stabiliser of τ_j, which gives the form $v(c_0,f,g,c_3)$. The involution $v\mapsto v^\vee$ and the integral are multiplicative over the three factors, with $\int_X\theta_1^2\theta_2^2\theta_3^2=8$; on one factor $B_j^\vee B_j=1-\tau_j(q)\theta_j^2$, $\theta_j^\vee\theta_j=-\theta_j^2$ and the mixed products are odd in θ_j. So $\chi(v,v)=-8\sum_a C_a^2\prod_{j\notin a}\tau_j(q)$, which is the formula.

(iv) For $q=1$, $s_j=\sqrt{-1}$ and $e_{(1,1,1)}=\prod_j e^{\sqrt{-1}\,\theta_j}=e^{\sqrt{-1}\,\theta}$. The polarisation is principal, so each $\theta^k/k!$ is integral, and so are the real and imaginary parts $\sum_k(-1)^k\theta^{2k}/(2k)!$ and $\sum_k(-1)^k\theta^{2k+1}/(2k+1)!$. They are $\frac12(e_{(1,1,1)}+e_{(-1,-1,-1)})$ and $\frac{1}{2\sqrt{-1}}(e_{(1,1,1)}-e_{(-1,-1,-1)})$, so $N_w=2$ and $\chi=-64\,(\frac14+\frac14)=-32$ by Proposition 5.32(ii); expanding $\prod_j(B_j+\sqrt{-1}\,\theta_j)$ gives the coordinates. ∎

The flatness of Proposition 5.32(iv) makes κ a Hodge class on every member of the family, but in degree four the Hodge classes of a very general member are the products of divisor classes and the Weil classes $W_F(A)$ ([I, Proposition 17.3(iii)]), and only a nonzero component of κ in $W_F(A)$ bears on $\mathbf{W}(F,2,1)$. We compute that component. Keep the notation of Proposition 5.32, with $t_1=t_2=0$, $q_1=q_2=q$ and trivial B-field, put $d_j=\tau_j(q)$, and let $\Psi(Z)=\mathrm{ch}(\Phi(Z))e^{\ell/2}$ as in Proposition 3.21, so that $\kappa(v_1,v_2)=\Psi(v_1\boxtimes v_2^\vee)$ for the characters v_i of F_i. As in the proof of Proposition 5.32(iv), $H^1(A,\mathbb{C})$ is the sum of three factors $U_j\oplus U_j^\vee$, on each of which $\sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta=\theta_j$ and $d=d_j$. On the j-th factor write $\widehat{\theta}_j$ and ℓ_j for the components of the dual polarisation and of the Poincaré class, and put $\gamma_j=d_j\theta_j-\widehat{\theta}_j$, as in [I, (33)], and $\eta_j=d_j\theta_j+\widehat{\theta}_j$, a positive multiple of the polarisation of the factor for which $\sqrt{-q}$ is anti-self-adjoint ([I, Proposition 15.19(i)]). For $\sigma=\pm1$ let $\tau_{j,\sigma}$ be the embedding of F over τ_j with $\tau_{j,\sigma}(\sqrt{-q})=\sigma\sqrt{-1}\,\sqrt{\tau_j(q)}=\sigma s_j$, and $V_{j,\sigma}\subset H^1(A,\mathbb{C})$ its eigenspace, of dimension four. By [I, Theorem 15.2(ii)] with $n=2$ and $d=d_j$, the line $\bigwedge^4 V_{j,\sigma}$ is spanned by

$$\omega_{j,\sigma}=-\frac{1}{2d_j^2}\,(\gamma_j-\sigma s_j\ell_j)^2, \tag{33}$$

so the six classes $\omega_{j,\sigma}$ span $W_F(A)\otimes\mathbb{C}$. Since η_j pairs $V_{j,+}$ with $V_{j,-}$, the products $\eta_i\eta_k$ lie in summands of $H^4(A,\mathbb{C})=\bigwedge^4(\bigoplus_{j,\sigma}V_{j,\sigma})$ other than the six lines $\bigwedge^4 V_{j,\sigma}$. Let $\kappa_{j,\sigma}(v_1,v_2)$ be the coefficient of $\omega_{j,\sigma}$ in the component of $\kappa(v_1,v_2)$ in $\bigwedge^4 V_{j,\sigma}$; the *F-Weil part* of κ is

$\sum_{j,\sigma}\kappa_{j,\sigma}\,\omega_{j,\sigma}$, its component in $W_F(A)\otimes\mathbb{C}$. For $\varepsilon\in\{\pm1\}^3$ let $\varepsilon^{(j)}$ be the sign vector that agrees with ε at j and only there.

Lemma 5.35 (The F-Weil part of an Orlov product)**.** *Let $v_i=\sum_\varepsilon w_\varepsilon^i e_\varepsilon\in S(0,q)$ for $i=1,2$.*

(i) *The degree-four part of $\kappa(v_1,v_2)$ is a quadratic form in the η_j plus $\sum_{j,\sigma}\kappa_{j,\sigma}\,\omega_{j,\sigma}$, with*

$$\kappa_{j,\sigma}(v_1,v_2)=-16\mathrm{N}(q)\,\tau_j(q)\sum_{\varepsilon_j=\sigma}w_\varepsilon^1\,w^2_{\varepsilon^{(j)}}.$$

(ii) *Let $v=v(c_0,f,g,c_3)$ be rational, and for $\{j,k,l\}=\{1,2,3\}$ write $f_j=\tau_j(f)$, $g_j=\tau_j(g)$ and $q_j=\tau_j(q)$. The numbers*

$$R_j=c_0^2+\frac{f_k^2}{q_k}+\frac{f_l^2}{q_l}+\frac{g_j^2}{q_kq_l}-\frac{1}{q_j}\Big(f_j^2+\frac{g_l^2}{q_k}+\frac{g_k^2}{q_l}+\frac{c_3^2}{q_kq_l}\Big),$$
$$I_j=c_0f_j+\frac{f_kg_l}{q_k}+\frac{f_lg_k}{q_l}+\frac{c_3g_j}{q_kq_l}$$

are $\tau_j(R)$ and $\tau_j(I)$ for elements $R,I\in F_0$. Put $\Omega(v)=R+2I/\sqrt{-q}\in F$. Then

$$\kappa_{j,\sigma}(v,v)=-\mathrm{N}(q)\,\tau_j(q)\,\tau_{j,\sigma}\big(\Omega(v)\big).$$

So the F-Weil part of $\kappa(v,v)$ is zero if $R=I=0$, and otherwise all six $\kappa_{j,\sigma}(v,v)$ are nonzero.

(iii) *If v_1 and v_2 are supported on one pair $\{\varepsilon_0,-\varepsilon_0\}$, the F-Weil part of $\kappa(v_1,v_2)$ is zero. For $q=1$, F contains $K=\mathbb{Q}(\sqrt{-1})$, the classes $\mathrm{Re}\,e^{\sqrt{-1}\,\theta}$ and $\mathrm{Im}\,e^{\sqrt{-1}\,\theta}$ span the plane P_θ of* [I, Definition 14.4] *with $d=1$, and for $v_i=a_i\,\mathrm{Re}\,e^{\sqrt{-1}\,\theta}+b_i\,\mathrm{Im}\,e^{\sqrt{-1}\,\theta}$ with $a_i,b_i\in\mathbb{Q}$ the class $\kappa(v_1,v_2)$ lies in $\mathbb{Q}[\eta]\oplus W_K(A)$, where $\eta=\theta+\widehat{\theta}$ and $W_K(A)\subseteq H^{12}(A,\mathbb{Q})$ are the Weil classes of K on the twelvefold (*[I, Proposition 17.10]*); its part of degree at most four is*

$$-32(a_1a_2+b_1b_2)+16(a_1b_2-a_2b_1)\,\eta+4(a_1a_2+b_1b_2)\,\eta^2.$$

Proof. (i) The classes $e_\varepsilon\boxtimes e^\vee_{\varepsilon'}=\bigotimes_j(e^{\varepsilon_j s_j\theta_j}\boxtimes e^{-\varepsilon'_j s_j\theta_j})$ and Ψ decompose over the three factors, as in the proof of Proposition 5.32(iv), and on the j-th factor the two identities of Proposition 3.21 at $n=2$ read, with their constants,

$$\Psi_j(e^{\sigma s_j\theta_j}\boxtimes e^{\sigma s_j\theta_j})=-4d_j\,e^{\sigma(s_j/2d_j)\eta_j},\qquad \Psi_j(e^{\sigma s_j\theta_j}\boxtimes e^{-\sigma s_j\theta_j})=\tfrac12(\gamma_j-\sigma s_j\ell_j)^2=-d_j^2\,\omega_{j,\sigma},$$

since $s_j^2=-d_j$. So $\Psi(e_\varepsilon\boxtimes e^\vee_{\varepsilon'})$ is the product over j of the first class where $\varepsilon'_j=-\varepsilon_j$ and of the second where $\varepsilon'_j=\varepsilon_j$. In degree four a product with two or three factors of the second kind vanishes, one with none is a quadratic form in the η_j, and one with exactly one, at j, has $\varepsilon'=\varepsilon^{(j)}$ and equals $-d_j^2\prod_{i\neq j}(-4d_i)\,\omega_{j,\varepsilon_j}=-16\mathrm{N}(q)\,d_j\,\omega_{j,\varepsilon_j}$. Summing $w^1_\varepsilon w^2_{\varepsilon'}\Psi(e_\varepsilon\boxtimes e^\vee_{\varepsilon'})$ over ε and ε' gives (i).

(ii) An element h of the Galois group that fixes the place j permutes the two others, so it fixes R_j and I_j, which are symmetric in k and l; hence $R_j,I_j\in\tau_j(F_0)$, and moving j by the Galois group shows that they are the conjugates of two elements of F_0. Write $\varepsilon=(\sigma,\mu)$ with μ the signs at k and l, so that $\varepsilon^{(j)}=(\sigma,-\mu)$. Since $e^{\pm s_i\theta_i}=B_i\pm s_i\theta_i$, the class $v=\sum_a C_a\prod_{i\in a}\theta_i\prod_{i\notin a}B_i$, with $C_\emptyset=c_0$, $C_{\{i\}}=f_i$, $C_{\{1,2,3\}\setminus\{i\}}=g_i$ and $C_{\{1,2,3\}}=c_3$, has $w_\varepsilon=\frac18\sum_a C_a\prod_{i\in a}\varepsilon_i/s_i$. Grouping $a=b$ and $a=b\cup\{j\}$ for $b\subseteq\{k,l\}$ gives $w_{(\sigma,\mu)}=\frac18\sum_b D_b\prod_{i\in b}\mu_i/s_i$ with $D_b=C_b+u\,C_{b\cup\{j\}}$ and $u=\sigma/s_j=\tau_{j,\sigma}(1/\sqrt{-q})$. Since

$\sum_\mu \prod_{i\in b}\mu_i \prod_{i\in b'}(-\mu_i) = 4(-1)^{|b|}\delta_{bb'}$ and $s_i^2 = -q_i$,

$$\sum_{\varepsilon_j=\sigma} w_\varepsilon w_{\varepsilon^{(j)}} = \frac{1}{16}\Big((c_0+uf_j)^2 + \frac{(f_k+ug_l)^2}{q_k} + \frac{(f_l+ug_k)^2}{q_l} + \frac{(g_j+uc_3)^2}{q_kq_l}\Big),$$

and expanding with $u^2 = -1/q_j$ gives $\frac{1}{16}(R_j + 2uI_j) = \frac{1}{16}\tau_{j,\sigma}(\Omega(v))$. With (i) this is the formula. The embedding $\tau_{j,\sigma}$ is injective and 1, $1/\sqrt{-q}$ are independent over F_0, so one $\kappa_{j,\sigma}(v,v)$ vanishes if and only if $R = I = 0$, and then all do.

(iii) If w^1 and w^2 vanish off $\{\varepsilon_0, -\varepsilon_0\}$, no term of (i) survives, since $\varepsilon^{(j)}$ is neither ε nor $-\varepsilon$. For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = e^{\sqrt{-1}\,\theta}$, whose real and imaginary parts are u_θ and v_θ of [I, (30)] with $d = 1$, supported on $\{\pm(1,1,1)\}$. On each factor $\sqrt{-1} = \sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta = \theta_j$ and $d = 1$, so K acts on A by the action [I, (29)] determined by θ with $d = 1$, and Proposition 3.21 with $n = 6$ and $d = 1$ gives $\kappa \in \mathbb{Q}[\eta] \oplus W_K(A)$. For the constants write $v_1 = w_1 e^{\sqrt{-1}\,\theta} + \overline{w}_1 e^{-\sqrt{-1}\,\theta}$ and $v_2^\vee = a_2 \operatorname{Re} e^{\sqrt{-1}\,\theta} - b_2 \operatorname{Im} e^{\sqrt{-1}\,\theta} = w_2 e^{\sqrt{-1}\,\theta} + \overline{w}_2 e^{-\sqrt{-1}\,\theta}$, with $w_1 = \frac{1}{2}(a_1 - \sqrt{-1}\,b_1)$ and $w_2 = \frac{1}{2}(a_2 + \sqrt{-1}\,b_2)$. By the first identity in the proof of (i), $\Psi(e^{\pm\sqrt{-1}\,\theta} \boxtimes e^{\pm\sqrt{-1}\,\theta}) = -64\, e^{\pm(\sqrt{-1}/2)\eta}$, while the two mixed terms have degree twelve; so in degrees below twelve κ is $-128 \operatorname{Re}\big(w_1 w_2\, e^{(\sqrt{-1}/2)\eta}\big)$, whose terms of degree 0, 2 and 4 are the stated ones. ∎

Markman states his reduction for the normalised class $\operatorname{ch}(E)\exp(-c_1(E)/\operatorname{rk} E)$, and it is the component of degree four of that class, a polynomial in the invariant divisor classes plus a Weil class, whose algebraicity on the deformations he needs [Mar25c, Proposition 10.2.1, Corollary 10.2.3]. When $\operatorname{rk} E \neq 0$ that class is $\kappa \exp(-\kappa_1/\operatorname{rk} E)$, with κ_1 the degree-two part of κ, which is a linear form in the η_j by the proof of Lemma 5.35(i); so it has the same F-Weil part as κ.

For $q = 1$ the Orlov products of complexes with characters in $\mathbb{Q} \operatorname{Re} e^{\sqrt{-1}\,\theta} \oplus \mathbb{Q} \operatorname{Im} e^{\sqrt{-1}\,\theta}$ are therefore the template of Proposition 3.22 at $(n, d) = (6, 1)$, seen on the twelvefold through the restriction of the action of F to K. Their twisted characters are flat for the family of K and have a nonzero K-Weil part in H^{12} when $F_1 = F_2 \neq 0$ (Proposition 3.21), and they are not an input to $\mathrm{P2}_{\mathrm{split}}$. By [I, Proposition 17.10], $\mathbf{W}(F, 2, 1)$ gives the K-Weil classes on the members of the F-family, and the converse is not available; a consequence for K through [III, Proposition 2.13] would still need a twisted form of the propagation of Theorem 2.3.

Remark 5.36 (Natural objects). Let X be very general with real multiplication by $\mathcal{O}_{F_0}$, so that $\mathrm{NS}(X)_\mathbb{Q} = \{\theta_t = \sum_j \tau_j(t)\theta_j : t \in F_0\}$. Call *natural* the line bundles, the simple semi-homogeneous bundles of [Muk78], the skyscraper sheaves of points, and the objects obtained from these by shifts, duals, twists by line bundles, pull-back by automorphisms commuting with $\mathcal{O}_{F_0}$, and $\Phi_\mathcal{P}$ followed by $\hat{X} \cong X$. On the j-th factor write classes in $\langle 1, \theta_j, \theta_j^2/2\rangle$ as (r, c, s), and put $\nu(t) = e^{t\theta_j} = (1, t, t^2)$ and $\nu(\infty) = (0, 0, 1)$. The character of a natural object is a multiple of $\bigotimes_j \nu(\tau_j(t))$ for one $t \in F_0 \cup \{\infty\}$, its slope: a simple semi-homogeneous bundle has character $r\, e^{\theta_t}$ [Muk78], a point has slope ∞, and the operations act on slopes by $t \mapsto -t$, $t + t'$, tu^2 with $u \in \mathcal{O}_{F_0}^\times$, and $-1/t$, the last by [I, Theorem 6.5] applied factor by factor. Then

(a) no natural character lies in $S(0, q)$, for any totally positive q;

(b) if a real combination of natural characters of k pairwise distinct slopes is a nonzero element of $S(0, q)$, then $k \geq 6$, and if $k = 6$ it has $N_w = 2$.

So no cone of a morphism between two natural objects has a nonzero character in $S(0,q)$, and by Lemma 5.35(iii) a class whose Orlov square has a nonzero F-Weil part needs at least seven slopes. Six are attained: $\operatorname{Re} e^{\sqrt{-1}\,\theta} = \frac{5}{2} - \frac{5}{6}(e^{\theta}+e^{-\theta}) + \frac{1}{12}(e^{2\theta}+e^{-2\theta}) - 10$[pt].

For (a): $S(0,q)$ is the tensor product of the planes $P_j = \langle B_j, \theta_j\rangle = \{s = -d_j r\}$, the form $c^2 - rs$, for which $\nu(t)$ and $\nu(\infty)$ are isotropic, is positive definite on P_j, and a decomposable tensor lies in $\bigotimes_j P_j$ only if each factor does. For (b), let $T = \sum_i a_i \bigotimes_j \nu(\tau_j(t_i))$ and $\lambda_1 = s + d_1 r$, which is positive on $\nu(t)$ and $\nu(\infty)$. Applying λ_1 to the first factor gives a relation with nonzero coefficients among the images, under the embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ by forms of bidegree $(2,2)$, of the k points $(\tau_2(t_i), \tau_3(t_i))$, no two of which share a coordinate since each τ_j is injective. Any five such points are independent, since the product of the two vertical and the two horizontal lines through four of them misses the fifth; so $k \geq 6$. If $k = 6$, a form of bidegree $(2,2)$ vanishing at five of the points vanishes at the sixth; applied to CD, with C the curve of bidegree $(1,1)$ through three of the points and D running over the pencil of such curves through two others, whose base locus is those two points, this puts the sixth point on C. So the six points lie on the graph of a real Möbius transformation, and likewise for the other pairs of factors. Moving the second and third factors by these transformations, which act through GL_2 on $\langle 1, \theta_j, \theta_j^2/2\rangle$, turns T into a binary sextic, embedded in the tensor cube by $l^6 \mapsto (l^2)^{\otimes 3}$, that is annihilated by three definite binary quadrics, the transforms of the λ_j. Two non-proportional definite binary quadrics generate every binary form of degree at least three, and so would annihilate the sextic; as $T \neq 0$, the three quadrics are proportional to one quadric Λ, and the sextic lies in the kernel of contraction with Λ, spanned by l_+^6 and l_-^6 with $l_\pm$ the isotropic lines of Λ. Moving back, T is a combination of two conjugate decomposable tensors in $S(0,q)$, which are e_ε and $e_{-\varepsilon}$ for one ε; so $N_w = 2$.

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Ethics approval and consent. This article reports no research involving human participants, human data or animals, so ethics approval and consent to participate or to publish were not required. The results are original and have not been published elsewhere except in the earlier paper named on the title page, which this paper and its companions replace, and the work of others is credited by citation throughout.

Data availability. The archive [BMB26] is deposited on Zenodo under the DOI `10.5281/zenodo.23225483`. It holds programs that repeat much of the finite arithmetic of this paper and its companions [I, III] as an independent check, a Lean 4 certificate of part of that arithmetic, the Macaulay2 computations of local Ext algebras with their output, and the generators and TikZ sources of the figures; its programs perform 2218

checks, all of which pass, and the Lean kernel checks four hundred and nine theorems, each depending on no axiom or on propositional extensionality alone. Deep Bhattacharjee computed all the calculations, in C, Python, Julia, Macaulay2, Lean 4 and shell scripts, with the assistance of Claude Code, an AI model developed by Anthropic. The archive is immutable at that DOI; the command `python3 code/verify_all.py` in it runs every check and ends with their count, and `lean HodgeObstruction.lean` in its folder `lean` checks the certificate and prints the axioms of each theorem. The Lean theorems certify finite arithmetic only: rank certificates, enumerations, lattice computations and polynomial identities; no geometric argument of the three papers is formalised, and every proof in the text was written and checked by the authors. The authors are responsible for the mathematical statements and their proofs. The LaTeX source of this paper is archived on Zenodo under the DOI 10.5281/zenodo.23885236. No other data were generated or analysed.

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